# HUAWEI USG6000E Series Hardware Guide

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# 1 Info-Finder

**Info-Finder** is a tool platform. It allows you to search for key product information by product series and model. The key product information includes basic information such as the software specifications, life cycles, and hardware information, and operation and maintenance information such as the licenses, alarms, logs, commands, and MIBs. The hardware-related tools are as follows:

- Product image gallery: provides product photos, Visio-format templates, and network element icons for you to produce design drawings and networking diagrams.
- Hardware configuration: automatically generates hardware configuration diagrams after you select components are required and calculates the weight, power consumption, and heat consumption.
- Hardware center: provides the technical specifications of devices and components, as well as the mapping between devices, components, and versions.
- 3D model: Using this function, you can query product images, product overview, and component insertion/removal videos, enabling you to quickly obtain product information in one-stop mode.

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# 2 Software Versions Compatible with Hardware

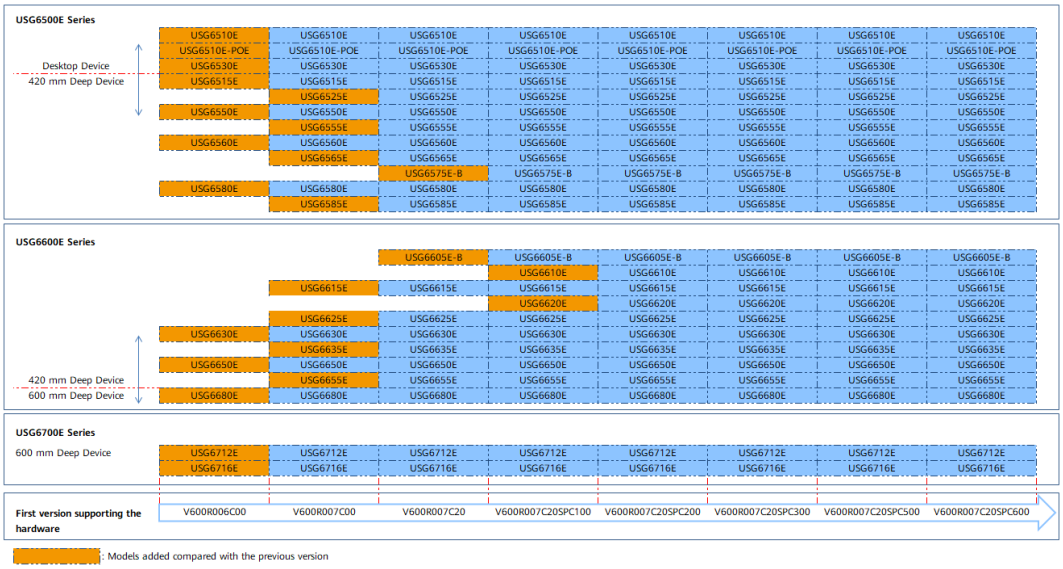
This section describes software versions compatible with the USG hardware.

The USG has its software versions constantly updated and optimized ever since its launch. **Figure 2-1** displays software versions compatible with the USG and version evolution.

NOTE

- As documentation and software versions are evolving, the *Hardware Guide* may describe a collection of all hardware of a product series, which does not mean that all hardware in this document is supported in this version. Before using hardware, carefully check the **First Supported Version** of the hardware in *Hardware Guide*.
- The *Hardware Guide* describes all the device models supported in a version, and these models may not be available for sales currently. To obtain accurate subscription information, regularly check the official **http://e.huawei.com** for product change notices (PCNs) and lifecycle management bulletins, or you can use the Info-Finder tool to query product EOX information.

**Figure 2-1** Software versions compatible with hardware



# 3 Hardware Overview

This chapter describes hardware information for all USG6000E series and main components.

3.1 Chassis

3.2 Power Modules

3.3 Fan Modules

3.4 Storage Devices

3.5 Optical/Electrical Modules

3.6 Cables

## 3.1 Chassis

This chapter describes hardware information for all USG6000E series, including hardware appearance, front and real panels, power supply system, heat dissipation system, and technical specifications.

NOTE

- The model shown on the panel of the USG is the series number USG6000E. For the specific model, see the nameplate attached on the bottom of the USG.
- The device series number can be provided using laser marking on the panel or film silkscreen. The device functions are the same regardless of the series number presentation mode.

### 3.1.1 USG6510E

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## Overview

**Table 3-1** Basic information about the USG6510E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6510E AC Host (10*GE RJ45+2*GE SFP,1*Adapter) | 02352PQH | USG6510E-AC | V600R007C00 |
| USG6510E AC Host (10*GE RJ45+2*GE SFP,1*Adapter) | 02352PQH-001 | USG6510E-AC | V600R007C20SPC300 |
| USG6510E AC Host(10*GE RJ45+2*GE SFP,1*Adapter) | 02352PQH-002 | USG6510E-AC | V600R007C20SPC600 |

## Appearance

**NOTE**

The standard USB port of the earlier-delivered models is USB 3.0, and that of the later-delivered models is USB 2.0. This document uses a model with a USB 3.0 port as an example.

**Figure 3-1** Appearance of the USG6510E (front view)

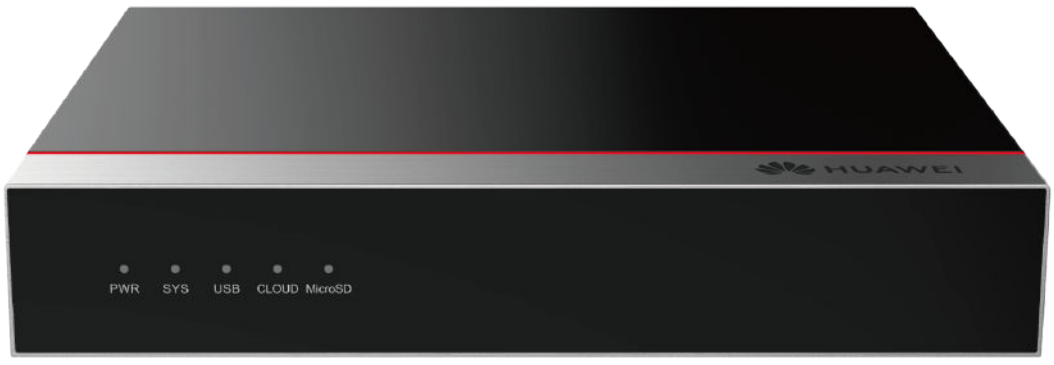
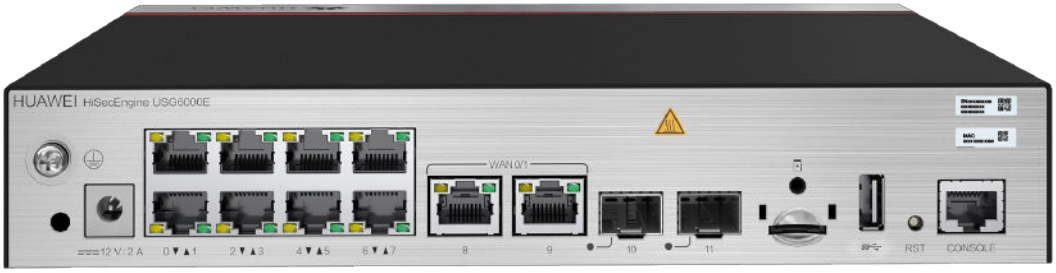


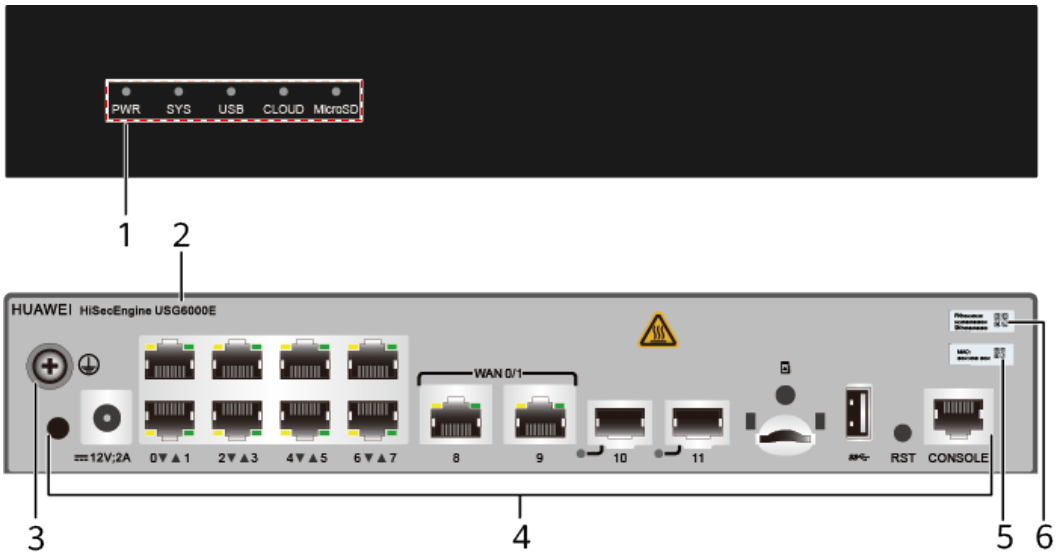

**Figure 3-2** Appearance of the USG6510E (rear view)




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## Components

**Figure 3-3** Components of the USG6510E




| | | | | |
|---|---|---|---|---|
| 1. Indicator area | 2. Product series ID | 3. Protective ground terminal | 4. Fixed interface area | 5. MAC label |
| 6. SN label | - | - | - | - |

**Table 3-2** Component functions

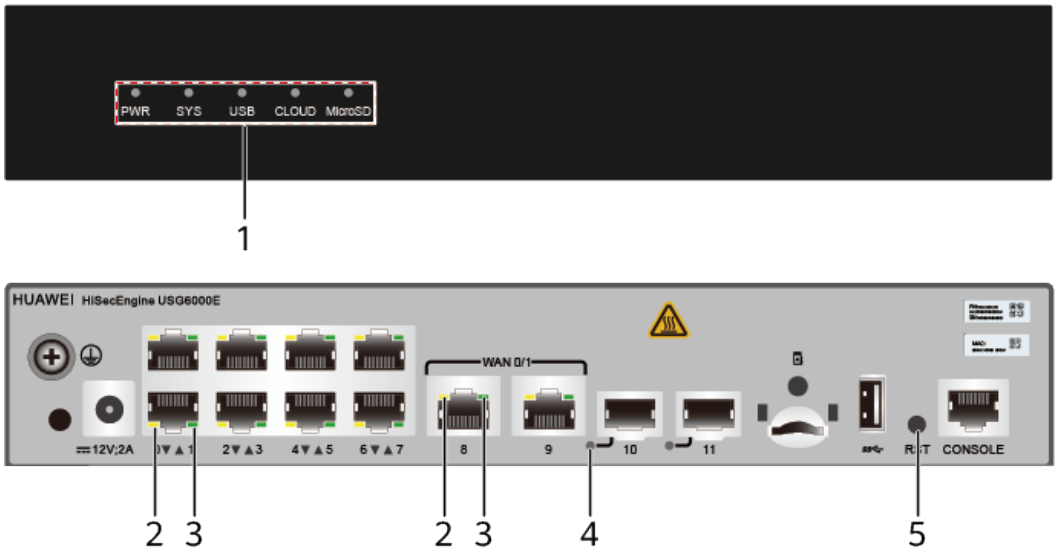
| Name | Description |
|---|---|
| Indicator area | Provides various indicators to display the device running status in real time. |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| Fixed interface area | • Provides clip hole and power receptacle.<br>• Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance.<br>• Houses a Micro SD card to store logs and reports. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |

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| Name | Description |
| --- | --- |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |

## Indicators and Buttons

**Figure 3-4** Indicators and buttons on the USG6510E




| 1. PWR, SYS, USB, CLOUD, and MicroSD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port indicator | 5. RST button |
| --- | --- | --- | --- | --- |

**Table 3-3** Indicators on the USG6510E

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | A system fault occurs. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The device is not connected to the cloud management platform. |
| MicroSD | Micro SD indicator | Green | Steady on | The micro SD card is present. |
| | | - | Off | The micro SD card is not detected. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Optical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

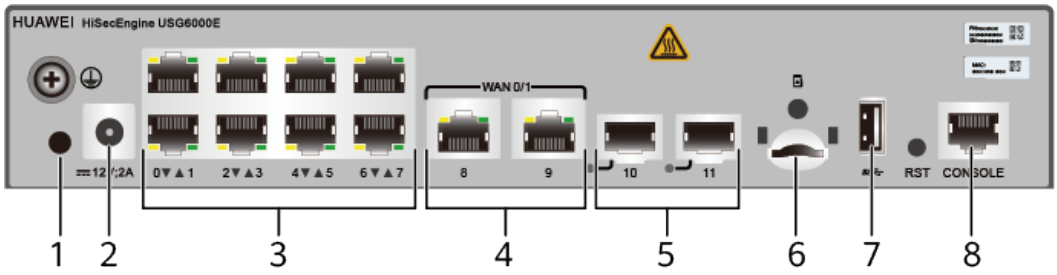
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**Table 3-4** Buttons on the USG6510E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-5** Ports on the USG6510E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. Clip hole | 2. Power receptacle | 3. GE electrical port | 4. WAN port | 5. GE optical port |

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| | | | | |
|---|---|---|---|---|
| 6. Micro SD card slot | 7. USB port<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB 2.0 | 8. Console port | - | - |

**Table 3-5** Ports on the USG6510E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. | Power cable clip |
| Power receptacle | - | The receptacle connects to the tuning fork plug of the power adapter. | • **36 W Power Adapter (02220947)**<br>• **36 W Power Adapter (02221468)** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 7) | RJ45 | 8 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7.<br>GigabitEthernet 0/0/0 is an inband management port and its default IP address is 192.168.0.1. After this port is connected to your PC through Ethernet cables, you can log in to the device using Telnet, STelnet, or web UI to configure or manage the device.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| WAN ports (8 to 9) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/8 to GigabitEthernet 0/0/9.<br>The device is connected to the WAN or cloud management platform using an Ethernet cable. | **Ethernet Cable** |
| GE optical port (10 to 11) | SFP | 2 GE Ethernet optical ports, numbered from GigabitEthernet 0/0/10 to GigabitEthernet 0/0/11. Compatible with GE Ethernet optical ports. | • **100Mbps SFP Optical Modules**<br>• **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, GE optical ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Micro SD card slot | Micro SD | The micro SD card slot allows you to insert a micro SD card to record logs and reports in real time. The micro SD card is optional. You can purchase one from Huawei if needed.<br>You are advised to install an anti-theft board delivered with the device to protect the micro SD card. | SDSDQAE-064G (BOM code: 06010308, capacity: 64GB, dimensions (H x W x D): 1 mm x 15.00 mm x 11.00 mm/0.04 in. x 0.59 in. x 0.43 in) |
| USB port | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

## Power Supply System

- Early delivery: The device has no built-in power modules and uses a 36 W Power Adapter (02220947) or 36 W Power Adapter (02221468) for power supply.
- Later delivery: The device has no built-in power modules and uses a 36 W Power Adapter (02221468) for power supply.

## Heat Dissipation System

The device has no fans and uses natural heat dissipation.

## Technical Specifications

**Table 3-6** Technical specifications of the USG6510E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench<br>• Against the wall |
| Cabinet installation standard | Cabinet with a depth of 300 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 250 mm x 210 mm (1.72 in. x 9.84 in. x 8.27 in.)<br>• Maximum dimensions (the depth is the distance from ports on the rear panel to the front panel): 43.6 mm x 250 mm x 220 mm (1.72 in. x 9.84 in. x 8.66 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 110 mm x 465 mm x 335 mm (4.33 in. x 18.31 in. x 13.19 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 2.3 kg (5.07 lb) |
| Weight without packaging [kg(lb)] | 1.57 kg (3.46 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.2 GHz |
| Memory | 1 GB DDR4 ECC memory |
| NOR Flash | • USG6510E-AC (02352PQH): 32 MB<br>• USG6510E-AC (02352PQH-001): 64MB<br>• USG6510E-AC (02352PQH-002): 64MB |
| NAND Flash | 1 GB |
| Hard disk | Optional. Purchase one 64-GB micro SD card from Huawei as required. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6510E-AC (02352PQH): 16.37 W<br>• USG6510E-AC (02352PQH-001): 16.37 W<br>• USG6510E-AC (02352PQH-002): 16.4 W |
| Typical heat dissipation [BTU/hour] | • USG6510E-AC (02352PQH): -<br>• USG6510E-AC (02352PQH-001): -<br>• USG6510E-AC (02352PQH-002): 55.96 BTU/hour |
| Maximum power consumption [W] | 21.6 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | 73.7 BTU/hour |
| MTBF [years] | 57.29 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999996 |
| Power supply mode | Power adapter external |
| Number of power modules | 1 |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 1 A/power module |
| Rated output power [W] | 36W/12V |
| Maximum output power [W] | 36 W/power module |
| Types of fans | None |
| Number of fan modules | 0 |
| Heat dissipation mode | Has no fan and uses natural cooling. |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

## 3.1.2 USG6510E-POE

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## Overview

**Table 3-7** Basic information about the USG6510E-POE

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6510E-POE AC Host (10*GE RJ45 + 2*GE SFP,with the AC/DC adapter) | 02352CSC | USG6510E-POE-AC | V600R006C00 |

## Appearance

**Figure 3-6** Appearance of the USG6510E-POE (front view)

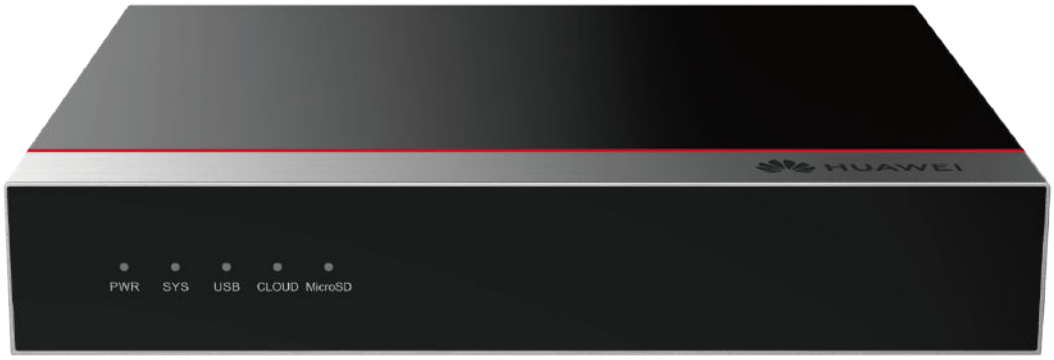
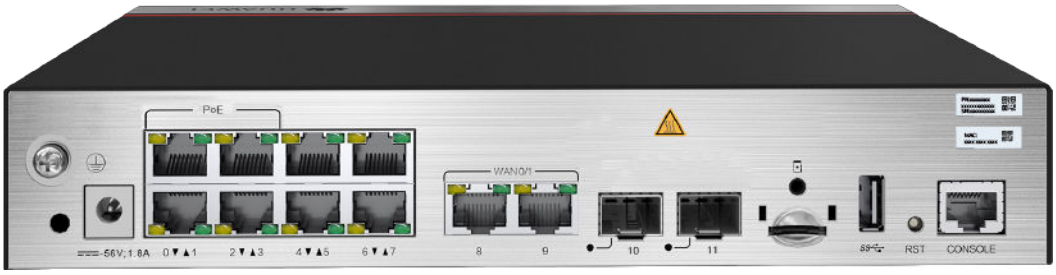


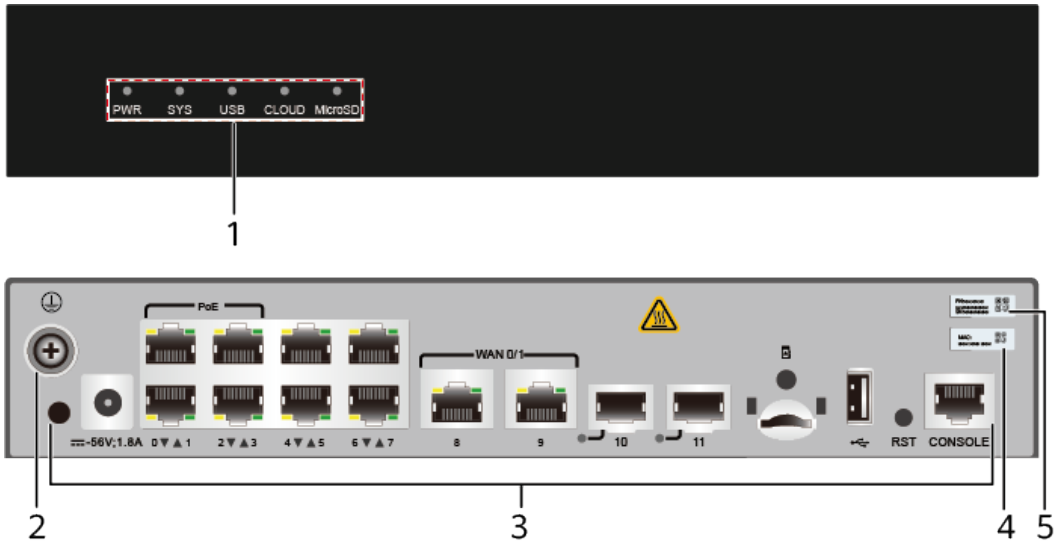

**Figure 3-7** Appearance of the USG6510E-POE (rear view)




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## Components

**Figure 3-8** Components of the USG6510E-POE




| 1. Indicator area | 2. Protective ground terminal | 3. Fixed interface area | 4. MAC label | 5. SN label |
|---|---|---|---|---|

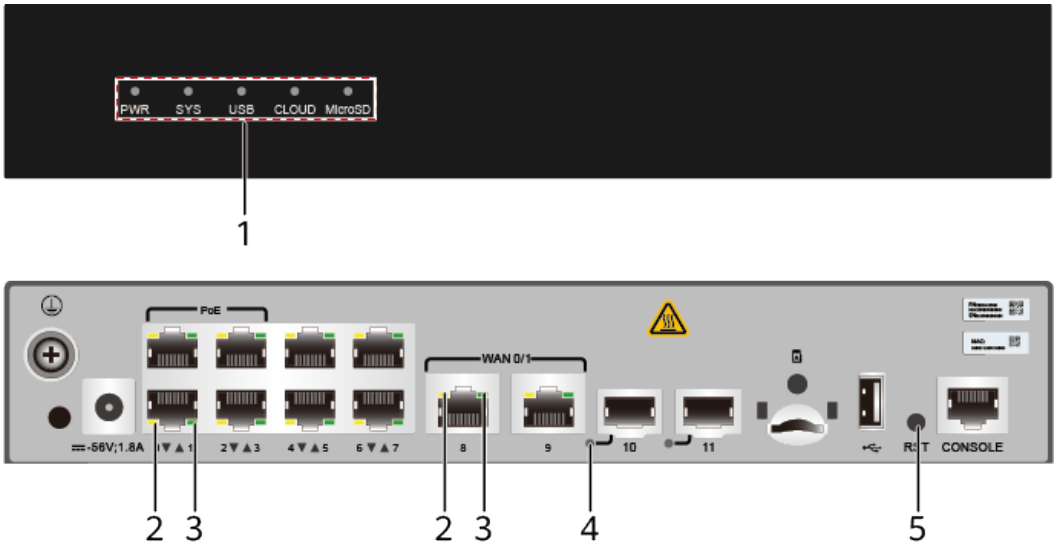
**Table 3-8** Component functions

| Name | Description |
|---|---|
| Indicator area | Provides various indicators to display the device running status in real time. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| Fixed interface area | • Provides clip hole and power receptacle.<br>• Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance.<br>• Houses a Micro SD card to store logs and reports. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |

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## Indicators and Buttons

**Figure 3-9** Indicators and buttons on the USG6510E-POE




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, CLOUD, and MicroSD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port indicator | 5. RST button |

**Table 3-9** Indicators on the USG6510E-POE

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | A system fault occurs. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| MicroSD | Micro SD indicator | Green | Steady on | The micro SD card is present. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The micro SD card is not detected. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Optical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

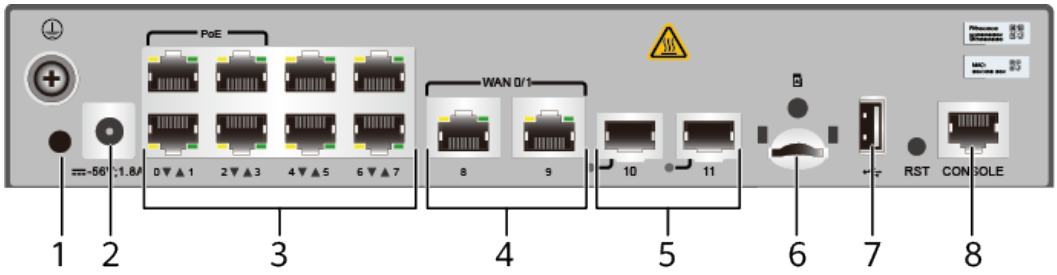
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**Table 3-10** Buttons on the USG6510E-POE

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-10** Ports on the USG6510E-POE




| | | | | |
| --- | --- | --- | --- | --- |
| 1. Clip hole | 2. Power receptacle | 3. GE electrical port | 4. WAN port | 5. GE optical port |

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| | | | | |
|---|---|---|---|---|
| 6. Micro SD card slot | 7. USB 3.0 port | 8. Console port | - | - |

**Table 3-11** Ports on the USG6510E-POE

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. | Power cable clip |
| Power receptacle | - | The receptacle connects to the tuning fork plug of the power adapter. | **150 W Power Adapter** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 7) | RJ45 | 8 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7. Among the interfaces, GigabitEthernet0/0/0 to GigabitEthernet0/0/3 support PoE output and can supply power to PDs (such as IP phones and surveillance cameras).<br>GigabitEthernet 0/0/0 is an inband management port and its default IP address is 192.168.0.1. After this port is connected to your PC through network cables, you can log in to the device using Telnet, STelnet, or web UI to configure or manage the device. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | **NOTE**<br>• When the device supplies power to PDs, the total power consumption of the PDs must not exceed 60 W.<br>• Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |
| WAN ports (8 to 9) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/8 to GigabitEthernet 0/0/9.<br>The device is connected to the WAN or cloud management platform using a network cable. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE optical port (10 to 11) | SFP | 2 GE Ethernet optical ports, numbered from GigabitEthernet 0/0/10 to GigabitEthernet 0/0/11. Compatible with GE Ethernet optical ports. | • **100Mbps SFP Optical Modules**<br>• **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, GE optical ports support 100Mbps SFP Optical Modules. |
| Micro SD card slot | Micro SD | The micro SD card slot allows you to insert a micro SD card to record logs and reports in real time. The micro SD card is optional. You can purchase one from Huawei if needed.<br>You are advised to install an anti-theft board delivered with the device to protect the micro SD card. | SDSDQAE-064G (BOM code: 06010308, capacity: 64GB, dimensions (H x W x D): 1 mm x 15.00 mm x 11.00 mm/0.04 in. x 0.59 in. x 0.43 in) |
| USB 2.0 port | USB | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

## Power Supply System

The device has no built-in power modules and uses a 150 W Power Adapter for power supply.

## Heat Dissipation System

The device has no fans and uses natural heat dissipation.

## Technical Specifications

**Table 3-12** Technical specifications of the USG6510E-POE-AC

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench<br>• Against the wall |
| Cabinet installation standard | Cabinet with a depth of 300 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 250 mm x 210 mm (1.72 in. x 9.84 in. x 8.27 in.)<br>• Maximum dimensions (the depth is the distance from ports on the rear panel to the front panel): 43.6 mm x 250 mm x 220 mm (1.72 in. x 9.84 in. x 8.66 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 110 mm x 470 mm x 335 mm (4.33 in. x 18.5 in. x 13.19 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 3.45 kg (7.61 lb) |
| Weight without packaging [kg(lb)] | 1.75 kg (3.86 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.2 GHz |
| Memory | 1 GB DDR4 ECC memory |
| NOR Flash | 32 MB |
| NAND Flash | 1 GB |
| Hard disk | Optional. Purchase one 64-GB micro SD card from Huawei as required. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 82.8 W |
| Maximum power consumption [W] | 85.1 W |
| Maximum heat dissipation [BTU/hour] | 291 BTU/hour |
| MTBF [years] | 57.29 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999996 |
| Power supply mode | Power adapter external |
| Number of power modules | 1 |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |

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| Item | Specification |
| --- | --- |
| Rated output power [W] | 150 W/55.5 V |
| Maximum output power [W] | 150 W/power module |
| Types of fans | None |
| Number of fan modules | 0 |
| Heat dissipation mode | Has no fan and uses natural cooling. |
| PoE | Supported |
| Maximum PoE output power [W] | 60 W |
| PoE power interfaces | 4 |
| Network cable type and longest transmission | • IEEE 802.3af: 100 m if there is no special requirement<br>• IEEE 802.3at: 100 m for CAT5e and higher |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

## 3.1.3 USG6515E

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## Overview

**Table 3-13** Basic information about the USG6515E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6515E AC Host (2*GE WAN + 8*GE Combo + 16*GE RJ45 + 2*10GE SFP+,1 AC power) | 02352CBP | USG6515E-AC | V600R006C00 |

## Appearance

**Figure 3-11** Appearance of the USG6515E (front view)

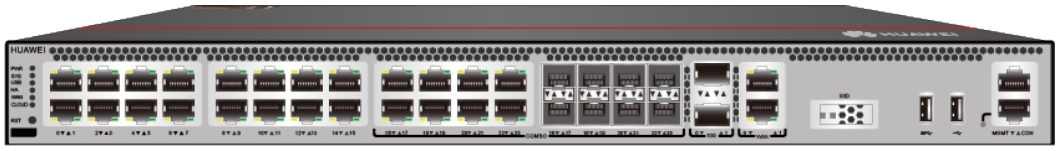
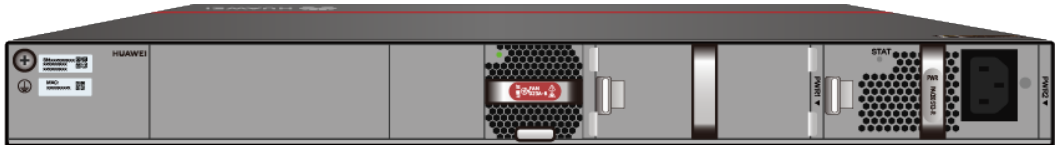
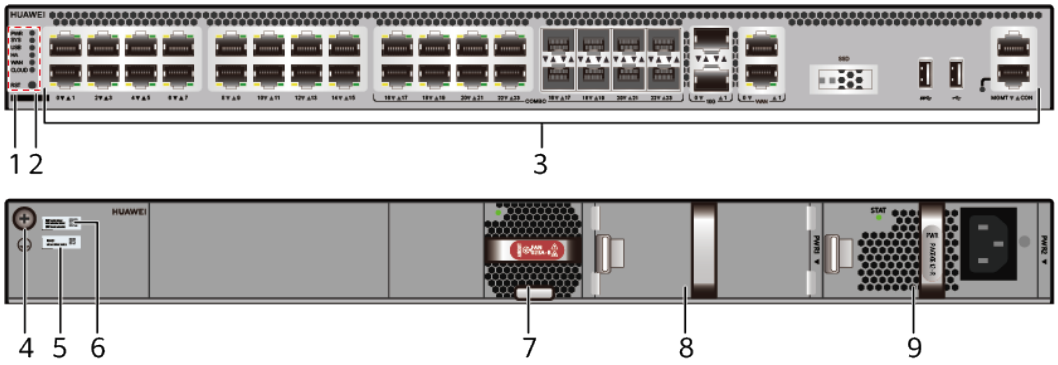


**Figure 3-12** Appearance of the USG6515E (rear view)



## Components

**Figure 3-13** Components of the USG6515E




| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| --- | --- | --- | --- | --- |

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| | | | | |
|---|---|---|---|---|
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

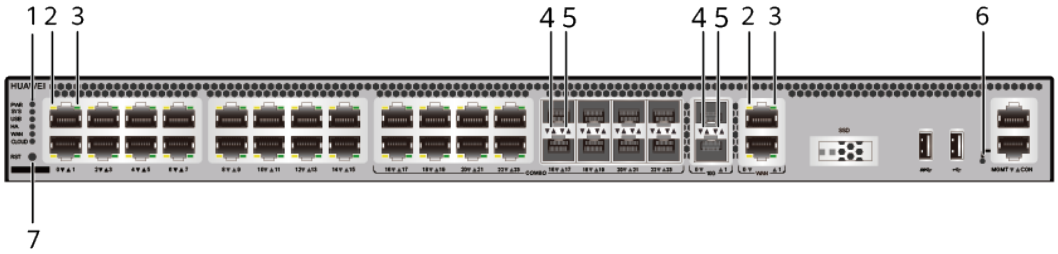
**Table 3-14** Component functions

| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-14** Indicators and buttons on the USG6515E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-15** Indicators on the USG6515E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

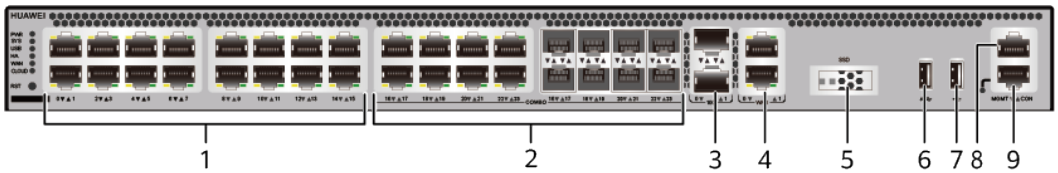
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**Table 3-16** Buttons on the USG6515E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-15** Ports on the USG6515E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. GE electrical port | 2. Combo port | 3. 10GE optical port | 4. WAN port | 5. SSD card slot |
| 6. USB 3.0 port | 7. USB 2.0 port | 8. Console port | 9. MGMT port | - |

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**Table 3-17** Ports on the USG6515E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **1Gbps eSFP Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1. Not compatible with GE optical modules. | **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA64G-A**<br>• **Hard Disk Unit M.2-SATA64G-B**<br>• **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

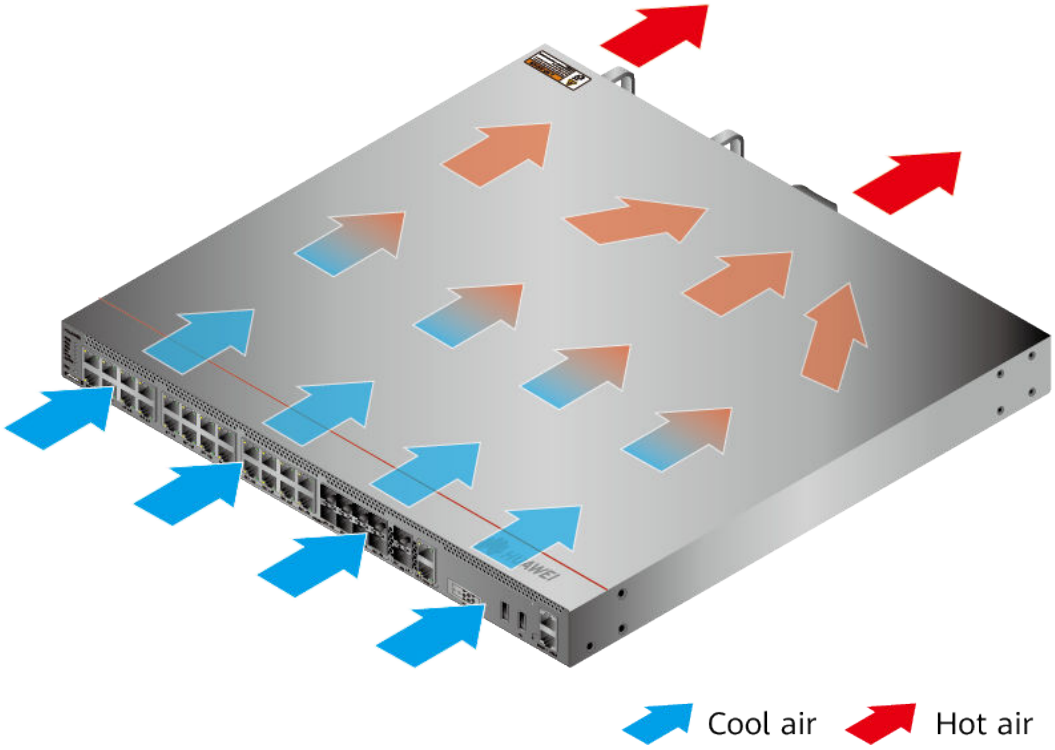
The power supply system has one PAC60S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-16** System air flow of the USG6515E




## Technical Specifications

**Table 3-18** Technical specifications of the USG6515E-AC

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.1 kg (17.86 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 5.71 kg (12.59 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (64GB/240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 32.4 W |
| Maximum power consumption [W] | 46.4 W |
| Maximum heat dissipation [BTU/hour] | 158.7 BTU/hour |
| MTBF [years] | 45.56 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.4 USG6525E

## Overview

**Table 3-19** Basic information about the USG6525E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6525E AC Host (2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AFX | USG6525E-AC | V600R007C00 |
| USG6525E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AFX-001 | USG6525E-AC | V600R007C20SPC500 |
| USG6525E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AFX-002 | USG6525E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The two USB ports of earlier-delivered models are USB 3.0 and USB 2.0 respectively, whereas those of later-delivered models are both USB 2.0. In this document, earlier-delivered models are used as examples.

**Figure 3-17** Appearance of the USG6525E (front view)

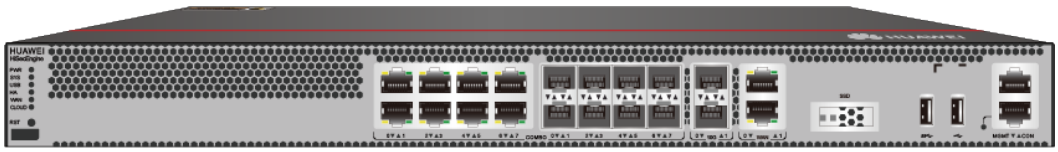
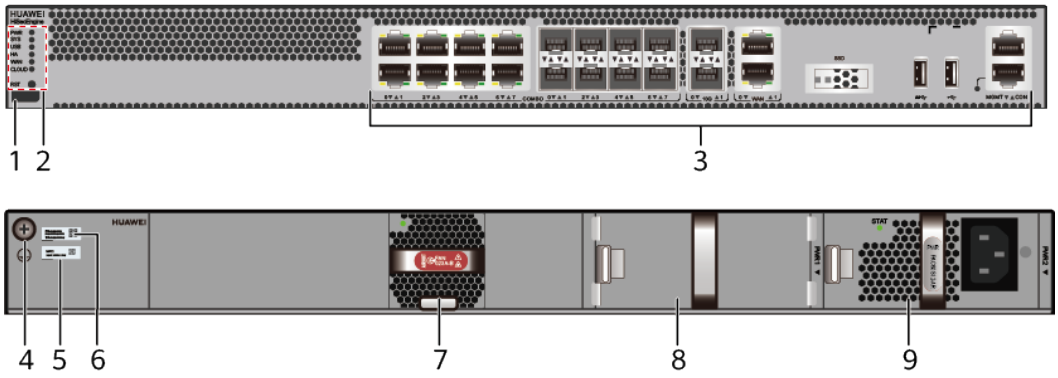


**Figure 3-18** Appearance of the USG6525E (rear view)



## Components

**Figure 3-19** Components of the USG6525E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

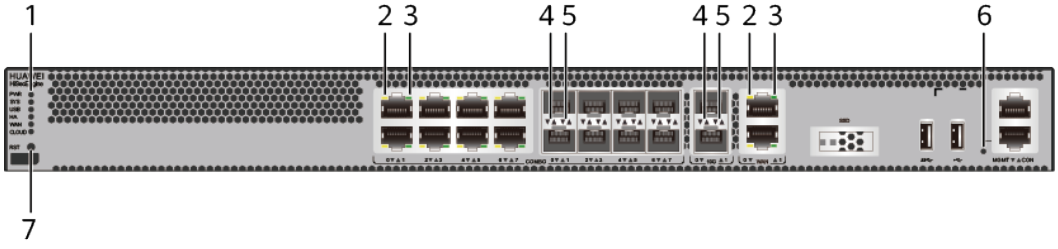
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**Table 3-20** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-20** Indicators and buttons on the USG6525E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-21** Indicators on the USG6525E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

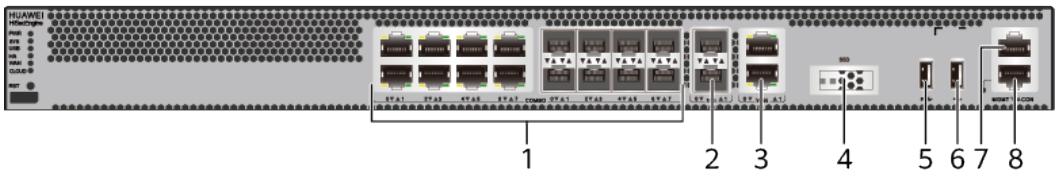
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**Table 3-22** Buttons on the USG6525E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-21** Ports on the USG6525E




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| | | | | |
|---|---|---|---|---|
| 1. Combo port | 2. 10GE optical port | 3. WAN port | 4. SSD card slot | 5. USB 3.0 port or USB0<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB0 (USB 2.0) |
| 6. USB 2.0 port or USB1<br>**NOTE**<br>• Earlier-delivered models: USB 2.0<br>• Later-delivered models: USB1 (USB 2.0) | 7. Console port | 8. MGMT port | - | - |

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**Table 3-23** Ports on the USG6525E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (0 to 7) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | **NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1.<br>**NOTE**<br>In V600R007C20 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. By default, the rate of 10GE optical ports are 10GE. You can run the set device port-config-mode [ 10ge \| ge ] command to set the rate of 10GE optical ports are to 10GE or GE as required. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA64G-A**<br>• **Hard Disk Unit M.2-SATA64G-B**<br>• **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port or USB0 | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| USB 2.0 port or USB1 | USB Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

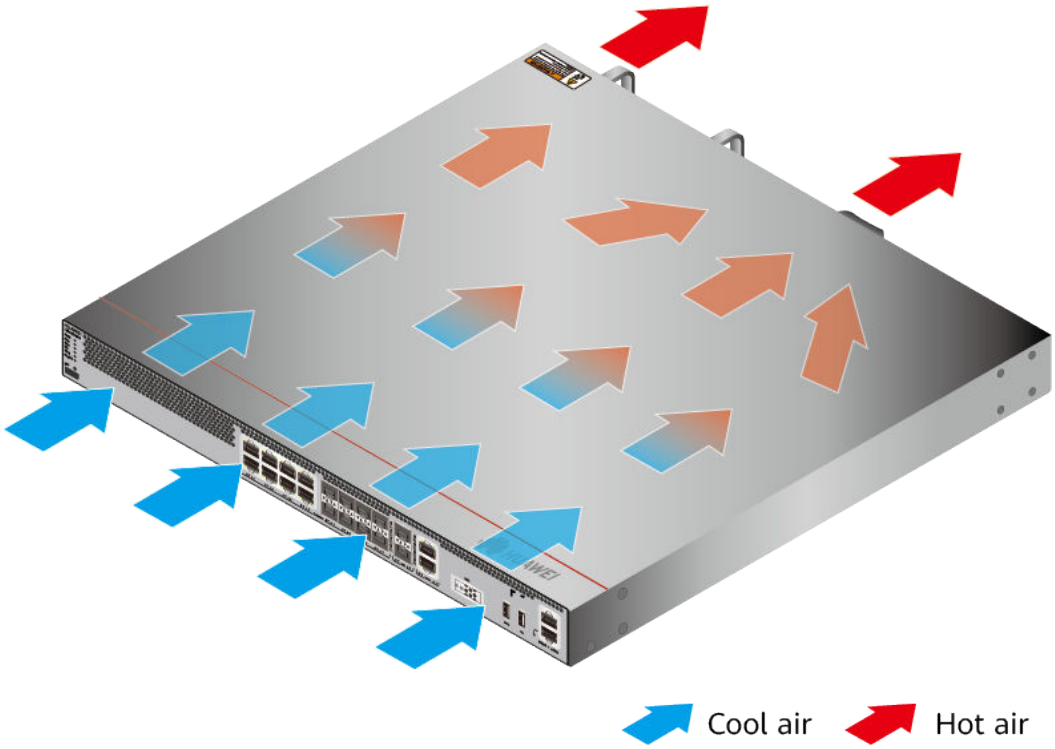
- Early delivery: The power supply system has one PAC60S12-AR, PAC60S12-R, or PAC600S12-CB AC power module. Two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC60S12-AR AC power module. Two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-22** System air flow of the USG6525E




## Technical Specifications

**Table 3-24** Technical specifications of the USG6525E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • USG6525E-AC (02353AFX):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6525E-AC (02353AFX-001):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6525E-AC (02353AFX-002):<br>– Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | • USG6525E-AC (02353AFX): 8.75 kg (19.29 lb)<br>• USG6525E-AC (02353AFX-001): 7.93 kg (17.48 lb)<br>• USG6525E-AC (02353AFX-002): 7.93 kg (17.48 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | • USG6525E-AC (02353AFX): 5.8 kg (12.79 lb)<br>• USG6525E-AC (02353AFX-001): 5.95 kg (13.12 lb)<br>• USG6525E-AC (02353AFX-002): 5.95 kg (13.12 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB<br>**NOTE**<br>The NOR flash memory size of devices delivered later is 64 MB. You can run the display version command to check the NOR flash memory size. |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (64GB/240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6525E-AC (02353AFX): 35 W<br>• USG6525E-AC (02353AFX-001): 25 W<br>• USG6525E-AC (02353AFX-002): 25 W |
| Typical heat dissipation [BTU/hour] | • USG6525E-AC (02353AFX): 119.42 BTU/hour<br>• USG6525E-AC (02353AFX-001): 85.4 BTU/hour<br>• USG6525E-AC (02353AFX-002): 85.4 BTU/hour |
| Maximum power consumption [W] | • USG6525E-AC (02353AFX): 40 W<br>• USG6525E-AC (02353AFX-001): 45.3 W<br>• USG6525E-AC (02353AFX-002): 45.3 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6525E-AC (02353AFX): 136.8 BTU/hour<br>• USG6525E-AC (02353AFX-001): 154.4 BTU/hour<br>• USG6525E-AC (02353AFX-002): 154.4 BTU/hour |
| MTBF [years] | 68.98 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999997 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6525E-AC (02353AFX): Two power modules are supported for 1+1 power redundancy.<br>• USG6525E-AC (02353AFX-001): Two power modules are supported for 1+1 power redundancy.<br>• USG6525E-AC (02353AFX-002): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6525E-AC (02353AFX): The fan module locates at the air exhaust of the system.<br>• USG6525E-AC (02353AFX-001): The fan module locates at the air exhaust of the system.<br>• USG6525E-AC (02353AFX-002): Absorbing cold air into the device |

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| Item | Specification |
| --- | --- |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.5 USG6530E

## Overview

**Table 3-25** Basic information about the USG6530E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6530E AC Host (10*GE RJ45 + 2*10GE SFP +,1*Adapter) | 02352QVD | USG6530E-AC | V600R007C00 |
| USG6530E AC Host (10*GE RJ45 + 2*10GE SFP +,1*Adapter) | 02352QVD-001 | USG6530E-AC | V600R007C20SPC300 |
| USG6530E AC Host(10*GE RJ45+2*10GE SFP +,1*Adapter) | 02352QVD-002 | USG6530E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The standard USB port of the earlier-delivered models is USB 3.0, and that of the later-delivered models is USB 2.0. This document uses a model with a USB 3.0 port as an example.

**Figure 3-23** Appearance of the USG6530E (front view)

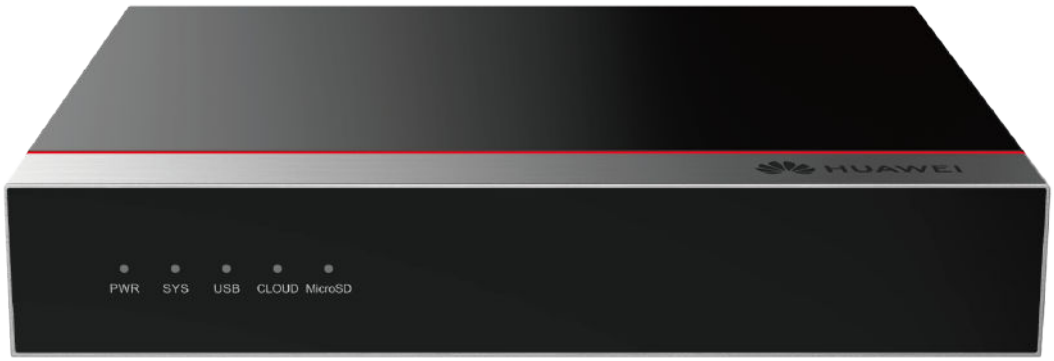
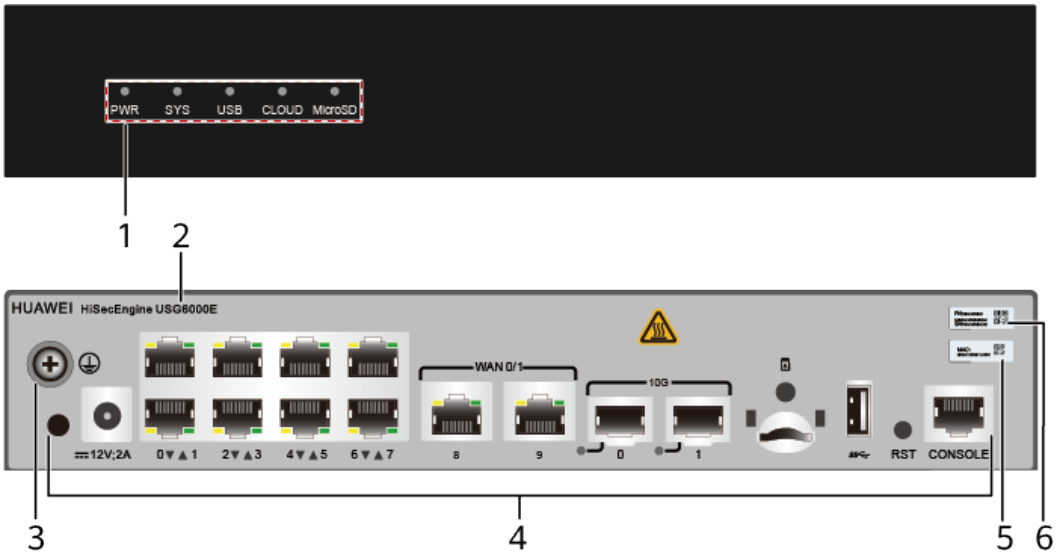



**Figure 3-24** Appearance of the USG6530E (rear view)




## Components

**Figure 3-25** Components of the USG6530E




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| | | | | |
|---|---|---|---|---|
| 1. Indicator area | 2. Product series ID | 3. Protective ground terminal | 4. Fixed interface area | 5. MAC label |
| 6. SN label | - | - | - | - |

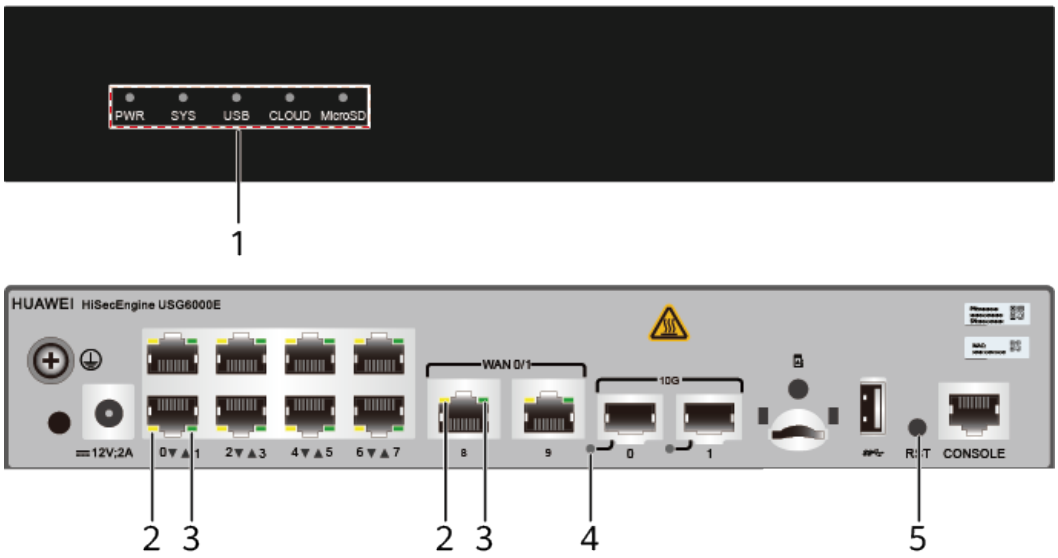
**Table 3-26** Component functions

| Name | Description |
|---|---|
| Indicator area | Provides various indicators to display the device running status in real time. |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| Fixed interface area | • Provides clip hole and power receptacle.<br>• Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance.<br>• Houses a Micro SD card to store logs and reports. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |

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## Indicators and Buttons

**Figure 3-26** Indicators and buttons on the USG6530E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, CLOUD, and MicroSD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port indicator | 5. RST button |

**Table 3-27** Indicators on the USG6530E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | A system fault occurs. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| MicroSD | Micro SD indicator | Green | Steady on | The micro SD card is present. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The micro SD card is not detected. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Optical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

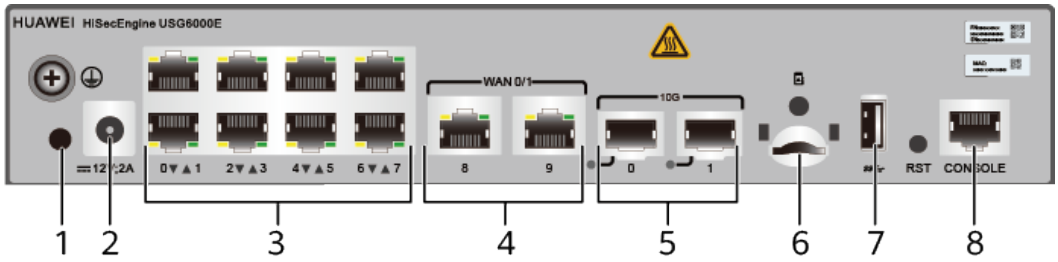
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**Table 3-28** Buttons on the USG6530E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-27** Ports on the USG6530E




| 1. Clip hole | 2. Power receptacle | 3. GE electrical port | 4. WAN port | 5. 10GE optical port |
| --- | --- | --- | --- | --- |

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| | | | | |
|---|---|---|---|---|
| 6. Micro SD card slot | 7. USB port<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB 2.0 | 8. Console port | - | - |

**Table 3-29** Ports on the USG6530E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. | Power cable clip |
| Power receptacle | - | The receptacle connects to the tuning fork plug of the power adapter. | • **36 W Power Adapter (02220947)**<br>• **36 W Power Adapter (02221468)** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 7) | RJ45 | 8 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7.<br>GigabitEthernet 0/0/0 is an inband management port and its default IP address is 192.168.0.1. After this port is connected to your PC through Ethernet cables, you can log in to the device using Telnet, STelnet, or web UI to configure or manage the device.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| WAN port (8 to 9) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/8 to GigabitEthernet 0/0/9.<br>The device is connected to the WAN or cloud management platform using an Ethernet cable.<br>**NOTE**<br>The WAN ports of the devices delivered earlier are vertically arranged, whose functions are the same as those of the WAN ports that are horizontally arranged. | **Ethernet Cable** |
| 10GE optical ports (0 to 1) | SFP+ | 2 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1. | • **100Mbps SFP Optical Modules**<br>• **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, 10GE optical ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Micro SD card slot | Micro SD | The micro SD card slot allows you to insert a micro SD card to record logs and reports in real time. The micro SD card is optional. You can purchase one from Huawei if needed.<br>You are advised to install an anti-theft board delivered with the device to protect the micro SD card. | SDSDQAE-064G (BOM code: 06010308, capacity: 64GB, dimensions (H x W x D): 1 mm x 15.00 mm x 11.00 mm/0.04 in. x 0.59 in. x 0.43 in) |
| USB port | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

## Power Supply System

- Early delivery: The device has no built-in power modules and uses a 36 W Power Adapter (02220947) or 36 W Power Adapter (02221468) for power supply.
- Later delivery: The device has no built-in power modules and uses a 36 W Power Adapter (02221468) for power supply.

## Heat Dissipation System

The device has no fans and uses natural heat dissipation.

## Technical Specifications

**Table 3-30** Technical specifications of the USG6530E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench<br>• Against the wall |
| Cabinet installation standard | Cabinet with a depth of 300 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 250 mm x 210 mm (1.72 in. x 9.84 in. x 8.27 in.)<br>• Maximum dimensions (the depth is the distance from ports on the rear panel to the front panel): 43.6 mm x 250 mm x 220 mm (1.72 in. x 9.84 in. x 8.66 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 110 mm x 470 mm x 335 mm (4.33 in. x 18.5 in. x 13.19 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 2.3 kg (5.07 lb) |
| Weight without packaging [kg(lb)] | 1.57 kg (3.46 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.2 GHz |
| Memory | 2 GB DDR4 ECC memory |
| NOR Flash | • USG6530E-AC (02352QVD): 32 MB<br>• USG6530E-AC (02352QVD-001): 64MB<br>• USG6530E-AC (02352QVD-002): 64MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. Purchase one 64-GB micro SD card from Huawei as required. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 17 W |
| Maximum power consumption [W] | 22.3 W |
| Maximum heat dissipation [BTU/hour] | 76.08 BTU/hour |
| MTBF [years] | 57.29 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999996 |
| Power supply mode | Power adapter external |
| Number of power modules | 1 |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |

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| Item | Specification |
| --- | --- |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 1 A/power module |
| Rated output power [W] | 36W/12V |
| Maximum output power [W] | 36 W/power module |
| Types of fans | None |
| Number of fan modules | 0 |
| Heat dissipation mode | Has no fan and uses natural cooling. |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.6 USG6550E

## Overview

**Table 3-31** Basic information about the USG6550E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6550E AC Host (2*GE WAN + 8*GE Combo + 16*GE RJ45 + 2*10GE SFP+,1 AC power) | 02352CBQ | USG6550E-AC | V600R006C00 |

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## Appearance

**Figure 3-28** Appearance of the USG6550E (front view)

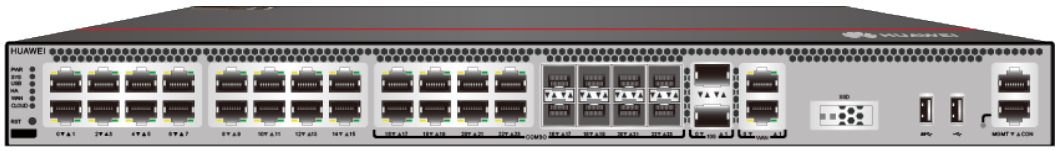
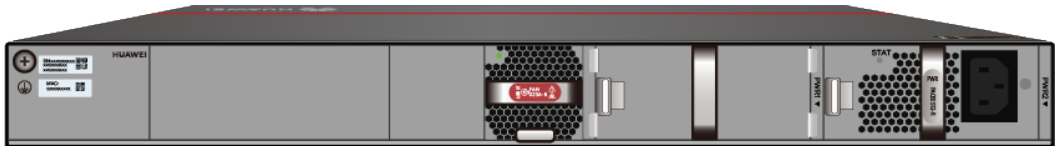
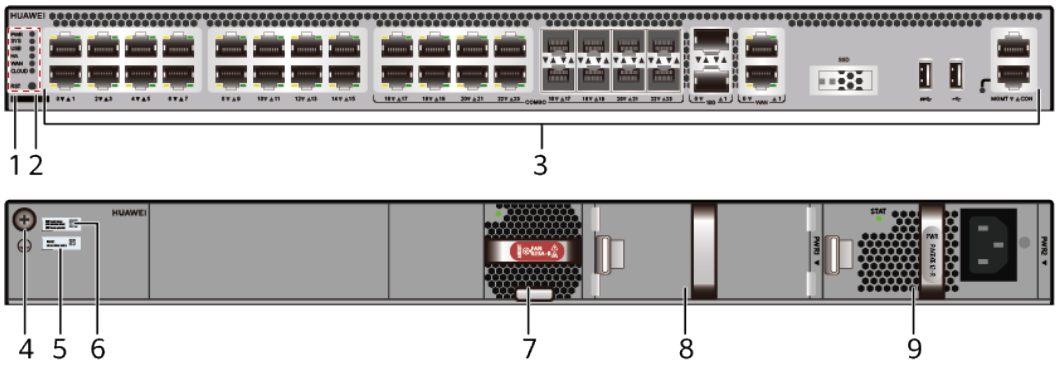


**Figure 3-29** Appearance of the USG6550E (rear view)



## Components

**Figure 3-30** Components of the USG6550E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

**Table 3-32** Component functions

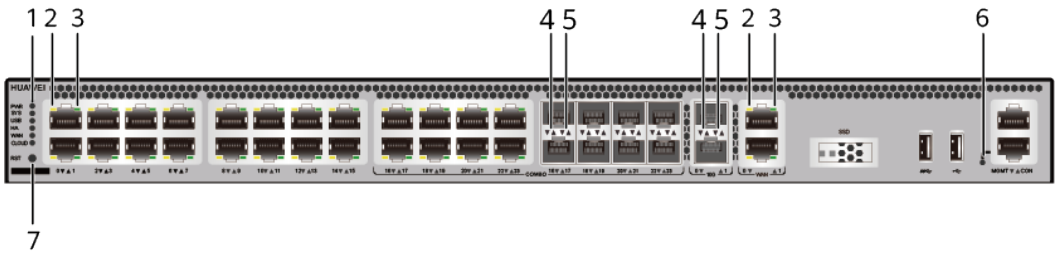
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |

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| Name | Description |
| --- | --- |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-31** Indicators and buttons on the USG6550E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-33** Indicators on the USG6550E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

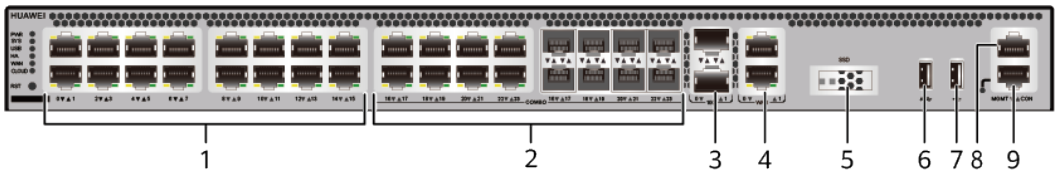
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**Table 3-34** Buttons on the USG6550E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-32** Ports on the USG6550E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. GE electrical port | 2. Combo port | 3. 10GE optical port | 4. WAN port | 5. SSD card slot |
| 6. USB 3.0 port | 7. USB 2.0 port | 8. Console port | 9. MGMT port | - |

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**Table 3-35** Ports on the USG6550E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **1Gbps eSFP Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1. Not compatible with GE optical modules. | **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

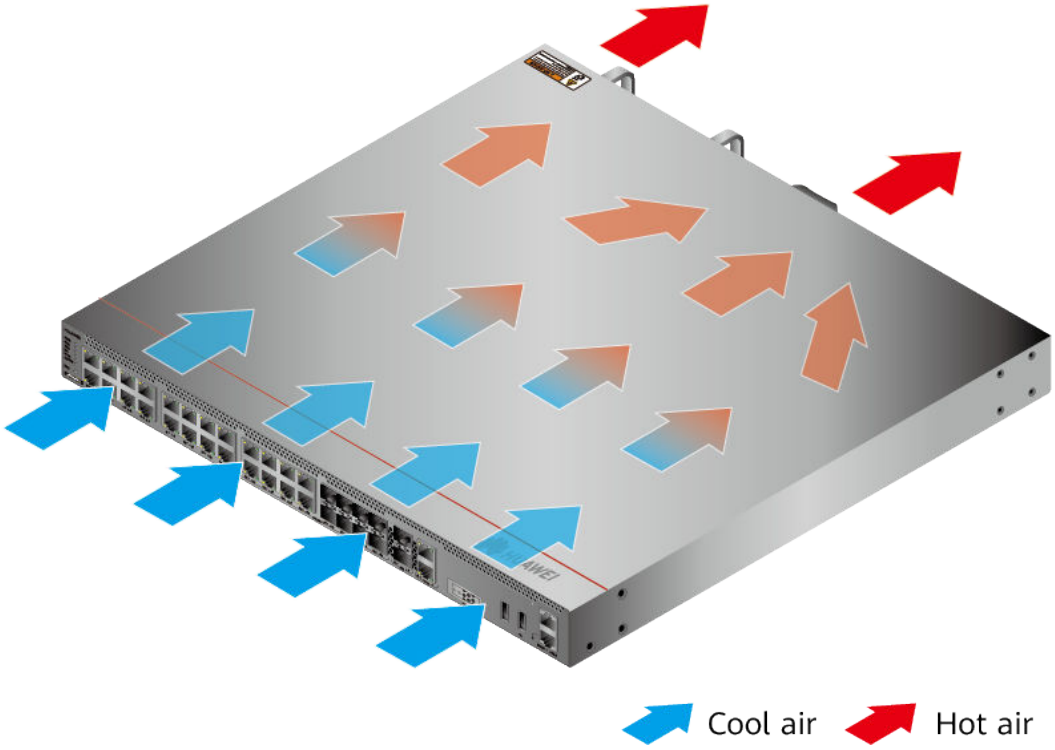
The power supply system has one PAC60S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-33** System air flow of the USG6550E




## Technical Specifications

**Table 3-36** Technical specifications of the USG6550E-AC

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.1 kg (17.86 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 5.71 kg (12.59 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 32.4 W |
| Maximum power consumption [W] | 46.4 W |
| Maximum heat dissipation [BTU/hour] | 158.7 BTU/hour |
| MTBF [years] | 45.56 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.7 USG6555E

## Overview

**Table 3-37** Basic information about the USG6555E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6555E AC Host (2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEK | USG6555E-AC | V600R007C00 |
| USG6555E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEK-001 | USG6555E-AC | V600R007C20SPC500 |
| USG6555E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEK-004 | USG6555E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The two USB ports of earlier-delivered models are USB 3.0 and USB 2.0 respectively, whereas those of later-delivered models are both USB 2.0. In this document, earlier-delivered models are used as examples.

**Figure 3-34** Appearance of the USG6555E (front view)

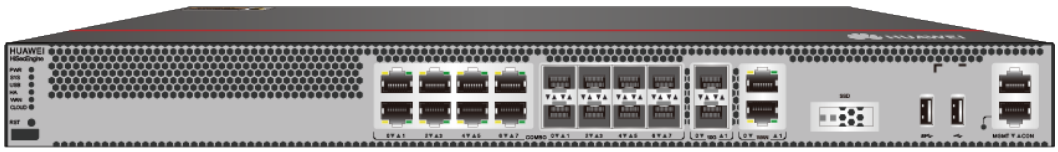
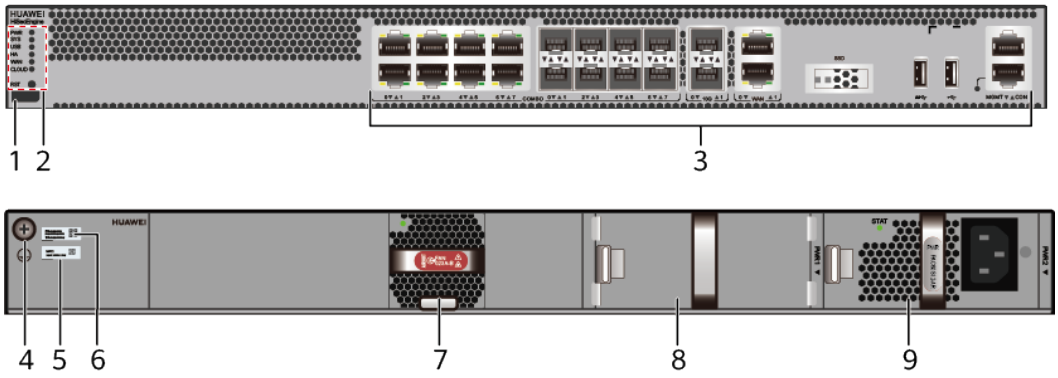


**Figure 3-35** Appearance of the USG6555E (rear view)



## Components

**Figure 3-36** Components of the USG6555E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

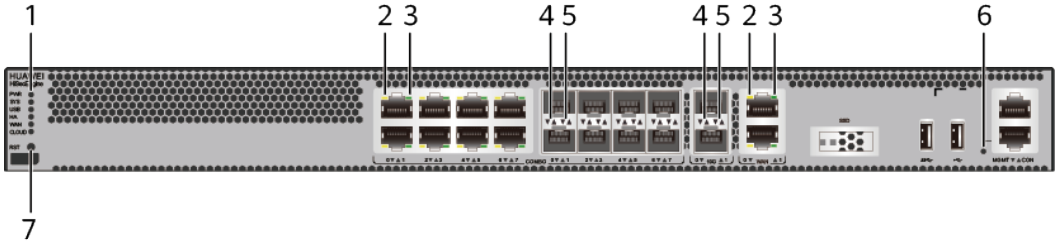
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**Table 3-38** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-37** Indicators and buttons on the USG6555E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-39** Indicators on the USG6555E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

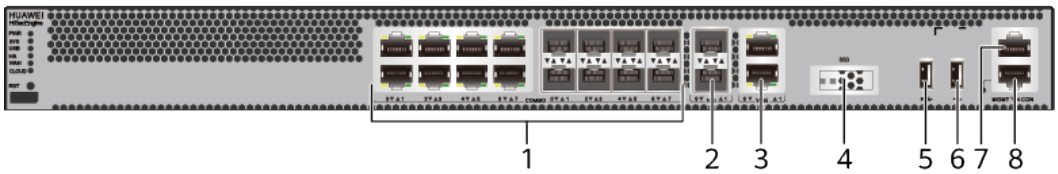
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**Table 3-40** Buttons on the USG6555E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-38** Ports on the USG6555E




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| | | | | |
|---|---|---|---|---|
| 1. Combo port | 2. 10GE optical port | 3. WAN port | 4. SSD card slot | 5. USB 3.0 port or USB0<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB0 (USB 2.0) |
| 6. USB 2.0 port or USB1<br>**NOTE**<br>• Earlier-delivered models: USB 2.0<br>• Later-delivered models: USB1 (USB 2.0) | 7. Console port | 8. MGMT port | - | - |

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**Table 3-41** Ports on the USG6555E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (0 to 7) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | **NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1.<br>**NOTE**<br>In V600R007C20 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. By default, the rate of 10GE optical ports are 10GE. You can run the set device port-config-mode [ 10ge \| ge ] command to set the rate of 10GE optical ports are to 10GE or GE as required. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port or USB0 | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| USB 2.0 port or USB1 | USB Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

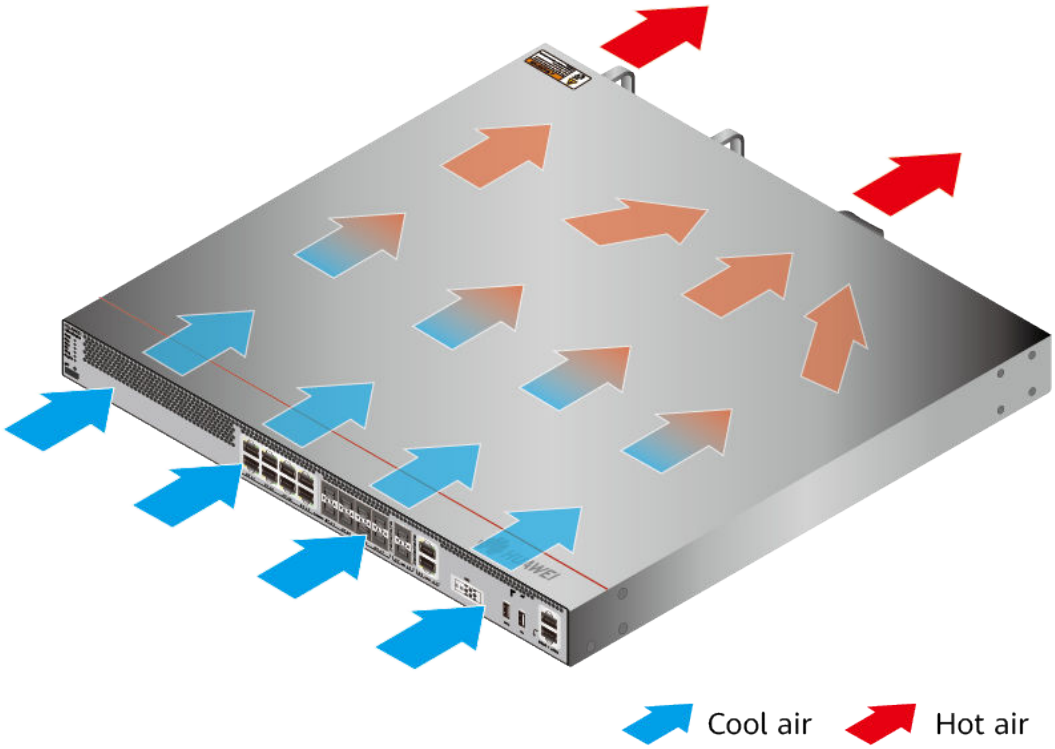
- Early delivery: The power supply system has one PAC60S12-AR, PAC60S12-R, or PAC600S12-CB AC power module. Two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC60S12-AR AC power module. Two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-39** System air flow of the USG6555E




## Technical Specifications

**Table 3-42** Technical specifications of the USG6555E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • USG6555E-AC (02353AEK):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6555E-AC (02353AEK-001):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6555E-AC (02353AEK-004):<br>– Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | • USG6555E-AC (02353AEK): 8.75 kg (19.29 lb)<br>• USG6555E-AC (02353AEK-001): 7.93 kg (17.48 lb)<br>• USG6555E-AC (02353AEK-004): 7.93 kg (17.48 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | • USG6555E-AC (02353AEK): 5.8 kg (12.79 lb)<br>• USG6555E-AC (02353AEK-001): 5.95 kg (13.12 lb)<br>• USG6555E-AC (02353AEK-004): 5.95 kg (13.12 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB<br>**NOTE**<br>The NOR flash memory size of devices delivered later is 64 MB. You can run the display version command to check the NOR flash memory size. |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6555E-AC (02353AEK): 35 W<br>• USG6555E-AC (02353AEK-001): 25 W<br>• USG6555E-AC (02353AEK-004): 25 W |
| Typical heat dissipation [BTU/hour] | • USG6555E-AC (02353AEK): 119.42 BTU/hour<br>• USG6555E-AC (02353AEK-001): 85.4 BTU/hour<br>• USG6555E-AC (02353AEK-004): 85.4 BTU/hour |
| Maximum power consumption [W] | • USG6555E-AC (02353AEK): 40 W<br>• USG6555E-AC (02353AEK-001): 45.3 W<br>• USG6555E-AC (02353AEK-004): 45.3 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6555E-AC (02353AEK): 136.8 BTU/hour<br>• USG6555E-AC (02353AEK-001): 154.4 BTU/hour<br>• USG6555E-AC (02353AEK-004): 154.4 BTU/hour |
| MTBF [years] | 68.98 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999997 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6555E-AC (02353AEK): Two power modules are supported for 1+1 power redundancy.<br>• USG6555E-AC (02353AEK-001): Two power modules are supported for 1+1 power redundancy.<br>• USG6555E-AC (02353AEK-004): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6555E-AC (02353AEK): The fan module locates at the air exhaust of the system.<br>• USG6555E-AC (02353AEK-001): The fan module locates at the air exhaust of the system.<br>• USG6555E-AC (02353AEK-004): Absorbing cold air into the device |

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| Item | Specification |
| --- | --- |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.8 USG6560E

## Overview

**Table 3-43** Basic information about the USG6560E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6560E AC Host (2*GE WAN + 8*GE Combo + 16*GE RJ45 + 2*10GE SFP+,1 AC power) | 02352CBR | USG6560E-AC | V600R006C00 |

## Appearance

**Figure 3-40** Appearance of the USG6560E (front view)

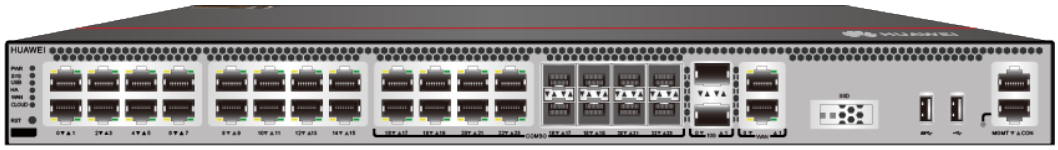
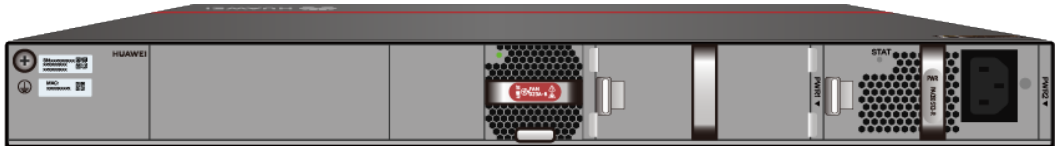
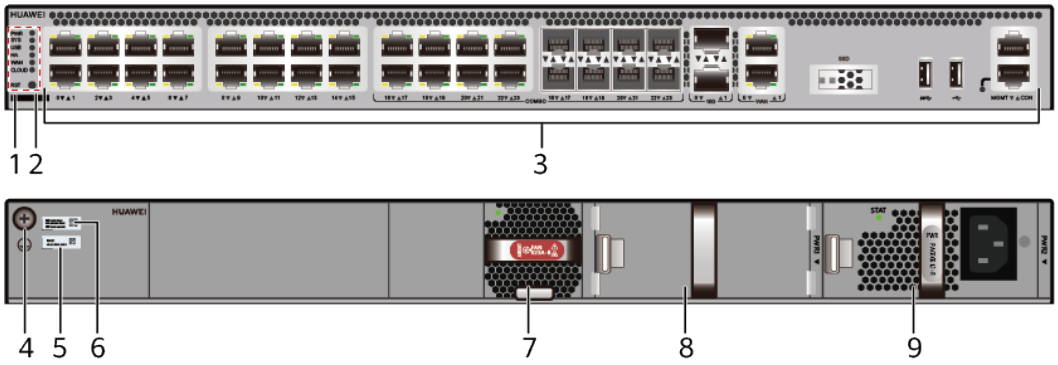


**Figure 3-41** Appearance of the USG6560E (rear view)



## Components

**Figure 3-42** Components of the USG6560E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

**Table 3-44** Component functions

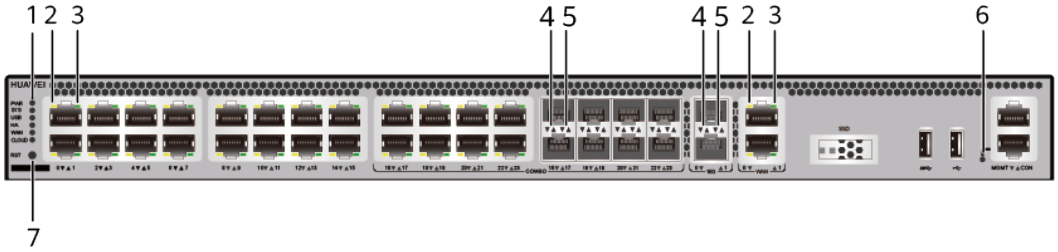
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |

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| Name | Description |
| --- | --- |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-43** Indicators and buttons on the USG6560E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

**NOTE**

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

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**Table 3-45** Indicators on the USG6560E

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

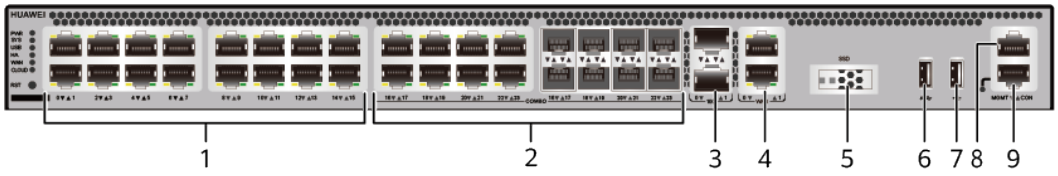
**Table 3-46** Buttons on the USG6560E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

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## Ports

**Figure 3-44** Ports on the USG6560E




| | | | | |
|---|---|---|---|---|
| 1. GE electrical port | 2. Combo port | 3. 10GE optical port | 4. WAN port | 5. SSD card slot |
| 6. USB 3.0 port | 7. USB 2.0 port | 8. Console port | 9. MGMT port | - |

**Table 3-47** Ports on the USG6560E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **1Gbps eSFP Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1. Not compatible with GE optical modules. | **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

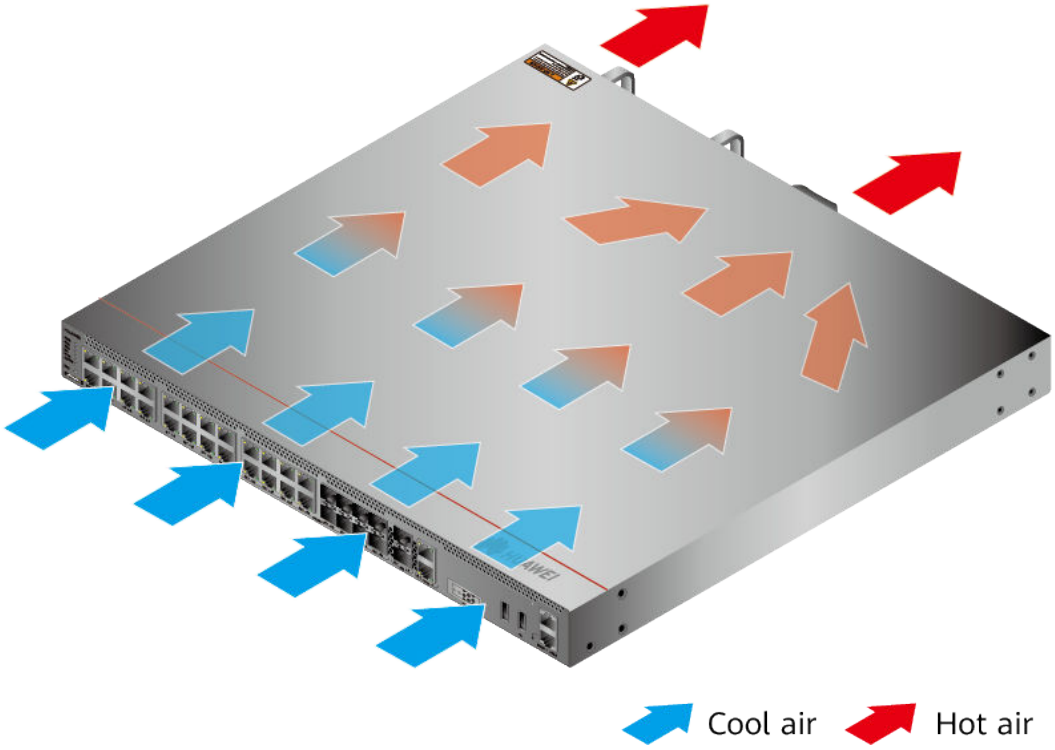
The power supply system has one PAC60S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-45** System air flow of the USG6560E




## Technical Specifications

**Table 3-48** Technical specifications of the USG6560E-AC

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.1 kg (17.86 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 5.71 kg (12.59 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 32.4 W |
| Maximum power consumption [W] | 46.4 W |
| Maximum heat dissipation [BTU/hour] | 158.7 BTU/hour |
| MTBF [years] | 45.56 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.9 USG6565E

## Overview

**Table 3-49** Basic information about the USG6565E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6565E AC Host (2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEJ | USG6565E-AC | V600R007C00 |
| USG6565E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEJ-001 | USG6565E-AC | V600R007C20SPC500 |
| USG6565E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353AEJ-002 | USG6565E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The two USB ports of earlier-delivered models are USB 3.0 and USB 2.0 respectively, whereas those of later-delivered models are both USB 2.0. In this document, earlier-delivered models are used as examples.

**Figure 3-46** Appearance of the USG6565E (front view)

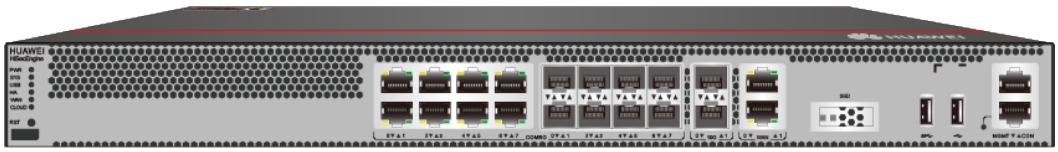
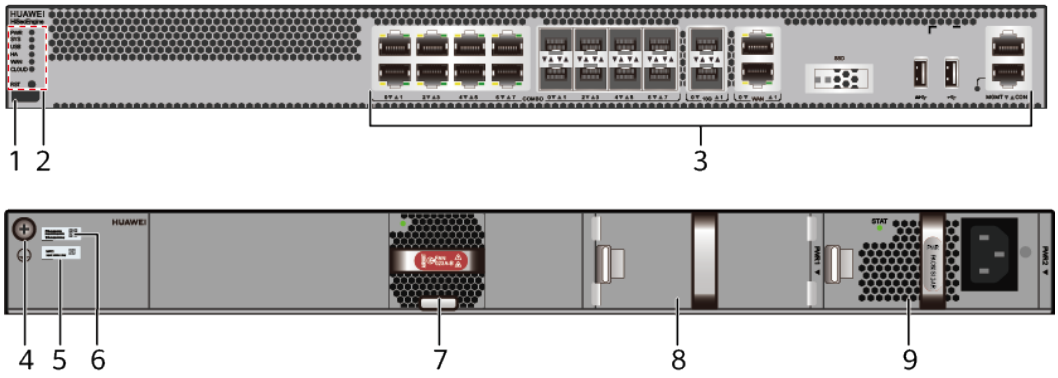


**Figure 3-47** Appearance of the USG6565E (rear view)



## Components

**Figure 3-48** Components of the USG6565E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

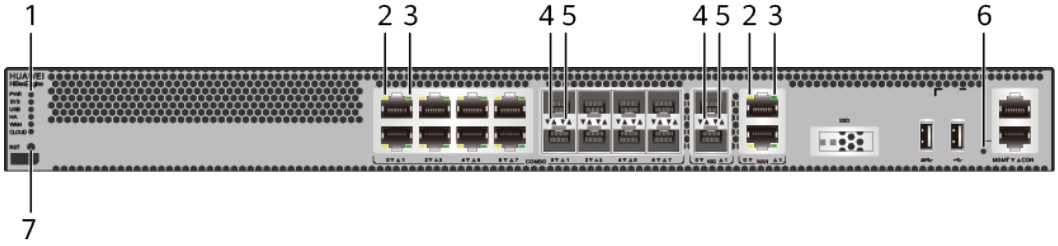
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**Table 3-50** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-49** Indicators and buttons on the USG6565E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-51** Indicators on the USG6565E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

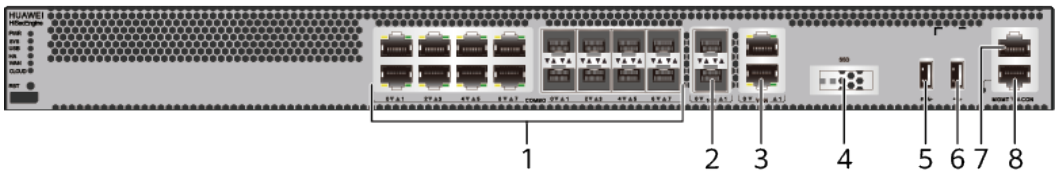
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**Table 3-52** Buttons on the USG6565E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-50** Ports on the USG6565E




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| | | | | |
|---|---|---|---|---|
| 1. Combo port | 2. 10GE optical port | 3. WAN port | 4. SSD card slot | 5. USB 3.0 port or USB0<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB0 (USB 2.0) |
| 6. USB 2.0 port or USB1<br>**NOTE**<br>• Earlier-delivered models: USB 2.0<br>• Later-delivered models: USB1 (USB 2.0) | 7. Console port | 8. MGMT port | - | - |

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**Table 3-53** Ports on the USG6565E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (0 to 7) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | **NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1.<br>**NOTE**<br>In V600R007C20 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. By default, the rate of 10GE optical ports are 10GE. You can run the set device port-config-mode [ 10ge \| ge ] command to set the rate of 10GE optical ports are to 10GE or GE as required. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port or USB0 | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| USB 2.0 port or USB1 | USB Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

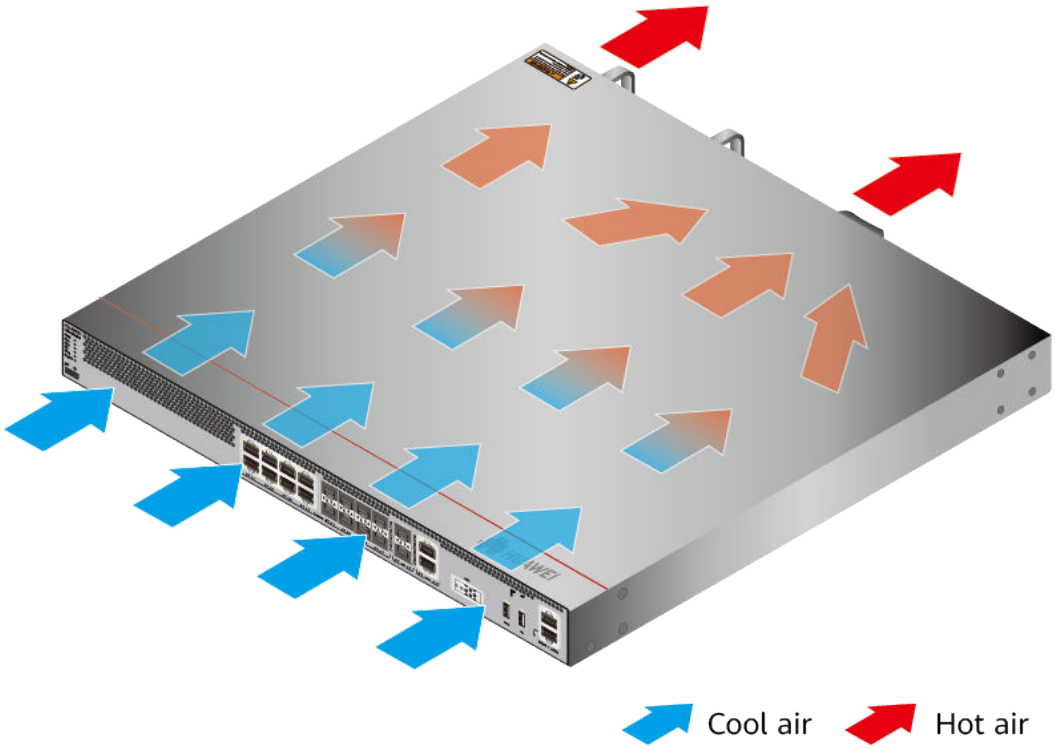
- Early delivery: The power supply system has one PAC60S12-AR, PAC60S12-R, or PAC600S12-CB AC power module. Two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC60S12-AR AC power module. Two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-51** System air flow of the USG6565E




## Technical Specifications

**Table 3-54** Technical specifications of the USG6565E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • USG6565E-AC (02353AEJ):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6565E-AC (02353AEJ-001):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6565E-AC (02353AEJ-002):<br>– Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | • USG6565E-AC (02353AEJ): 8.75 kg (19.29 lb)<br>• USG6565E-AC (02353AEJ-001): 7.93 kg (17.48 lb)<br>• USG6565E-AC (02353AEJ-002): 7.93 kg (17.48 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | • USG6565E-AC (02353AEJ): 5.8 kg (12.79 lb)<br>• USG6565E-AC (02353AEJ-001): 5.95 kg (13.12 lb)<br>• USG6565E-AC (02353AEJ-002): 5.95 kg (13.12 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB<br>**NOTE**<br>The NOR flash memory size of devices delivered later is 64 MB. You can run the display version command to check the NOR flash memory size. |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6565E-AC (02353AEJ): 35 W<br>• USG6565E-AC (02353AEJ-001): 25 W<br>• USG6565E-AC (02353AEJ-002): 25 W |
| Typical heat dissipation [BTU/hour] | • USG6565E-AC (02353AEJ): 119.42 BTU/hour<br>• USG6565E-AC (02353AEJ-001): 85.4 BTU/hour<br>• USG6565E-AC (02353AEJ-002): 85.4 BTU/hour |
| Maximum power consumption [W] | • USG6565E-AC (02353AEJ): 40 W<br>• USG6565E-AC (02353AEJ-001): 45.3 W<br>• USG6565E-AC (02353AEJ-002): 45.3 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6565E-AC (02353AEJ): 136.8 BTU/hour<br>• USG6565E-AC (02353AEJ-001): 154.4 BTU/hour<br>• USG6565E-AC (02353AEJ-002): 154.4 BTU/hour |
| MTBF [years] | 68.98 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999997 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6565E-AC (02353AEJ): Two power modules are supported for 1+1 power redundancy.<br>• USG6565E-AC (02353AEJ-001): Two power modules are supported for 1+1 power redundancy.<br>• USG6565E-AC (02353AEJ-002): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6565E-AC (02353AEJ): The fan module locates at the air exhaust of the system.<br>• USG6565E-AC (02353AEJ-001): The fan module locates at the air exhaust of the system.<br>• USG6565E-AC (02353AEJ-002): Absorbing cold air into the device |

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| Item | Specification |
| --- | --- |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.10 USG6575E-B

## Overview

**Table 3-55** Basic information about the USG6575E-B

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6575E-B AC Host (16*GE RJ45+8*GE Combo +2*10GE SFP+,1 AC Power) | 02353LKD | USG6575E-B-AC | V600R007C20 |
| USG6575E-B AC Host(16*GE RJ45+8*GE Combo +2*10GE SFP+,1 AC Power) | 02353LKD-002 | USG6575E-B-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The standard USB port of the earlier-delivered models is USB 3.0, and that of the later-delivered models is USB 2.0. This document uses a model with a USB 3.0 port as an example.

**Figure 3-52** Appearance of the USG6575E-B (front view)

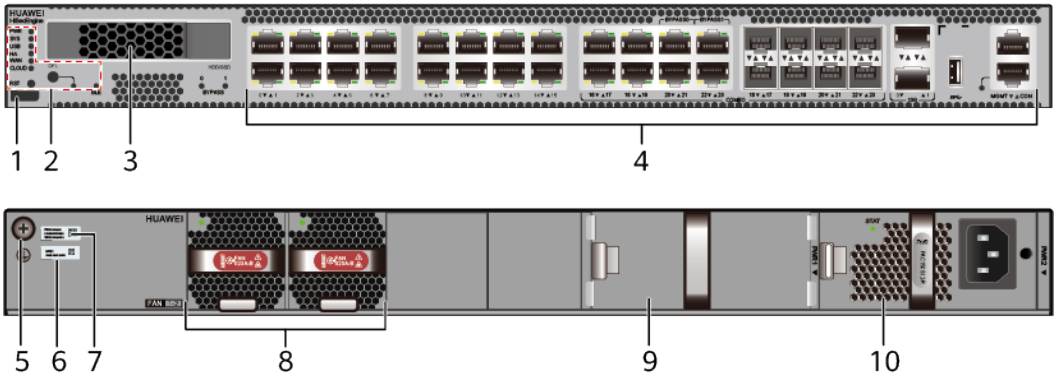


**Figure 3-53** Appearance of the USG6575E-B (rear view)



## Components

**Figure 3-54** Components of the USG6575E-B




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

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**Table 3-56** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN2, and the corresponding slot numbers are 4 and 5 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-55** Indicators and buttons on the USG6575E-B




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. BLE indicator | 4. BYPASS 0 indicator | 5. BYPASS 1 indicator |
| 6. GE electrical port ACT indicator | 7. GE electrical port LINK indicator | 8. Optical port ACT indicator | 9. Optical port LINK indicator | 10. MGMT port indicator |
| 11. RST button | 12. OFL button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-57** Indicators on the USG6575E-B

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The hard disk cannot be removed. |
| BLE | Bluetooth indicator | - | Off | Reserved function. This function is not enabled. |
| Bypass | Bypass indicator 0 | Green | Steady on | Device is powered on. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in protection state. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Device is powered on. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in working state. |
| | | - | Off | Device is powered off. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in protection state. |
| Bypass | Bypass indicator 1 | Green | Steady on | Device is powered on. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in protection state. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Device is powered on. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in working state. |
| | | - | Off | Device is powered off. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in protection state. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

**Table 3-58** Buttons on the USG6575E-B

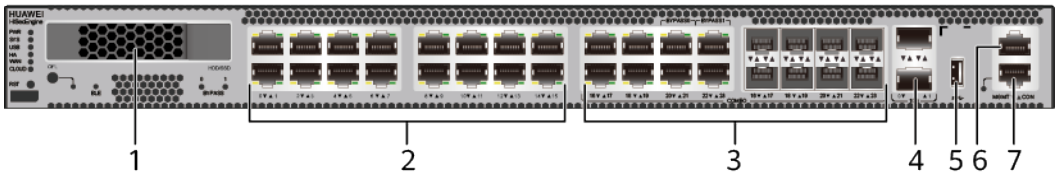
| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

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| Silkscreen | Name | Description |
| --- | --- | --- |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-56** Ports on the USG6575E-B




| | | | | |
| --- | --- | --- | --- | --- |
| 1. HDD/SSD slot | 2. GE electrical port | 3. Combo port | 4. 10GE optical port | 5. USB port<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB 2.0 |
| 6. Console port | 7. MGMT port | - | - | - |

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**Table 3-59** Ports on the USG6575E-B

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports are logic ports. One combo port can work as a GE electrical port or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23.<br>Among the four GE ports from GigabitEthernet0/0/20 to GigabitEthernet0/0/23, GigabitEthernet0/0/20 and GigabitEthernet0/0/21 are a pair of two bypass interfaces, and GigabitEthernet0/0/22 and GigabitEthernet0/0/23 are another pair. When the interfaces of each bypass interface pair work at Layer | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | 2, they can form an electrical bypass link.By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements.<br>**NOTE**<br>A bypass link can be established only when GigabitEthernet0/0/20 to GigabitEthernet0/0/23 work as electrical ports.<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/1. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules**<br>**NOTE**<br>In V600R007C20SPC300 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. |
| USB port | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

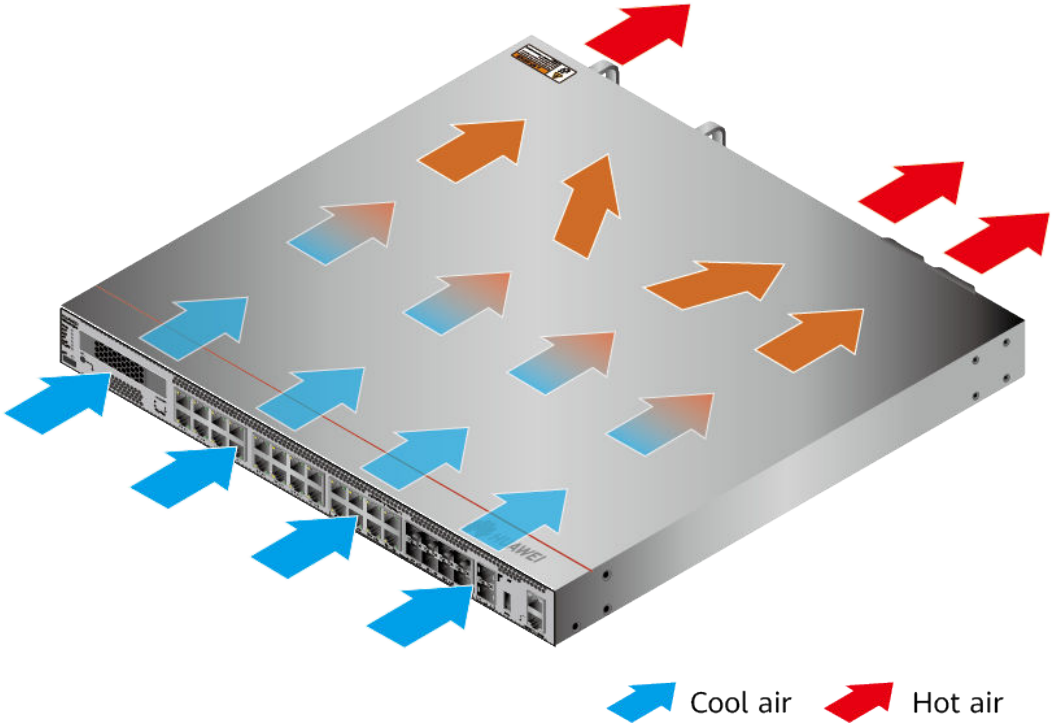
The power supply system has one PAC150S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of two FAN-023A-B modules working in 1+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-57** System air flow of the USG6575E-B




## Technical Specifications

**Table 3-60** Technical specifications of the USG6575E-B

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 175 mm x 650 mm x 550 mm (6.89 in. x 25.59 in. x 21.65 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.25 kg (18.19 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 6.25 kg (13.78 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 8 GB DDR4 ECC memory |
| NOR Flash | 64MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 61.7 W |
| Typical heat dissipation [BTU/hour] | 210.52 BTU/hour |
| Maximum power consumption [W] | 76.9 W |
| Maximum heat dissipation [BTU/hour] | 262.38 BTU/hour |
| MTBF [years] | 41.89 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6575E-B-AC (02353LKD): Two power modules are supported for 1+1 power redundancy.<br>• USG6575E-B-AC (02353LKD-002): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 3 A/power module |
| Rated output power [W] | 150W/12V |
| Maximum output power [W] | 150 W/power module |
| Types of fans | Pluggable |

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| Item | Specification |
| --- | --- |
| Number of fan modules | 2 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6575E-B-AC (02353LKD): The fan module locates at the air exhaust of the system.<br>• USG6575E-B-AC (02353LKD-002): Absorbing cold air into the device |
| Airflow direction | Air inlet from front and air outlet from rear |
| Redundant fans | 1+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

**NOTE**

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

## 3.1.11 USG6580E

## Overview

**Table 3-61** Basic information about the USG6580E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6580E AC Host (2*GE WAN + 8*GE Combo + 16*GE RJ45 + 2*10GE SFP+,1 AC power) | 02352CBS | USG6580E-AC | V600R006C00 |

## Appearance

**Figure 3-58** Appearance of the USG6580E (front view)

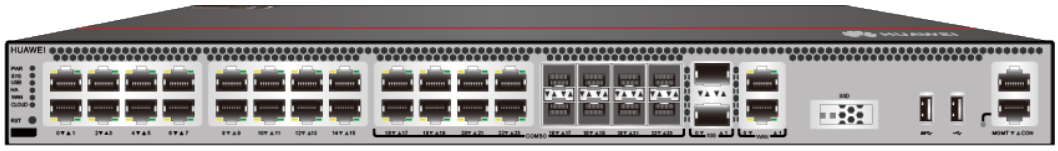
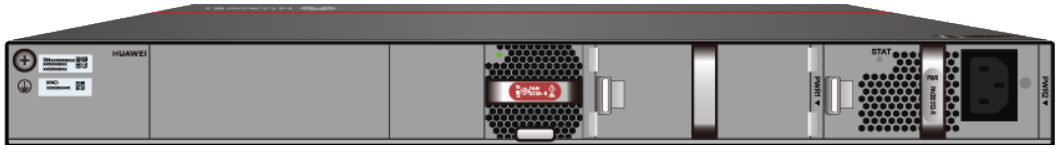
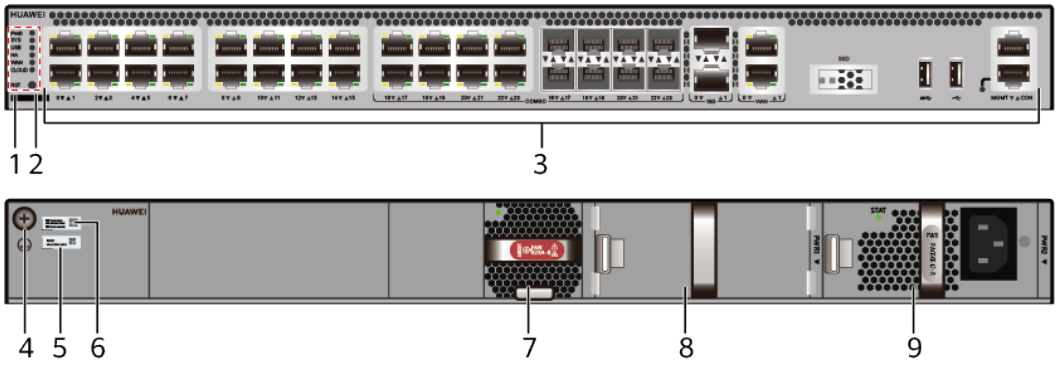


**Figure 3-59** Appearance of the USG6580E (rear view)



## Components

**Figure 3-60** Components of the USG6580E




| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| --- | --- | --- | --- | --- |

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| | | | | |
|---|---|---|---|---|
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

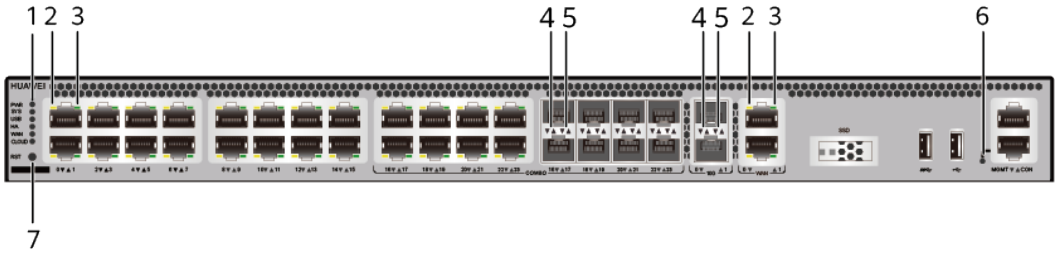
**Table 3-62** Component functions

| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-61** Indicators and buttons on the USG6580E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-63** Indicators on the USG6580E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

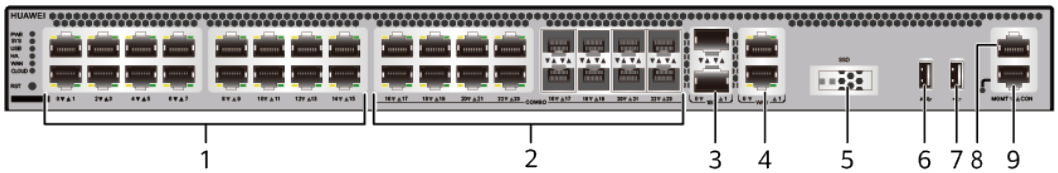
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**Table 3-64** Buttons on the USG6580E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-62** Ports on the USG6580E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. GE electrical port | 2. Combo port | 3. 10GE optical port | 4. WAN port | 5. SSD card slot |
| 6. USB 3.0 port | 7. USB 2.0 port | 8. Console port | 9. MGMT port | - |

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**Table 3-65** Ports on the USG6580E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **1Gbps eSFP Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1. Not compatible with GE optical modules. | **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

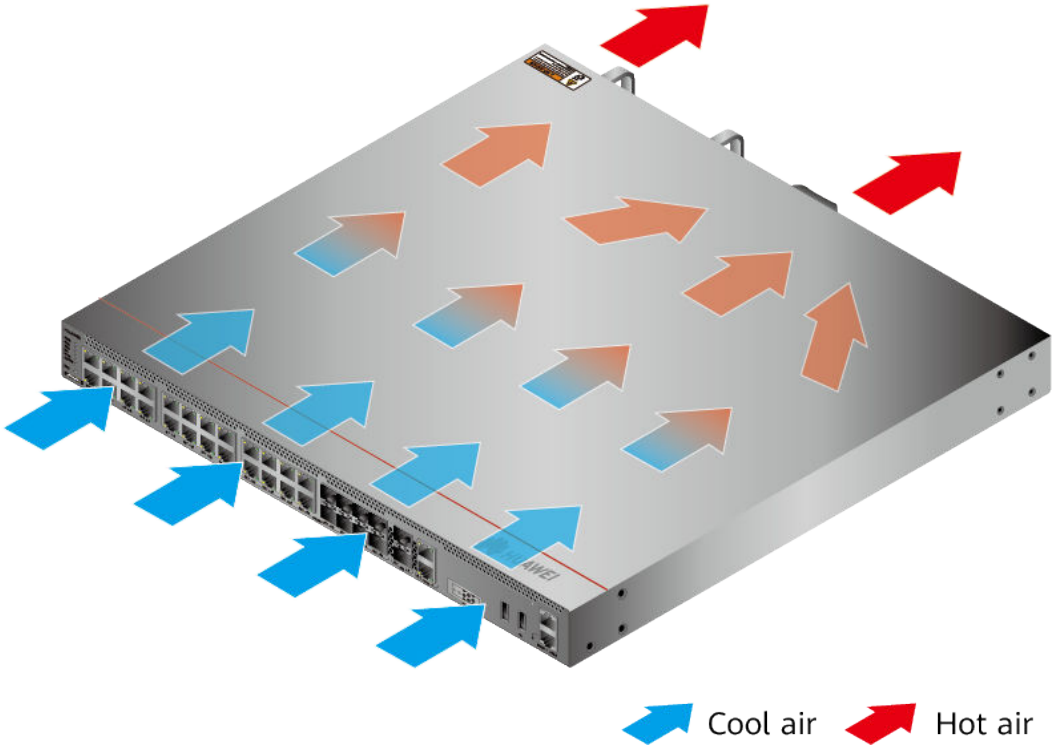
The power supply system has one PAC60S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-63** System air flow of the USG6580E




## Technical Specifications

**Table 3-66** Technical specifications of the USG6580E-AC

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.1 kg (17.86 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 5.71 kg (12.59 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 32.4 W |
| Maximum power consumption [W] | 46.4 W |
| Maximum heat dissipation [BTU/hour] | 158.7 BTU/hour |
| MTBF [years] | 45.56 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.12 USG6585E

## Overview

**Table 3-67** Basic information about the USG6585E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6585E AC Host (2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353ADA | USG6585E-AC | V600R007C00 |
| USG6585E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353ADA-001 | USG6585E-AC | V600R007C20SPC500 |
| USG6585E AC Host(2*GE WAN +8*GE Combo +2*10GE SFP+,1 AC power) | 02353ADA-002 | USG6585E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The two USB ports of earlier-delivered models are USB 3.0 and USB 2.0 respectively, whereas those of later-delivered models are both USB 2.0. In this document, earlier-delivered models are used as examples.

**Figure 3-64** Appearance of the USG6585E (front view)

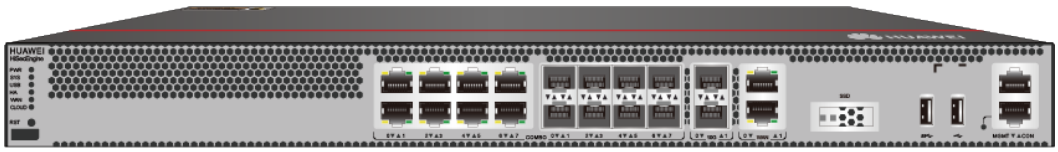
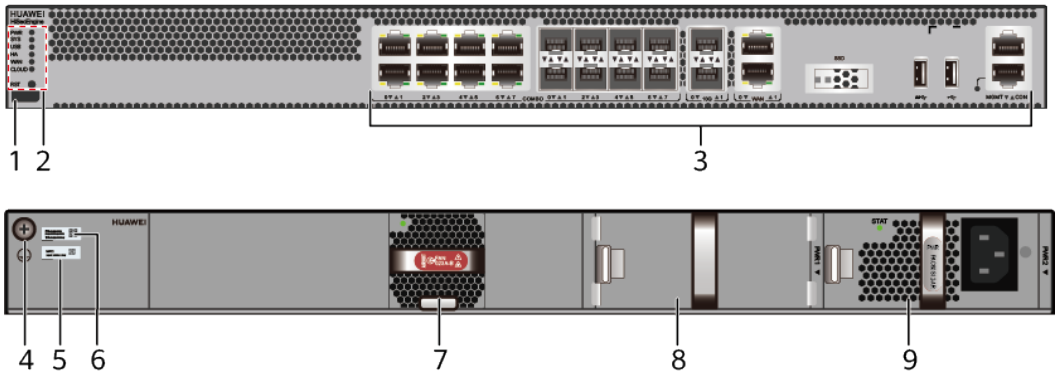


**Figure 3-65** Appearance of the USG6585E (rear view)



## Components

**Figure 3-66** Components of the USG6585E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. Fixed interface area | 4. Protective ground terminal | 5. MAC label |
| 6. SN label | 7. Fan module | 8. Optional power module slot | 9. Power module | - |

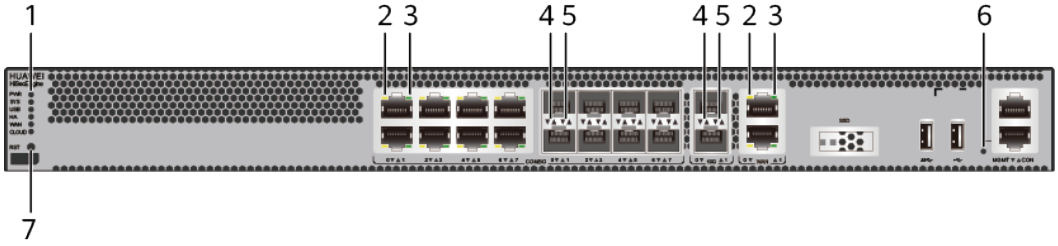
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**Table 3-68** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The slot number of the fan module is 4. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-67** Indicators and buttons on the USG6585E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. GE electrical port ACT indicator | 3. GE electrical port LINK indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-69** Indicators on the USG6585E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

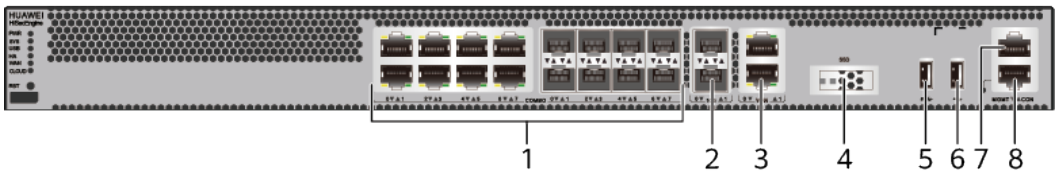
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**Table 3-70** Buttons on the USG6585E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

## Ports

**Figure 3-68** Ports on the USG6585E




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| | | | | |
|---|---|---|---|---|
| 1. Combo port | 2. 10GE optical port | 3. WAN port | 4. SSD card slot | 5. USB 3.0 port or USB0<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB0 (USB 2.0) |
| 6. USB 2.0 port or USB1<br>**NOTE**<br>• Earlier-delivered models: USB 2.0<br>• Later-delivered models: USB1 (USB 2.0) | 7. Console port | 8. MGMT port | - | - |

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**Table 3-71** Ports on the USG6585E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (0 to 7) | RJ45 + SFP | Combo ports. Combo ports are logic ports. One combo port can work as a GE electrical interface or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/7. By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements. | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | **NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |
| 10GE optical ports (0 to 1) | SFP+ | 2 10GE optical ports, numbered from XGigabitEthernet 0/0/0 to XGigabitEthernet 0/0/1.<br>**NOTE**<br>In V600R007C20 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. By default, the rate of 10GE optical ports are 10GE. You can run the set device port-config-mode [ 10ge \| ge ] command to set the rate of 10GE optical ports are to 10GE or GE as required. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| WAN ports (0 to 1) | RJ45 | 2 10/100/1000M autosensing Ethernet electrical ports, numbered from WAN 0/0/0 to WAN 0/0/1.<br>The device is connected to the WAN or cloud management center using a network cable. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| SSD card slot | - | The M.2 module is inserted to record logs and reports in real time. The M.2 module is optional. You can purchase the M.2 module as required. | • **Hard Disk Unit M.2-SATA240G-A**<br>• **Hard Disk Unit M.2-SATA960G-A** |
| USB 3.0 port or USB0 | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| USB 2.0 port or USB1 | USB Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

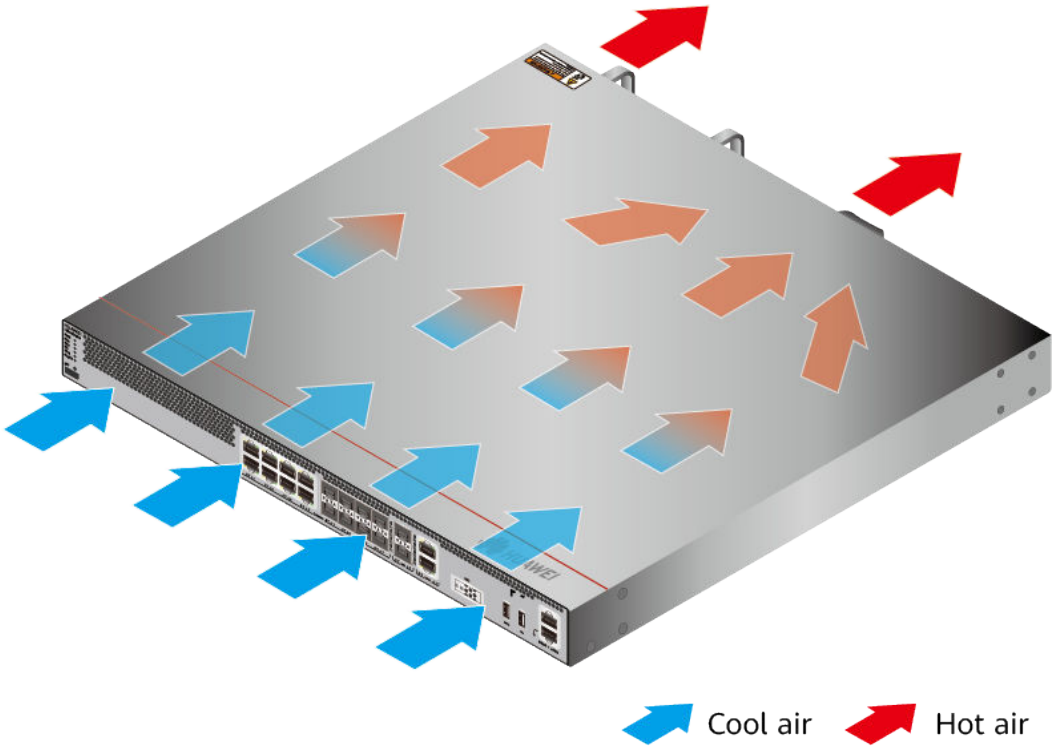
- Early delivery: The power supply system has one PAC60S12-AR, PAC60S12-R, or PAC600S12-CB AC power module. Two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC60S12-AR AC power module. Two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system uses one FAN-023A-B module to dissipate heat for the system. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-69** System air flow of the USG6585E




## Technical Specifications

**Table 3-72** Technical specifications of the USG6585E

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • USG6585E-AC (02353ADA):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6585E-AC (02353ADA-001):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6585E-AC (02353ADA-002):<br>– Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 220 mm x 550 mm x 530 mm (8.66 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | • USG6585E-AC (02353ADA): 8.75 kg (19.29 lb)<br>• USG6585E-AC (02353ADA-001): 7.93 kg (17.48 lb)<br>• USG6585E-AC (02353ADA-002): 7.93 kg (17.48 lb) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | • USG6585E-AC (02353ADA): 5.8 kg (12.79 lb)<br>• USG6585E-AC (02353ADA-001): 5.95 kg (13.12 lb)<br>• USG6585E-AC (02353ADA-002): 5.95 kg (13.12 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 4 GB DDR4 ECC memory |
| NOR Flash | 32 MB<br>**NOTE**<br>The NOR flash memory size of devices delivered later is 64 MB. You can run the display version command to check the NOR flash memory size. |
| NAND Flash | 2 GB |
| Hard disk | Optional. M.2 SSD (240GB/960GB), hot-swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6585E-AC (02353ADA): 35 W<br>• USG6585E-AC (02353ADA-001): 25 W<br>• USG6585E-AC (02353ADA-002): 25 W |
| Typical heat dissipation [BTU/hour] | • USG6585E-AC (02353ADA): 119.42 BTU/hour<br>• USG6585E-AC (02353ADA-001): 85.4 BTU/hour<br>• USG6585E-AC (02353ADA-002): 85.4 BTU/hour |
| Maximum power consumption [W] | • USG6585E-AC (02353ADA): 40 W<br>• USG6585E-AC (02353ADA-001): 45.3 W<br>• USG6585E-AC (02353ADA-002): 45.3 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6585E-AC (02353ADA): 136.8 BTU/hour<br>• USG6585E-AC (02353ADA-001): 154.4 BTU/hour<br>• USG6585E-AC (02353ADA-002): 154.4 BTU/hour |
| MTBF [years] | 68.98 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999997 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6585E-AC (02353ADA): Two power modules are supported for 1+1 power redundancy.<br>• USG6585E-AC (02353ADA-001): Two power modules are supported for 1+1 power redundancy.<br>• USG6585E-AC (02353ADA-002): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 2 A/power module |
| Rated output power [W] | 60W/12V |
| Maximum output power [W] | 60 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 1 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6585E-AC (02353ADA): The fan module locates at the air exhaust of the system.<br>• USG6585E-AC (02353ADA-001): The fan module locates at the air exhaust of the system.<br>• USG6585E-AC (02353ADA-002): Absorbing cold air into the device |

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| Item | Specification |
| --- | --- |
| Airflow direction | Front-to-back airflow |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |
| Long-term operating temperature [°C(°F)] | 0°C to 45°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | 0 m to 5000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.

# 3.1.13 USG6605E-B

## Overview

**Table 3-73** Basic information about the USG6605E-B

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6605E-B AC Host (16*GE RJ45 + 8*GE Combo +2*10GE SFP+,1 AC POWER) | 02353PSK | USG6605E-B-AC | V600R007C20 |
| USG6605E-B AC Host(16*GE RJ45+8*GE Combo +2*10GE SFP+,1 AC Power) | 02353PSK-001 | USG6605E-B-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The standard USB port of the earlier-delivered models is USB 3.0, and that of the later-delivered models is USB 2.0. This document uses a model with a USB 3.0 port as an example.

**Figure 3-70** Appearance of the USG6605E-B (front view)

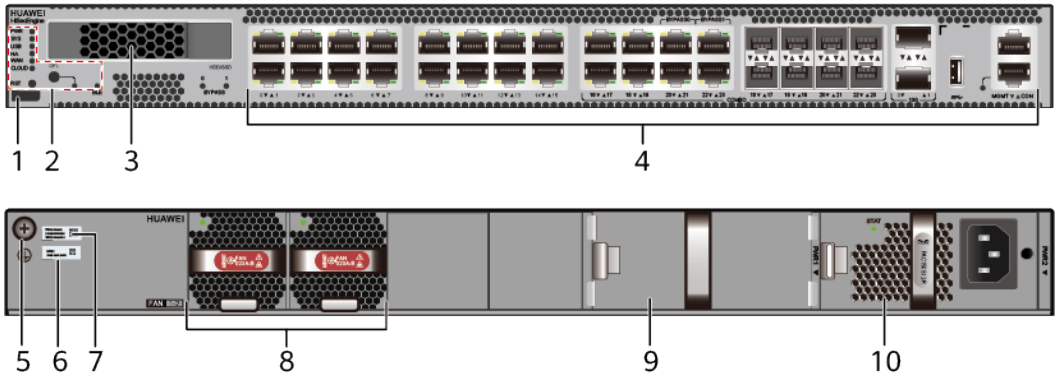


**Figure 3-71** Appearance of the USG6605E-B (rear view)



## Components

**Figure 3-72** Components of the USG6605E-B




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

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**Table 3-74** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN2, and the corresponding slot numbers are 4 and 5 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-73** Indicators and buttons on the USG6605E-B




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. BLE indicator | 4. BYPASS 0 indicator | 5. BYPASS 1 indicator |
| 6. GE electrical port ACT indicator | 7. GE electrical port LINK indicator | 8. Optical port ACT indicator | 9. Optical port LINK indicator | 10. MGMT port indicator |
| 11. RST button | 12. OFL button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-75** Indicators on the USG6605E-B

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Hot backup, managing the master device |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Hot backup, managing the slave device |
| | | Red | Steady on | Dual-system hot backup is faulty. |
| | | - | Off | The dual-system hot backup function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The hard disk cannot be removed. |
| BLE | Bluetooth indicator | - | Off | Reserved function. This function is not enabled. |
| Bypass | Bypass indicator 0 | Green | Steady on | Device is powered on. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in protection state. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Device is powered on. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in working state. |
| | | - | Off | Device is powered off. The bypass formed by GigabitEthernet0/0/20 and GigabitEthernet0/0/21 is in protection state. |
| Bypass | Bypass indicator 1 | Green | Steady on | Device is powered on. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in protection state. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Device is powered on. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in working state. |
| | | - | Off | Device is powered off. The bypass formed by GigabitEthernet0/0/22 and GigabitEthernet0/0/23 is in protection state. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

**Table 3-76** Buttons on the USG6605E-B

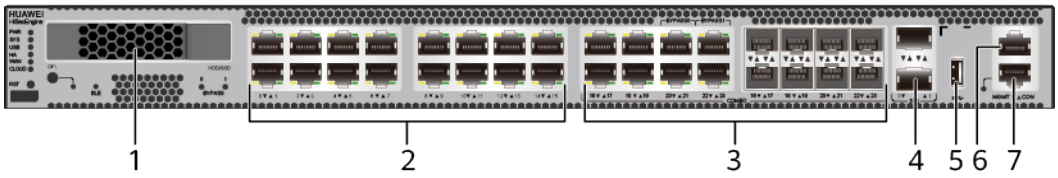
| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |

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| Silkscreen | Name | Description |
| --- | --- | --- |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-74** Ports on the USG6605E-B




| | | | | |
| --- | --- | --- | --- | --- |
| 1. HDD/SSD slot | 2. GE electrical port | 3. Combo port | 4. 10GE optical port | 5. USB port<br>**NOTE**<br>• Earlier-delivered models: USB 3.0<br>• Later-delivered models: USB 2.0 |
| 6. Console port | 7. MGMT port | - | - | - |

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**Table 3-77** Ports on the USG6605E-B

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Combo ports (16 to 23) | RJ45 + SFP | Combo ports are logic ports. One combo port can work as a GE electrical port or a GE optical port. Each combo port has only one internal forwarding port. When the electrical port is enabled, the optical port is disabled. When the optical port is enabled, the electrical port is disabled. The electrical and optical ports of a combo port use the same interface view, numbered from GigabitEthernet 0/0/16 to GigabitEthernet 0/0/23.<br>Among the four GE ports from GigabitEthernet0/0/20 to GigabitEthernet0/0/23, GigabitEthernet0/0/20 and GigabitEthernet0/0/21 are a pair of two bypass interfaces, and GigabitEthernet0/0/22 and GigabitEthernet0/0/23 are another pair. When the interfaces of each bypass interface pair work at Layer | • **Ethernet Cable**<br>• **100Mbps SFP Optical Modules**<br>• **1Gbps eSFP Optical Modules**<br>**NOTE**<br>In V600R007C20SPC500 and later versions, Combo ports support 100Mbps SFP Optical Modules. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| | | 2, they can form an electrical bypass link.By default, the combo port is used as an electrical port. You can use the combo enable { copper \| fiber } command to set the working mode of combo ports according to network requirements.<br>**NOTE**<br>A bypass link can be established only when GigabitEthernet0/0/20 to GigabitEthernet0/0/23 work as electrical ports.<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | |

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 1) | SFP+ | 2 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/1. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules**<br>**NOTE**<br>In V600R007C20SPC300 and later versions, 10GE optical ports support 1Gbps SFP Copper Modules and 1Gbps eSFP Optical Modules. |
| USB port | USB Type A | USB port allows you to insert an USB flash drive for system software upgrades. For details on upgrades through USB flash drives, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

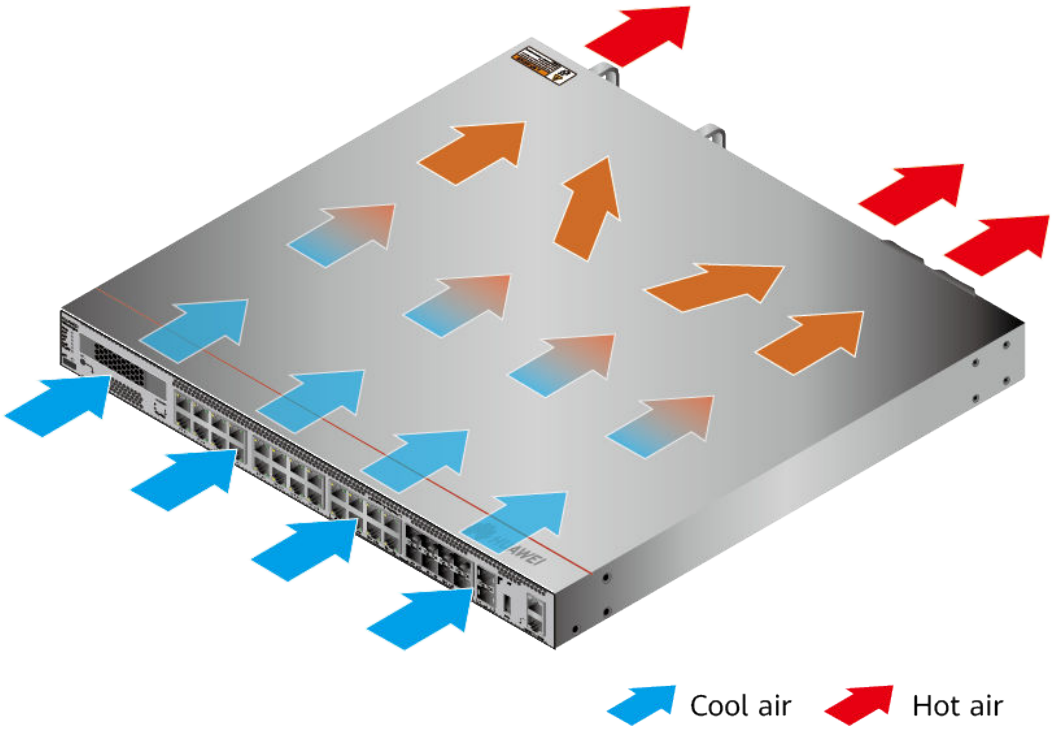
The power supply system has one PAC150S12-R AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of two FAN-023A-B modules working in 1+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-75** System air flow of the USG6605E-B




## Technical Specifications

**Table 3-78** Technical specifications of the USG6605E-B

| Item | Specification |
| --- | --- |
| Installation Type | • Rack<br>• Work bench |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • USG6605E-B-AC (02353PSK):<br>– Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.)<br>• USG6605E-B-AC (02353PSK-001):<br>– Typical dimensions (the depth excludes the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>– Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 175 mm x 650 mm x 550 mm (6.89 in. x 25.59 in. x 21.65 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 8.25 kg (18.19 lb) |
| Weight without packaging [kg(lb)] | 6.25 kg (13.78 lb) |
| CPU | 1 CPU, 4 cores/CPU, up to 1.4 GHz |
| Memory | 8 GB DDR4 ECC memory |
| NOR Flash | 64MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 61.7 W |

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| Item | Specification |
| --- | --- |
| Typical heat dissipation [BTU/hour] | 210.52 BTU/hour |
| Maximum power consumption [W] | 76.9 W |
| Maximum heat dissipation [BTU/hour] | 262.38 BTU/hour |
| MTBF [years] | 41.89 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999995 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | • USG6605E-B-AC (02353PSK): Two power modules are supported for 1+1 power redundancy.<br>• USG6605E-B-AC (02353PSK-001): Dual power modules can be purchased to form 1+1 redundancy backup. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 264 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 3 A/power module |
| Rated output power [W] | 150W/12V |
| Maximum output power [W] | 150 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 2 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6605E-B-AC (02353PSK): The fan module locates at the air exhaust of the system.<br>• USG6605E-B-AC (02353PSK-001): Absorbing cold air into the device |
| Airflow direction | Air inlet from front and air outlet from rear |
| Redundant fans | 1+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤55dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.14 USG6610E

## Overview

**Table 3-79** Basic information about the USG6610E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6610E AC Host (12*GE RJ45 + 8*GE SFP+ 4*10GE SFP+,1 AC power) | 02352HMD | USG6610E-AC | V600R007C20SPC100 |
| USG6610E AC Host (12*GE RJ45 + 8*GE SFP+ 4*10GE SFP+,1 AC power) | 02352HMD-001 | USG6610E-AC | V600R007C20SPC300 |

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| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6610E AC Host(12*GE RJ45+8*GE SFP +4*10GE SFP+,1 AC power) | 02352HMD-002 | USG6610E-AC | V600R007C20SPC600 |

## Appearance

NOTE

The model with the BOM number containing "-001" and "-002" have one more OFL indicator than the model with the BOM number not containing "-001" or "-002". The silkscreen of the device contains "HiSecEngine", and the distribution of heat dissipation holes on the front of one model slightly differ from that of another model. This document uses the model whose BOM number contains "-001" as an example.

**Figure 3-76** Appearance of the USG6610E (front view)

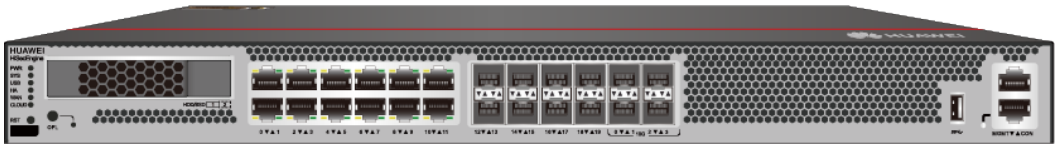
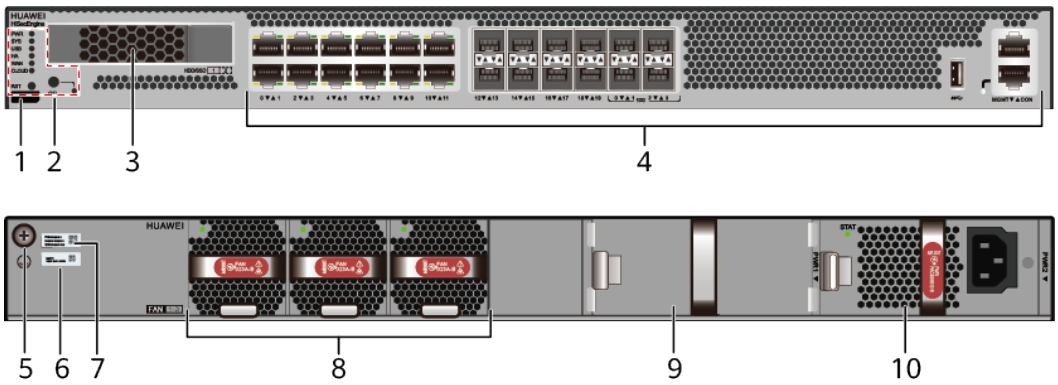


**Figure 3-77** Appearance of the USG6610E (rear view)



## Components

**Figure 3-78** Components of the USG6610E




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| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

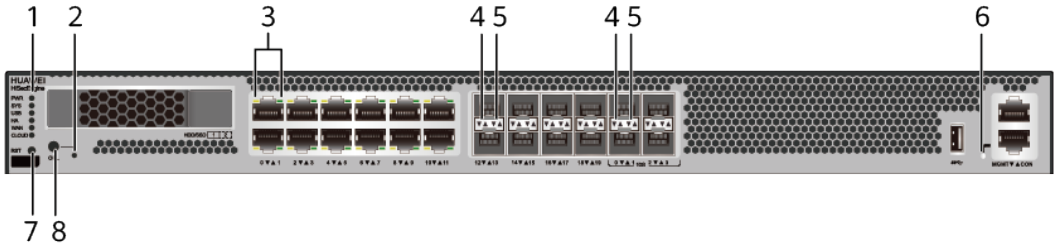
**Table 3-80** Component functions

| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-79** Indicators and buttons on the USG6610E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator<br>**NOTE**<br>Only the models whose BOM number contain "-001" and "-002" support the OFL indicator. | 3. GE electrical port indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | 8. OFL button | - | - |

**NOTE**

- GE electrical interface indicators from left to right correspond to GE electrical interfaces in ascending order. For example, the first indicator on the panel corresponds to GigabitEthernet0/0/0 and the second indicator corresponds to GigabitEthernet0/0/1.
- Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-81** Indicators on the USG6610E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | GE electrical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

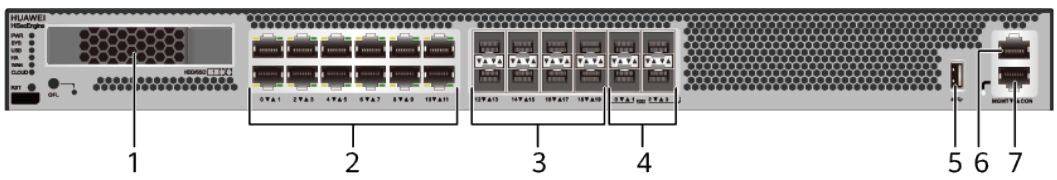
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**Table 3-82** Buttons on the USG6610E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-80** Ports on the USG6610E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. GE optical port | 4. 10GE optical port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-83** Ports on the USG6610E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 11) | RJ45 | 12 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/11.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| GE optical ports (12 to 19) | SFP | 8 GE autosensing Ethernet optical ports, numbered from GigabitEthernet0/0/12 to GigabitEthernet0/0/19. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules** |
| 10GE optical ports (0 to 3) | SFP+ | 4 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

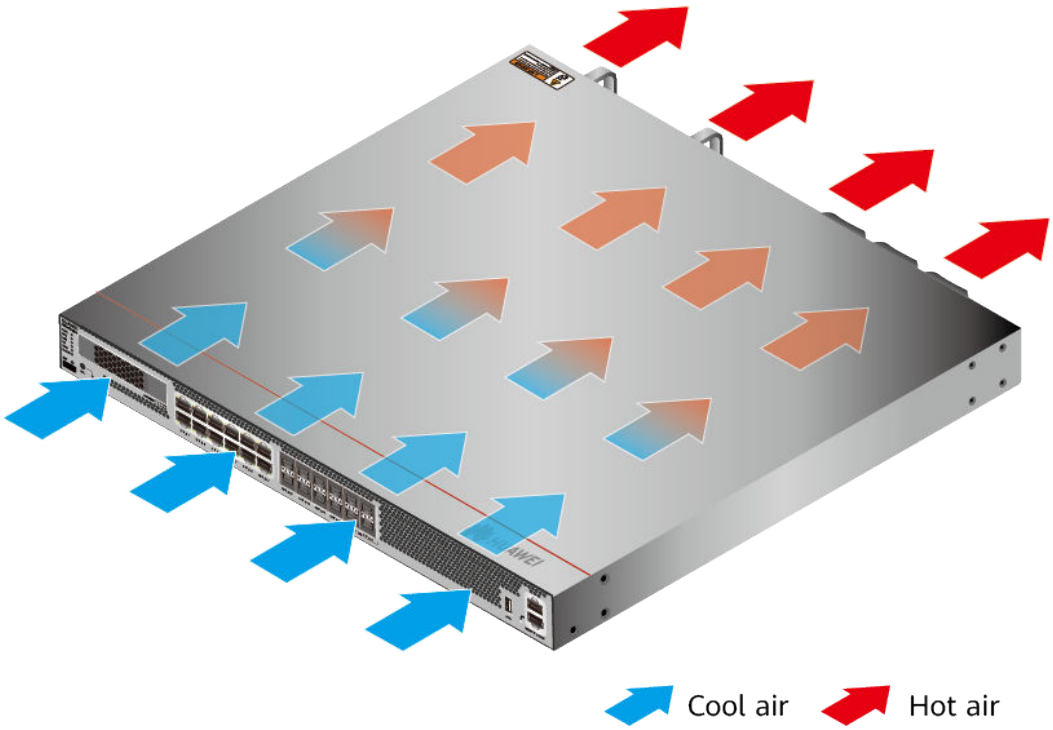
- Early delivery: The power supply system has one PAC600S12-B AC power module, but two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-81** System air flow of the USG6610E




## Technical Specifications

**Table 3-84** Technical specifications of the USG6610E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |

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| Item | Specification |
| --- | --- |
| Weight with packaging [kg(lb)] | • USG6610E-AC (02352HMD): 10.1 kg (22.27 lb)<br>• USG6610E-AC (02352HMD-001): 9.4 kg (20.72 lb)<br>• USG6610E-AC (02352HMD-002): 9.4 kg (20.72 lb) |
| Weight without packaging [kg(lb)] | • USG6610E-AC (02352HMD): 7.1 kg (15.65 lb)<br>• USG6610E-AC (02352HMD-001): 6.7 kg (14.77 lb)<br>• USG6610E-AC (02352HMD-002): 6.7 kg (14.77 lb) |
| CPU | 1 CPU, 8 cores/CPU, up to 2.3 GHz |
| Memory | 8 GB DDR4 ECC memory |
| Storage | 32GB |
| NOR Flash | • USG6610E-AC (02352HMD): 32 MB<br>• USG6610E-AC (02352HMD-001): 64MB<br>• USG6610E-AC (02352HMD-002): 64MB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 116.49 W |
| Typical heat dissipation [BTU/hour] | 397.46 BTU/hour |
| Maximum power consumption [W] | • USG6610E-AC (02352HMD): 161.79 W<br>• USG6610E-AC (02352HMD-001): 139 W<br>• USG6610E-AC (02352HMD-002): 139 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6610E-AC (02352HMD): 552.02 BTU/hour<br>• USG6610E-AC (02352HMD-001): 474.27 BTU/hour<br>• USG6610E-AC (02352HMD-002): 474.27 BTU/hour |
| MTBF [years] | 29.34 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999992 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 9 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |

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| Item | Specification |
| --- | --- |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.15 USG6615E

## Overview

**Table 3-85** Basic information about the USG6615E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6615E AC Host (16*GE RJ45 + 6*GE SFP + 6*10GE SFP+,16G Memory,1 AC POWER) | 02352RQL | USG6615E-AC | V600R007C00 |

## Appearance

**Figure 3-82** Appearance of the USG6615E (front view)

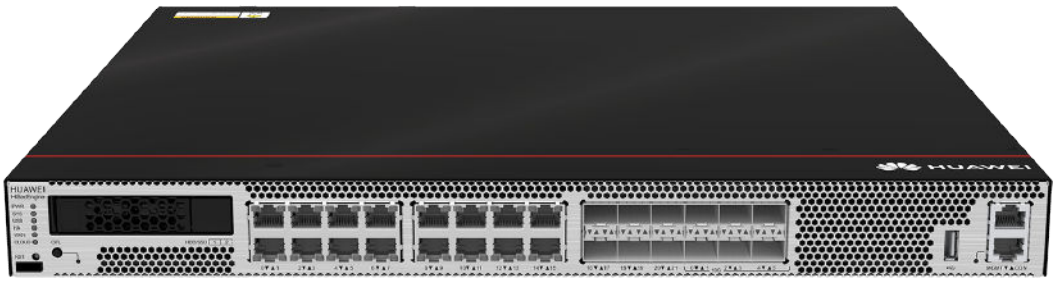
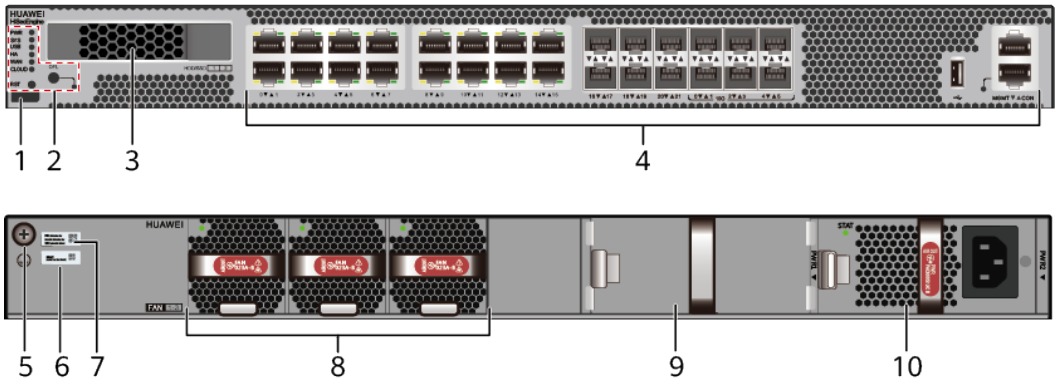


**Figure 3-83** Appearance of the USG6615E (rear view)



## Components

**Figure 3-84** Components of the USG6615E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

**Table 3-86** Component functions

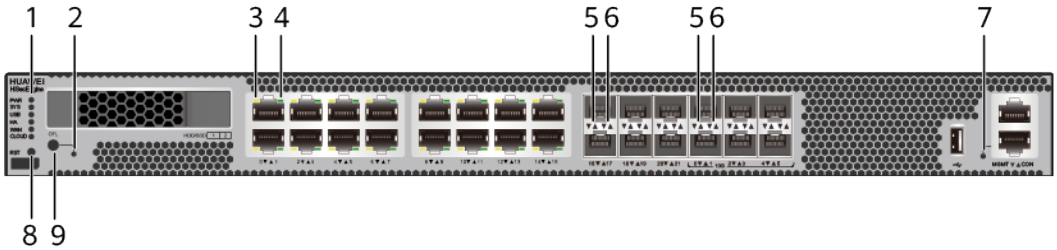
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |

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| Name | Description |
| --- | --- |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-85** Indicators and buttons on the USG6615E



| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. GE electrical port ACT indicator | 4. GE electrical port LINK indicator | 5. Optical port ACT indicator |
| 6. Optical port LINK indicator | 7. MGMT port indicator | 8. RST button | 9. OFL button | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-87** Indicators on the USG6615E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

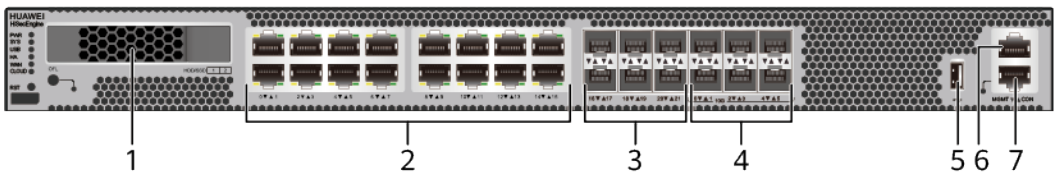
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**Table 3-88** Buttons on the USG6615E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-86** Ports on the USG6615E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. GE optical port | 4. 10GE optical port | 5. USB 2.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-89** Ports on the USG6615E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| GE optical ports (16 to 21) | SFP | 6 GE autosensing Ethernet optical ports, numbered from GigabitEthernet0/0/16 to GigabitEthernet0/0/21. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules** |
| 10GE optical ports (0 to 5) | SFP+ | 6 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/5. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

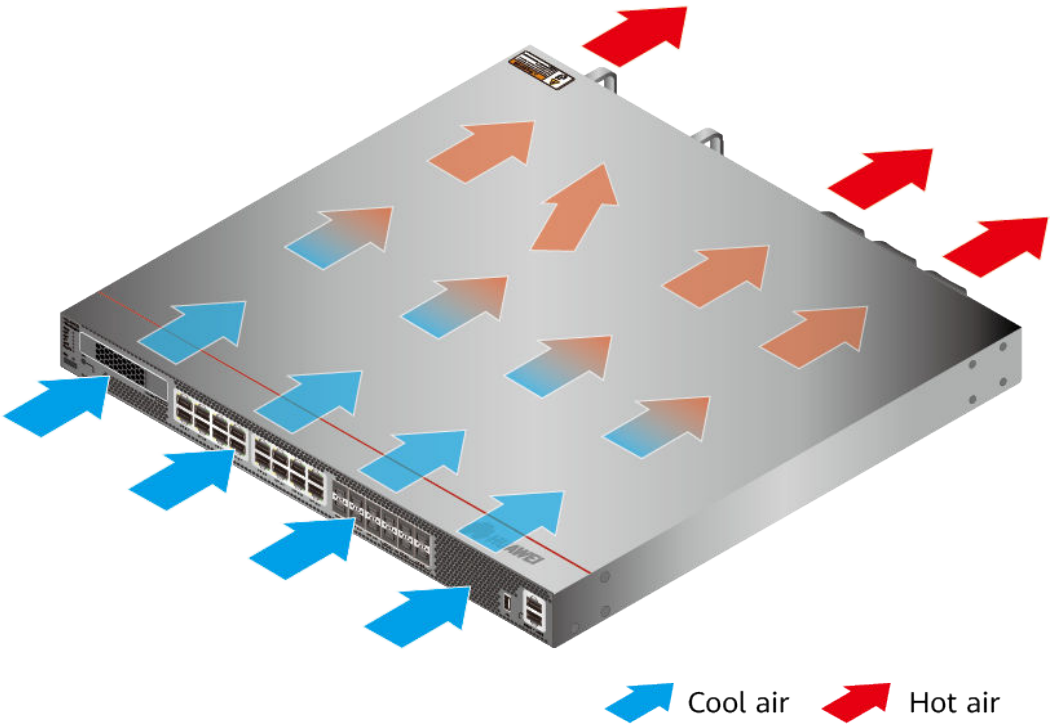
The power supply system has one PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-87** System air flow of the USG6615E




## Technical Specifications

**Table 3-90** Technical specifications of the USG6615E-AC

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 10.9 kg (24.03 lb) |
| Weight without packaging [kg(lb)] | 7.75 kg (17.09 lb) |

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| Item | Specification |
| --- | --- |
| CPU | 1 CPU, 16 cores/CPU, up to 1.85 GHz |
| Memory | 2 x 8 GB DDR4 ECC memory |
| NOR Flash | 64MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 104.5 W |
| Maximum power consumption [W] | 118 W |
| Maximum heat dissipation [BTU/hour] | 403.56 BTU/hour |
| MTBF [years] | 37.76 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999994 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 8 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |

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| Item | Specification |
| --- | --- |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.16 USG6620E

## Overview

**Table 3-91** Basic information about the USG6620E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6620E AC Host (12*GE RJ45 + 8*GE SFP+ 4*10GE SFP+,1 AC power) | 02352HMF | USG6620E-AC | V600R007C20SPC100 |

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| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6620E AC Host (12*GE RJ45 + 8*GE SFP+ 4*10GE SFP+,1 AC power) | 02352HMF-001 | USG6620E-AC | V600R007C20SPC300 |
| USG6620E AC Host(12*GE RJ45+8*GE SFP +4*10GE SFP+ ,1 AC power) | 02352HMF-002 | USG6620E-AC | V600R007C20SPC600 |

## Appearance

NOTE

The model with the BOM number containing "-001" and "-002" have one more OFL indicator than the model with the BOM number not containing "-001" or "-002". The silkscreen of the device contains "HiSecEngine", and the distribution of heat dissipation holes on the front of one model slightly differ from that of another model. This document uses the model whose BOM number contains "-001" as an example.

**Figure 3-88** Appearance of the USG6620E (front view)

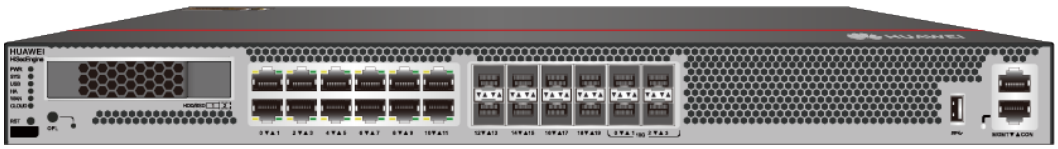
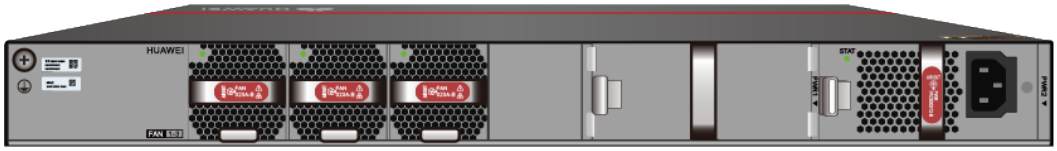
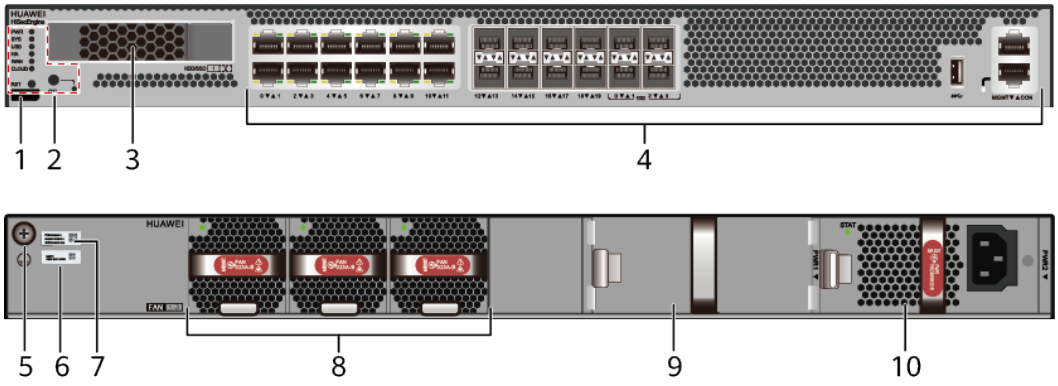


**Figure 3-89** Appearance of the USG6620E (rear view)



## Components

**Figure 3-90** Components of the USG6620E




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| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

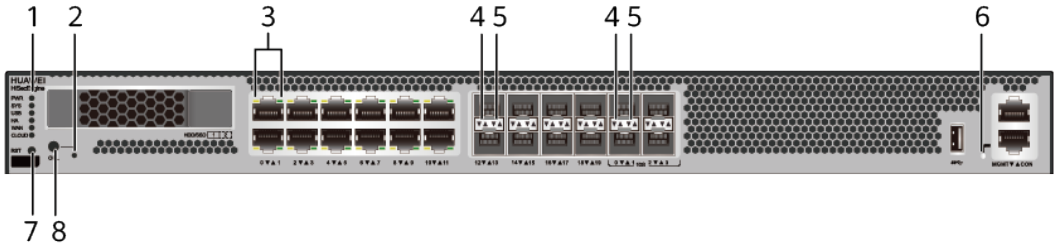
**Table 3-92** Component functions

| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-91** Indicators and buttons on the USG6620E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator<br>**NOTE**<br>Only the models whose BOM number contain "-001" and "-002" support the OFL indicator. | 3. GE electrical port indicator | 4. Optical port ACT indicator | 5. Optical port LINK indicator |
| 6. MGMT port indicator | 7. RST button | 8. OFL button | - | - |

**NOTE**

- GE electrical interface indicators from left to right correspond to GE electrical interfaces in ascending order. For example, the first indicator on the panel corresponds to GigabitEthernet0/0/0 and the second indicator corresponds to GigabitEthernet0/0/1.
- Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-93** Indicators on the USG6620E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | GE electrical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

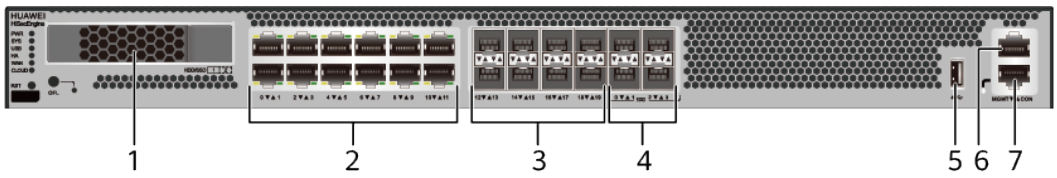
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**Table 3-94** Buttons on the USG6620E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-92** Ports on the USG6620E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. GE optical port | 4. 10GE optical port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-95** Ports on the USG6620E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 11) | RJ45 | 12 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/11.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| GE optical ports (12 to 19) | SFP | 8 GE autosensing Ethernet optical ports, numbered from GigabitEthernet0/0/12 to GigabitEthernet0/0/19. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules** |
| 10GE optical ports (0 to 3) | SFP+ | 4 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

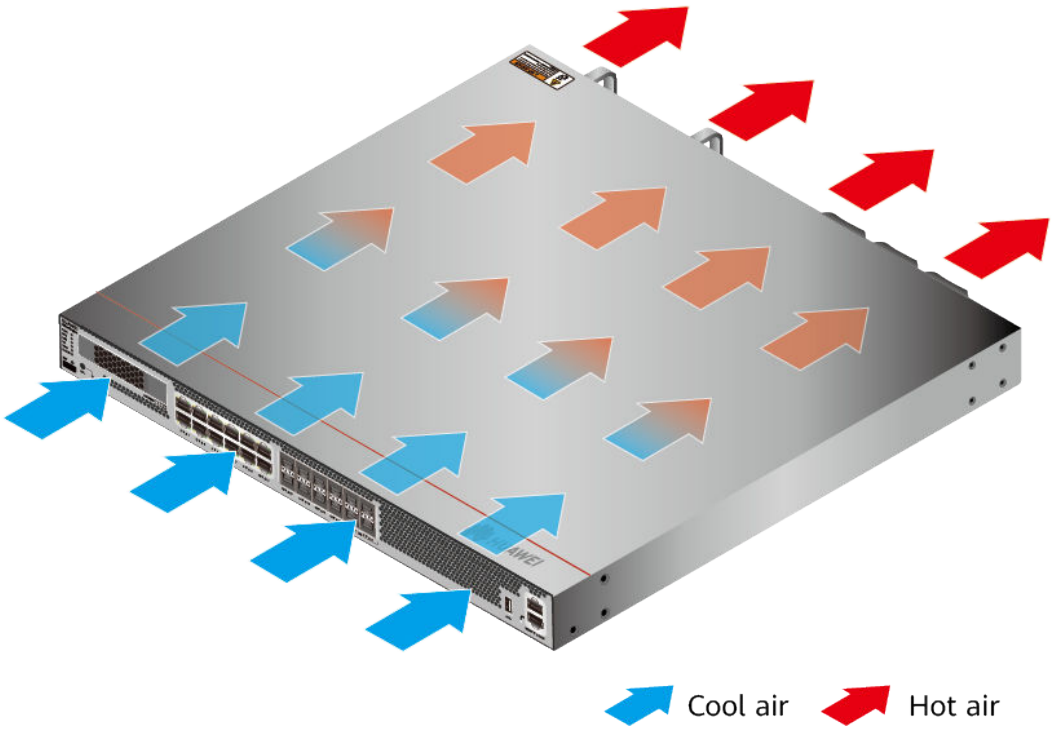
- Early delivery: The power supply system has one PAC600S12-B AC power module, but two power modules can be configured for 1+1 power redundancy.
- Later delivery: The power supply system has one PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-93** System air flow of the USG6620E




## Technical Specifications

**Table 3-96** Technical specifications of the USG6620E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |

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| Item | Specification |
| --- | --- |
| Weight with packaging [kg(lb)] | • USG6620E-AC (02352HMF): 10.1 kg (22.27 lb)<br>• USG6620E-AC (02352HMF-001): 9.4 kg (20.72 lb)<br>• USG6620E-AC (02352HMF-002): 9.4 kg (20.72 lb) |
| Weight without packaging [kg(lb)] | • USG6620E-AC (02352HMF): 7.1 kg (15.65 lb)<br>• USG6620E-AC (02352HMF-001): 6.7 kg (14.77 lb)<br>• USG6620E-AC (02352HMF-002): 6.7 kg (14.77 lb) |
| CPU | 1 CPU, 8 cores/CPU, up to 2.3 GHz |
| Memory | 8 GB DDR4 ECC memory |
| Storage | 32GB |
| NOR Flash | • USG6620E-AC (02352HMF): 32 MB<br>• USG6620E-AC (02352HMF-001): 64MB<br>• USG6620E-AC (02352HMF-002): 64MB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 116.49 W |
| Typical heat dissipation [BTU/hour] | 397.46 BTU/hour |
| Maximum power consumption [W] | • USG6620E-AC (02352HMF): 161.79 W<br>• USG6620E-AC (02352HMF-001): 139 W<br>• USG6620E-AC (02352HMF-002): 139 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6620E-AC (02352HMF): 552.02 BTU/hour<br>• USG6620E-AC (02352HMF-001): 474.27 BTU/hour<br>• USG6620E-AC (02352HMF-002): 474.27 BTU/hour |
| MTBF [years] | 29.34 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999992 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 9 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |

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| Item | Specification |
| --- | --- |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.17 USG6625E

## Overview

**Table 3-97** Basic information about the USG6625E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6625E AC Host (16*GE RJ45 + 6*GE SFP + 6*10GE SFP+,16G Memory,1 AC POWER) | 02352RQN | USG6625E-AC | V600R007C00 |

## Appearance

**Figure 3-94** Appearance of the USG6625E (front view)

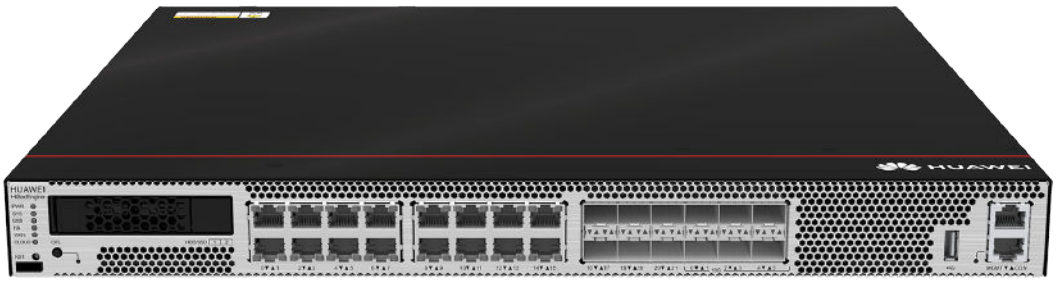
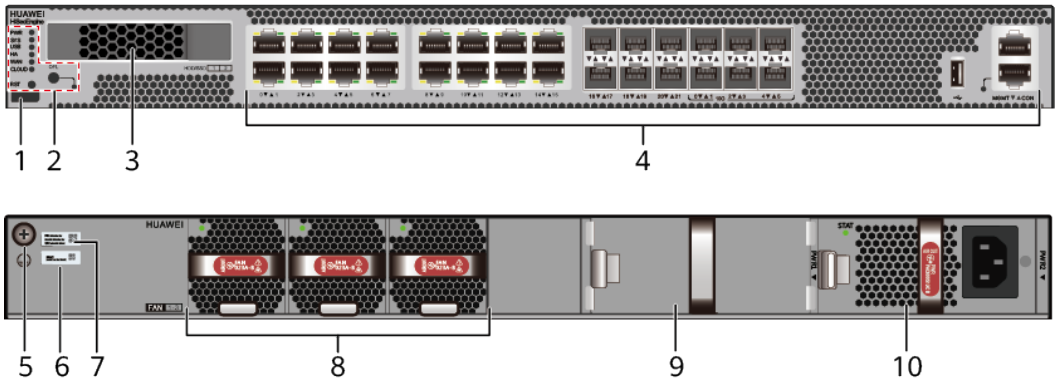


**Figure 3-95** Appearance of the USG6625E (rear view)



## Components

**Figure 3-96** Components of the USG6625E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Optional power module slot | 10. Power module |

**Table 3-98** Component functions

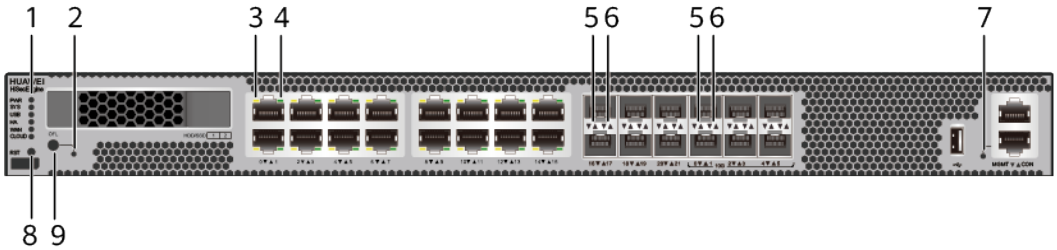
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |

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| Name | Description |
| --- | --- |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. One power module is included in the standard configuration. Two power modules can be configured to achieve 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-97** Indicators and buttons on the USG6625E



| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. GE electrical port ACT indicator | 4. GE electrical port LINK indicator | 5. Optical port ACT indicator |
| 6. Optical port LINK indicator | 7. MGMT port indicator | 8. RST button | 9. OFL button | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-99** Indicators on the USG6625E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | LINK indicator of the optical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of an optical port | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

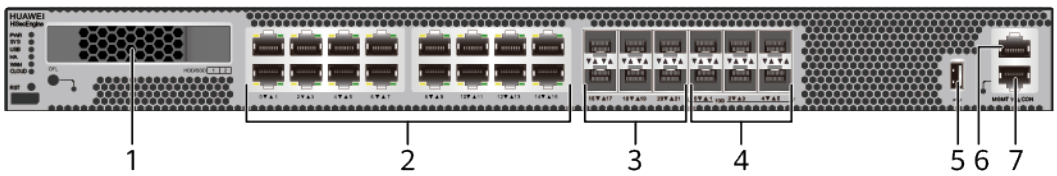
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**Table 3-100** Buttons on the USG6625E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-98** Ports on the USG6625E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. GE optical port | 4. 10GE optical port | 5. USB 2.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-101** Ports on the USG6625E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| GE optical ports (16 to 21) | SFP | 6 GE autosensing Ethernet optical ports, numbered from GigabitEthernet0/0/16 to GigabitEthernet0/0/21. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules** |
| 10GE optical ports (0 to 5) | SFP+ | 6 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/5. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

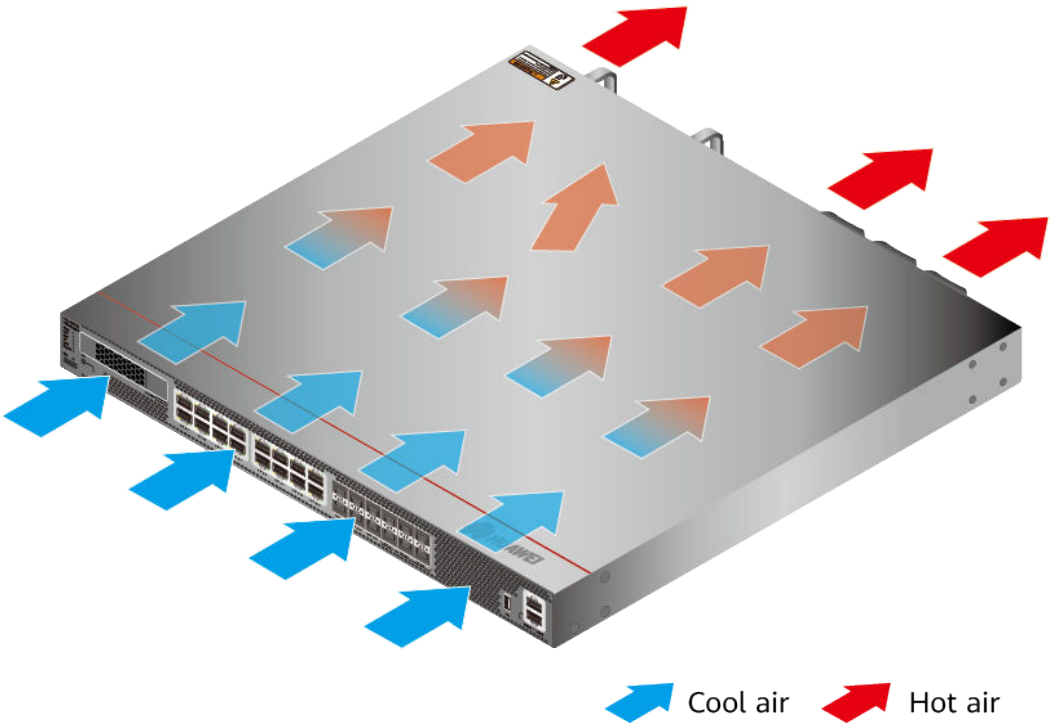
The power supply system has one PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power module, but two power modules can be configured for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-99** System air flow of the USG6625E




## Technical Specifications

**Table 3-102** Technical specifications of the USG6625E-AC

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 10.9 kg (24.03 lb) |
| Weight without packaging [kg(lb)] | 7.75 kg (17.09 lb) |

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| Item | Specification |
| --- | --- |
| CPU | 1 CPU, 16 cores/CPU, up to 1.85 GHz |
| Memory | 2 x 8 GB DDR4 ECC memory |
| NOR Flash | 64MB |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 104.5 W |
| Maximum power consumption [W] | 118 W |
| Maximum heat dissipation [BTU/hour] | 403.56 BTU/hour |
| MTBF [years] | 37.76 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999994 |
| Power supply mode | AC pluggable |
| Number of power modules | 1 |
| Redundant power supply | Two power modules are supported for 1+1 power redundancy. |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 8 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |

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| Item | Specification |
| --- | --- |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.18 USG6630E

## Overview

**Table 3-103** Basic information about the USG6630E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6630E AC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02352FQA | USG6630E-AC | V600R006C00 |

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6630E AC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02352FQA-001 | USG6630E-AC | V600R007C20SPC300 |
| USG6630E AC Host(12*GE RJ45+12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02352FQA-002 | USG6630E-AC | V600R007C20SPC600 |
| USG6630E DC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 DC power) | 02352CKV | USG6630E-DC | V600R006C00 |
| USG6630E DC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 DC power) | 02352CKV-001 | USG6630E-DC | V600R007C20SPC300 |
| USG6630E DC Host(12*GE RJ45+12*10GE SFP+ + 2*40GE QSFP+,2 DC power) | 02352CKV-002 | USG6630E-DC | V600R007C20SPC600 |

## Appearance

NOTE

- The appearance difference between the DC model and AC model is the power module installed on the rear of the device. The following figure uses the AC model as an example.
- The model with the BOM number containing "-001" and "-002" have one more OFL indicator than the model with the BOM number not containing "-001" or "-002". The silkscreen of the device contains "HiSecEngine", and the distribution of heat dissipation holes on the front of one model slightly differ from that of another model. This document uses the model whose BOM number contains "-001" as an example.

**Figure 3-100** Appearance of the USG6630E (front view)

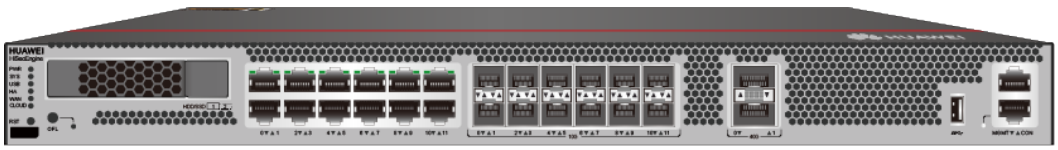
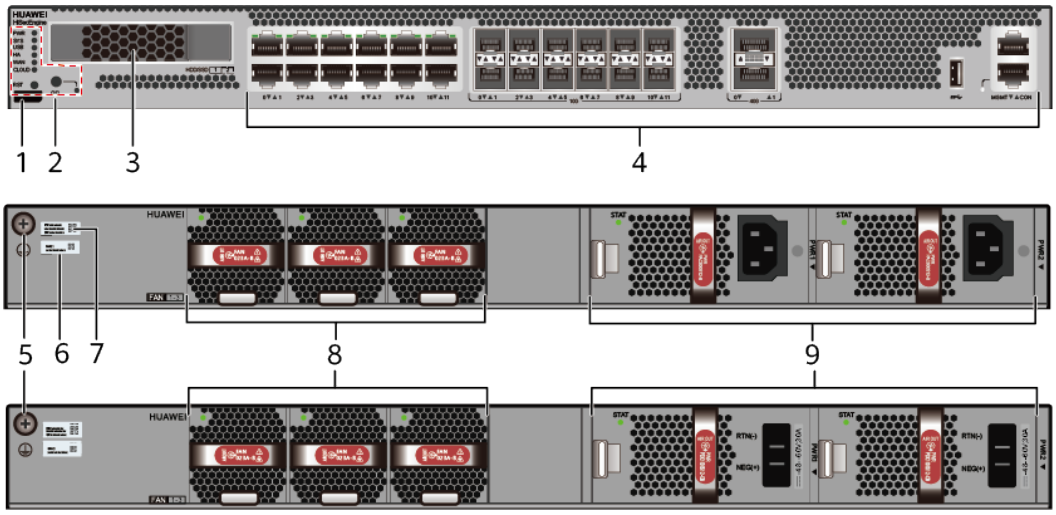


**Figure 3-101** Appearance of the USG6630E (rear view)



## Components

**Figure 3-102** Components of the USG6630E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

**Table 3-104** Component functions

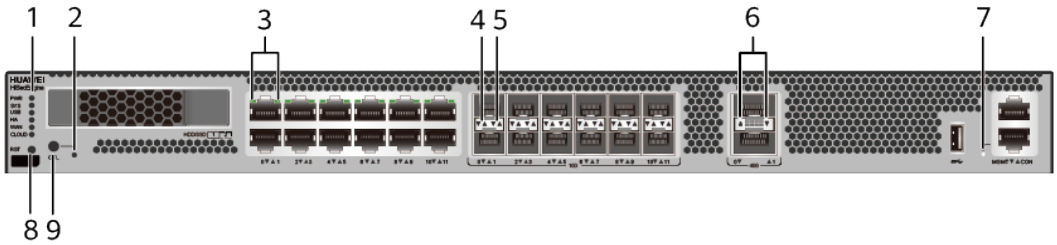
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |

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| Name | Description |
| --- | --- |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | The fan module provides air flow for heat dissipation. The power module supports hot-swapping and can be replaced without interrupting device operation. However, to prevent overheating, do not operate the device without a functioning fan module for more than one minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-103** Indicators and buttons on the USG6630E



| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator<br>**NOTE**<br>Only the models whose BOM number contain "-001" and "-002" support the OFL indicator. | 2. GE electrical port indicator | 3. 10GE optical port ACT indicator | 4. 10GE optical port LINK indicator |
| 5. 40GE optical port indicator | 6. MGMT port indicator | 7. RST button | 8. OFL button | - |

**NOTE**

- GE electrical interface indicators from left to right correspond to GE electrical interfaces in ascending order. For example, the first indicator on the panel corresponds to GigabitEthernet0/0/0 and the second indicator corresponds to GigabitEthernet0/0/1.
- Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-105** Indicators on the USG6630E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | GE electrical port indicator | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

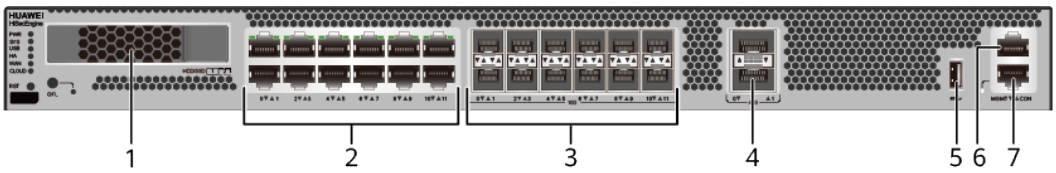
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**Table 3-106** Buttons on the USG6630E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-104** Ports on the USG6630E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. 10GE optical port | 4. 40GE optical port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-107** Ports on the USG6630E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 11) | RJ45 | 12 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/11.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| 10GE optical ports (0 to 11) | SFP+ | 12 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| 40GE optical ports (0 to 1) | QSFP+ | Two 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/1. | **40Gbps QSFP+ Optical Modules** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

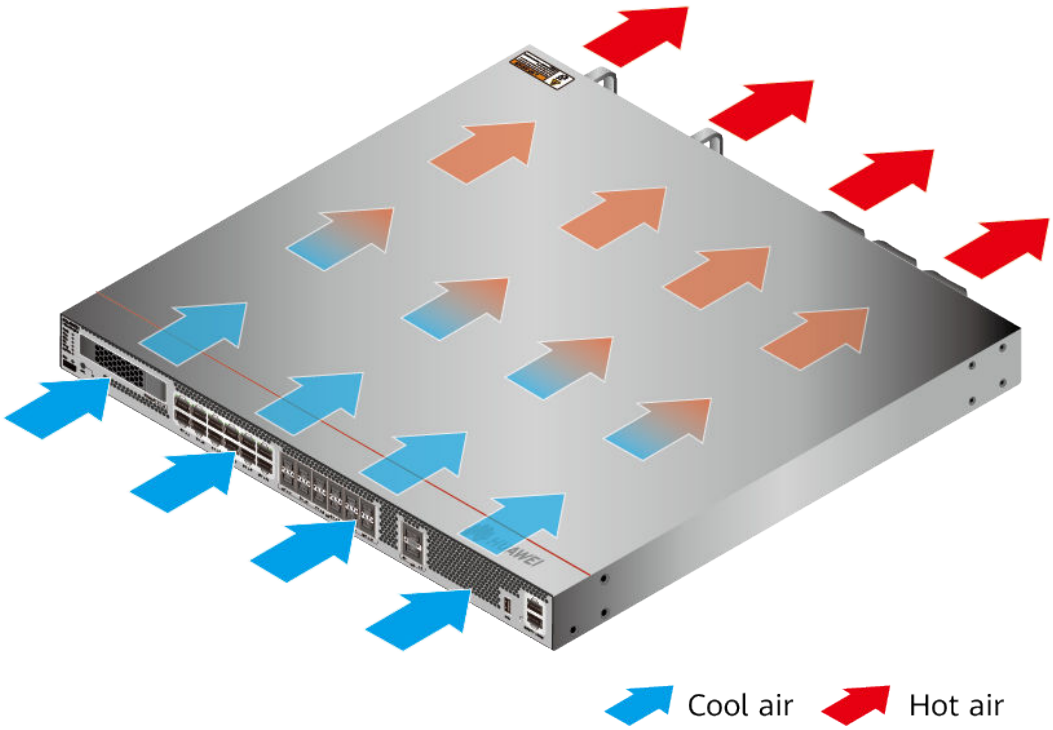
- Early delivery: The power supply system has two PAC600S12-B AC power modules or PDC350S12-B DC power modules for 1+1 power redundancy.
- Later delivery: The power supply system has two PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power modules or PDC1000S12-CB DC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-105** System air flow of the USG6630E




## Technical Specifications

**Table 3-108** Technical specifications of the USG6630E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 215 mm x 550 mm x 530 mm (8.46 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |

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| Item | Specification |
| --- | --- |
| Weight with packaging [kg(lb)] | • USG6630E-AC (02352FQA): 10.9 kg (24.03 lb)<br>• USG6630E-AC (02352FQA-001): 10.2 kg (22.49 lb)<br>• USG6630E-AC (02352FQA-002): 10.2 kg (22.49 lb)<br>• USG6630E-DC (02352CKV): 10.3 kg (22.71 lb)<br>• USG6630E-DC (02352CKV-001): 10.2 kg (22.49 lb)<br>• USG6630E-DC (02352CKV-002): 10.2 kg (22.49 lb) |
| Weight without packaging [kg(lb)] | • USG6630E-AC (02352FQA): 7.9 kg (17.42 lb)<br>• USG6630E-AC (02352FQA-001): 7.5 kg (16.53 lb)<br>• USG6630E-AC (02352FQA-002): 7.5 kg (16.53 lb)<br>• USG6630E-DC (02352CKV): 7.3 kg (16.09 lb)<br>• USG6630E-DC (02352CKV-001): 7.5 kg (16.53 lb)<br>• USG6630E-DC (02352CKV-002): 7.5 kg (16.53 lb) |
| CPU | 1 CPU, 8 cores/CPU, up to 2.3 GHz |
| Memory | 16 GB DDR4, supporting ECC |
| Storage | 32GB |
| NOR Flash | • USG6630E-AC (02352FQA): 32 MB<br>• USG6630E-AC (02352FQA-001): 64MB<br>• USG6630E-AC (02352FQA-002): 64MB<br>• USG6630E-DC (02352CKV): 32 MB<br>• USG6630E-DC (02352CKV-001): 64MB<br>• USG6630E-DC (02352CKV-002): 64MB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |

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| Item | Specification |
| --- | --- |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6630E-AC (02352FQA): 134.34 W<br>• USG6630E-AC (02352FQA-001): 134.34 W<br>• USG6630E-AC (02352FQA-002): 134.34 W<br>• USG6630E-DC (02352CKV): 121.39 W<br>• USG6630E-DC (02352CKV-001): 121.39 W<br>• USG6630E-DC (02352CKV-002): 121.39 W |
| Typical heat dissipation [BTU/hour] | • USG6630E-AC (02352FQA): 458.37 BTU/hour<br>• USG6630E-AC (02352FQA-001): 458.37 BTU/hour<br>• USG6630E-AC (02352FQA-002): 458.37 BTU/hour<br>• USG6630E-DC (02352CKV): 414.18 BTU/hour<br>• USG6630E-DC (02352CKV-001): 414.18 BTU/hour<br>• USG6630E-DC (02352CKV-002): 414.18 BTU/hour |
| Maximum power consumption [W] | • USG6630E-AC (02352FQA): 191 W<br>• USG6630E-AC (02352FQA-001): 171.7 W<br>• USG6630E-AC (02352FQA-002): 171.7 W<br>• USG6630E-DC (02352CKV): 172.14 W<br>• USG6630E-DC (02352CKV-001): 192.4 W<br>• USG6630E-DC (02352CKV-002): 192.4 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6630E-AC (02352FQA): 651.7 BTU/hour<br>• USG6630E-AC (02352FQA-001): 585.84 BTU/hour<br>• USG6630E-AC (02352FQA-002): 585.84 BTU/hour<br>• USG6630E-DC (02352CKV): 587.34 BTU/hour<br>• USG6630E-DC (02352CKV-001): 656.47 BTU/hour<br>• USG6630E-DC (02352CKV-002): 656.47 BTU/hour |
| MTBF [years] | 29.34 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999992 |
| Power supply mode | • USG6630E-AC (02352FQA): AC pluggable<br>• USG6630E-AC (02352FQA-001): AC pluggable<br>• USG6630E-AC (02352FQA-002): AC pluggable<br>• USG6630E-DC (02352CKV): DC pluggable<br>• USG6630E-DC (02352CKV-001): DC pluggable<br>• USG6630E-DC (02352CKV-002): DC pluggable |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | • USG6630E-AC (02352FQA): 100 V to 240 V, 50 Hz/60 Hz<br>• USG6630E-AC (02352FQA-001): 100 V to 240 V, 50 Hz/60 Hz<br>• USG6630E-AC (02352FQA-002): 100 V to 240 V, 50 Hz/60 Hz<br>• USG6630E-DC (02352CKV): -48 V to -60 V<br>• USG6630E-DC (02352CKV-001): -48 V to -60 V<br>• USG6630E-DC (02352CKV-002): -48 V to -60 V |

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| Item | Specification |
| --- | --- |
| Input voltage range [V] | • USG6630E-AC (02352FQA): 90 V to 290 V, 47 Hz to 63 Hz<br>• USG6630E-AC (02352FQA-001): 90 V to 290 V, 47 Hz to 63 Hz<br>• USG6630E-AC (02352FQA-002): 90 V to 290 V, 47 Hz to 63 Hz<br>• USG6630E-DC (02352CKV): -38.4 V to -72 V<br>• USG6630E-DC (02352CKV-001): -38.4 V to -72 V<br>• USG6630E-DC (02352CKV-002): -38.4 V to -72 V |
| Maximum input current [A] | • USG6630E-AC (02352FQA): 9 A/power module<br>• USG6630E-AC (02352FQA-001): 9 A/power module<br>• USG6630E-AC (02352FQA-002): 9 A/power module<br>• USG6630E-DC (02352CKV): 11 A/power module<br>• USG6630E-DC (02352CKV-001): 11 A/power module<br>• USG6630E-DC (02352CKV-002): 11 A/power module |
| Rated output power [W] | • USG6630E-AC (02352FQA): 600W/12V<br>• USG6630E-AC (02352FQA-001): 600W/12V<br>• USG6630E-AC (02352FQA-002): 600W/12V<br>• USG6630E-DC (02352CKV): 350 W/12 V<br>• USG6630E-DC (02352CKV-001): 1000 W/12 V<br>• USG6630E-DC (02352CKV-002): 1000 W/12 V |

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| Item | Specification |
| --- | --- |
| Maximum output power [W] | • USG6630E-AC (02352FQA): 600 W/power module<br>• USG6630E-AC (02352FQA-001): 600 W/power module<br>• USG6630E-AC (02352FQA-002): 600 W/power module<br>• USG6630E-DC (02352CKV): 350 W/power module<br>• USG6630E-DC (02352CKV-001): 1000 W/power module<br>• USG6630E-DC (02352CKV-002): 1000 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72db(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

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NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.19 USG6635E

## Overview

**Table 3-109** Basic information about the USG6635E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6635E AC Host (16*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP +,32G Memory,2 AC POWER) | 02352RQD | USG6635E-AC | V600R007C00 |
| USG6635E DC Host (16*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP +,32G Memory,2 DC POWER) | 02352RQE | USG6635E-DC | V600R007C20 |

## Appearance

NOTE

The appearance difference between the DC model and AC model is the power module installed on the rear of the device. The following figure uses the AC model as an example.

**Figure 3-106** Appearance of the USG6635E (front view)

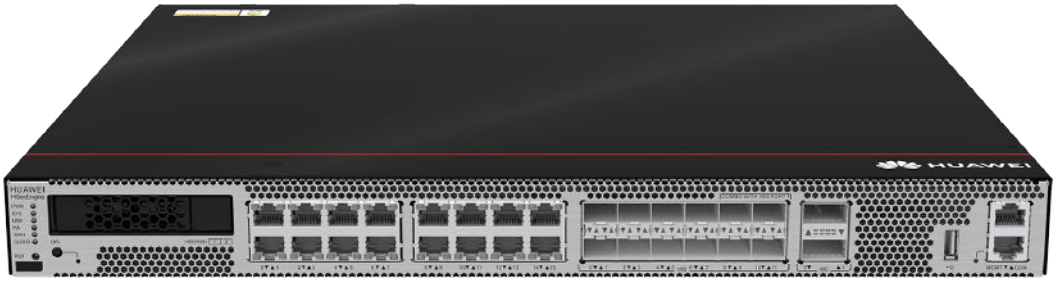
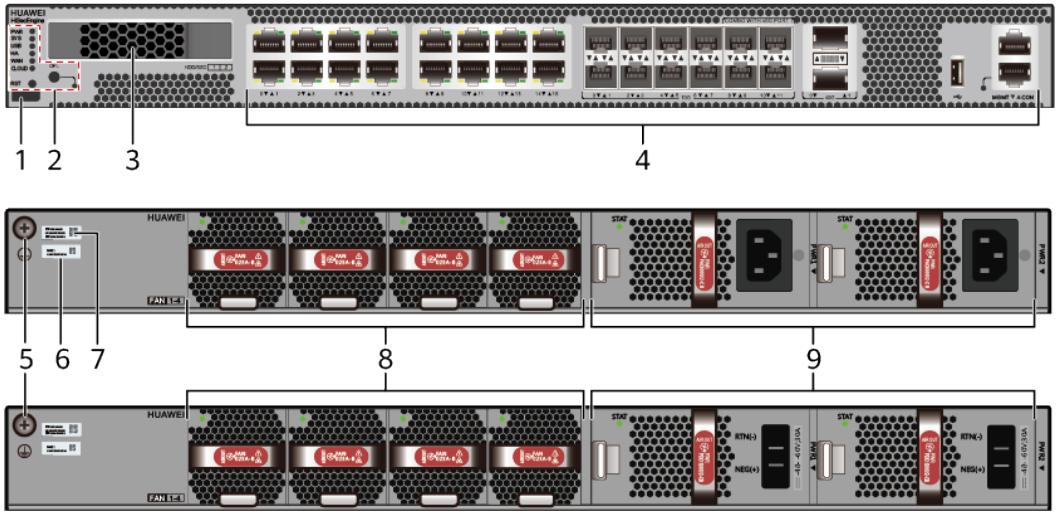


**Figure 3-107** Appearance of the USG6635E (rear view)



## Components

**Figure 3-108** Components of the USG6635E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

**Table 3-110** Component functions

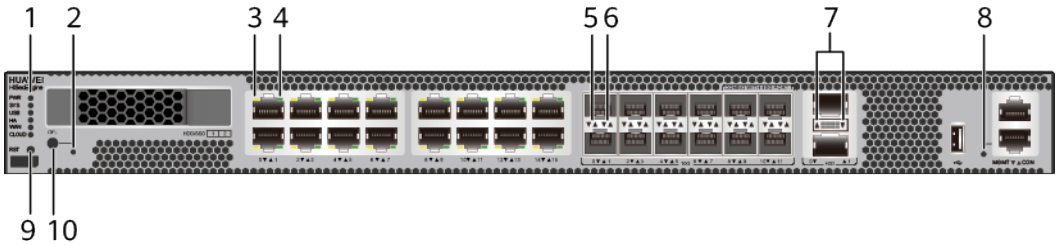
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |

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| Name | Description |
| --- | --- |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN4, and the corresponding slot numbers are 4, 5, 6, and 7 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-109** Indicators and buttons on the USG6635E




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| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. GE electrical port ACT indicator | 4. GE electrical port LINK indicator | 5. 10GE optical port ACT indicator |
| 6. 10GE optical port LINK indicator | 7. 40GE optical port indicator | 8. MGMT port indicator | 9. RST button | 10. OFL button |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-111** Indicators on the USG6635E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

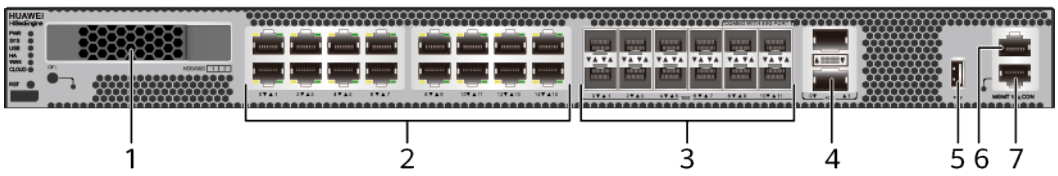
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**Table 3-112** Buttons on the USG6635E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-110** Ports on the USG6635E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. 10GE optical port | 4. 40GE optical port | 5. USB 2.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-113** Ports on the USG6635E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| 10GE optical ports (0 to 11) | SFP+ | 12 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11.<br>When 40GE0/0/1 optical port on the device is enabled, the 10GE optical ports numbered from XGigabitEthernet0/0/8 to XGigabitEthernet0/0/11 cannot be used. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 40GE optical ports (0 to 1) | QSFP+ | Two 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/1.<br>By default, the 40GE0/0/1 optical port mode is disabled. To enable this mode, run the set device port-config-mode 40g-port enable command. | **40Gbps QSFP+ Optical Modules** |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

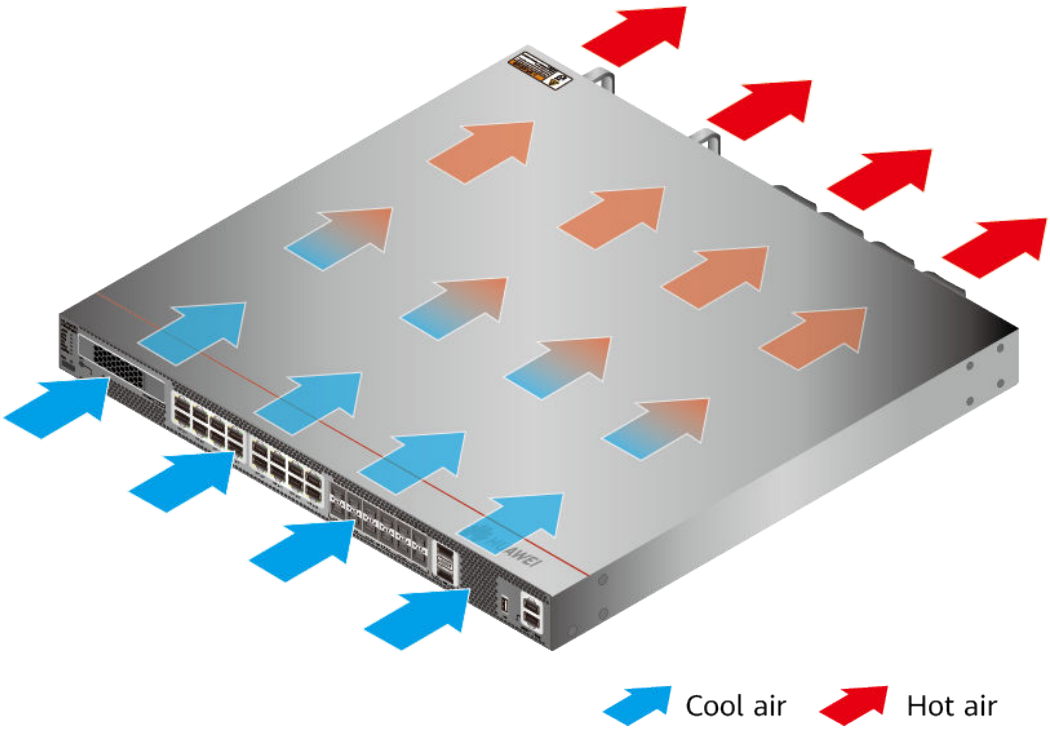
The power supply system has two PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power modules or PDC1000S12-CB DC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of four FAN-023A-B modules working in 3+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-111** System air flow of the USG6635E




## Technical Specifications

**Table 3-114** Technical specifications of the USG6635E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 11.88 kg (26.19 lb) |
| Weight without packaging [kg(lb)] | 7.75 kg (17.09 lb) |

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| Item | Specification |
| --- | --- |
| CPU | 2 CPU, 16 cores/CPU, up to 1.85 GHz |
| Memory | 32 GB DDR4 ECC memory, 16 GB/CPU |
| NOR Flash | 128 MB, 64 MB/CPU |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | • USG6635E-AC (02352RQD): 136.1 W<br>• USG6635E-DC (02352RQE): 142.5 W |
| Typical heat dissipation [BTU/hour] | • USG6635E-AC (02352RQD): 464.37 BTU/hour<br>• USG6635E-DC (02352RQE): 486.21 BTU/hour |
| Maximum power consumption [W] | • USG6635E-AC (02352RQD): 156 W<br>• USG6635E-DC (02352RQE): 169.7 W |
| Maximum heat dissipation [BTU/hour] | • USG6635E-AC (02352RQD): 533.4 BTU/hour<br>• USG6635E-DC (02352RQE): 579.02 BTU/hour |
| MTBF [years] | 31.63 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999993 |
| Power supply mode | • USG6635E-AC (02352RQD): AC pluggable<br>• USG6635E-DC (02352RQE): DC pluggable |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |

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| Item | Specification |
| --- | --- |
| Rated input voltage [V] | • USG6635E-AC (02352RQD): 100 V to 240 V, 50 Hz/60 Hz<br>• USG6635E-DC (02352RQE): -48 V to -60 V |
| Input voltage range [V] | • USG6635E-AC (02352RQD): 90 V to 290 V, 47 Hz to 63 Hz<br>• USG6635E-DC (02352RQE): -38.4 V to -72 V |
| Maximum input current [A] | • USG6635E-AC (02352RQD): 8 A/ power module<br>• USG6635E-DC (02352RQE): 30 A/ power module |
| Rated output power [W] | • USG6635E-AC (02352RQD): 600W/12V<br>• USG6635E-DC (02352RQE): 1000 W/12 V |
| Maximum output power [W] | • USG6635E-AC (02352RQD): 600 W/ power module<br>• USG6635E-DC (02352RQE): 1000 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 4 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | • USG6635E-AC (02352RQD): Front-to-back airflow<br>• USG6635E-DC (02352RQE): Air inlet from front and air outlet from rear |
| Redundant fans | 3+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |

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| Item | Specification |
| --- | --- |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.20 USG6650E

## Overview

**Table 3-115** Basic information about the USG6650E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6650E AC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02351YRY | USG6650E-AC | V600R006C00 |
| USG6650E AC Host (12*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02351YRY-001 | USG6650E-AC | V600R007C20SPC300 |
| USG6650E AC Host(12*GE RJ45+12*10GE SFP+ + 2*40GE QSFP+,2 AC power) | 02351YRY-002 | USG6650E-AC | V600R007C20SPC600 |

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## Appearance

NOTE

The model with the BOM number containing "-001" and "-002" have one more OFL indicator than the model with the BOM number not containing "-001" or "-002". The silkscreen of the device contains "HiSecEngine", and the distribution of heat dissipation holes on the front of one model slightly differ from that of another model. This document uses the model whose BOM number contains "-001" as an example.

**Figure 3-112** Appearance of the USG6650E (front view)

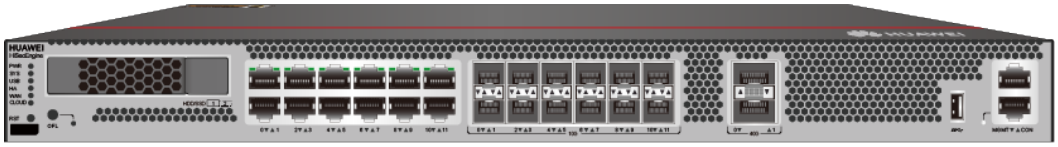
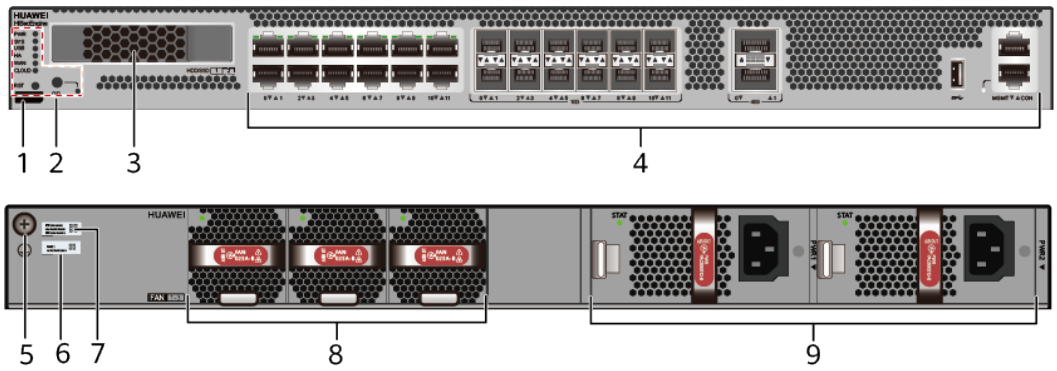


**Figure 3-113** Appearance of the USG6650E (rear view)



## Components

**Figure 3-114** Components of the USG6650E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

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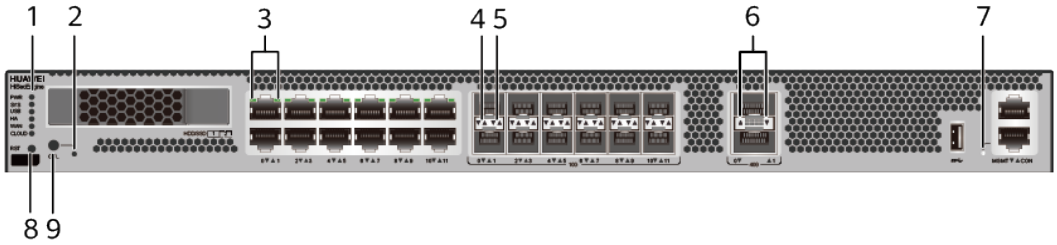
**Table 3-116** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | The fan module provides air flow for heat dissipation. The power module supports hot-swapping and can be replaced without interrupting device operation. However, to prevent overheating, do not operate the device without a functioning fan module for more than one minute.<br>The silkscreens of the fan module slots are FAN1 to FAN3, and the corresponding slot numbers are 4, 5, and 6 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-115** Indicators and buttons on the USG6650E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator<br>**NOTE**<br>Only the models whose BOM number contain "-001" and "-002" support the OFL indicator. | 2. GE electrical port indicator | 3. 10GE optical port ACT indicator | 4. 10GE optical port LINK indicator |
| 5. 40GE optical port indicator | 6. MGMT port indicator | 7. RST button | 8. OFL button | - |

**NOTE**

- GE electrical interface indicators from left to right correspond to GE electrical interfaces in ascending order. For example, the first indicator on the panel corresponds to GigabitEthernet0/0/0 and the second indicator corresponds to GigabitEthernet0/0/1.
- Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-117** Indicators on the USG6650E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | GE electrical port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

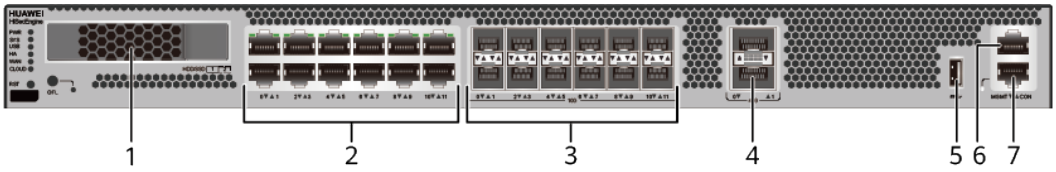
**Table 3-118** Buttons on the USG6650E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

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## Ports

**Figure 3-116** Ports on the USG6650E




| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. 10GE optical port | 4. 40GE optical port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

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**Table 3-119** Ports on the USG6650E

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 11) | RJ45 | 12 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/11.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| 10GE optical ports (0 to 11) | SFP+ | 12 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| 40GE optical ports (0 to 1) | QSFP+ | Two 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/1. | **40Gbps QSFP+ Optical Modules** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | Console Cable |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

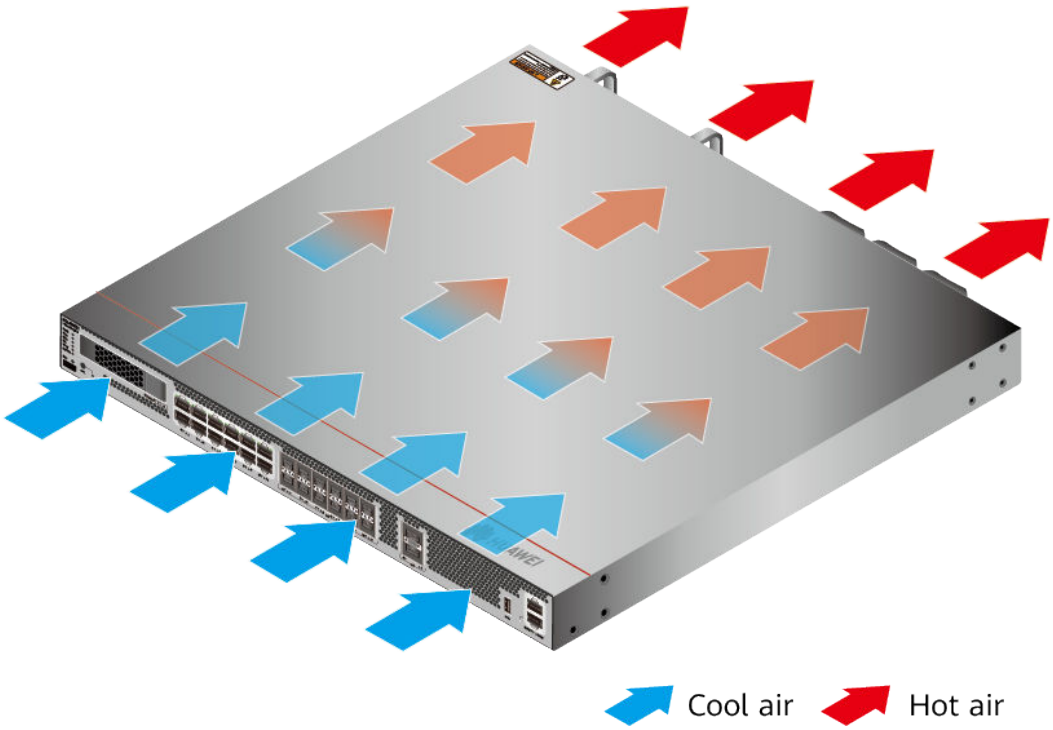
- Early delivery: The power supply system has two PAC600S12-B AC power modules for 1+1 power redundancy.
- Later delivery: The power supply system has two PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of three FAN-023A-B modules working in 2+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-117** System air flow of the USG6650E




## Technical Specifications

**Table 3-120** Technical specifications of the USG6650E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 215 mm x 550 mm x 530 mm (8.46 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |

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| Item | Specification |
| --- | --- |
| Weight with packaging [kg(lb)] | • USG6650E-AC (02351YRY): 10.9 kg (24.03 lb)<br>• USG6650E-AC (02351YRY-001): 10.2 kg (22.49 lb)<br>• USG6650E-AC (02351YRY-002): 10.2 kg (22.49 lb) |
| Weight without packaging [kg(lb)] | • USG6650E-AC (02351YRY): 7.9 kg (17.42 lb)<br>• USG6650E-AC (02351YRY-001): 7.5 kg (16.53 lb)<br>• USG6650E-AC (02351YRY-002): 7.5 kg (16.53 lb) |
| CPU | 1 CPU, 8 cores/CPU, up to 2.3 GHz |
| Memory | 16 GB DDR4, supporting ECC |
| Storage | 32GB |
| NOR Flash | • USG6650E-AC (02351YRY): 32 MB<br>• USG6650E-AC (02351YRY-001): 64MB<br>• USG6650E-AC (02351YRY-002): 64MB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 134.34 W |
| Typical heat dissipation [BTU/hour] | 458.37 BTU/hour |
| Maximum power consumption [W] | • USG6650E-AC (02351YRY): 191 W<br>• USG6650E-AC (02351YRY-001): 171.7 W<br>• USG6650E-AC (02351YRY-002): 171.7 W |

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| Item | Specification |
| --- | --- |
| Maximum heat dissipation [BTU/hour] | • USG6650E-AC (02351YRY): 651.7 BTU/hour<br>• USG6650E-AC (02351YRY-001): 585.84 BTU/hour<br>• USG6650E-AC (02351YRY-002): 585.84 BTU/hour |
| MTBF [years] | 29.34 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999992 |
| Power supply mode | AC pluggable |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 9 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 3 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 2+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72db(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |

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| Item | Specification |
| --- | --- |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.21 USG6655E

## Overview

**Table 3-121** Basic information about the USG6655E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6655E AC Host (16*GE RJ45 + 12*10GE SFP+ + 2*40GE QSFP +,32G Memory,2 AC POWER) | 02352RPS | USG6655E-AC | V600R007C00 |

## Appearance

**Figure 3-118** Appearance of the USG6655E (front view)

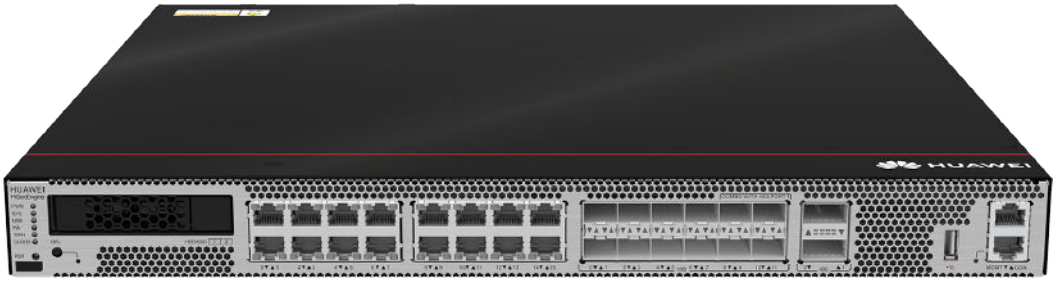
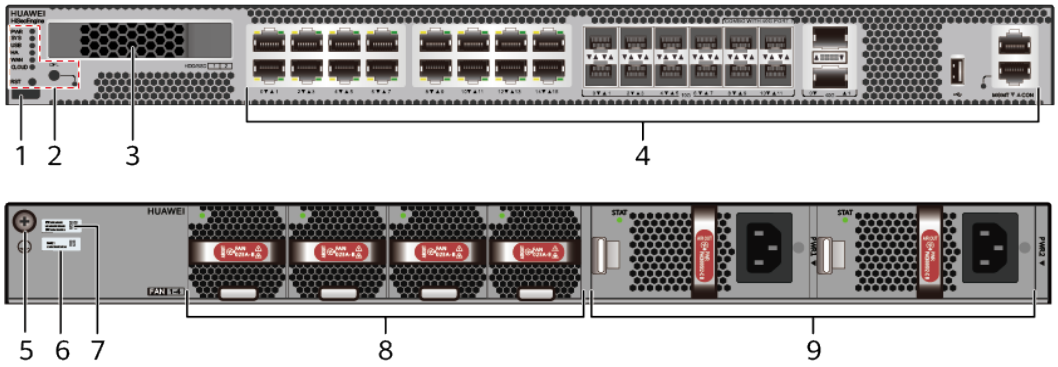


**Figure 3-119** Appearance of the USG6655E (rear view)



## Components

**Figure 3-120** Components of the




| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
|---|---|---|---|---|
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

**Table 3-122** Component functions

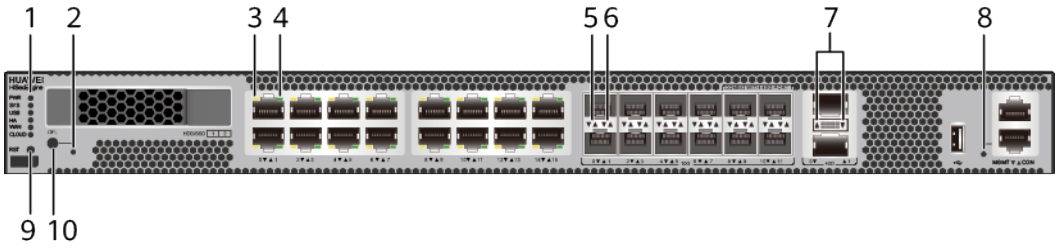
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |

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| Name | Description |
| --- | --- |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN4, and the corresponding slot numbers are 4, 5, 6, and 7 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-121** Indicators and buttons on the USG6655E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. OFL indicator | 3. GE electrical port ACT indicator | 4. GE electrical port LINK indicator | 5. 10GE optical port ACT indicator |
| 6. 10GE optical port LINK indicator | 7. 40GE optical port indicator | 8. MGMT port indicator | 9. RST button | 10. OFL button |

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NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-123** Indicators on the USG6655E

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | OFL indicator | Green | Steady on | The hard disk can be removed. |
| | | - | Off | The hard disk cannot be removed. |
| - | ACT indicator of the GE electrical port | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator of the GE electrical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

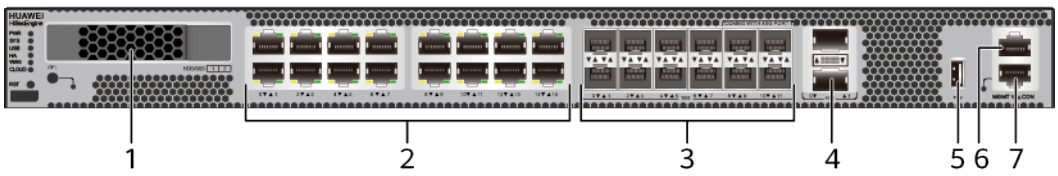
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**Table 3-124** Buttons on the USG6655E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for four seconds until the OFL indicator turns on. |

## Ports

**Figure 3-122** Ports on the USG6655E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. GE electrical port | 3. 10GE optical port | 4. 40GE optical port | 5. USB 2.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-125** Ports on the USG6655E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| GE electrical ports (0 to 15) | RJ45 | 16 10/100/1000M autosensing Ethernet electrical ports, numbered from GigabitEthernet 0/0/0 to GigabitEthernet 0/0/15.<br>**NOTE**<br>Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top. | **Ethernet Cable** |
| 10GE optical ports (0 to 11) | SFP+ | 12 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/11.<br>When 40GE0/0/1 optical port on the device is enabled, the 10GE optical ports numbered from XGigabitEthernet0/0/8 to XGigabitEthernet0/0/11 cannot be used. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 40GE optical ports (0 to 1) | QSFP+ | Two 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/1.<br>By default, the 40GE0/0/1 optical port mode is disabled. To enable this mode, run the set device port-config-mode 40g-port enable command. | **40Gbps QSFP+ Optical Modules** |
| USB 2.0 port | USB 2.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

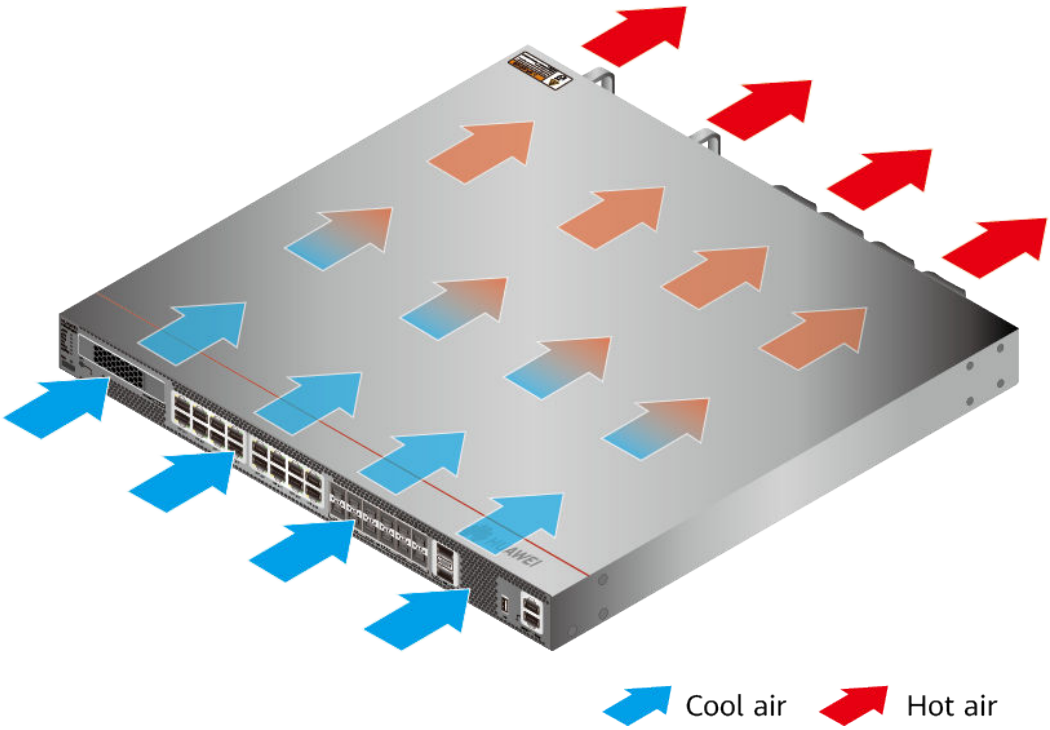
The power supply system has two PAC600S12-CB, PAC600S12-DB, or PAC600S12-EB AC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of four FAN-023A-B modules working in 3+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-123** System air flow of the USG6655E




## Technical Specifications

**Table 3-126** Technical specifications of the USG6655E-AC

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 600 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 420 mm (1.72 in. x 17.4 in. x 16.54 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the handle on the rear panel): 43.6 mm x 442 mm x 453 mm (1.72 in. x 17.4 in. x 17.83 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 550 mm x 530 mm (8.86 in. x 21.65 in. x 20.87 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 11.88 kg (26.19 lb) |
| Weight without packaging [kg(lb)] | 7.75 kg (17.09 lb) |

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| Item | Specification |
| --- | --- |
| CPU | 2 CPU, 16 cores/CPU, up to 1.85 GHz |
| Memory | 32 GB DDR4 ECC memory, 16 GB/CPU |
| NOR Flash | 128 MB, 64 MB/CPU |
| NAND Flash | 2 GB |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 136.1 W |
| Typical heat dissipation [BTU/hour] | 464.37 BTU/hour |
| Maximum power consumption [W] | 156 W |
| Maximum heat dissipation [BTU/hour] | 533.4 BTU/hour |
| MTBF [years] | 31.63 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999993 |
| Power supply mode | AC pluggable |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | 100 V to 240 V, 50 Hz/60 Hz |
| Input voltage range [V] | 90 V to 290 V, 47 Hz to 63 Hz |
| Maximum input current [A] | 8 A/power module |
| Rated output power [W] | 600W/12V |
| Maximum output power [W] | 600 W/power module |
| Types of fans | Pluggable |
| Number of fan modules | 4 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | The fan module locates at the air exhaust of the system. |
| Airflow direction | Front-to-back airflow |

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| Item | Specification |
| --- | --- |
| Redundant fans | 3+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.22 USG6680E

## Overview

**Table 3-127** Basic information about the USG6680E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6680E AC Host (28* (SFP+) + 4*QSFP + 2*HA,2AC power supply) | 02351YRR | USG6680E-AC | V600R006C00 |

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| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6680E AC Host(28*(SFP+) +4*QSFP +2*HA,2AC power supply) | 02351YRR-001 | USG6680E-AC | V600R007C20SPC600 |

## Appearance

**Figure 3-124** Appearance of the USG6680E (front view)

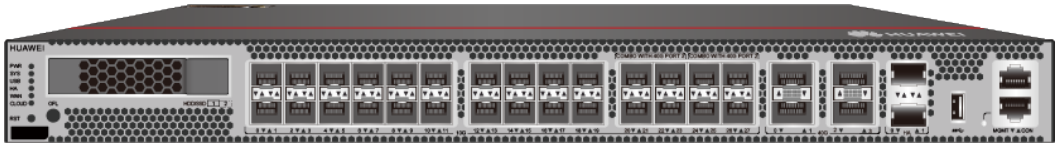
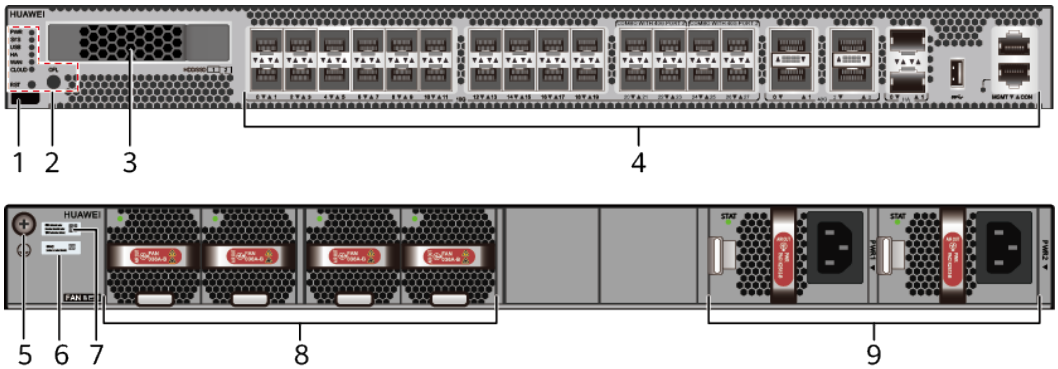


**Figure 3-125** Appearance of the USG6680E (rear view)



## Components

**Figure 3-126** Components of the USG6680E




| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| --- | --- | --- | --- | --- |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

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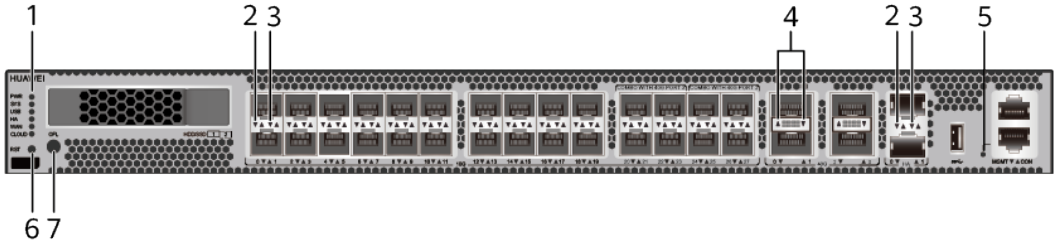
**Table 3-128** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN4, and the corresponding slot numbers are 4, 5, 6, and 7 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-127** Indicators and buttons on the USG6680E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. 10GE optical port ACT indicator | 3. 10GE optical port LINK indicator | 4. 40GE Optical port indicator | 5. MGMT port indicator |
| 6. RST button | 7. OFL button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-129** Indicators on the USG6680E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

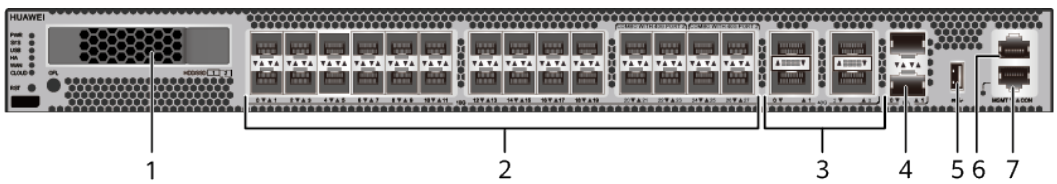
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**Table 3-130** Buttons on the USG6680E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-128** Ports on the USG6680E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. 10GE optical port | 3. 40GE optical port | 4. HA port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | - | - | - |

**Table 3-131** Ports on the USG6680E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 27) | SFP+ | 28 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/27.<br>When 40GE0/0/2 to 40GE0/0/3 optical ports on the device are enabled, the 10GE optical ports numbered from XGigabitEthernet0/0/20 to XGigabitEthernet0/0/27 cannot be used. | • **1Gbps SFP Copper Modules**<br>• **1Gbps eSFP Optical Modules**<br>• **10Gbps SFP+ Optical Modules** |
| 40GE optical ports (0 to 3) | QSFP+ | Four 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/3.<br>By default, the 40GE0/0/2 to 40GE0/0/3 optical ports mode are disabled. To enable this mode, run the set device port-config-mode 40g-port enable command. | 40 Gbps QSFP+ Optical Module |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| HA ports(0 to 1) | SFP+ | Heartbeat interfaces for hot standby or the cluster. Two devices are connected through independent links to form hot standby or the cluster. numbered from HA0/0/0 to HA0/0/1. | **10Gbps SFP+ Optical Modules** |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |

## Power Supply System

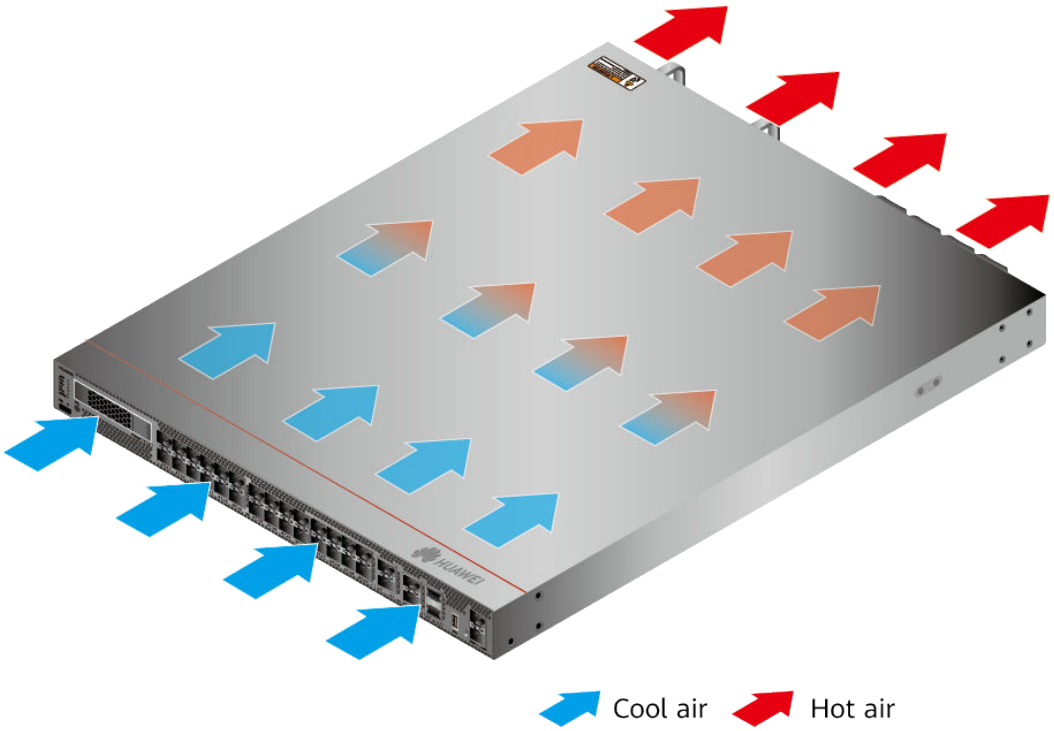
- Early delivery: The power supply system has two PAC1K2S12-B AC power modules for 1+1 power redundancy.
- Later delivery: The power supply system has two PAC1K2S12-CB AC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of four FAN-036A-B modules working in 3+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

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**Figure 3-129** System air flow of the USG6680E




## Technical Specifications

**Table 3-132** Technical specifications of the USG6680E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 800 mm or above |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 600 mm (1.72 in. x 17.40 in. x 23.62 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the parts protruding from the rear panel): 43.6 mm x 442 mm x 633 mm (1.72 in. x 17.40 in. x 24.92 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 590 mm x 790 mm (8.86 in. x 23.23 in. x 31.1 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 19.12 kg (42.15 lb) |
| Weight without packaging [kg(lb)] | 14.8 kg (32.63 lb) |

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| Item | Specification |
| --- | --- |
| Weight with packaging (full configuration) [kg(lb)] | 19.12 kg (42.15 lb) |
| CPU | 2CPU, 8 cores/CPU, up to 2.3 GHz |
| Memory | 32 GB DDR4 ECC memory, 16 GB/CPU |
| Storage | 32GB |
| NOR Flash | 64MB, 32MB/CPU |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 346 W |
| Typical heat dissipation [BTU/hour] | 1183.04 BTU/hour |
| Maximum power consumption [W] | 488.3 W |
| Maximum heat dissipation [BTU/hour] | 1666.1 BTU/hour |
| MTBF [years] | 25 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999991 |
| Power supply mode | AC pluggable |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | • USG6680E-AC (02351YRR):<br>– AC power input: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC power input: 240V<br>• USG6680E-AC (02351YRR-001):<br>– AC: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC: 240 V DC |
| Input voltage range [V] | AC power input: 90 V AC to 290 V AC, 47 Hz to 63 Hz |

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| Item | Specification |
| --- | --- |
| Maximum input current [A] | 11 A/power module |
| Rated output power [W] | • 800 W (100 V AC to 130 V AC)<br>• 1200 W (200 V AC to 240 V AC)<br>• 1200 W (240 V DC) |
| Maximum output power [W] | • 800 W (100 V AC to 130 V AC)/ power module<br>• 1200 W (200 V AC to 240 V AC)/ power module<br>• 1200 W (240 V DC)/power module |
| Types of fans | Pluggable |
| Number of fan modules | 4 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6680E-AC (02351YRR): The fan module locates at the air exhaust of the system.<br>• USG6680E-AC (02351YRR-001): Absorbing cold air into the device |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 3+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

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NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.23 USG6712E

## Overview

**Table 3-133** Basic information about the USG6712E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6712E-AC Host (20* (SFP+) + 2*QSFP +2*QSFP28 + 2*HA,2AC power supply) | 02352CNL | USG6712E-AC | V600R006C00 |
| USG6712E-AC Host(20*(SFP+) +2*QSFP +2*QSFP28+2*HA, 2AC power supply) | 02352CNL-001 | USG6712E-AC | V600R007C20SPC600 |

## Appearance

**Figure 3-130** Appearance of the USG6712E (front view)

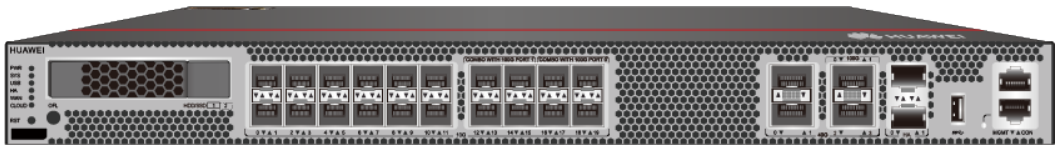
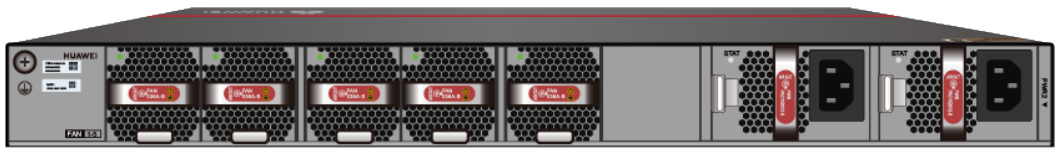


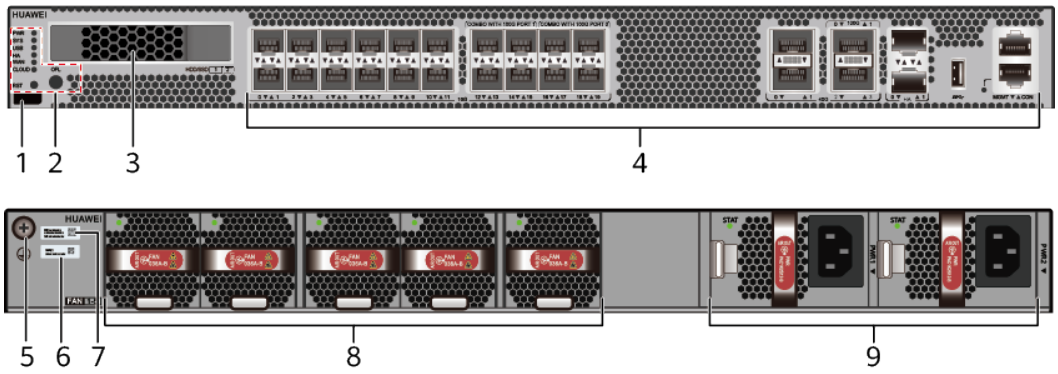
**Figure 3-131** Appearance of the USG6712E (rear view)



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## Components

**Figure 3-132** Components of the USG6712E




| | | | | |
|---|---|---|---|---|
| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

**Table 3-134** Component functions

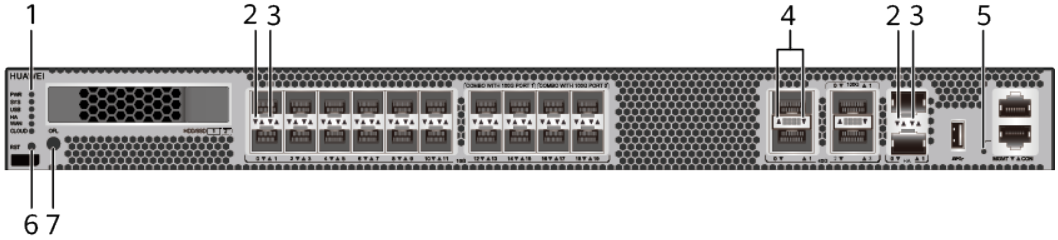
| Name | Description |
|---|---|
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |

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| Name | Description |
| --- | --- |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN5, and the corresponding slot numbers are 4, 5, 6, 7, and 8 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

## Indicators and Buttons

**Figure 3-133** Indicators and buttons on the USG6712E




| | | | | |
| --- | --- | --- | --- | --- |
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. 10GE optical port ACT indicator | 3. 10GE optical port LINK indicator | 4. 40GE/100GE optical port indicator | 5. MGMT port indicator |
| 6. RST button | 7. OFL button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

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**Table 3-135** Indicators on the USG6712E

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE/100GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

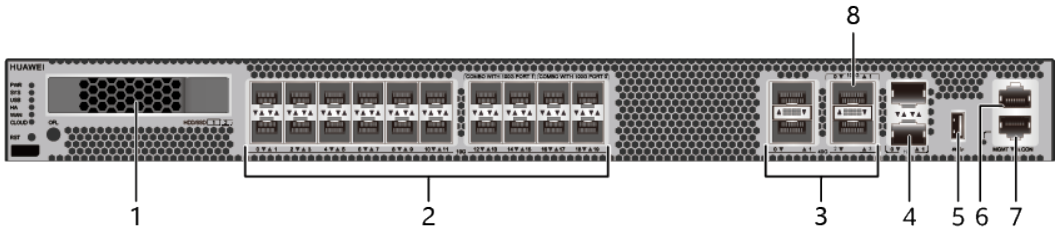
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**Table 3-136** Buttons on the USG6712E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-134** Ports on the USG6712E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. 10GE optical port | 3. 40GE optical port | 4. HA port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | 8. 100GE optical port | - | - |

**Table 3-137** Ports on the USG6712E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 19) | SFP+ | 20 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/19.<br>When 100GE0/0/0 to 100GE0/0/1 optical ports on the device are enabled, the 10GE optical ports numbered from XGigabitEthernet0/0/12 to XGigabitEthernet0/0/19 cannot be used. | • 1Gbps SFP Copper Modules<br>• 1Gbps eSFP Optical Modules<br>• 10Gbps SFP+ Optical Modules |
| 40GE optical ports (0 to 3) | QSFP+ | Four 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/3.<br>40GE optical ports 40GE0/0/2 and 40GE0/0/3 can be switched to 100GE optical ports. | 40Gbps QSFP+ Optical Modules |
| HA ports(0 to 1) | SFP+ | Heartbeat interfaces for hot standby or the cluster. Two devices are connected through independent links to form hot standby or the cluster. numbered from HA0/0/0 to HA0/0/1. | 10Gbps SFP+ Optical Modules |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |
| 100GE optical ports (0 to 1) | QSFP28 | Two 100GE Ethernet optical ports, numbered 100GE0/0/0 and 100GE0/0/1.<br>By default, the 100GE optical port mode is disabled. To enable this mode, run the set device port-config-mode 100g-port enable command. | **100Gbps QSFP28 Optical Modules** |

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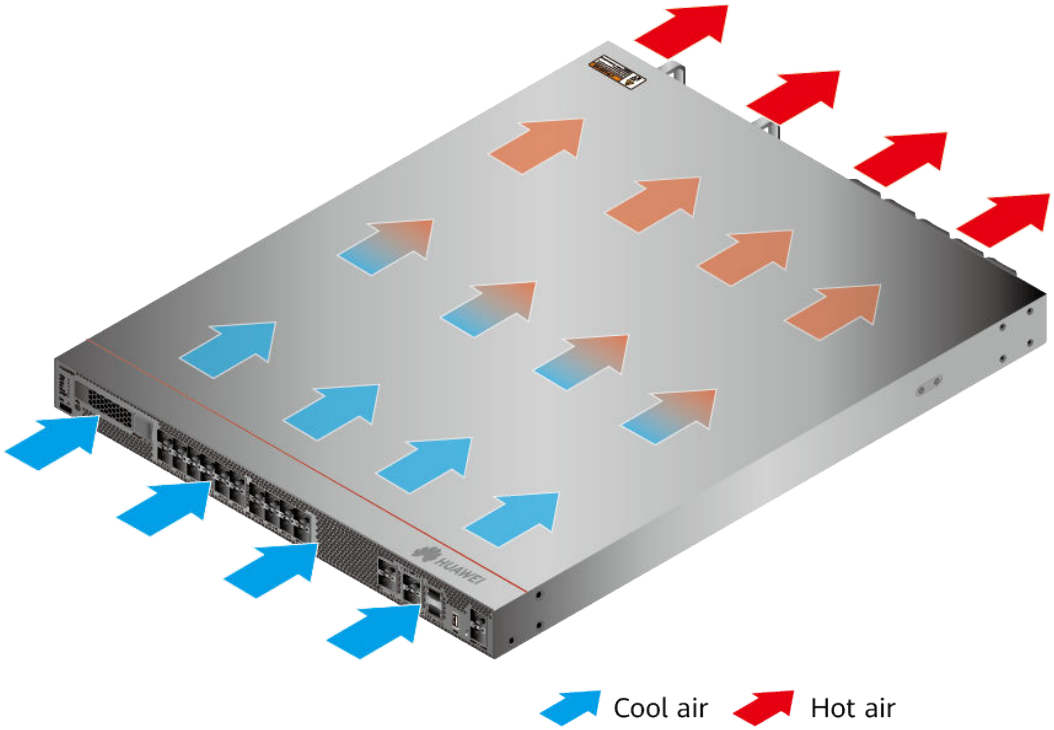
## Power Supply System

- Early delivery: The power supply system has two PAC1K2S12-B AC power modules for 1+1 power redundancy.
- Later delivery: The power supply system has two PAC1K2S12-CB AC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of five FAN-036A-B modules working in 4+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

**Figure 3-135** System air flow of the USG6712E




## Technical Specifications

**Table 3-138** Technical specifications of the USG6712E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 800 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 600 mm (1.72 in. x 17.40 in. x 23.62 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the parts protruding from the rear panel): 43.6 mm x 442 mm x 633 mm (1.72 in. x 17.40 in. x 24.92 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 590 mm x 790 mm (8.86 in. x 23.23 in. x 31.1 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 17.6 kg (38.8 lb) |
| Weight without packaging [kg(lb)] | 12 kg (26.46 lb) |
| Weight with packaging (full configuration) [kg(lb)] | 17.6 kg (38.8 lb) |
| CPU | 2 CPU, 14 cores/CPU, up to 1.9 GHz |
| Memory | 64 GB DDR4 ECC memory, 32 GB/CPU |
| Storage | 32GB |
| NOR Flash | 64MB, 32MB/CPU |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 382.9 W |
| Typical heat dissipation [BTU/hour] | 1309.21 BTU/hour |
| Maximum power consumption [W] | 566 W |
| Maximum heat dissipation [BTU/hour] | 1935.27 BTU/hour |
| MTBF [years] | 25 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999991 |
| Power supply mode | AC pluggable |

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| Item | Specification |
| --- | --- |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | • USG6712E-AC (02352CNL):<br>– AC power input: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC power input: 240V<br>• USG6712E-AC (02352CNL-001):<br>– AC: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC: 240 V DC |
| Input voltage range [V] | AC power input: 90 V AC to 290 V AC, 47 Hz to 63 Hz |
| Maximum input current [A] | 11 A/power module |
| Rated output power [W] | • 800 W (100 V AC to 130 V AC)<br>• 1200 W (200 V AC to 240 V AC)<br>• 1200 W (240 V DC) |
| Maximum output power [W] | • 800 W (100 V AC to 130 V AC)/ power module<br>• 1200 W (200 V AC to 240 V AC)/ power module<br>• 1200 W (240 V DC)/power module |
| Types of fans | Pluggable |
| Number of fan modules | 5 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6712E-AC (02352CNL): The fan module locates at the air exhaust of the system.<br>• USG6712E-AC (02352CNL-001): Absorbing cold air into the device |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 4+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.1.24 USG6716E

## Overview

**Table 3-139** Basic information about the USG6716E

| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6716E-AC Host (20* (SFP+) + 2*QSFP +2*QSFP28 + 2*HA,2AC power supply) | 02351YRS | USG6716E-AC | V600R006C00 |

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| Description | Part Number | Model | First supported version |
| --- | --- | --- | --- |
| USG6716E-AC Host(20*(SFP+) +2*QSFP +2*QSFP28+2*HA, 2AC power supply) | 02351YRS-001 | USG6716E-AC | V600R007C20SPC600 |

## Appearance

**Figure 3-136** Appearance of the USG6716E (front view)

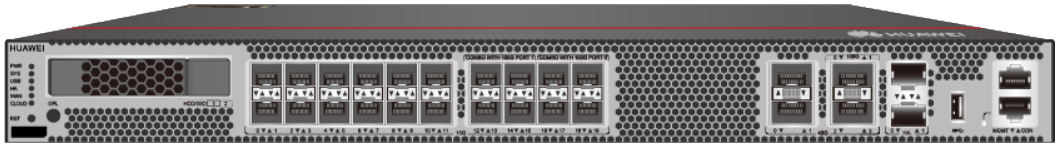
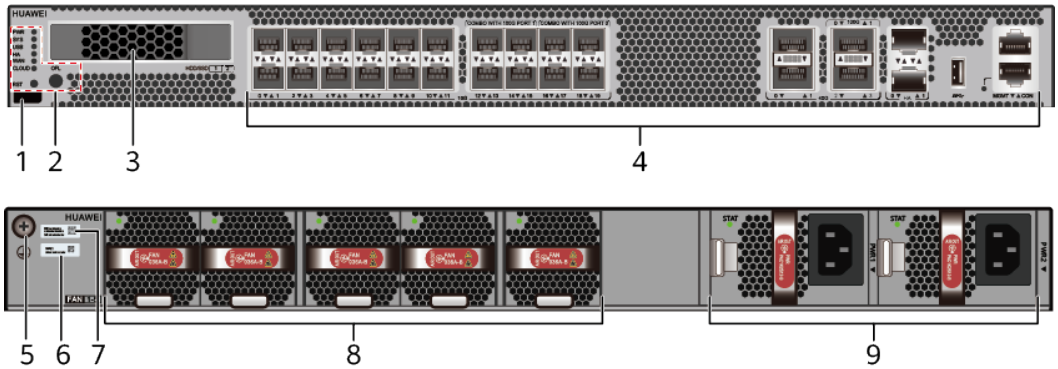


**Figure 3-137** Appearance of the USG6716E (rear view)



## Components

**Figure 3-138** Components of the USG6716E




| 1. Product series ID | 2. Indicator and button area | 3. HDD/SSD slot | 4. Fixed interface area | 5. Protective ground terminal |
| --- | --- | --- | --- | --- |
| 6. MAC label | 7. SN label | 8. Fan module | 9. Power module | - |

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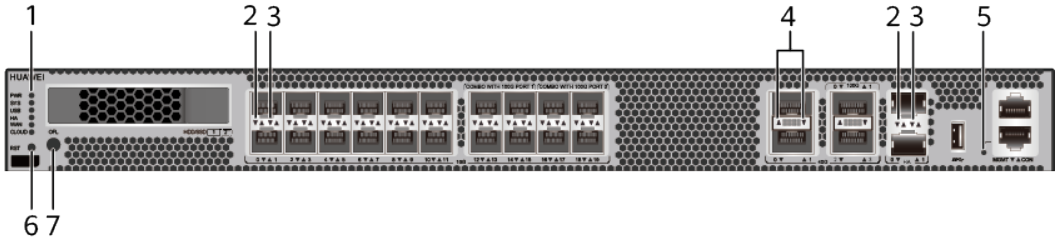
**Table 3-140** Component functions

| Name | Description |
| --- | --- |
| Product series ID | Indicates the product series of the device. For details about the product model, see the nameplate in the lower part of the device. |
| Indicator and button area | Provides multiple indicators to display the running status of the device in real time and provides the RST button for emergency maintenance. |
| HDD/SSD slot | Houses hard disks for storing log and report data. |
| Fixed interface area | Provides service ports, USB ports, console ports, and out-of-band management ports for device configuration and maintenance. |
| Protective ground terminal | Connects the M4 OT terminal of a PGND cable to the cabinet or the ground bar in the equipment room. |
| MAC label | Uniquely identifies the MAC address of the device, which is required during network forwarding configuration. |
| SN label | Uniquely identifies the device, which needs to be provided for the local technical support personnel to apply for a license. |
| Fan module | Dissipates heat for the device and is swappable. The fan module can be removed for no more than 1 minute.<br>The silkscreens of the fan module slots are FAN1 to FAN5, and the corresponding slot numbers are 4, 5, 6, 7, and 8 respectively. |
| Power module | Provides power input and distribution for the device. Two power modules are included in the standard configuration to provide 1+1 power redundancy. When one power module is running properly, the other one is hot swappable.<br>The silkscreens of the power module slots are PWR1 and PWR2, and the corresponding slot numbers are 2 and 3 respectively. |

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## Indicators and Buttons

**Figure 3-139** Indicators and buttons on the USG6716E




| | | | | |
|---|---|---|---|---|
| 1. PWR, SYS, USB, HA, WAN and CLOUD indicators | 2. 10GE optical port ACT indicator | 3. 10GE optical port LINK indicator | 4. 40GE/100GE optical port indicator | 5. MGMT port indicator |
| 6. RST button | 7. OFL button | - | - | - |

NOTE

Arrowheads of optical ports show the positions of the ports. A down arrowhead indicates a port in the upper part, and an up arrowhead indicates a port in the lower part.

**Table 3-141** Indicators on the USG6716E

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| PWR | Power indicator | Green | Steady on | The power module is working properly. |
| | | - | Off | The power module is faulty or the device is not powered on. |
| SYS | SYS indicator | Green | Steady on | The system is being powered on or restarted. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | The system is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking four times every second (4 Hz) | The system is starting. |
| | | Red | Steady on | • The system is faulty.<br>• The power supply is abnormal.<br>• The fan module is abnormal.<br>**NOTE**<br>If the system starts with two power modules and one power module is not powered on, the SYS indicator is steady red, but the system is running properly. |
| | | - | Off | The system is not running. |
| USB | USB indicator | Green | Steady on | USB-based deployment has been completed. |
| | | Green | Blinking four times every second (4 Hz) | The system is reading data from the USB flash drive. |
| | | Red | Steady on | USB-based deployment fails. |
| | | - | Off | USB-based deployment is disabled (default state). |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| HA | HA indicator | Green | Steady on | Management master device in hot standby or cluster deployment. |
| | | Green | Blinks once every 2 seconds (0.5 Hz). | Management backup device in hot standby or cluster deployment. |
| | | Red | Steady on | Hot standby or the cluster is faulty. |
| | | - | Off | The hot standby or the cluster function is disabled. |
| WAN | WAN indicator | - | Off | Reserved function. This function is not enabled. |
| CLOUD | CLOUD indicator | Green | Steady on | Connected to the cloud management platform. |
| | | Green | Blinking four times every second (4 Hz) | Connecting to the cloud management platform, transmitting or receiving data. |
| | | - | Off | The device is not connected to the cloud management platform. |
| - | ACT indicator for 10GE optical ports | Yellow | Blinking (12 Hz) | The port is sending or receiving data. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The port is not sending or receiving data. |
| - | LINK indicator for 10GE optical ports | Green | Steady on | The port link is connected. |
| | | - | Off | No link is established on the port. |
| - | Indicator for 40GE/100GE optical ports | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |
| - | MGMT port indicator | Green | Steady on | The port link is connected. |
| | | Green | Blinking (12 Hz) | The port is sending or receiving data. |
| | | - | Off | No link is established on the port. |

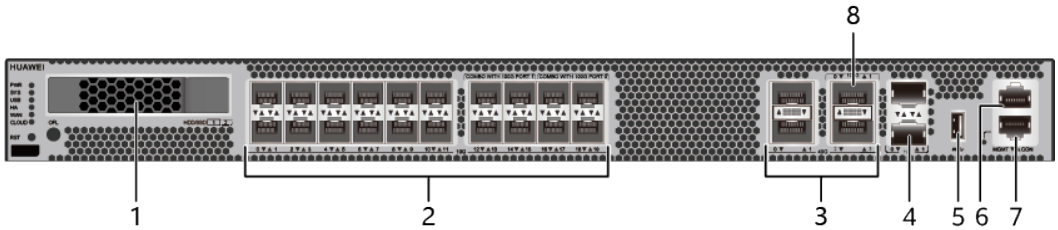
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**Table 3-142** Buttons on the USG6716E

| Silkscreen | Name | Description |
| --- | --- | --- |
| RST | RST button | When the device is running properly, you can press the RST button to restart the device. You are advised to save the current configuration before pressing the RST button.<br>This button can also be used to restore the default settings with one click. If you press and hold the RST button for 5 seconds and then release the RST button, the device restores its default settings and restarts.<br>**NOTE**<br>If needed, you can run the factory-configuration prohibit command in the system view to disable the function of restoring the factory settings by holding down the RST button. To enable this function again, run the undo factory-configuration prohibit command. |
| OFL | OFL button | To remove a hard disk, press and hold the OFL button for 4 seconds until the OFL indicator turns on. |

## Ports

**Figure 3-140** Ports on the USG6716E




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| | | | | |
|---|---|---|---|---|
| 1. HDD/SSD slot | 2. 10GE optical port | 3. 40GE optical port | 4. HA port | 5. USB 3.0 port |
| 6. Console port | 7. MGMT port | 8. 100GE optical port | - | - |

**Table 3-143** Ports on the USG6716E

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| HDD/SSD Hard disk slot | - | The hard disks are optional. You can purchase hard disk from Huawei if needed. | • **Hard Disk Unit SSD-Sata240G-A**<br>• **Hard Disk Unit SSD-SATA240G-C**<br>• **Hard Disk Unit SSD-SATA960G-B**<br>• **Hard Disk Unit SSD-SATA960G-C**<br>• **Hard Disk Unit SSD-SATA1920G-A**<br>• **Hard Disk Unit HDD-SATA1000G-A**<br>**NOTE**<br>• From V600R007C20, SSD-SATA960G-B disks are supported.<br>• From V600R007C20SPC600, SSD-SATA240G-C disks are supported.<br>• From V600R007C20SPC602, SSD-SATA960G-C disks are supported. |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| 10GE optical ports (0 to 19) | SFP+ | 20 GE/10GE autosensing Ethernet optical ports, numbered from XGigabitEthernet0/0/0 to XGigabitEthernet0/0/19.<br>When 100GE0/0/0 to 100GE0/0/1 optical ports on the device are enabled, the 10GE optical ports numbered from XGigabitEthernet0/0/12 to XGigabitEthernet0/0/19 cannot be used. | • 1Gbps SFP Copper Modules<br>• 1Gbps eSFP Optical Modules<br>• 10Gbps SFP+ Optical Modules |
| 40GE optical ports (0 to 3) | QSFP+ | Four 40GE Ethernet optical ports, numbered from 40GE0/0/0 to 40GE0/0/3.<br>40GE optical ports 40GE0/0/2 and 40GE0/0/3 can be switched to 100GE optical ports. | 40Gbps QSFP+ Optical Modules |
| HA ports(0 to 1) | SFP+ | Heartbeat interfaces for hot standby or the cluster. Two devices are connected through independent links to form hot standby or the cluster. numbered from HA0/0/0 to HA0/0/1. | 10Gbps SFP+ Optical Modules |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| USB 3.0 port | USB 3.0 Type A | USB ports allow you to insert USB devices for system software upgrades. For details on upgrades through USB devices, refer to the Upgrade Guide delivered with the device. | USB flash drive |
| Console port | RJ45 | Console ports allow you to locally connect a PC to the device.<br>You can use a console cable to connect the console port (RJ45) on the device to the COM port on your PC and use a serial port terminal program on your PC to access, configure, and manage the device. | **Console Cable** |

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| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| MGMT port | RJ45 | Out-of-band 10/100/1000M RJ45 autosensing Ethernet management port. The interface number is METH 0/0/0 and the default IP address of the interface is 192.168.0.1.<br>You can connect this port to the network port or any reachable port on a PC through a network cable. Then, you can use Telnet to access the CLI or use a web browser to access the web UI to configure, manage, and maintain the device.<br>**NOTE**<br>The MGMT port cannot be used as a service port. | **Ethernet Cable** |
| 100GE optical ports (0 to 1) | QSFP28 | Two 100GE Ethernet optical ports, numbered 100GE0/0/0 and 100GE0/0/1.<br>By default, the 100GE optical port mode is disabled. To enable this mode, run the set device port-config-mode 100g-port enable command. | **100Gbps QSFP28 Optical Modules** |

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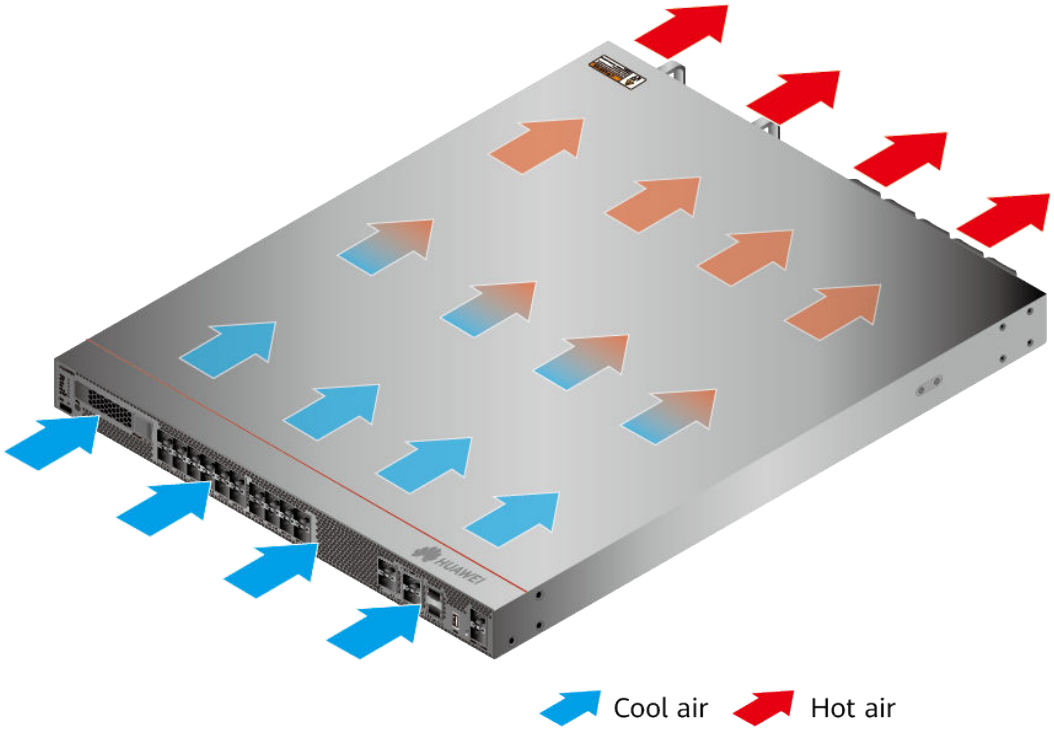
## Power Supply System

- Early delivery: The power supply system has two PAC1K2S12-B AC power modules for 1+1 power redundancy.
- Later delivery: The power supply system has two PAC1K2S12-CB AC power modules for 1+1 power redundancy.

## Heat Dissipation System

The heat dissipation system consists of five FAN-036A-B modules working in 4+1 backup mode. From the front panel, the device provides a front-to-rear air flow. The fan module locates at the air exhaust of the system.

**Figure 3-141** System air flow of the USG6716E




## Technical Specifications

**Table 3-144** Technical specifications of the USG6716E

| Item | Specification |
| --- | --- |
| Installation Type | Rack |
| Cabinet installation standard | Cabinet with a depth of 800 mm or above |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | • Typical dimensions (excluding the parts protruding from the body): 43.6 mm x 442 mm x 600 mm (1.72 in. x 17.40 in. x 23.62 in.)<br>• Maximum dimensions (the depth is the distance from ports on the front panel to the parts protruding from the rear panel): 43.6 mm x 442 mm x 633 mm (1.72 in. x 17.40 in. x 24.92 in.) |
| Dimensions with packaging (H x W x D) [mm(in.)] | 225 mm x 590 mm x 790 mm (8.86 in. x 23.23 in. x 31.1 in.) |
| Chassis height [U] | 1 U |
| Weight with packaging [kg(lb)] | 17.6 kg (38.8 lb) |
| Weight without packaging [kg(lb)] | 12 kg (26.46 lb) |
| Weight with packaging (full configuration) [kg(lb)] | 17.6 kg (38.8 lb) |
| CPU | 2 CPU, 14 cores/CPU, up to 1.9 GHz |
| Memory | 64 GB DDR4 ECC memory, 32 GB/CPU |
| Storage | 32GB |
| NOR Flash | 64MB, 32MB/CPU |
| Hard disk | Optional. You can purchase a 2.5-inch SATA hard disk (240 GB/960 GB/1000 GB) from Huawei based on the site requirements. The hard disk is hot swappable. |
| Console port | RJ45 |
| Eth Management port | RJ45 |
| Typical power consumption [W] | 382.9 W |
| Typical heat dissipation [BTU/hour] | 1309.21 BTU/hour |
| Maximum power consumption [W] | 566 W |
| Maximum heat dissipation [BTU/hour] | 1935.27 BTU/hour |
| MTBF [years] | 25 years |
| MTTR [hours] | 2 hours |
| Availability | 0.999991 |
| Power supply mode | AC pluggable |

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| Item | Specification |
| --- | --- |
| Number of power modules | 2 |
| Redundant power supply | 1+1 |
| Rated input voltage [V] | • USG6716E-AC (02351YRS):<br>– AC power input: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC power input: 240V<br>• USG6716E-AC (02351YRS-001):<br>– AC: 100 V AC to 130 V AC/200 V AC to 240 V AC, 50/60 Hz<br>– High-voltage DC: 240 V DC |
| Input voltage range [V] | AC power input: 90 V AC to 290 V AC, 47 Hz to 63 Hz |
| Maximum input current [A] | 11 A/power module |
| Rated output power [W] | • 800 W (100 V AC to 130 V AC)<br>• 1200 W (200 V AC to 240 V AC)<br>• 1200 W (240 V DC) |
| Maximum output power [W] | • 800 W (100 V AC to 130 V AC)/ power module<br>• 1200 W (200 V AC to 240 V AC)/ power module<br>• 1200 W (240 V DC)/power module |
| Types of fans | Pluggable |
| Number of fan modules | 5 |
| Automatic fan speed adjustment | Supported |
| Heat dissipation mode | • USG6716E-AC (02351YRS): The fan module locates at the air exhaust of the system.<br>• USG6716E-AC (02351YRS-001): Absorbing cold air into the device |
| Airflow direction | Front-to-back airflow |
| Redundant fans | 4+1 |
| Noise at normal temperature (acoustic power) [db(A)] | ≤72dB(A) |

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| Item | Specification |
| --- | --- |
| Long-term operating temperature [°C(°F)] | • Without hard disk driver: 0°C to 45°C<br>• With hard disk driver: 5°C to 40°C |
| Storage temperature [°C(°F)] | -40°C to 70°C |
| Long-term operating relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Storage relative humidity [RH] | 5% RH to 95% RH, non-condensing |
| Long-term operating altitude [m(ft.)] | • Without hard disk driver: 0 m to 5000 m<br>• With hard disk driver: 0 m to 3000 m |
| Storage altitude [m(ft.)] | 0 m to 5000 m |

NOTE

- The width does not include the size of mounting ears.
- The height is 1U (1U = 1.75 inches, or about 44.45 mm), which is a height unit defined in International Electrotechnical Commission (IEC) 60297 standards.
- Temperature and humidity are measured 1.5 m above the floor and 0.4 m in front of the rack when no protection plate exists before or after the rack.
- The operating temperature change rate of a device with HDDs must be less than or equal to 20°C/hour.

# 3.2 Power Modules

This chapter describes hardware information for power modules of the USG6000E, including hardware appearance, functions, and technical specifications.

## 3.2.1 HW-120300D1D (02220947: 36W AC Power Module)

### Overview

**Table 3-145** Basic information about the HW-120300D1D

| Item | Details |
| --- | --- |
| Description | 36W AC Power Module |
| Part Number | 02220947 |
| Model | HW-120300D1D |

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## Appearance

**Figure 3-142** Appearance of the HW-120300D1D



## Version Mapping

**Table 3-146** Mappings between HW-120300D1D and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6530E | USG6530E-AC (02352QVD) | V600R007C00 |
| USG6510E | USG6510E-AC (02352PQH) | V600R007C00 |

## Functions and Features

**Table 3-147** Functions and features of the HW-120300D1D

| Functions and Features | Description |
| --- | --- |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |

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| Functions and Features | Description |
| --- | --- |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Heat dissipation | Fanless design and natural heat dissipation |

### Technical Specifications

**Table 3-148** Technical specifications of the HW-120300D1D

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 32 mm x 53.95 mm x 99.45 mm (1.26 in. x 2.12 in. x 3.92 in.) |
| Weight without packaging [kg(lb)] | 0.205 kg (0.45 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 1 A |
| Rated output voltage [V] | 12V DC |
| Rated output current [A] | 3 A |
| Rated output power [W] | 36 W |
| Type of power cables | C7 |

## 3.2.2 HW-120300D1D (02221468: 36W AC Power Module)

### Overview

**Table 3-149** Basic information about the HW-120300D1D

| Item | Details |
| --- | --- |
| Description | 36W AC Power Module |
| Part Number | 02221468 |
| Model | HW-120300D1D |

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## Appearance

**Figure 3-143** Appearance of the HW-120300D1D



## Version Mapping

**Table 3-150** Mappings between HW-120300D1D and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6530E | USG6530E-AC (02352QVD) | V600R007C00 |
| USG6530E | USG6530E-AC (02352QVD-001) | V600R007C20SPC300 |
| USG6530E | USG6530E-AC (02352QVD-002) | V600R007C20SPC600 |
| USG6510E | USG6510E-AC (02352PQH) | V600R007C00 |
| USG6510E | USG6510E-AC (02352PQH-001) | V600R007C20SPC300 |
| USG6510E | USG6510E-AC (02352PQH-002) | V600R007C20SPC600 |

## Functions and Features

**Table 3-151** Functions and features of the HW-120300D1D

| Functions and Features | Description |
| --- | --- |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |

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| Functions and Features | Description |
| --- | --- |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Heat dissipation | Fanless design and natural heat dissipation |

### Technical Specifications

**Table 3-152** Technical specifications of the HW-120300D1D

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 32 mm x 53.95 mm x 99.45 mm (1.26 in. x 2.12 in. x 3.92 in.) |
| Weight without packaging [kg(lb)] | 0.205 kg (0.45 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 1 A |
| Rated output voltage [V] | 12V DC |
| Rated output current [A] | 3 A |
| Rated output power [W] | 36 W |
| Type of power cables | C7 |

## 3.2.3 HW-560268D0D (150W AC Power Module)

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## Overview

**Table 3-153** Basic information about the HW-560268D0D

| Item | Details |
| --- | --- |
| Description | 150W AC Power Module |
| Part Number | 02221024 |
| Model | HW-560268D0D |

## Appearance

**Figure 3-144** Appearance of the HW-560268D0D



## Version Mapping

**Table 3-154** Mappings between HW-560268D0D and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6510E-POE | USG6510E-POE-AC (02352CSC) | V600R006C00 |

## Functions and Features

**Table 3-155** Functions and features of the HW-560268D0D

| Functions and Features | Description |
| --- | --- |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |

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| Functions and Features | Description |
| --- | --- |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Heat dissipation | Fanless design and natural heat dissipation |

### Technical Specifications

**Table 3-156** Technical specifications of the HW-560268D0D

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 42 mm x 80 mm x 180 mm (1.65 in. x 3.15 in. x 7.09 in.) |
| Weight without packaging [kg(lb)] | 0.8 kg (1.76 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 290 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 2 A |
| Rated output voltage [V] | 55.5V DC |
| Rated output current [A] | 2.68 A |
| Rated output power [W] | 150 W |
| Type of power cables | C7 |

## 3.2.4 PAC150S12-R (150W AC Power Module)

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## Overview

**Table 3-157** Basic information about the PAC150S12-R

| Item | Details |
| --- | --- |
| Description | 150W AC Power Module |
| Part Number | 02312DUY |
| Model | PAC150S12-R |

## Appearance

**Figure 3-145** Appearance of the PAC150S12-R



## Version Mapping

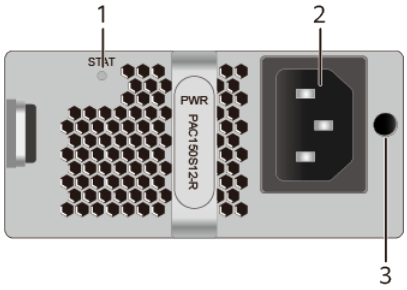
**Table 3-158** Mappings between PAC150S12-R and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |

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## Panel

**Figure 3-146** Panel of the PAC150S12-R




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
|---|---|---|

**Table 3-159** Indicators on the PAC150S12-R

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | Green | Blinking | The output power is out of range. For example, overvoltage, overcurrent, or short circuit has occurred. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-160** Ports on the PAC150S12-R

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

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## Functions and Features

**Table 3-161** Functions and features of the PAC150S12-R

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | The power module does not have fans. The heat dissipation is provided by the fan module of the device. |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

## Technical Specifications

**Table 3-162** Technical specifications of the PAC150S12-R

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 214.3 mm (1.57 in. x 3.54 in. x 8.43 in.) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 0.761 kg (1.68 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 3 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 12.5 A |
| Rated output power [W] | 150 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

## 3.2.5 PAC1K2S12-B (1200W AC Power Module(Power panel side exhaust))

### Overview

**Table 3-163** Basic information about the PAC1K2S12-B

| Item | Details |
| --- | --- |
| Description | 1200W AC Power Module(Power panel side exhaust) |
| Part Number | 02312DUM |
| Model | PAC1K2S12-B |

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## Appearance

**Figure 3-147** Appearance of the PAC1K2S12-B



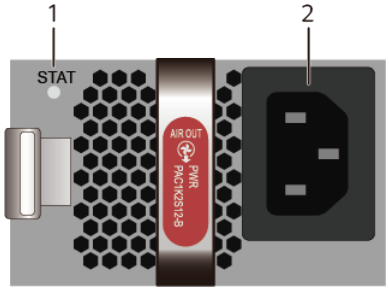

## Version Mapping

**Table 3-164** Mappings between PAC1K2S12-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6716E | USG6716E-AC (02351YRS) | V600R006C00 |
| USG6680E | USG6680E-AC (02351YRR) | V600R006C00 |
| USG6712E | USG6712E-AC (02352CNL) | V600R006C00 |

## Panel

**Figure 3-148** Panel of the PAC1K2S12-B




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| | |
|---|---|
| 1. Power status indicator | 2. Power receptacle |

**Table 3-165** Indicators on the PAC1K2S12-B

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| STAT | Power status indicator | Green | Steady on | The power module is working properly. |
| | | Green | Blinking | The power module is in loading or standby state, or the power cable has been connected but the power module is not installed in the device. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • Fans of the power module fail.<br>• The power module is in overtemperature protection state.<br>• The power input is abnormal (input undervoltage or input overvoltage).<br>• The power output is abnormal (output overcurrent, output short-circuit, or output overvoltage). |
| | | - | Off | There is no power input. |

**Table 3-166** Ports on the PAC1K2S12-B

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |

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## Functions and Features

**Table 3-167** Functions and features of the PAC1K2S12-B

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

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## Technical Specifications

**Table 3-168** Technical specifications of the PAC1K2S12-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.6 mm x 66 mm x 374 mm (1.56 in. x 2.6 in. x 14.72 in.) |
| Weight without packaging [kg(lb)] | 1.4 kg (3.09 lb) |
| Rated input voltage [V] | - Rated AC input voltage range:<br>• 100 V AC to 130 V AC, 50/60 Hz<br>• 200 V AC to 240 V AC, 50/60 Hz<br>- Rated voltage of 240 V high-voltage DC power input: 240 V DC |
| Input voltage range [V] | • Maximum AC input voltage range: 90 V AC to 290 V AC, 47 Hz to 63 Hz<br>• Maximum voltage range of 240 V high-voltage DC power input: 188 V DC to 290 V DC |
| Rated input current [A] | 10 A (100 V AC to 130 V AC), 8 A (200 V AC to 240 V AC), 8 A (240 V DC) |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | • 10 A (100 V AC to 130 V AC)<br>• 8 A (200 V AC to 240 V AC)<br>• 8 A (240 V DC) |
| Rated output power [W] | • 800 W (100 V AC to 130 V AC)<br>• 1200 W (200 V AC to 240 V AC)<br>• 1200 W (240 V DC) |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.6 PAC1K2S12-CB (1200W AC&240V DC Power Module (Back to Front,Power panel side air-out))

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## Overview

**Table 3-169** Basic information about the PAC1K2S12-CB

| Item | Details |
| --- | --- |
| Description | 1200W AC&240V DC Power Module (Back to Front,Power panel side air-out) |
| Part Number | 02312VLE |
| Model | PAC1K2S12-CB |

## Appearance

**Figure 3-149** Appearance of the PAC1K2S12-CB




## Version Mapping

**Table 3-170** Mappings between PAC1K2S12-CB and product models

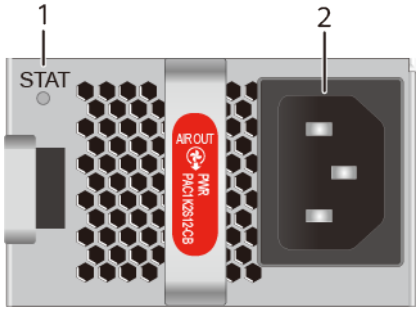
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6716E | USG6716E-AC (02351YRS) | V600R007C20SPC300 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6680E | USG6680E-AC (02351YRR) | V600R007C20SPC300 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6712E | USG6712E-AC (02352CNL) | V600R007C20SPC300 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |

## Panel

**Figure 3-150** Panel of the PAC1K2S12-CB




| 1. Power status indicator | 2. Power receptacle |
| --- | --- |

**Table 3-171** Indicators on the PAC1K2S12-CB

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | The power module is working properly. |
| | | Green | Blinking | The power module is in loading or standby state, or the power cable has been connected but the power module is not installed in the device. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | • Fans of the power module fail.<br>• The power module is in overtemperature protection state.<br>• The power input is abnormal (input undervoltage or input overvoltage).<br>• The power output is abnormal (output overcurrent, output short-circuit, or output overvoltage). |
| | | - | Off | There is no power input. |

**Table 3-172** Ports on the PAC1K2S12-CB

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connect the C13 plug of the AC power cable. For details about AC power cables, see AC Power Cables. | - |

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## Functions and Features

**Table 3-173** Functions and features of the PAC1K2S12-CB

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

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## Technical Specifications

**Table 3-174** Technical specifications of the PAC1K2S12-CB

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.6 mm x 66 mm x 374 mm (1.56 in. x 2.6 in. x 14.72 in.) |
| Weight without packaging [kg(lb)] | 1.2 kg (2.65 lb) |
| Rated input voltage [V] | - Rated AC input voltage range:<br>• 100 V AC to 130 V AC, 50/60 Hz<br>• 200 V AC to 240 V AC, 50/60 Hz<br>- Rated voltage of 240 V high-voltage DC power input: 240 V DC |
| Input voltage range [V] | • Maximum AC input voltage range: 90 V AC to 290 V AC, 47 Hz to 63 Hz<br>• Maximum voltage range of 240 V high-voltage DC power input: 188 V DC to 290 V DC |
| Rated input current [A] | 10 A (100 V AC to 130 V AC), 8 A (200 V AC to 240 V AC), 8 A (240 V DC) |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | • 10 A (100 V AC to 130 V AC)<br>• 8 A (200 V AC to 240 V AC)<br>• 8 A (240 V DC) |
| Rated output power [W] | • 800 W (100 V AC to 130 V AC)<br>• 1200 W (200 V AC to 240 V AC)<br>• 1200 W (240 V DC) |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.7 PAC600S12-B (600W AC Power Module(Back to Front, Power panel side exhaust))

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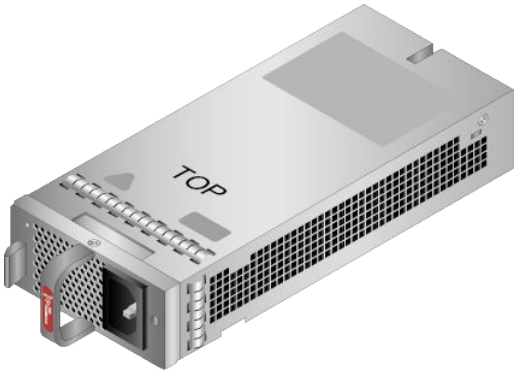
## Overview

**Table 3-175** Basic information about the PAC600S12-B

| Item | Details |
| --- | --- |
| Description | 600W AC Power Module(Back to Front, Power panel side exhaust) |
| Part Number | 02312DUP |
| Model | PAC600S12-B |

## Appearance

**Figure 3-151** Appearance of the PAC600S12-B




## Version Mapping

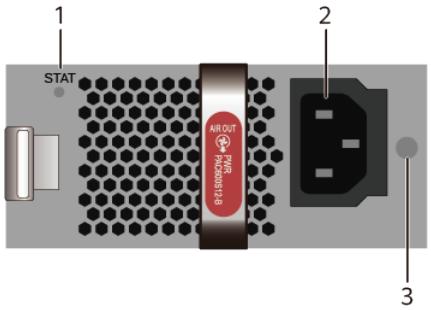
**Table 3-176** Mappings between PAC600S12-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |

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## Panel

**Figure 3-152** Panel of the PAC600S12-B




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
|---|---|---|

**Table 3-177** Indicators on the PAC600S12-B

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

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**Table 3-178** Ports on the PAC600S12-B

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

## Functions and Features

**Table 3-179** Functions and features of the PAC600S12-B

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |

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| Functions and Features | Description |
| --- | --- |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

### Technical Specifications

**Table 3-180** Technical specifications of the PAC600S12-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 215 mm (1.57 in. x 3.54 in. x 8.46 in.) |
| Weight without packaging [kg(lb)] | 0.97 kg (2.14 lb) |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 290 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 9 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 50 A |
| Rated output power [W] | 600 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

## 3.2.8 PAC600S12-CB (600W AC Power Module(Back to Front, Power panel side exhaust))

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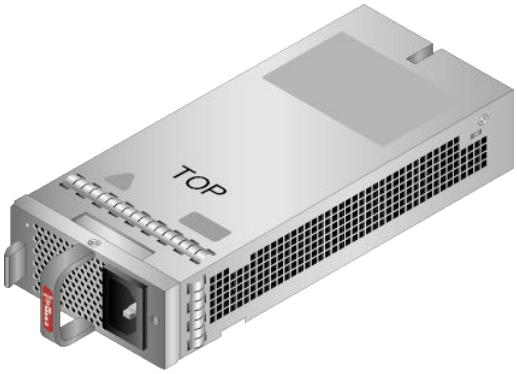
## Overview

**Table 3-181** Basic information about the PAC600S12-CB

| Item | Details |
| --- | --- |
| Description | 600W AC Power Module(Back to Front, Power panel side exhaust) |
| Part Number | 02312FFU |
| Model | PAC600S12-CB |

## Appearance

**Figure 3-153** Appearance of the PAC600S12-CB




## Version Mapping

**Table 3-182** Mappings between PAC600S12-CB and product models

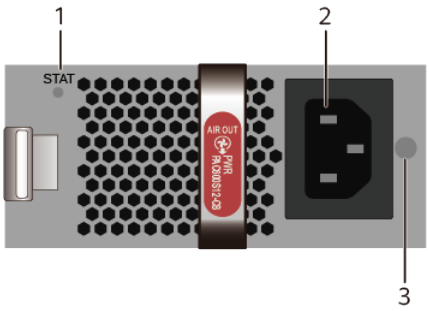
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

## Panel

**Figure 3-154** Panel of the PAC600S12-CB




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
| --- | --- | --- |

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**Table 3-183** Indicators on the PAC600S12-CB

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-184** Ports on the PAC600S12-CB

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

## Functions and Features

**Table 3-185** Functions and features of the PAC600S12-CB

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |

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| Functions and Features | Description |
| --- | --- |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

## Technical Specifications

**Table 3-186** Technical specifications of the PAC600S12-CB

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.6 mm x 90 mm x 214.5 mm (1.56 in. x 3.54 in. x 8.44 in.) |
| Weight without packaging [kg(lb)] | 0.95 kg (2.09 lb) |
| Rated input voltage [V] | 100 V AC to 240 V AC, 50 Hz/60Hz<br>240 V DC |
| Input voltage range [V] | 90 V AC to 290 V AC, 47 Hz to 63 Hz<br>190 V DC to 290 V DC |
| Maximum input current [A] | 8 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 50 A |
| Rated output power [W] | 600 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.9 PAC600S12-DB (600W AC Power Module(Back to Front, Power panel side exhaust))

## Overview

**Table 3-187** Basic information about the PAC600S12-DB

| Item | Details |
| --- | --- |
| Description | 600W AC Power Module(Back to Front, Power panel side exhaust) |

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| Item | Details |
| --- | --- |
| Part Number | 02131740 |
| Model | PAC600S12-DB |

## Appearance

**Figure 3-155** Appearance of the PAC600S12-DB




## Version Mapping

**Table 3-188** Mappings between PAC600S12-DB and product models

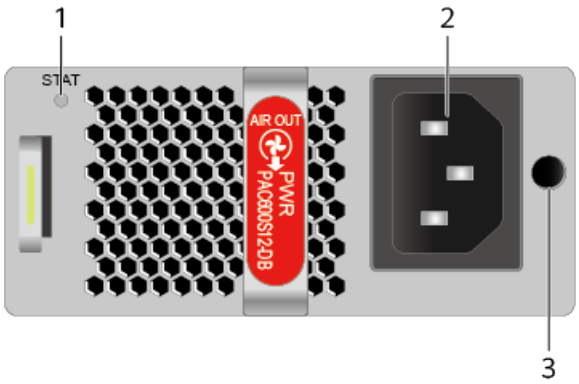
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C20SPC300 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C20SPC300 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R007C20SPC300 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C20SPC300 |

## Panel

**Figure 3-156** Panel of the PAC600S12-DB




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
| --- | --- | --- |

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**Table 3-189** Indicators on the PAC600S12-DB

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-190** Ports on the PAC600S12-DB

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

## Functions and Features

**Table 3-191** Functions and features of the PAC600S12-DB

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |

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| Functions and Features | Description |
| --- | --- |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

## Technical Specifications

**Table 3-192** Technical specifications of the PAC600S12-DB

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.6 mm x 90 mm x 214.5 mm (1.56 in. x 3.54 in. x 8.44 in.) |
| Weight without packaging [kg(lb)] | 0.95 kg (2.09 lb) |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 290 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 8 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 50 A |
| Rated output power [W] | 600 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.10 PAC600S12-EB (600W AC Power Module(Back to Front, Power panel side exhaust))

## Overview

**Table 3-193** Basic information about the PAC600S12-EB

| Item | Details |
| --- | --- |
| Description | 600W AC Power Module(Back to Front, Power panel side exhaust) |
| Part Number | 02312FFU-002 |

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| Item | Details |
| --- | --- |
| Model | PAC600S12-EB |

## Appearance

**Figure 3-157** Appearance of the PAC600S12-EB




## Version Mapping

**Table 3-194** Mappings between PAC600S12-EB and product models

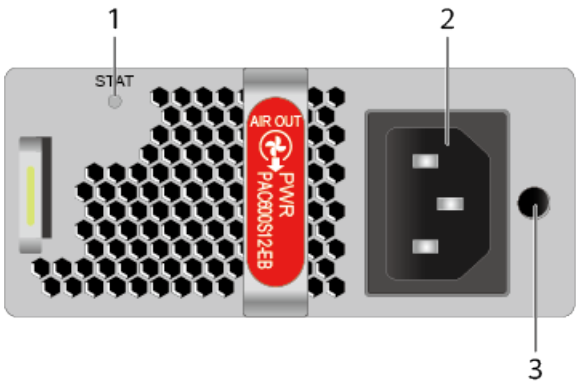
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC500 |
| USG6650E | USG6650E-AC (02351YRY) | V600R007C20SPC500 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C20SPC500 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C20SPC500 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C20SPC500 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C20SPC500 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC500 |
| USG6630E | USG6630E-AC (02352FQA) | V600R007C20SPC500 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC500 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC500 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C20SPC500 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC500 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC500 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C20SPC500 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C20SPC500 |

## Panel

**Figure 3-158** Panel of the PAC600S12-EB




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| | | |
|---|---|---|
| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |

**Table 3-195** Indicators on the PAC600S12-EB

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-196** Ports on the PAC600S12-EB

| Port | Connector Type | Description | Available Components |
|---|---|---|---|
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

## Functions and Features

**Table 3-197** Functions and features of the PAC600S12-EB

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |

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| Functions and Features | Description |
| --- | --- |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

## Technical Specifications

**Table 3-198** Technical specifications of the PAC600S12-EB

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.6 mm x 90 mm x 214.5 mm (1.56 in. x 3.54 in. x 8.44 in.) |
| Weight without packaging [kg(lb)] | 0.985 kg (2.17 lb) |
| Rated input voltage [V] | 100 V AC to 240 V AC, 50 Hz/60Hz<br>240 V DC |
| Input voltage range [V] | 90 V AC to 290 V AC, 47 Hz to 63 Hz<br>190 V DC to 290 V DC |
| Maximum input current [A] | 8 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 50 A |
| Rated output power [W] | 600 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.11 PAC60S12-AR (02312SLE-001: 60W AC Power Module)

## Overview

**Table 3-199** Basic information about the PAC60S12-AR

| Item | Details |
| --- | --- |
| Description | 60W AC Power Module |
| Part Number | 02312SLE-001 |
| Model | PAC60S12-AR |

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## Appearance

**Figure 3-159** Appearance of the PAC60S12-AR



## Version Mapping

**Table 3-200** Mappings between PAC60S12-AR and product models

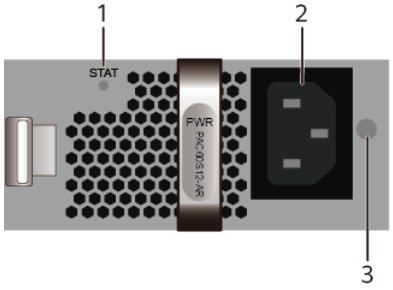
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C20SPC500 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |

## Panel

**Figure 3-160** Panel of the PAC60S12-AR




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
| --- | --- | --- |

**Table 3-201** Indicators on the PAC60S12-AR

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | Green | Blinking | The output power is out of range. For example, overvoltage, overcurrent, or short circuit has occurred. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-202** Ports on the PAC60S12-AR

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | C13 straight female | Connect the C13 plug of the AC power cable. For details about AC power cables, see AC Power Cables. | C13 AC power cable |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

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## Functions and Features

**Table 3-203** Functions and features of the PAC60S12-AR

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | The power module does not have fans. The heat dissipation is provided by the fan module of the device. |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a 60 W power module without interrupting device operation. |

## Technical Specifications

**Table 3-204** Technical specifications of the PAC60S12-AR

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 214.3 mm (1.57 in. x 3.54 in. x 8.43 in.) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 0.68 kg (1.5 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 2 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 5 A |
| Rated output power [W] | 60 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

## 3.2.12 PAC60S12-AR (02312SLE: 60W AC Power Module)

### Overview

**Table 3-205** Basic information about the PAC60S12-AR

| Item | Details |
| --- | --- |
| Description | 60W AC Power Module |
| Part Number | 02312SLE |
| Model | PAC60S12-AR |

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## Appearance

**Figure 3-161** Appearance of the PAC60S12-AR




## Version Mapping

**Table 3-206** Mappings between PAC60S12-AR and product models

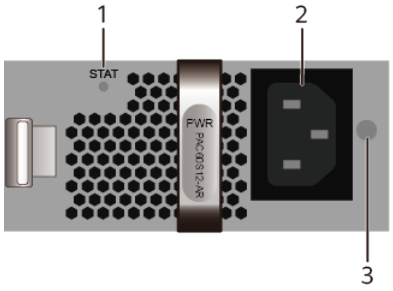
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |

## Panel

**Figure 3-162** Panel of the PAC60S12-AR




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
| --- | --- | --- |

**Table 3-207** Indicators on the PAC60S12-AR

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | Green | Blinking | The output power is out of range. For example, overvoltage, overcurrent, or short circuit has occurred. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

**Table 3-208** Ports on the PAC60S12-AR

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | C13 straight female | Connect the C13 plug of the AC power cable. For details about AC power cables, see AC Power Cables. | C13 AC power cable |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

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## Functions and Features

**Table 3-209** Functions and features of the PAC60S12-AR

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | The power module does not have fans. The heat dissipation is provided by the fan module of the device. |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a 60 W power module without interrupting device operation. |

## Technical Specifications

**Table 3-210** Technical specifications of the PAC60S12-AR

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 214.3 mm (1.57 in. x 3.54 in. x 8.43 in.) |

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| Item | Specification |
| --- | --- |
| Weight without packaging [kg(lb)] | 0.68 kg (1.5 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 2 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 5 A |
| Rated output power [W] | 60 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

## 3.2.13 PAC60S12-R (60W AC Power Module)

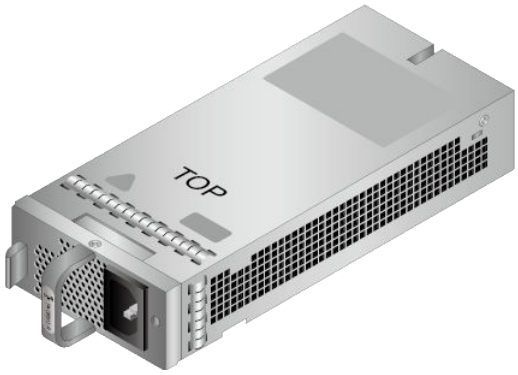
### Overview

**Table 3-211** Basic information about the PAC60S12-R

| Item | Details |
| --- | --- |
| Description | 60W AC Power Module |
| Part Number | 02312DVA |
| Model | PAC60S12-R |

### Appearance

**Figure 3-163** Appearance of the PAC60S12-R




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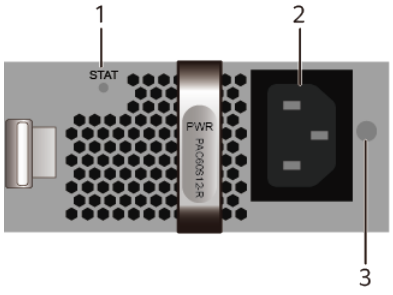
## Version Mapping

**Table 3-212** Mappings between PAC60S12-R and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6560E | USG6560E-AC (02352CBR) | V600R006C00 |
| USG6550E | USG6550E-AC (02352CBQ) | V600R006C00 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6580E | USG6580E-AC (02352CBS) | V600R006C00 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |

## Panel

**Figure 3-164** Panel of the PAC60S12-R




| 1. Power status indicator | 2. Power receptacle | 3. Clip hole |
| --- | --- | --- |

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**Table 3-213** Indicators on the PAC60S12-R

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | Power output of the AC power module is normal. |
| | | Green | Blinking | The output power is out of range. For example, overvoltage, overcurrent, or short circuit has occurred. |
| | | - | Off | • Power input of the AC power module is abnormal (no input, input overvoltage, or input undervoltage).<br>• Power output of the AC power module is abnormal (undervoltage or overtemperature). |

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**Table 3-214** Ports on the PAC60S12-R

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power receptacle | - | Connects the C13 plug of the AC power cable. For details, see AC Power Cables. | - |
| Clip hole | - | The hole is used to install the power cable clip, which is used to bind and fix the power cable. The power cable clip is installed before shipment. | - |

## Functions and Features

**Table 3-215** Functions and features of the PAC60S12-R

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |

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| Functions and Features | Description |
| --- | --- |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | The power module does not have fans. The heat dissipation is provided by the fan module of the device. |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a 60 W power module without interrupting device operation. |

## Technical Specifications

**Table 3-216** Technical specifications of the PAC60S12-R

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 214.3 mm (1.57 in. x 3.54 in. x 8.43 in.) |
| Weight without packaging [kg(lb)] | 0.65 kg (1.43 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | 100 V AC to 240 V AC (50 Hz/60 Hz) |
| Input voltage range [V] | 90 V AC to 264 V AC (47 Hz to 63 Hz) |
| Maximum input current [A] | 2 A |
| Rated output voltage [V] | 12 V |
| Rated output current [A] | 5 A |
| Rated output power [W] | 60 W |
| Hot swapping | Supported |
| Type of power cables | C13 |

# 3.2.14 PDC1000S12-CB (1000W DC Power Module(Back to Front,Power panel side exhaust))

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## Overview

**Table 3-217** Basic information about the PDC1000S12-CB

| Item | Details |
| --- | --- |
| Description | 1000W DC Power Module(Back to Front,Power panel side exhaust) |
| Part Number | 02312JVG |
| Model | PDC1000S12-CB |

## Appearance

**Figure 3-165** Appearance of the PDC1000S12-CB




## Version Mapping

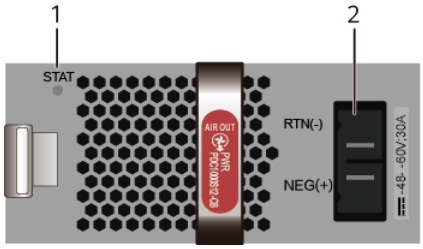
**Table 3-218** Mappings between PDC1000S12-CB and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |

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## Panel

**Figure 3-166** Panel of the PDC1000S12-CB




| | |
|---|---|
| 1. Power status indicator | 2. Power receptacle |

**Table 3-219** Indicators on the PDC1000S12-CB

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| STAT | Power status indicator | Green | Steady on | The output of the DC power module is normal. |
| | | Green | Blinking | The output power is out of range. For example, overvoltage, overcurrent, or short circuit has occurred. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | • The input of the DC power module is out of range. For example, no DC input, DC input overvoltage, or DC input undervoltage has occurred.<br>• The output of the DC power module is out of range. For example, undervoltage, or overtemperature has occurred. |

**Table 3-220** Ports on the PDC1000S12-CB

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power cable terminal | - | Connect the black wire to the RTN (+) terminal and the blue wire to the NEG (-) terminal. For details, see DC Power Cables. | - |

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## Functions and Features

**Table 3-221** Functions and features of the PDC1000S12-CB

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently stops output and automatically restores output after the overvoltage condition is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

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### Technical Specifications

**Table 3-222** Technical specifications of the PDC1000S12-CB

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 215 mm (1.57 in. x 3.54 in. x 8.46 in.) |
| Weight without packaging [kg(lb)] | 1 kg (2.2 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | -48V DC to -60V DC |
| Input voltage range [V] | -38.4V DC to -72V DC |
| Maximum input current [A] | 30 A |
| Rated output voltage [V] | 12 V |
| Rated output power [W] | 1000 W |
| Hot swapping | Supported |

## 3.2.15 PDC350S12-B (350W DC Power Module)

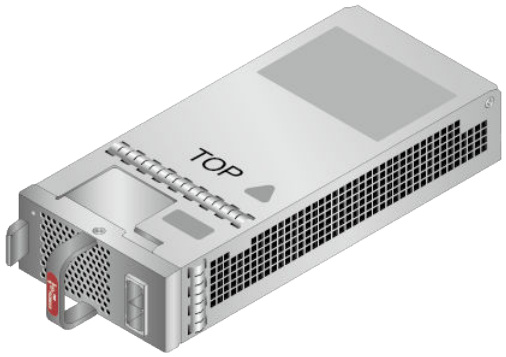
### Overview

**Table 3-223** Basic information about the PDC350S12-B

| Item | Details |
| --- | --- |
| Description | 350W DC Power Module |
| Part Number | 02312DVB |
| Model | PDC350S12-B |

### Appearance

**Figure 3-167** Appearance of the PDC350S12-B




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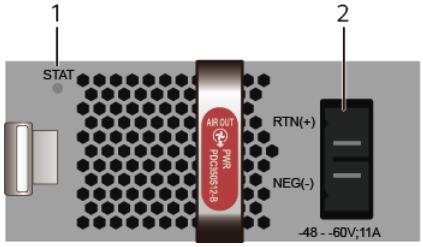
## Version Mapping

**Table 3-224** Mappings between PDC350S12-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |

## Panel

**Figure 3-168** Panel of the PDC350S12-B




| | |
| --- | --- |
| 1. Power status indicator | 2. Power receptacle |

**Table 3-225** Indicators on the PDC350S12-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| STAT | Power status indicator | Green | Steady on | The output of the DC power module is normal. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | • The input of the DC power module is out of range. For example, no DC input, DC input overvoltage, or DC input undervoltage has occurred.<br>• The output of the DC power module is out of range. For example, undervoltage, or overtemperature has occurred. |

**Table 3-226** Ports on the PDC350S12-B

| Port | Connector Type | Description | Available Components |
| --- | --- | --- | --- |
| Power cable terminal | - | Connect the black wire to the RTN (+) terminal and the blue wire to the NEG (-) terminal. For details, see DC Power Cables. | - |

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## Functions and Features

**Table 3-227** Functions and features of the PDC350S12-B

| Functions and Features | Description |
| --- | --- |
| Input undervoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overvoltage protection | Stops power output and automatically restores power output after the input voltage becomes normal. |
| Input overcurrent protection | Stops power output and does not automatically restore power output after the input current becomes normal. |
| Output current limiting protection | Intermittently provides output and automatically restores normal output after the output current falls within a normal range. |
| Output overvoltage protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Output short-circuit protection | Intermittently provides output and automatically restores normal output after the output short circuit is removed. |
| Overtemperature protection | When the temperature of the power module reaches a preset threshold, the power module stops power output and will automatically restore power output after the temperature drops back to the normal range. |
| Heat dissipation | Heat dissipation through fans |
| Hot swap | The device has 1+1 power module redundancy. You can hot-swap a power module without interrupting device operation. |

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### Technical Specifications

**Table 3-228** Technical specifications of the PDC350S12-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 39.8 mm x 90 mm x 215 mm (1.57 in. x 3.54 in. x 8.46 in.) |
| Weight without packaging [kg(lb)] | 0.689 kg (1.52 lb) |
| Number of inputs | 1 |
| Rated input voltage [V] | -48V DC to -60V DC |
| Input voltage range [V] | -38.4V DC to -72V DC |
| Maximum input current [A] | 11 A |
| Rated output voltage [V] | 12 V |
| Rated output power [W] | 350 W |
| Hot swapping | Supported |

# 3.3 Fan Modules

This chapter describes hardware information for all fan modules of the USG6000E, including hardware appearance, and technical specifications.

## 3.3.1 FAN-023A-B (Fan Box,FAN-023A-B,Fan Box,Fan box(B,FAN panel side exhaust))

### Overview

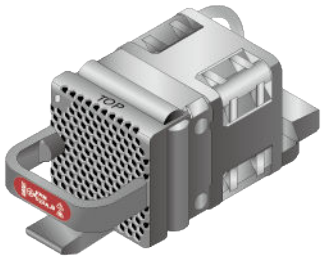
**Table 3-229** Basic information about the FAN-023A-B

| Item | Details |
| --- | --- |
| Description | Fan Box,FAN-023A-B,Fan Box,Fan box(B,FAN panel side exhaust) |
| Part Number | 02312DKW |
| Model | FAN-023A-B |

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## Appearance

**Figure 3-169** Appearance of the FAN-023A-B



## Version Mapping

**Table 3-230** Mappings between FAN-023A-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6560E | USG6560E-AC (02352CBR) | V600R006C00 |
| USG6550E | USG6550E-AC (02352CBQ) | V600R006C00 |
| USG6580E | USG6580E-AC (02352CBS) | V600R006C00 |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |

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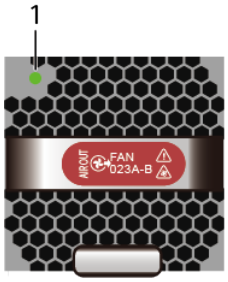
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

## Panel

**Figure 3-170** Panel of the FAN-023A-B




| |
| --- |
| 1. FAN indicator |

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**Table 3-231** Indicators on the FAN-023A-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | FAN indicator | Green | Blink every two seconds (0.5 Hz) | The fan module is running normally. |
| | | Red | Steady on | The fan module is faulty. For example, the fans stop running or are blocked, the power supply of the fan is abnormal, the rotation speed of the fan does not match, or the fan module is not in position. For details, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.1 hwFanRemove_137216, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139264, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139267. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Blink every two seconds (0.5 Hz) | The fan module generates an alarm, for example, indicating the air extraction and blowing mode is incorrect. s, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139265. |
| | | - | Off | The fan module is not installed. |

## Functions and Features

**Table 3-232** Functions and features of the FAN-023A-B

| Functions and Features | Description |
| --- | --- |
| Basic function | The fan module supports hot swapping. It consists of the fan module frame, fan, and light pipe. |

## Technical Specifications

**Table 3-233** Technical specifications of the FAN-023A-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 40 mm x 40 mm x 100.3 mm (1.57 in. x 1.57 in. x 3.95 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Number of fans | 1 |
| Typical power consumption [W] | 7.2 W |

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| Item | Specification |
| --- | --- |
| Maximum power consumption (40°C) [W] | 8.64 W |
| Operating voltage range [V] | 7 V DC to 15 V DC |
| Maximum airflow [CFM] | 25.29 CFM |
| Maximum noise level [dB(A)] | 59 dB(A) |
| Maximum wind pressure [Pa] | 531.94 Pa |

## 3.3.2 FAN-023A-B (Fan Box,FAN-023A-B,Fan Box,Fan box(B,FAN panel side exhaust))

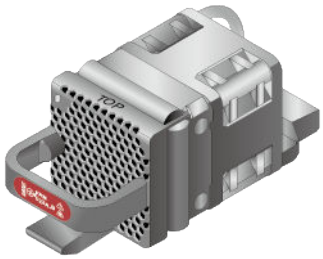
### Overview

**Table 3-234** Basic information about the FAN-023A-B

| Item | Details |
| --- | --- |
| Description | Fan Box,FAN-023A-B,Fan Box,Fan box(B,FAN panel side exhaust) |
| Part Number | 02312DKW |
| Model | FAN-023A-B |

### Appearance

**Figure 3-171** Appearance of the FAN-023A-B




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## Version Mapping

**Table 3-235** Mappings between FAN-023A-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6560E | USG6560E-AC (02352CBR) | V600R006C00 |
| USG6550E | USG6550E-AC (02352CBQ) | V600R006C00 |
| USG6580E | USG6580E-AC (02352CBS) | V600R006C00 |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

## Panel

**Figure 3-172** Panel of the FAN-023A-B

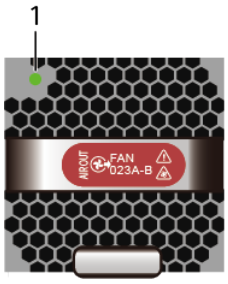



**Table 3-236** Indicators on the FAN-023A-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | FAN indicator | Green | Blink every two seconds (0.5 Hz) | The fan module is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | The fan module is faulty. For example, the fans stop running or are blocked, the power supply of the fan is abnormal, the rotation speed of the fan does not match, or the fan module is not in position. For details, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.1 hwFanRemove_137216, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139264, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139267. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Blink every two seconds (0.5 Hz) | The fan module generates an alarm, for example, indicating the air extraction and blowing mode is incorrect. s, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139265. |
| | | - | Off | The fan module is not installed. |

## Functions and Features

**Table 3-237** Functions and features of the FAN-023A-B

| Functions and Features | Description |
| --- | --- |
| Basic function | The fan module supports hot swapping. It consists of the fan module frame, fan, and light pipe. |

## Technical Specifications

**Table 3-238** Technical specifications of the FAN-023A-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 40 mm x 40 mm x 100.3 mm (1.57 in. x 1.57 in. x 3.95 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Number of fans | 1 |
| Typical power consumption [W] | 7.2 W |

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| Item | Specification |
| --- | --- |
| Maximum power consumption (40°C) [W] | 8.64 W |
| Operating voltage range [V] | 7 V DC to 15 V DC |
| Maximum airflow [CFM] | 25.29 CFM |
| Maximum noise level [dB(A)] | 59 dB(A) |
| Maximum wind pressure [Pa] | 531.94 Pa |

## 3.3.3 FAN-036A-B (FAN-036A-B,Fan Box,Fan box(B,FAN panel side exhaust))

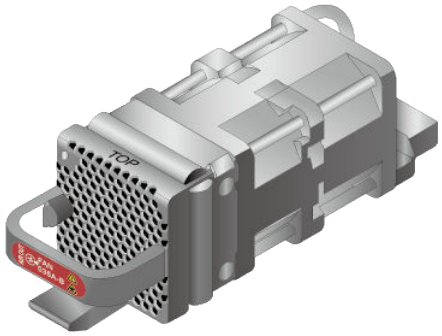
### Overview

**Table 3-239** Basic information about the FAN-036A-B

| Item | Details |
| --- | --- |
| Description | FAN-036A-B,Fan Box,Fan box(B,FAN panel side exhaust) |
| Part Number | 02312EYW |
| Model | FAN-036A-B |

### Appearance

**Figure 3-173** Appearance of the FAN-036A-B




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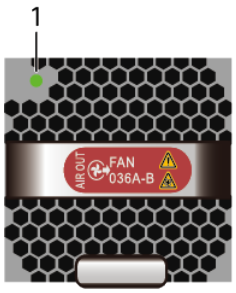
## Version Mapping

**Table 3-240** Mappings between FAN-036A-B and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6716E | USG6716E-AC (02351YRS) | V600R006C00 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6680E | USG6680E-AC (02351YRR) | V600R006C00 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6712E | USG6712E-AC (02352CNL) | V600R006C00 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |

## Panel

**Figure 3-174** Panel of the FAN-036A-B




1. FAN indicator

**Table 3-241** Indicators on the FAN-036A-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | FAN indicator | Green | Blink every two seconds (0.5 Hz) | The fan module is running normally. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Steady on | The fan module is faulty. For example, the fans stop running or are blocked, the power supply of the fan is abnormal, the rotation speed of the fan does not match, or the fan module is not in position. For details, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.1 hwFanRemove_137216, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139264, SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139267. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Red | Blink every two seconds (0.5 Hz) | The fan module generates an alarm, for example, indicating the air extraction and blowing mode is incorrect. s, see SRM_ENTITY_1.3.6.1.4.1.2011.5.25.219.2.6.5 hwFanInvalid_139265. |
| | | - | Off | The fan module is not installed. |

## Functions and Features

**Table 3-242** Functions and features of the FAN-036A-B

| Functions and Features | Description |
| --- | --- |
| Basic function | The fan module supports hot swapping. It consists of the fan module frame, fan, and light pipe. |

## Technical Specifications

**Table 3-243** Technical specifications of the FAN-036A-B

| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 40 mm x 40 mm x 128.3 mm |
| Weight without packaging [kg(lb)] | 0.7 kg (1.54 lb) |
| Number of fans | 1 |
| Typical power consumption [W] | 28 W |

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| Item | Specification |
| --- | --- |
| Maximum power consumption (40°C) [W] | 33.6 W |
| Operating voltage range [V] | 10.8 V DC to 13.2 V DC |
| Maximum airflow [CFM] | 36 CFM |
| Maximum noise level [dB(A)] | 69.8 dB(A) |
| Maximum wind pressure [Pa] | 950 Pa |

# 3.4 Storage Devices

This chapter describes hardware information for hard disk modules of the USG6000E, including hardware appearance, functions, and technical specifications.

## 3.4.1 Hard Disk

### 3.4.1.1 HDD-SATA1000G-A (HardDisk-1000GB-SATA-7200rpm-2.5"-64M-hot-swap-built-in-Front Panel-Silver)

#### Overview

**Table 3-244** Basic information about the HDD-SATA1000G-A

| Item | Details |
| --- | --- |
| Description | HardDisk-1000GB-SATA-7200rpm-2.5"-64M-hot-swap-built-in-Front Panel-Silver |
| Part Number | 02312DLF |
| Model | HDD-SATA1000G-A |
| Silkscreen | SATA<br>1000GB 7.2K<br>02312DLF |

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## Appearance

**Figure 3-175** Appearance of the HDD-SATA1000G-A




## Version Mapping

**Table 3-245** Mappings between HDD-SATA1000G-A and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6680E | USG6680E-AC (02351YRR) | V600R006C00 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS) | V600R006C00 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6712E | USG6712E-AC (02352CNL) | V600R006C00 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

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## Panel

**Figure 3-176** Panel of the HDD-SATA1000G-A




| 1. ALM indicator | 2. RUN indicator |
|---|---|

**Table 3-246** Indicators on the HDD-SATA1000G-A

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-247** Functions and features of the HDD-SATA1000G-A

| Functions and Features | Description |
|---|---|
| Basic functions | Storage log. |

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| Functions and Features | Description |
| --- | --- |
| Hot swap | Hot swap is supported, but hard disks cannot be directly swapped or removed. Run the disk offline command. Remove the hard disk about 30 seconds after the system displays a message indicating that the hard disk stops working. Alternatively, press and hold the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, and the system may crash. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-248** Technical specifications of the HDD-SATA1000G-A

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 1000 |
| Port type | SATA |
| Maximum power consumption [W] | 16.72 W |
| Maximum heat dissipation [BTU/hour] | 57.03 BTU/hour |

## 3.4.1.2 M.2-SATA240G-A (M.2 SSD,SATA 6Gb/s-240GB,Hot-Swappable)

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## Overview

**Table 3-249** Basic information about the M.2-SATA240G-A

| Item | Details |
| --- | --- |
| Description | M.2 SSD,SATA 6Gb/s-240GB,Hot-Swappable |
| Part Number | 02312DLK |
| Model | M.2-SATA240G-A |

## Appearance

**Figure 3-177** Appearance of the M.2-SATA240G-A




## Version Mapping

**Table 3-250** Mappings between M.2-SATA240G-A and product models

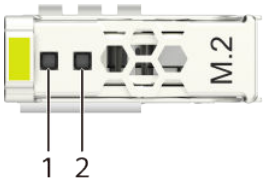
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6560E | USG6560E-AC (02352CBR) | V600R006C00 |
| USG6550E | USG6550E-AC (02352CBQ) | V600R006C00 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6580E | USG6580E-AC (02352CBS) | V600R006C00 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |

## Panel

**Figure 3-178** Panel of the M.2-SATA240G-A




| | |
| --- | --- |
| 1. ALM indicator | 2. RUN indicator |

**Table 3-251** Indicators on the M.2-SATA240G-A

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-252** Functions and features of the M.2-SATA240G-A

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Supports hot swap. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-253** Technical specifications of the M.2-SATA240G-A

| Item | Specification |
| --- | --- |
| Form factor | M.2 |
| Dimensions without packaging (H x W x D) [mm(in.)] | 10 mm x 25 mm x 110 mm (0.39 in. x 0.98 in. x 4.33 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Storage capacity [GB] | 240 |
| Port type | SATA |
| Maximum power consumption [W] | 5 W |
| Maximum heat dissipation [BTU/hour] | 5 BTU/hour |

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### 3.4.1.3 M.2-SATA64G-A (M.2 SSD,SATA 6Gb/s-64GB,Hot-Swappable)

## Overview

**Table 3-254** Basic information about the M.2-SATA64G-A

| Item | Details |
| --- | --- |
| Description | M.2 SSD,SATA 6Gb/s-64GB,Hot-Swappable |
| Part Number | 02312DLJ |
| Model | M.2-SATA64G-A |

## Appearance

**Figure 3-179** Appearance of the M.2-SATA64G-A



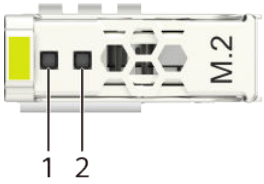

## Version Mapping

**Table 3-255** Mappings between M.2-SATA64G-A and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |

## Panel

**Figure 3-180** Panel of the M.2-SATA64G-A




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| | |
|---|---|
| 1. ALM indicator | 2. RUN indicator |

**Table 3-256** Indicators on the M.2-SATA64G-A

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-257** Functions and features of the M.2-SATA64G-A

| Functions and Features | Description |
|---|---|
| Basic functions | Storage log. |
| Hot swap | Supports hot swap. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-258** Technical specifications of the M.2-SATA64G-A

| Item | Specification |
|---|---|
| Form factor | M.2 |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 10 mm x 25 mm x 110 mm (0.39 in. x 0.98 in. x 4.33 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Storage capacity [GB] | 64 |
| Port type | SATA |
| Maximum power consumption [W] | 2.7 W |
| Maximum heat dissipation [BTU/hour] | 2.7 BTU/hour |

### 3.4.1.4 SSD-Sata240G-A (SSD,240GB,SATA 6Gb/s,Read Intensive,2.5inch)

## Overview

**Table 3-259** Basic information about the SSD-Sata240G-A

| Item | Details |
| --- | --- |
| Description | SSD,240GB,SATA 6Gb/s,Read Intensive,2.5inch |
| Part Number | 02312DLH |
| Model | SSD-Sata240G-A |
| Silkscreen | SATA<br>240GB SSD<br>02312DLH |

## Appearance

**Figure 3-181** Appearance of the SSD-Sata240G-A




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## Version Mapping

**Table 3-260** Mappings between SSD-Sata240G-A and product models

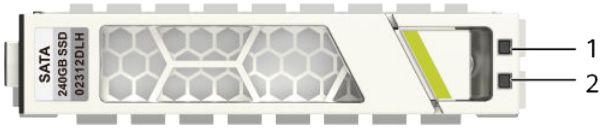
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6680E | USG6680E-AC (02351YRR) | V600R006C00 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS) | V600R006C00 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6712E | USG6712E-AC (02352CNL) | V600R006C00 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

## Panel

**Figure 3-182** Panel of the SSD-Sata240G-A




| | |
| --- | --- |
| 1. ALM indicator | 2. RUN indicator |

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**Table 3-261** Indicators on the SSD-Sata240G-A

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-262** Functions and features of the SSD-Sata240G-A

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Hot swap is supported, but hard disks cannot be directly swapped or removed. Run the disk offline command. Remove the hard disk about 30 seconds after the system displays a message indicating that the hard disk stops working. Alternatively, press and hold the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, and the system may crash. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

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## Technical Specifications

**Table 3-263** Technical specifications of the SSD-Sata240G-A

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 240 |
| Port type | SATA |
| Maximum power consumption [W] | 4.36 W |
| Maximum heat dissipation [BTU/hour] | 14.86 BTU/hour |

### 3.4.1.5 SSD-SATA960G-B (SSD 960GB,SATA 6Gb/s,Read Intensive,2.5inch)

## Overview

**Table 3-264** Basic information about the SSD-SATA960G-B

| Item | Details |
| --- | --- |
| Description | SSD 960GB,SATA 6Gb/s,Read Intensive,2.5inch |
| Part Number | 02312YNW |
| Model | SSD-SATA960G-B |
| Silkscreen | SATA<br>960GB SSD<br>02312YNW |

## Appearance

**Figure 3-183** Appearance of the SSD-SATA960G-B




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## Version Mapping

**Table 3-265** Mappings between SSD-SATA960G-B and product models

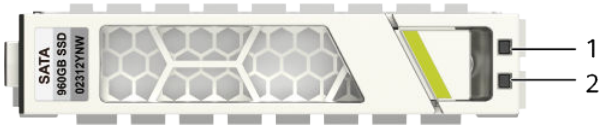
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R007C20 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R007C20 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R007C20 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6680E | USG6680E-AC (02351YRR) | V600R007C20 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS) | V600R007C20 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C20 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6712E | USG6712E-AC (02352CNL) | V600R007C20 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C20 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C20 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C20 |

## Panel

**Figure 3-184** Panel of the SSD-SATA960G-B




| | |
| --- | --- |
| 1. ALM indicator | 2. RUN indicator |

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**Table 3-266** Indicators on the SSD-SATA960G-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-267** Functions and features of the SSD-SATA960G-B

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Hot swap is supported, but hard disks cannot be directly swapped or removed. Run the disk offline command. Remove the hard disk about 30 seconds after the system displays a message indicating that the hard disk stops working. Alternatively, press and hold the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, and the system may crash. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

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## Technical Specifications

**Table 3-268** Technical specifications of the SSD-SATA960G-B

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 960 |
| Port type | SATA |
| Maximum power consumption [W] | 5.23 W |
| Maximum heat dissipation [BTU/hour] | 17.85 BTU/hour |

### 3.4.1.6 SSD-SATA240G-C (SSD-SATA240G-C-Solid State Drives,2.5inch)

## Overview

**Table 3-269** Basic information about the SSD-SATA240G-C

| Item | Details |
| --- | --- |
| Description | SSD-SATA240G-C-Solid State Drives,2.5inch |
| Part Number | 02313XED |
| Model | SSD-SATA240G-C |
| Silkscreen | M.2 |

## Appearance

**Figure 3-185** Appearance of the SSD-SATA240G-C




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## Version Mapping

**Table 3-270** Mappings between SSD-SATA240G-C and product models

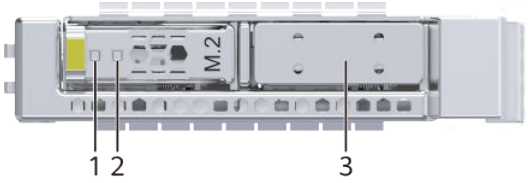
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20SPC600 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6680E | USG6680E-AC (02351YRR) | V600R007C20SPC600 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C20SPC600 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20SPC600 |
| USG6712E | USG6712E-AC (02352CNL) | V600R007C20SPC600 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C20SPC600 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C20SPC600 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C20SPC600 |

## Panel

**Figure 3-186** Panel of the SSD-SATA240G-C




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| | | |
|---|---|---|
| 1. ALM indicator | 2. RUN indicator | 3. Filler panel of the M.2 hard disk slot<br>**NOTE**<br>The hard disk slot function is reserved and is not enabled. |

**Table 3-271** Indicators on the SSD-SATA240G-C

| Silkscreen | Name | Color | Status | Description |
|---|---|---|---|---|
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-272** Functions and features of the SSD-SATA240G-C

| Functions and Features | Description |
|---|---|
| Basic functions | Storage log. |

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| Functions and Features | Description |
| --- | --- |
| Hot swap | Hot swap is supported, but hard disks cannot be directly swapped or removed. Run the disk offline command. Remove the hard disk about 30 seconds after the system displays a message indicating that the hard disk stops working. Alternatively, press and hold the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, and the system may crash. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-273** Technical specifications of the SSD-SATA240G-C

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 240 |
| Port type | SATA |
| Maximum power consumption [W] | 5.23 W |
| Maximum heat dissipation [BTU/hour] | 17.85 BTU/hour |

## 3.4.1.7 SSD-SATA960G-C (SSD-SATA960G-C-Solid State Drives,2.5inch)

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## Overview

**Table 3-274** Basic information about the SSD-SATA960G-C

| Item | Details |
| --- | --- |
| Description | SSD-SATA960G-C-Solid State Drives,2.5inch |
| Part Number | 02313XEE |
| Model | SSD-SATA960G-C |
| Silkscreen | M.2 |

## Appearance

**Figure 3-187** Appearance of the SSD-SATA960G-C




## Version Mapping

**Table 3-275** Mappings between SSD-SATA960G-C and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20SPC602 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC602 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC602 |
| USG6630E | USG6630E-DC (02352CKV) | V600R007C20SPC602 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC602 |

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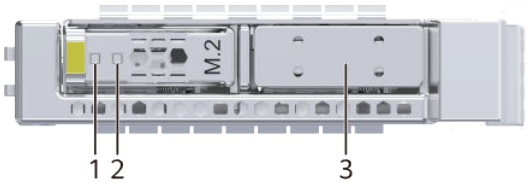
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-AC (02352FQA) | V600R007C20SPC602 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC602 |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC602 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC602 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC602 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC602 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC602 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC602 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC602 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC602 |
| USG6650E | USG6650E-AC (02351YRY) | V600R007C20SPC602 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC602 |
| USG6680E | USG6680E-AC (02351YRR) | V600R007C20SPC602 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC602 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20SPC602 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC602 |
| USG6716E | USG6716E-AC (02351YRS) | V600R007C20SPC602 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC602 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C20SPC602 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20SPC602 |
| USG6712E | USG6712E-AC (02352CNL) | V600R007C20SPC602 |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC602 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C20SPC602 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C20SPC602 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C20SPC602 |

## Panel

**Figure 3-188** Panel of the SSD-SATA960G-C




| | | |
| --- | --- | --- |
| 1. ALM indicator | 2. RUN indicator | 3. Filler panel of the M.2 hard disk slot<br>**NOTE**<br>The hard disk slot function is reserved and is not enabled. |

**Table 3-276** Indicators on the SSD-SATA960G-C

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | Alarm indicator | Orange | Steady on | The hard disk is faulty. |
| | | - | Off | The hard disk is running properly. |
| - | Running indicator | Green | Steady on | The hard disk is in position. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-277** Functions and features of the SSD-SATA960G-C

| Functions and Features | Description |
| --- | --- |
| Basic function | Store logs. |
| Hot swapping | Hot swapping is supported, but hard disks cannot be forcibly inserted or removed. Instead, run the disk offline command and wait for about 30 seconds until the system displays a message indicating that the hard disk stops working. Then you can remove the hard disk. Alternatively, press and hold down the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, or the system may break down. |
| Restrictions and remarks | Do not perform low-level formatting on the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-278** Technical specifications of the SSD-SATA960G-C

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |

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| Item | Specification |
| --- | --- |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 960 |
| Port type | SATA |
| Maximum power consumption [W] | 5.23 W |
| Maximum heat dissipation [BTU/hour] | 17.85 BTU/hour |

### 3.4.1.8 M.2-SATA64G-B (M.2 SSD,SATA 6Gb/s-64GB,Hot-Swappable)

## Overview

**Table 3-279** Basic information about the M.2-SATA64G-B

| Item | Details |
| --- | --- |
| Description | M.2 SSD,SATA 6Gb/s-64GB,Hot-Swappable |
| Part Number | 02314JAM |
| Model | M.2-SATA64G-B |

## Appearance

**Figure 3-189** Appearance of the M.2-SATA64G-B




## Version Mapping

**Table 3-280** Mappings between M.2-SATA64G-B and product models

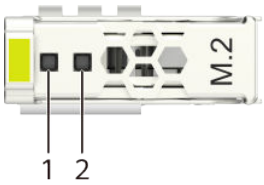
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |

## Panel

**Figure 3-190** Panel of the M.2-SATA64G-B




| 1. ALM indicator | 2. RUN indicator |
| --- | --- |

**Table 3-281** Indicators on the M.2-SATA64G-B

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

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## Functions and Features

**Table 3-282** Functions and features of the M.2-SATA64G-B

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Supports hot swap. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-283** Technical specifications of the M.2-SATA64G-B

| Item | Specification |
| --- | --- |
| Form factor | M.2 |
| Dimensions without packaging (H x W x D) [mm(in.)] | 10 mm x 25 mm x 110 mm (0.39 in. x 0.98 in. x 4.33 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Storage capacity [GB] | 64 |
| Port type | SATA |
| Maximum power consumption [W] | 2.7 W |
| Maximum heat dissipation [BTU/hour] | 2.7 BTU/hour |

# 3.4.1.9 M.2-SATA960G-A (M.2 SSD,SATA 6Gb/s-960GB,Hot-Swappable)

## Overview

**Table 3-284** Basic information about the M.2-SATA960G-A

| Item | Details |
| --- | --- |
| Description | M.2 SSD,SATA 6Gb/s-960GB,Hot-Swappable |
| Part Number | 02313XEF |
| Model | M.2-SATA960G-A |

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## Appearance

**Figure 3-191** Appearance of the M.2-SATA960G-A




## Version Mapping

**Table 3-285** Mappings between M.2-SATA960G-A and product models

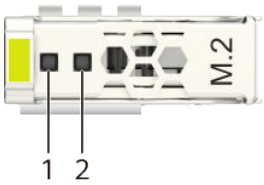
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6560E | USG6560E-AC (02352CBR) | V600R006C00 |
| USG6550E | USG6550E-AC (02352CBQ) | V600R006C00 |
| USG6585E | USG6585E-AC (02353ADA) | V600R007C00 |
| USG6585E | USG6585E-AC (02353ADA-001) | V600R007C20SPC500 |
| USG6585E | USG6585E-AC (02353ADA-002) | V600R007C20SPC600 |
| USG6580E | USG6580E-AC (02352CBS) | V600R006C00 |
| USG6525E | USG6525E-AC (02353AFX) | V600R007C00 |
| USG6525E | USG6525E-AC (02353AFX-001) | V600R007C20SPC500 |
| USG6525E | USG6525E-AC (02353AFX-002) | V600R007C20SPC600 |
| USG6515E | USG6515E-AC (02352CBP) | V600R006C00 |
| USG6565E | USG6565E-AC (02353AEJ) | V600R007C00 |
| USG6565E | USG6565E-AC (02353AEJ-001) | V600R007C20SPC500 |
| USG6565E | USG6565E-AC (02353AEJ-002) | V600R007C20SPC600 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6555E | USG6555E-AC (02353AEK) | V600R007C00 |
| USG6555E | USG6555E-AC (02353AEK-001) | V600R007C20SPC500 |
| USG6555E | USG6555E-AC (02353AEK-004) | V600R007C20SPC600 |

## Panel

**Figure 3-192** Panel of the M.2-SATA960G-A




| | |
| --- | --- |
| 1. ALM indicator | 2. RUN indicator |

**Table 3-286** Indicators on the M.2-SATA960G-A

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

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## Functions and Features

**Table 3-287** Functions and features of the M.2-SATA960G-A

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Supports hot swap. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-288** Technical specifications of the M.2-SATA960G-A

| Item | Specification |
| --- | --- |
| Form factor | M.2 |
| Dimensions without packaging (H x W x D) [mm(in.)] | 10 mm x 25 mm x 110 mm (0.39 in. x 0.98 in. x 4.33 in.) |
| Weight without packaging [kg(lb)] | 0.1 kg (0.22 lb) |
| Storage capacity [GB] | 960 |
| Port type | SATA |
| Maximum power consumption [W] | 5 W |
| Maximum heat dissipation [BTU/hour] | 5 BTU/hour |

# 3.4.1.10 SSD-SATA1920G-A (SSD,1920GB,SATA 6Gb/s,Read Intensive,2.5inch)

## Overview

**Table 3-289** Basic information about the SSD-SATA1920G-A

| Item | Details |
| --- | --- |
| Description | SSD,1920GB,SATA 6Gb/s,Read Intensive,2.5inch |
| Part Number | 02314PBR |
| Model | SSD-SATA1920G-A |

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| Item | Details |
| --- | --- |
| Silkscreen | SATA<br>1920GB SSD<br>02314PBR |

## Appearance

**Figure 3-193** Appearance of the SSD-SATA1920G-A




## Version Mapping

**Table 3-290** Mappings between SSD-SATA1920G-A and product models

| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6575E-B | USG6575E-B-AC (02353LKD) | V600R007C20 |
| USG6575E-B | USG6575E-B-AC (02353LKD-002) | V600R007C20SPC600 |
| USG6630E | USG6630E-AC (02352FQA-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-DC (02352CKV) | V600R006C00 |
| USG6630E | USG6630E-DC (02352CKV-001) | V600R007C20SPC300 |
| USG6630E | USG6630E-AC (02352FQA) | V600R006C00 |
| USG6630E | USG6630E-AC (02352FQA-002) | V600R007C20SPC600 |

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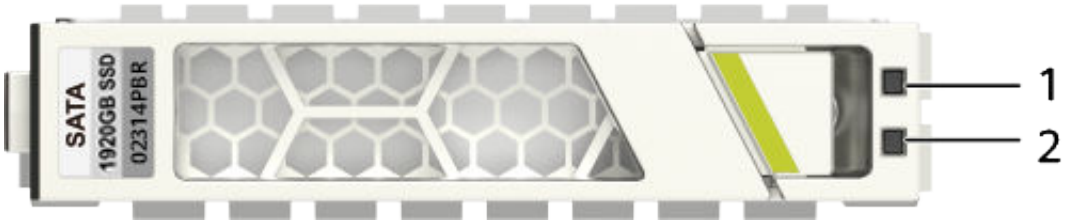
| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6630E | USG6630E-DC (02352CKV-002) | V600R007C20SPC600 |
| USG6620E | USG6620E-AC (02352HMF-001) | V600R007C20SPC300 |
| USG6620E | USG6620E-AC (02352HMF) | V600R007C20SPC100 |
| USG6620E | USG6620E-AC (02352HMF-002) | V600R007C20SPC600 |
| USG6610E | USG6610E-AC (02352HMD-001) | V600R007C20SPC300 |
| USG6610E | USG6610E-AC (02352HMD) | V600R007C20SPC100 |
| USG6610E | USG6610E-AC (02352HMD-002) | V600R007C20SPC600 |
| USG6650E | USG6650E-AC (02351YRY-001) | V600R007C20SPC300 |
| USG6650E | USG6650E-AC (02351YRY) | V600R006C00 |
| USG6650E | USG6650E-AC (02351YRY-002) | V600R007C20SPC600 |
| USG6680E | USG6680E-AC (02351YRR) | V600R006C00 |
| USG6680E | USG6680E-AC (02351YRR-001) | V600R007C20SPC600 |
| USG6605E-B | USG6605E-B-AC (02353PSK) | V600R007C20 |
| USG6605E-B | USG6605E-B-AC (02353PSK-001) | V600R007C20SPC600 |
| USG6716E | USG6716E-AC (02351YRS) | V600R006C00 |
| USG6716E | USG6716E-AC (02351YRS-001) | V600R007C20SPC600 |
| USG6635E | USG6635E-AC (02352RQD) | V600R007C00 |
| USG6635E | USG6635E-DC (02352RQE) | V600R007C20 |
| USG6712E | USG6712E-AC (02352CNL) | V600R006C00 |

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| Product | Product Model | First Supported Version |
| --- | --- | --- |
| USG6712E | USG6712E-AC (02352CNL-001) | V600R007C20SPC600 |
| USG6625E | USG6625E-AC (02352RQN) | V600R007C00 |
| USG6615E | USG6615E-AC (02352RQL) | V600R007C00 |
| USG6655E | USG6655E-AC (02352RPS) | V600R007C00 |

## Panel

**Figure 3-194** Panel of the SSD-SATA1920G-A




| 1. ALM indicator | 2. RUN indicator |
| --- | --- |

**Table 3-291** Indicators on the SSD-SATA1920G-A

| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| - | ALM indicator | Orange | Steady on | The hard disk fails. |
| | | - | Off | The hard disk is running properly. |
| - | RUN indicator | Green | Steady on | The hard disk is in position. |
| | | Green | Blinking | Data is being read from or written to the hard disk. |

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| Silkscreen | Name | Color | Status | Description |
| --- | --- | --- | --- | --- |
| | | - | Off | The hard disk is not detected or the device is not powered on. |

## Functions and Features

**Table 3-292** Functions and features of the SSD-SATA1920G-A

| Functions and Features | Description |
| --- | --- |
| Basic functions | Storage log. |
| Hot swap | Hot swap is supported, but hard disks cannot be directly swapped or removed. Run the disk offline command. Remove the hard disk about 30 seconds after the system displays a message indicating that the hard disk stops working. Alternatively, press and hold the OFL button for about 4 seconds until the OFL indicator is on, and then remove the hard disk unit. Otherwise, the hard disk may be damaged, data may be lost, and the system may crash. |
| Restrictions and limitations | Do not low-level format the hard disk. Otherwise, the hard disk cannot be used. |

## Technical Specifications

**Table 3-293** Technical specifications of the SSD-SATA1920G-A

| Item | Specification |
| --- | --- |
| Form factor | 2.5 inch |
| Dimensions without packaging (H x W x D) [mm(in.)] | 15 mm x 75 mm x 130 mm (0.59 in. x 2.95 in. x 5.12 in.) |
| Weight without packaging [kg(lb)] | 0.135 kg (0.3 lb) |
| Storage capacity [GB] | 1920 |

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| Item | Specification |
| --- | --- |
| Port type | SATA |
| Maximum power consumption [W] | 4.36 W |
| Maximum heat dissipation [BTU/hour] | 14.86 BTU/hour |

# 3.5 Optical/Electrical Modules

This chapter describes hardware information for optical/electrical modules of the USG6000E.

## 3.5.1 Before You Start

Before using the optical module, please understand the risk of using the non-certified optical module and how to How to Identify Huawei-Certified optical modules.

**NOTICE**

- A USG must use optical modules that have been certified for use. Non-certified optical modules cannot ensure transmission reliability and may affect service stability. Huawei is not liable for any problem caused by the use of non-certified optical modules and will not fix such problems.
- The methods provided here are only for reference. To confirm whether optical modules you are using have been certified for use on Huawei USGs, contact Huawei technical support.

### Risks of Using Non-Huawei-Certified Optical Modules

During certification of optical modules for USG, Huawei completes comprehensive functionality verification to ensure quality of optical modules. The verified items include optical module plug/unplug, transmit optical power, receive optical power, signal transmission quality, data reading, error tolerance, compatibility, electromagnetic compatibility (EMC), and environmental parameters.

Non-Huawei-certified optical modules may cause the following problems:

- Non-standard structure and size cause failures to install optical modules on adjacent optical interfaces.

  Structures or sizes of some non-Huawei-certified optical modules do not comply with the Multi-Source Agreement (MSA). When such an optical module is installed on an optical interface, the size of this optical module hinders optical module installation on adjacent optical interfaces.
- Data bus defects cause suspension of a USG's data bus.

  Some non-Huawei-certified optical modules have defects in data bus designs. Using such an optical module on a USG causes suspension of the connected data bus on the USG. As a result, data on the suspended bus cannot be read.

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- Improper edge connector size damages electronic devices of optical interfaces.

  If a non-Huawei-certified USG optical module with improper edge connector size is used on an optical interface, electronic devices of the optical interface will be damaged by short circuits.
- Unnormalized temperature monitoring causes incorrect alarms.

  The temperature monitoring systems of some non-Huawei-certified USG optical modules do not comply with industry standards and report temperature values higher than the real temperature. When such optical modules are used on a USG, the system will report incorrect temperature alarms.
- Improper register settings cause errors or failures in reading parameters or diagnostic information.

  Some non-Huawei-certified USG optical modules have improper register values on page A0, which can cause errors or failures when the system attempts to read parameters or diagnostic information from a data bus.
- Some non-Huawei-certified USG optical modules are not designed in compliance with EMC standards and have low anti-interference capability. Additionally, they bring electromagnetic interference to nearby devices.
- The operating temperature ranges of non-Huawei-certified USG optical modules cannot meet service requirements. When they are used under relatively high temperature, the optical power decreases, resulting in service interruption.

## How to Identify Huawei-Certified Optical Modules

**Method 1: Check for "HUAWEI" on the label**

If an optical module has been certified by Huawei, its label contains "HUAWEI", as shown in **Figure 3-195**.

**Figure 3-195** "HUAWEI" on the label of a Huawei-certified optical module




**Method 2: Run the display esn interface command**

If **Certified** is **YES**, and **VendorName** is **huawei** in the **display esn interface** command output, the optical module has been certified by Huawei. Otherwise, it is not a Huawei-certified optical module.

# 3.5.2 100Mbps SFP Optical Modules

### 3.5.2.1 S-SFP-FE-LH40-SM1310

**Table 3-294** S-SFP-FE-LH40-SM1310 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | S-SFP-FE-LH40-SM1310 |
| Part Number | 02317344 |
| Model | S-SFP-FE-LH40-SM1310 |
| Form factor | eSFP |
| Application standard | STM-1 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155Mbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 0 dBm |
| Minimum Tx optical power [dBm] | -5 dBm |
| Minimum extinction ratio [dB] | 10.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -37 dBm |
| Overload power (AVG) [dBm] | -10 dBm |

### 3.5.2.2 S-SFP-FE-LH80-SM1550

**Table 3-295** S-SFP-FE-LH80-SM1550 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | S-SFP-FE-LH80-SM1550 |
| Part Number | 02317345 |
| Model | S-SFP-FE-LH80-SM1550 |

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| Item | Value |
| --- | --- |
| Form factor | eSFP |
| Application standard | STM-1 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155Mbit/s |
| Target transmission distance [km] | 80 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |
| Maximum Tx optical power [dBm] | 0 dBm |
| Minimum Tx optical power [dBm] | -5 dBm |
| Minimum extinction ratio [dB] | 10.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -37 dBm |
| Overload power (AVG) [dBm] | -10 dBm |

### 3.5.2.3 SFP-FE-LX-SM1310-BIDI

**Table 3-296** SFP-FE-LX-SM1310-BIDI specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-FE-LX-SM1310-BIDI |
| Part Number | 02315203 |
| Model | SFP-FE-LX-SM1310-BIDI |
| Form factor | eSFP |
| Application standard | 100BASE-BX |
| Connector type | LC/PC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155Mbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | 15 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1550/RX1310 |
| Maximum Tx optical power [dBm] | -8 dBm |
| Minimum Tx optical power [dBm] | -15 dBm |
| Minimum extinction ratio [dB] | 8.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -32 dBm |
| Overload power (AVG) [dBm] | -8 dBm |
| **NOTE**<br>BIDI optical modules must be used in pairs. For example, SFP-FE-LX-SM1310-BIDI must be used with SFP-FE-LX-SM1550-BIDI. | |

### 3.5.2.4 SFP-FE-LX-SM1550-BIDI

**Table 3-297** SFP-FE-LX-SM1550-BIDI specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-FE-LX-SM1550-BIDI |
| Part Number | 02315202 |
| Model | SFP-FE-LX-SM1550-BIDI |
| Form factor | eSFP |
| Application standard | 100BASE-BX |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155Mbit/s |
| Target transmission distance [km] | 15 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1550/RX1310 |
| Maximum Tx optical power [dBm] | -8 dBm |

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| Item | Value |
| --- | --- |
| Minimum Tx optical power [dBm] | -15 dBm |
| Minimum extinction ratio [dB] | 8.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -32 dBm |
| Overload power (AVG) [dBm] | -8 dBm |
| **NOTE**<br>BIDI optical modules must be used in pairs. For example, SFP-FE-LX-SM1550-BIDI must be used with SFP-FE-LX-SM1310-BIDI. | |

### 3.5.2.5 SFP-FE-SX-MM1310

**Table 3-298** SFP-FE-SX-MM1310 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-FE-SX-MM1310 |
| Part Number | 02315233 |
| Model | SFP-FE-SX-MM1310 |
| Form factor | SFP |
| Application standard | 100BASE-FX |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C (32°F to 158°F) |
| Transmission rate [bit/s] | 100 Mbit/s |
| Target transmission distance [km] | Multimode fiber (50 μm or 62.5 μm diameter): 2 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | -14.0 dBm |
| Minimum Tx optical power [dBm] | -19.0 dBm |
| Minimum extinction ratio [dB] | 10 dB |
| **Receiver Optical Characteristics** | |

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| Item | Value |
| --- | --- |
| Rx sensitivity (AVG) [dBm] | -30.0 dBm |
| Overload power (AVG) [dBm] | -14.0 dBm |

### 3.5.2.6 eSFP-FE-LX-SM1310

**Table 3-299** eSFP-FE-LX-SM1310 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | eSFP-FE-LX-SM1310 |
| Part Number | 02315205 |
| Model | eSFP-FE-LX-SM1310 |
| Form factor | eSFP |
| Application standard | STM-1 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155Mbit/s |
| Target transmission distance [km] | 15 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | -8 dBm |
| Minimum Tx optical power [dBm] | -15 dBm |
| Minimum extinction ratio [dB] | 8.2 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -28 dBm |
| Overload power (AVG) [dBm] | -8 dBm |

## 3.5.3 1Gbps SFP Copper Modules

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### 3.5.3.1 SFP-1000BaseT

**Table 3-300** SFP-1000BaseT specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-1000BaseT |
| Part Number | 02314171 |
| Model | SFP-1000BaseT |
| Form factor | SFP |
| Application standard | 1000BASE-T |
| Connector type | RJ45 |
| Optical fiber type | - |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 10-1000M |
| Target transmission distance [km] | 0.1 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | - |
| Maximum Tx optical power [dBm] | - |
| Minimum Tx optical power [dBm] | - |
| Minimum extinction ratio [dB] | - |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | - |
| Overload power (AVG) [dBm] | - |

## 3.5.4 1Gbps eSFP Optical Modules

### 3.5.4.1 OSU015N00

**Table 3-301** OSU015N00 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |

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| Item | Value |
| --- | --- |
| Module name | OSU015N00 |
| Part Number | 02310CPX |
| Model | OSU015N00 |
| Form factor | eSFP |
| Application standard | - |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 155M-2.67G |
| Target transmission distance [km] | 15 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 0 dBm |
| Minimum Tx optical power [dBm] | -5 dBm |
| Minimum extinction ratio [dB] | 8.2 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -21 dBm |
| Overload power (AVG) [dBm] | 0 dBm |

### 3.5.4.2 S-SFP-GE-LH40-SM1310

**Table 3-302** S-SFP-GE-LH40-SM1310 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | S-SFP-GE-LH40-SM1310 |
| Part Number | 02317346 |
| Model | S-SFP-GE-LH40-SM1310 |
| Form factor | eSFP |
| Application standard | 1000base-LX/LH |
| Connector type | LC |

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| Item | Value |
| --- | --- |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 0 dBm |
| Minimum Tx optical power [dBm] | -5 dBm |
| Minimum extinction ratio [dB] | 9 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -23 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

### 3.5.4.3 S-SFP-GE-LH40-SM1550

**Table 3-303** S-SFP-GE-LH40-SM1550 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | S-SFP-GE-LH40-SM1550 |
| Part Number | 02317347 |
| Model | S-SFP-GE-LH40-SM1550 |
| Form factor | eSFP |
| Application standard | 1000BASE-LX |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |

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| Item | Value |
| --- | --- |
| Maximum Tx optical power [dBm] | 0 dBm |
| Minimum Tx optical power [dBm] | -5 dBm |
| Minimum extinction ratio [dB] | 9.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -22 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

### 3.5.4.4 S-SFP-GE-LH80-SM1550

**Table 3-304** S-SFP-GE-LH80-SM1550 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | S-SFP-GE-LH80-SM1550 |
| Part Number | 02317348 |
| Model | S-SFP-GE-LH80-SM1550 |
| Form factor | eSFP |
| Application standard | 1000BASE-ZX |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 80 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |
| Maximum Tx optical power [dBm] | 5 dBm |
| Minimum Tx optical power [dBm] | -2 dBm |
| Minimum extinction ratio [dB] | 9 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -23 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

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### 3.5.4.5 SFP-GE-LX-SM1310

**Table 3-305** SFP-GE-LX-SM1310 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-GE-LX-SM1310 |
| Part Number | 02315200 |
| Model | SFP-GE-LX-SM1310 |
| Form factor | eSFP |
| Application standard | 1000BASE-LX10/LH |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | -3 dBm |
| Minimum Tx optical power [dBm] | -9 dBm |
| Minimum extinction ratio [dB] | 9 dB |
| **Receiver Optical Characteristics** | |
| Rx operating wavelength range [nm] | - |
| Rx sensitivity (AVG) [dBm] | -20 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

### 3.5.4.6 SFP-GE-LX-SM1310-BIDI (02315285)

**Table 3-306** SFP-GE-LX-SM1310-BIDI specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-GE-LX-SM1310-BIDI |
| Part Number | 02315285 |

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| Item | Value |
| --- | --- |
| Model | SFP-GE-LX-SM1310-BIDI |
| Form factor | eSFP |
| Application standard | 1000base-BX |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1310/RX1490 |
| Maximum Tx optical power [dBm] | -3 dBm |
| Minimum Tx optical power [dBm] | -9 dBm |
| Minimum extinction ratio [dB] | 6 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -19.5 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

### 3.5.4.7 SFP-GE-LX-SM1490-BIDI (02315286)

**Table 3-307** SFP-GE-LX-SM1490-BIDI specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-GE-LX-SM1490-BIDI |
| Part Number | 02315286 |
| Model | SFP-GE-LX-SM1490-BIDI |
| Form factor | eSFP |
| Application standard | 1000base-BX |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |

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| Item | Value |
| --- | --- |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1490/RX1310 |
| Maximum Tx optical power [dBm] | -3 dBm |
| Minimum Tx optical power [dBm] | -9 dBm |
| Minimum extinction ratio [dB] | 6 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -19.5 dBm |
| Overload power (AVG) [dBm] | -3 dBm |

### 3.5.4.8 eSFP-GE-SX-MM850

**Table 3-308** eSFP-GE-SX-MM850 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | eSFP-GE-SX-MM850 |
| Part Number | 02315204 |
| Model | eSFP-GE-SX-MM850 |
| Form factor | eSFP |
| Application standard | 1000base-SX |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | -20°C to 85°C |
| Transmission rate [bit/s] | 2.125Gbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | - Multimode fiber (with diameter of 62.5 μm): 220 m<br>- Multimode fiber (OM1) (with diameter of 62.5 μm): 275 m<br>- Multimode fiber (with diameter of 50 μm): 500 m<br>- Multimode fiber (OM2) (with diameter of 50 μm): 550 m |
| Modal bandwidth [MHz*km] | - Multimode fiber: 160 MHz*km<br>- Multimode fiber (OM1): 200 MHz*km<br>- Multimode fiber: 400 MHz*km<br>- Multimode fiber (OM2): 500 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | -2.5 dBm |
| Minimum Tx optical power [dBm] | -9.5 dBm |
| Minimum extinction ratio [dB] | 9 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -17 dBm |
| Overload power (AVG) [dBm] | 0 dBm |

## 3.5.4.9 eSFP-GE-ZX100-SM1550 (02315206)

**Table 3-309** eSFP-GE-ZX100-SM1550 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | eSFP-GE-ZX100-SM1550 |
| Part Number | 02315206 |
| Model | eSFP-GE-ZX100-SM1550 |
| Form factor | eSFP |
| Application standard | 1000base-ZX |
| Connector type | LC |

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| Item | Value |
| --- | --- |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 1.25Gbit/s |
| Target transmission distance [km] | 100 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |
| Maximum Tx optical power [dBm] | 5 dBm |
| Minimum Tx optical power [dBm] | 0 dBm |
| Minimum extinction ratio [dB] | 8 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -30 dBm |
| Overload power (AVG) [dBm] | -9 dBm |

## 3.5.5 10Gbps SFP+ Optical Modules

### 3.5.5.1 OMXD30000

**Table 3-310** OMXD30000 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | OMXD30000 |
| Part Number | 02318169 |
| Model | OMXD30000 |
| Form factor | SFP+ |
| Application standard | 10GBASE-SR |
| Connector type | LC |
| Optical fiber type | MMF |
| Bit error ratio (BER) | - |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 10Gbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | 0.3 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | -1 dBm |
| Minimum Tx optical power [dBm] | -7.3 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -9.9 dBm |
| Overload power (AVG) [dBm] | -1 dBm |

### 3.5.5.2 OSX010000

**Table 3-311** OSX010000 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | OSX010000 |
| Part Number | 02318170 |
| Model | OSX010000 |
| Form factor | SFP+ |
| Application standard | 10GBASE-LR |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 10Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 0.5 dBm |
| Minimum Tx optical power [dBm] | -8.2 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |

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| Item | Value |
| --- | --- |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -12.6 dBm |
| Overload power (AVG) [dBm] | 0.5 dBm |

### 3.5.5.3 OSX040N01

**Table 3-312** OSX040N01 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | OSX040N01 |
| Part Number | 02310CNF |
| Model | OSX040N01 |
| Form factor | SFP+ |
| Application standard | 10GBASE-ER |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 10Gbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |
| Maximum Tx optical power [dBm] | 4 dBm |
| Minimum Tx optical power [dBm] | -4.7 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -14.1 dBm |
| Overload power (AVG) [dBm] | 0.5 dBm |

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### 3.5.5.4 SFP-10G-BXD1

**Table 3-313** SFP-10G-BXD1 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-10G-BXD1 |
| Part Number | 02310QDT |
| Model | SFP-10G-BXD1 |
| Form factor | SFP+ |
| Application standard | 10GBase-BIDI |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 2.5Gbit/s-11.3Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1330/RX1270 |
| Maximum Tx optical power [dBm] | 0.5 dBm |
| Minimum Tx optical power [dBm] | -8.2 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -14.4 dBm |
| Overload power (AVG) [dBm] | 0.5 dBm |

### 3.5.5.5 SFP-10G-BXU1

**Table 3-314** SFP-10G-BXU1 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-10G-BXU1 |
| Part Number | 02310QBJ |
| Model | SFP-10G-BXU1 |

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| Item | Value |
| --- | --- |
| Form factor | SFP+ |
| Application standard | 10GBase-BIDI |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 2.5Gbit/s-11.3Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | TX1270/RX1330 |
| Maximum Tx optical power [dBm] | 0.5 dBm |
| Minimum Tx optical power [dBm] | -8.2 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -14.4 dBm |
| Overload power (AVG) [dBm] | 0.5 dBm |

### 3.5.5.6 SFP-10G-USR

**Table 3-315** SFP-10G-USR specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-10G-USR |
| Part Number | 02310MNW |
| Model | SFP-10G-USR |
| Form factor | SFP+ |
| Application standard | 10Gbase-USR |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 10.31Gbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | 0.1 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | -1 dBm |
| Minimum Tx optical power [dBm] | -7.3 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10.7 dBm |
| Overload power (AVG) [dBm] | 0.5 dBm |

### 3.5.5.7 SFP-10G-ZR

**Table 3-316** SFP-10G-ZR specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | SFP-10G-ZR |
| Part Number | 02310SNN |
| Model | SFP-10G-ZR |
| Form factor | SFP+ |
| Application standard | 10Gbase-ZR |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 9.95Gbit/s-10.31Gbit/s |
| Target transmission distance [km] | 80 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1550 nm |
| Maximum Tx optical power [dBm] | 4 dBm |
| Minimum Tx optical power [dBm] | 0 dBm |
| Minimum extinction ratio [dB] | 9 dB |

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| Item | Value |
| --- | --- |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -24 dBm |
| Overload power (AVG) [dBm] | -7 dBm |

## 3.5.6 40Gbps QSFP+ Optical Modules

### 3.5.6.1 QSFP-40G-LR4

**Table 3-317** QSFP-40G-LR4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-LR4 |
| Part Number | 02310MHS |
| Model | QSFP-40G-LR4 |
| Form factor | QSFP+ |
| Application standard | 40GBASE-LR4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2.3 dBm |
| Minimum Tx optical power [dBm] | -7 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -11.5 dBm |
| Overload power (AVG) [dBm] | 2.3 dBm |

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### 3.5.6.2 QSFP-40G-SDLC-PAM

**Table 3-318** QSFP-40G-SDLC-PAM specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-SDLC-PAM |
| Part Number | 02311PUU |
| Model | QSFP-40G-SDLC-PAM |
| Form factor | QSFP+ |
| Application standard | 40GBase-SDLC |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | - Multimode fiber (OM3) (with diameter of 50 μm): 0.1 km<br>- Multimode fiber (OM4) (with diameter of 50 μm): 0.15 km |
| Modal bandwidth [MHz*km] | - Multimode fiber (OM3): 2000 MHz*km<br>- Multimode fiber (OM4): 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 2.4 dBm |
| Minimum Tx optical power [dBm] | -2.5 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -8 dBm |
| Overload power (AVG) [dBm] | 2.4 dBm |

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### 3.5.6.3 QSFP-40G-SR-BD (02311FPA)

**Table 3-319** QSFP-40G-SR-BD specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-SR-BD |
| Part Number | 02311FPA |
| Model | QSFP-40G-SR-BD |
| Form factor | QSFP+ |
| Application standard | 40GBase-BD |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 10°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | 0.1 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm,900 nm |
| Maximum Tx optical power [dBm] | 5 dBm |
| Minimum Tx optical power [dBm] | -4 dBm |
| Minimum extinction ratio [dB] | 4.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -4.5 dBm |
| Overload power (AVG) [dBm] | 5 dBm |

### 3.5.6.4 QSFP-40G-eSR4

**Table 3-320** QSFP-40G-eSR4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-eSR4 |
| Part Number | 02310RMB |
| Model | QSFP-40G-eSR4 |

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| Item | Value |
| --- | --- |
| Form factor | QSFP+ |
| Application standard | 40GBase-eSR4 |
| Connector type | MPO |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | - Multimode fiber (OM2) (with diameter of 50 μm): 82 m<br>- Multimode fiber (OM3) (with diameter of 50 μm): 300 m<br>- Multimode fiber (OM4) (with diameter of 50 μm): 400 m |
| Modal bandwidth [MHz*km] | - Multimode fiber (OM2): 500 MHz*km<br>- Multimode fiber (OM3): 2000 MHz*km<br>- Multimode fiber (OM4): 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 0.5 dBm |
| Minimum Tx optical power [dBm] | -7.6 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -5.4 dBm |
| Overload power (AVG) [dBm] | 2.4 dBm |

### 3.5.6.5 QSFP-40G-iSR4

**Table 3-321** QSFP-40G-iSR4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-iSR4 |

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| Item | Value |
| --- | --- |
| Part Number | 02310MHR |
| Model | QSFP-40G-iSR4 |
| Form factor | QSFP+ |
| Application standard | 40GBASE-SR4 |
| Connector type | MPO |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | - Multimode fiber (OM3) (with diameter of 50 μm): 100 m<br>- Multimode fiber (OM4) (with diameter of 50 μm): 150 m |
| Modal bandwidth [MHz*km] | - Multimode fiber (OM3): 2000 MHz*km<br>- Multimode fiber (OM4): 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 0.5 dBm |
| Minimum Tx optical power [dBm] | -7.6 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -9.5 dBm |
| Overload power (AVG) [dBm] | 2.4 dBm |

### 3.5.6.6 QSFP-40G-LX4

**Table 3-322** QSFP-40G-LX4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-LX4 |
| Part Number | 02311HNP |

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| Item | Value |
| --- | --- |
| Model | QSFP-40G-LX4 |
| Form factor | QSFP+ |
| Application standard | 40G-LX4 |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25 bit/s |
| Target transmission distance [km] | 0.15 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1271nm,1291nm,1311nm,1331nm |
| Maximum Tx optical power [dBm] | 2.3 dBm |
| Minimum Tx optical power [dBm] | -7 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -11.5 dBm |
| Overload power (AVG) [dBm] | 2.3 dBm |

### 3.5.6.7 QSFP-40G-ER4

**Table 3-323** QSFP-40G-ER4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-ER4 |
| Part Number | 02311BKT |
| Model | QSFP-40G-ER4 |
| Form factor | QSFP+ |
| Application standard | 40GBASE-ER4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |

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| Item | Value |
| --- | --- |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1271nm, 1291nm, 1311nm, 1331nm |
| Maximum Tx optical power [dBm] | 4.5 dBm |
| Minimum Tx optical power [dBm] | -2.7 dBm |
| Minimum extinction ratio [dB] | 5.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -16.8 dBm |
| Overload power (AVG) [dBm] | 2.3 dBm |

### 3.5.6.8 QSFP-40G-LR4-Lite

**Table 3-324** QSFP-40G-LR4-Lite specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-LR4-Lite |
| Part Number | 02311YVB |
| Model | QSFP-40G-LR4-Lite |
| Form factor | QSFP+ |
| Application standard | 40GBASE-LR4 Lite |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 40Gbit/s |
| Target transmission distance [km] | 2 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1271nm, 1291nm, 1311nm, 1331nm |
| Maximum Tx optical power [dBm] | 2.3 dBm |
| Minimum Tx optical power [dBm] | -9 dBm |

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| Item | Value |
| --- | --- |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -13.5 dBm |
| Overload power (AVG) [dBm] | 2.3 dBm |

### 3.5.6.9 QSFP-40G-LX4-MM

**Table 3-325** QSFP-40G-LX4-MM specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-LX4-MM |
| Part Number | 02313NUG |
| Model | QSFP-40G-LX4-MM |
| Form factor | QSFP+ |
| Application standard | 40GBASE-LX4 |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 40Gbit/s |
| Target transmission distance [km] | Multimode OM3 optical fiber (diameter: 50 μm): 150 m<br>Multimode OM4 optical fiber (diameter: 50 μm): 150 m |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1271nm, 1291nm, 1311nm, 1331nm |
| Maximum Tx optical power [dBm] | 2.3 dBm |
| Minimum Tx optical power [dBm] | -7.0 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10.5 dBm |
| Overload power (AVG) [dBm] | 3.5 dBm |

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### 3.5.6.10 QSFP-40G-eSDLC-PAM

**Table 3-326** QSFP-40G-eSDLC-PAM specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-40G-eSDLC-PAM |
| Part Number | 02311QTR |
| Model | QSFP-40G-eSDLC-PAM |
| Form factor | QSFP+ |
| Application standard | 40GBase-eSDLC-PAM4 |
| Connector type | MPO |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 41.25Gbit/s |
| Target transmission distance [km] | - Multimode fiber (OM3) (with diameter of 50 μm): 100 m<br>- Multimode fiber (OM4) (with diameter of 50 μm): 300 m |
| Modal bandwidth [MHz*km] | - Multimode fiber (OM3): 2000 MHz*km<br>- Multimode fiber (OM4): 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 2.4 dBm |
| Minimum Tx optical power [dBm] | -2 dBm |
| Minimum extinction ratio [dB] | 3 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -8 dBm |
| Overload power (AVG) [dBm] | 2 dBm |

## 3.5.7 100Gbps QSFP28 Optical Modules

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### 3.5.7.1 QSFP-100G-CWDM4

**Table 3-327** QSFP-100G-CWDM4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-CWDM4 |
| Part Number | 02311MNN |
| Model | QSFP-100G-CWDM4 |
| Form factor | QSFP28 |
| Application standard | 100GBase-CWDM4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 2 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2.5 dBm |
| Minimum Tx optical power [dBm] | -6.5 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10 dBm |
| Overload power (AVG) [dBm] | 2.5 dBm |

### 3.5.7.2 QSFP28-100G-LR4

**Table 3-328** QSFP28-100G-LR4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP28-100G-LR4 |
| Part Number | 02311KNU |
| Model | QSFP28-100G-LR4 |

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| Item | Value |
| --- | --- |
| Form factor | QSFP28 |
| Application standard | 100GBase-LR4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 10 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 4.5 dBm |
| Minimum Tx optical power [dBm] | -4.3 dBm |
| Minimum extinction ratio [dB] | 4 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -8.6 dBm |
| Overload power (AVG) [dBm] | 4.5 dBm |

### 3.5.7.3 QSFP28-100G-SR4

**Table 3-329** QSFP28-100G-SR4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP28-100G-SR4 |
| Part Number | 02311GBW |
| Model | QSFP28-100G-SR4 |
| Form factor | QSFP28 |
| Application standard | 100GBase-SR4 |
| Connector type | MPO |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | 0.1 km |
| Modal bandwidth [MHz*km] | 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 2.4 dBm |
| Minimum Tx optical power [dBm] | -8.4 dBm |
| Minimum extinction ratio [dB] | 2 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10.3 dBm |
| Overload power (AVG) [dBm] | 2.4 dBm |

### 3.5.7.4 QSFP-100G-4WDM-40

**Table 3-330** QSFP-100G-4WDM-40 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-4WDM-40 |
| Part Number | 02312QTL |
| Model | QSFP-100G-4WDM-40 |
| Form factor | QSFP28 |
| Application standard | 100GBASE-4WDM |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2.9 dBm |
| Minimum Tx optical power [dBm] | -2.5 dBm |

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| Item | Value |
| --- | --- |
| Minimum extinction ratio [dB] | 8 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -21.4 dBm |
| Overload power (AVG) [dBm] | -3.5 dBm |

### 3.5.7.5 QSFP-100G-CWDM4-ISP

**Table 3-331** QSFP-100G-CWDM4-ISP specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-CWDM4-ISP |
| Part Number | 02312DYA |
| Model | QSFP-100G-CWDM4-ISP |
| Form factor | QSFP28 |
| Application standard | 100GBase-CWDM4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 2 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2.5 dBm |
| Minimum Tx optical power [dBm] | -6.5 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10 dBm |
| Overload power (AVG) [dBm] | 2.5 dBm |

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### 3.5.7.6 QSFP-100G-CWDM4-Lite

**Table 3-332** QSFP-100G-CWDM4-Lite specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-CWDM4-Lite |
| Part Number | 02312UJN |
| Model | QSFP-100G-CWDM4-Lite |
| Form factor | QSFP28 |
| Application standard | 100G-CWDM4-Lite |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 2 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310nm |
| Maximum Tx optical power [dBm] | 2.5 dBm |
| Minimum Tx optical power [dBm] | -6.5 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10 dBm |
| Overload power (AVG) [dBm] | 2.5 dBm |

### 3.5.7.7 QSFP-100G-ER4

**Table 3-333** QSFP-100G-ER4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-ER4 |
| Part Number | 02313HLU |
| Model | QSFP-100G-ER4 |

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| Item | Value |
| --- | --- |
| Form factor | QSFP28 |
| Application standard | 100GBASE-ER4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C (32°F to 158°F) |
| Transmission rate [bit/s] | 100 Gbit/s |
| Target transmission distance [km] | Single-mode fiber (G.652): 40 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1295 nm,1300 nm,1304 nm,1309 nm |
| Maximum Tx optical power [dBm] | 2.9 dBm |
| Minimum Tx optical power [dBm] | -2.9 dBm |
| Minimum extinction ratio [dB] | 8 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -20.9 dBm |
| Overload power (AVG) [dBm] | -3.5 dBm |

### 3.5.7.8 QSFP-100G-SWDM4

**Table 3-334** QSFP-100G-SWDM4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP-100G-SWDM4 |
| Part Number | 02311QUK |
| Model | QSFP-100G-SWDM4 |
| Form factor | QSFP28 |
| Application standard | 100G-SWDM4 |
| Connector type | LC |
| Optical fiber type | MMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |

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| Item | Value |
| --- | --- |
| Target transmission distance [km] | Multimode fiber (OM3) (with diameter of 50 μm): 75 m<br>Multimode fiber (OM4) (with diameter of 50 μm): 100 m |
| Modal bandwidth [MHz*km] | - Multimode fiber (OM3): 2000 MHz*km<br>- Multimode fiber (OM4): 4700 MHz*km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 850 nm |
| Maximum Tx optical power [dBm] | 3.4 dBm |
| Minimum Tx optical power [dBm] | -7.5 dBm |
| Minimum extinction ratio [dB] | 2 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -10.5 dBm |
| Overload power (AVG) [dBm] | 2.4 dBm |

### 3.5.7.9 QSFP28-100G-1310-40km-SM

**Table 3-335** QSFP28-100G-1310-40km-SM specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP28-100G-1310-40km-SM |
| Part Number | 02312NVQ |
| Model | QSFP28-100G-1310-40km-SM |
| Form factor | QSFP28 |
| Application standard | IEEE 802.3 100GBASE-ER4 |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 40 km |

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| Item | Value |
| --- | --- |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2.9 dBm |
| Minimum Tx optical power [dBm] | -2.9 dBm |
| Minimum extinction ratio [dB] | 8 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -20.9 dBm |
| Overload power (AVG) [dBm] | -3.5 dBm |

### 3.5.7.10 QSFP28-100G-DR

**Table 3-336** QSFP28-100G-DR specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP28-100G-DR |
| Part Number | 02312VSP |
| Model | QSFP28-100G-DR |
| Form factor | QSFP28 |
| Application standard | 100GBase-DR |
| Connector type | LC |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 0.5 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1311 nm |
| Maximum Tx optical power [dBm] | 4 dBm |
| Minimum Tx optical power [dBm] | -2.9 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |

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| Item | Value |
| --- | --- |
| Rx sensitivity (AVG) [dBm] | -5.9 dBm |
| Overload power (AVG) [dBm] | 4 dBm |

### 3.5.7.11 QSFP28-100G-PSM4

**Table 3-337** QSFP28-100G-PSM4 specifications

| Item | Value |
| --- | --- |
| **Basic Information** | |
| Module name | QSFP28-100G-PSM4 |
| Part Number | 02311MNM |
| Model | QSFP28-100G-PSM4 |
| Form factor | QSFP28 |
| Application standard | 100GBASE-PSM4 |
| Connector type | MPO |
| Optical fiber type | SMF |
| Working case temperature [°C(°F)] | 0°C to 70°C |
| Transmission rate [bit/s] | 100Gbit/s |
| Target transmission distance [km] | 0.5 km |
| **Transmitter Optical Characteristics** | |
| Center wavelength [nm] | 1310 nm |
| Maximum Tx optical power [dBm] | 2 dBm |
| Minimum Tx optical power [dBm] | -9.4 dBm |
| Minimum extinction ratio [dB] | 3.5 dB |
| **Receiver Optical Characteristics** | |
| Rx sensitivity (AVG) [dBm] | -11.35 dBm |
| Overload power (AVG) [dBm] | 2.2 dBm |

# 3.6 Cables

This section describes the PGND cable, power cable, configuration cable, Ethernet cable, and optical fiber of the USG.

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# 3.6.1 PGND Cable

## 3.6.1.1 PGND Cable

### Connection

A ground cable is connected as follows:

- The M4 lug is connected to the ground point on the USG.
- The M6 lug is connected to the ground point or equipotential terminal on a cabinet.

### Appearance and Structure

A PGND cable has two OT terminals and a copper cable with yellow/green plastic encapsulation. **Figure 3-196** shows the appearance of a type of PGND cables.

NOTE

The figure is for reference only.

**Figure 3-196** Appearance of the PGND cable



**Figure 3-197** shows the appearance of the OT terminal.

**Figure 3-197** OT terminal



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## Technical Specifications

**Table 3-338** Technical specifications of the PGND Cable

| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Power Cable,1.5 m,6mm^2,G&Y, (OT6-4), H07Z-K-6^2G&Y, (OT6-6), S6505-LSZH | 04150328 | CPGND0104 | 14170017: OT6-4 | 14170023: OT6-6 | 1.5 m | - |
| Power Cable,0.4 m,4mm^2,OT6-4, 227IEC02-4^2G&Y, OT6-6 | 04150617 | IEC401 | 14170017: OT6-4 | 14170023: OT6-6 | 0.4 m | - |
| Power Cable,4m ,OT2-16-8,H07Z-K-16^2G &Y,OT16-6,LSZH | 04152748-001 | - | OT2-16-8 | OT16-6 | 4 m | 30 mm |

# 3.6.2 AC Power Cable

## 3.6.2.1 AC Power Cable

## Connection

An AC power cable is connected as follows:

- The C7 straight female or C13 straight female connector is connected to the AC power socket of the USG.
- The PI straight male, PB straight male, PD angle male, or PG angle male connector is connected to an external power source.

Use a single-phase 3-line electrical outlet that has been properly grounded. The grounding point of the power supply must be reliably grounded in the building.

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Generally, the grounding point of the power supply system of the building has been buried in the grounding during the construction and wiring of the building.

**NOTICE**

Make sure that the power system of the building has been grounded before you connect the USG with AC power cables.

## Appearance and Structure

The USG provides both the European-standard and North American-standard AC power cables. **Figure 3-198** and **Figure 3-199** list the reference specifications. You can select the desired power cables based on your site conditions. For details about the power cable appearance, see **A.1 Quick Reference Tables of Power Cables**. You can view the photo of a power cable according to the corresponding BOM.

**Figure 3-198** Appearance of the England AC power cable




**Figure 3-199** Appearance of the North American-standard AC power cable

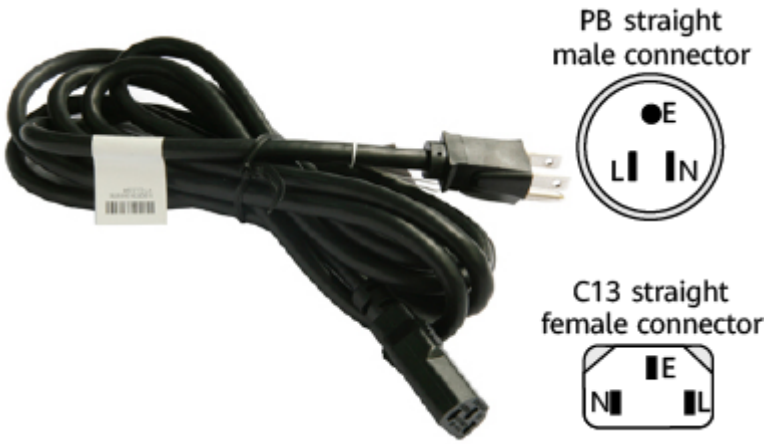



**Table 3-339** lists the connectors of AC power cables.

**Table 3-339** Connectors of AC power cables

| Connector | Description | Wire Color |
| --- | --- | --- |
| L | Live wire | Brown |

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| Connector | Description | Wire Color |
| --- | --- | --- |
| N | Null wire | Blue |
| E | PGND cable | Yellow/Green |

## Technical Specifications

**Table 3-340** Technical specifications of the AC Power Cable

| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
| --- | --- | --- | --- | --- | --- |
| Power Cable,America AC Power Cable,125V 10A,3.0m,PBSM,18SJT(3C),C13SF,Black | 04020728 | CC13NA300 | PB straight male | C13 straight female | 3 m |
| Power Cable,Japan AC Power Cable 125V12A,3.0m,PBSM,HVCTF-1.25mm^2(3C),C13SF,Black | 04040887 | CC13JP300 | PB straight male | C13 straight female | 3 m |
| Power Cords Cable,Australia AC Power Cable,250V 10A,3.0m,PISM,H05VV-F-1.0mm^2(3C),C13SF,Black | 04040888 | CC13AU300 | PI straight male | C13 straight female | 3 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power cord,BS546 250V10A,3.0m,PM-IAM,H05VV-F-1.5mm^2(3C),C13SF,250V,10A,Black | 04040889 | CC13HK300 | PM-I angle male | C13 straight female | 3 m |
| Power Cable,Britain AC Power Cable 250V10A,3.0m,PGAM , H05VV-F-1.0mm^2(3C),C13SF,Black | 04040890 | CC13UK300 | PG angle male | C13 straight female | 3 m |
| Power cord,Europe AC Power Cable,250V 10A,3.0m,PFSM, (H05VVF 1.0^2(3C)), C13SF,250V,10A,BLack | 04041056 | CC13EU300 | PF straight male | C13 straight female | 3 m |
| Power Cords Cable,China AC Power 250V10A,3.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black | 04041104 | CPACH0301 | PI straight male | C13 straight female | 3 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cable,Britain AC Power Cable 250V10A,3.0m,PGAM , H05VV-F-1.0mm^2(3C),C13AF-L,Black | 04041117 | CC13UK301 | PG angle male | C13 left angle female connector | 3 m |
| Power Cable,Switzerland AC Power Cable 250V10A,3.0m,PJSM , H05VV-F-1.0mm^2(3C),C13SF,Black | 04041119 | CC13CH300 | PJ straight male | C13 straight female | 3 m |
| Power Cable,Italy AC Power Cable 250V10A,3.0m,PLSM,H05VV-F-1.0mm^2(3C),C13SF,Black | 04041120 | CC13IT300 | PL straight male | C13 straight female | 3 m |
| Power Cords Cable,China AC Power 250V10A,10.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black | 04042697 | C3X1CHN00 | PI straight male | C13 straight female | 10 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cords Cable,China AC Power 250V10A,15.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black | 04042698 | CPWR00102 | PI straight male | C13 straight female | 15 m |
| Power Cords Cable,China AC Power 250V10A,20.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black | 04042699 | CC13CN200 | PI straight male | C13 straight female | 20 m |
| External Power Cable, Power Cable 250V2.5A,2m,2X0.75mm^2,Black,PASM,227IEC53-0.75^2(2C)-I,C7SF | 04043491 | CPC7SF201 | PA straight male | C7 straight female | 2 m |
| Power cord,Europe AC Power 250V2,2.0m,PCSM,(H03VVH2F-0.5^2(2C)),C7SF,Black | 04044216 | C0C7EU200 | PC straight male | C7 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cords Cable,Japan AC Power 125V7A,2.0m,PASM,VCTFK-0.75^2(2C),C7SF,Black | 04044496 | C0C7JP200 | PA straight male | C7 straight female | 2 m |
| Power Cable,America AC Power Cable 125V7A,2.0m,PASM,18NISPT-1(2C),C7SF,Black | 04044497 | C0C7NA200 | PA straight male | C7 straight female | 2 m |
| Power Cable, Britain AC Power Cable 250V2.5A,2.0m,PGAM,H05VVH2-F-0.75mm^2(2C),C7SF,Black | 04044533 | C0C7UK200 | PG angle male | C7 straight female | 2 m |
| Power Cords Cable,Australia AC Power 250V2.5A,2.0m,PISM-I,H03VVH2F-0.75^2(2C),C7SF,Black | 04044577 | C0C7AU200 | PI-I straight male | C7 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cords Cable,India AC Power 250V2.5A,2.0m,PC-IISM,IS 694-0.75^2(2C),C7S,250V,2.5A,Black | 04045094 | C0C7IN200 | PC-II straight male | C7 straight female | 2 m |
| Power cord,Korea AC Power 250V2.5A,2.0m,PCSM-I,K60227 IEC 52 0.75mm^2(2C),C7SF,250V,2.5A,Black | 04045785 | C0C7KO200 | PC-I straight male | C7 straight female | 2 m |
| Power Cords Cable,Argentina AC Power 250V10A,3.0m,PISM,H05VV-F-1.0mm^2(3C),C13SF,Black | 04047785 | CC13AR300 | PI straight male | C13 straight female | 3 m |
| Power Cords Cable,China AC Power Cable,250V 10A,3m,PISM,227IEC53-1.0^2(3C),C13SF,Black | 04050139 | CC13CN300 | PI straight male | C13 straight female | 3 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
| --- | --- | --- | --- | --- | --- |
| Power Cords Cable,China AC Power 250V10A,1.5m,C14SM,227IEC53(RVV)1.0mm^2(3C),C13SF,PDU Cable | 04050188 | IDS2PWRCBL00 | C14 straight male | C13 straight female | 1.5 m |
| Power Cable,China AC Power Cable 250V10A,1.0m,PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,Black | 04050206 | CC13SA300 | PI straight male | C13 straight female | 1 m |
| Power Cable,China AC Power Cable 250V10A,5.0m,PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,Black | 04050206-001 | W-AC250-5 | PI straight male | C13 straight female | 5 m |
| Power Cable,China AC Power Cable 250V10A,2.0m,PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,Black | 04050517 | CPC13SF00 | PI straight male | C13 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power cord,China AC Power Cable,250V 10A,3.0m,C14SM, (227IEC53-1.0^2(3C)), C13SF,250V,10A,Black, PDU Cable | 04050846 | - | C14 straight male | C13 straight female | 3 m |
| Power cord,China AC Power Cable,250V 10A,6.0m,C14SM, (227IEC53-1.0^2(3C)), C13SF,250V,10A,Black, PDU Cable | 04050847 | W-AC250-6 | C14 straight male | C13 straight female | 6 m |
| Power Cords Cable,China AC Power,250V10A,1.5m, PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,250V,10A,Black | 04050955 | IDSV2DKBM55 | PI straight male | C13 straight female | 1.5 m |
| Power Cord Cable,Brazil AC Power Cable,250V 2.5A,2m,PC-IIISM,H05VVH2-F 2*0.75^2(2C),C7SF,250V,2.5A,Black | 04050956 | C0C7BR200 | PC-III straight male | C7 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power cord,India AC Power 250V6A,3m,PM-IIAM,IS 694-1.0^2(3C),C13 SF,250V,6A, Black | 04051035 | CC13ID300 | PM-II angle male | C13 straight female | 3 m |
| Power cord,South Africa AC Power 250V10A,3m,PMAM,H05VV-F-1.0mm^2(3C),C13SF,250V,10A, Black | 04051080 | CC13SA300 | PM angle male | C13 straight female | 3 m |
| Power cord,BS546 250V2.5A,2m,PD-IAM,H03VVH2-F-0.75mm^2(2C),C7SF,250V,2.5A, Black | 04051081 | C0C7OT200 | PD-I angle male | C7 straight female | 2 m |
| Power cord,2m,IRAM2063 Straight Male,H05VVH2-F 2*0.75^2(2C),C7 Straight Female,250V,2.5A,BLACK | 04051997 | - | IRAM2063 straight male | C7 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power cord,Taiwan AC 125V11A,3.0m,PBSM,HVCTF 3*1.25mm2,C13SF,125V,11A,Black,BSMI | 04052137 | CC13TW300 | PB straight male | C13 straight female | 3 m |
| Power cord,Taiwan AC 125V7A,2.0m,PASM,VCTFK2*0.75mm2(2C),C7SF,125V,7A,Black,BSMI | 04052138 | - | PA straight male | C7 straight female | 2 m |
| Power cord,AC Power Cable 250V2.5A,2.0m,C14SM,H05VVH2-F-0.75mm^2(2C),C7SF,250V,2.5A,Black,Non-standard AC power cable | 04052181 | - | C14 straight male | C7 straight female | 2 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power cord,Britain AC Power 250V10A,3.0m,PGAM,H05VV-F-1.0mm^2(3C),C13SF,250V,10A,Black-Only for Saudi Arabia,Split from 04040890 | 04053153 | CC13UK302 | PG angle male | C13 straight female | 3 m |
| Power Cords Cable,Europe AC 250V10A,1.8m,C14SM,H05VV-F-3*1.00^2,C13SF,PDU Cable | 0405G019 | CC13EU180 | C14 straight male | C13 straight female | 1.8 m |
| Power Cords Cable,South Africa AC Power 250V2.5A,2m,PDAM,H03VVH2-F-0.75^2(2C),C7SF | 0405G025 | C0C7SA200 | PD angle male | C7 straight female | 2 m |
| Power Cords Cable,Korea AC Power 250V10A,3m,PFSM,H05VV-F 3*1.0^2(3C),C13SF,Black | 0405G028 | CC13KO300 | PF straight male | C13 straight female | 3 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cords Cable,North America AC Power 250V10A,1.8m,C14SM,SJT 18AWG(3C),C13SF,PDU Cable | 0405G029 | IDS2PWRCBL0A | C14 straight male | C13 straight female | 1.8 m |
| Power Cords Cable,Japan AC Power 250V12A,1.8m,C14SM,HVCTF 1.25^2(3C),C13SF,PDU Cable | 0405G02D | IDS2PWRCBL09 | C14 straight male | C13 straight female | 1.8 m |
| Power Cords Cable,Australia AC Power 250V10A,1.8m,C14SM,H05VV-F-1.0^2(3C),C13SF,PDU Cable | 0405G02F | IDS2PWRCBL08 | C14 straight male | C13 straight female | 1.8 m |
| Power Cords Cable,Korea AC Power 250V10A,1.8m,C14SM,H05VV-F-1.0^2(3C),C13SF,PDU Cable | 0405G02H | IDS2PWRCBL07 | C14 straight male | C13 straight female | 1.8 m |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length |
|---|---|---|---|---|---|
| Power Cords Cable,Denmark AC Power 250V10A,3m,PKSM,H05VV-F-3*1.0^2(3C),C13SF,Black | 0405G02K | CC13DE300 | PK straight male | C13 straight female | 3 m |
| Power Cable,Brazil AC Power Cable 250V10A,3.0m,PNSM ,H05VV-F-1.0mm^2(3C),C13SF,Black | 04150258 | CC13BR300 | PN straight male | C13 straight female | 3 m |

**NOTICE**

You must connect the AC power cable in compliance with local standards and requirements.

**NOTE**

- The cable with the C7 straight female X2 connector only to the desktop devices USG6510E, USG6510E-POE, USG6530E.
- The cable with the C13 straight female X2 connector and non-C14 straight male X1 connector applies to non-desktop-mounted scenarios.
- The cable with the C13 straight female X2 connector and C14 straight male X1 connector applies to non-desktop models in PDU installation scenarios.

## 3.6.3 DC Power Cable

### 3.6.3.1 DC Power Cable

### Connection

A DC power cable is connected as follows:

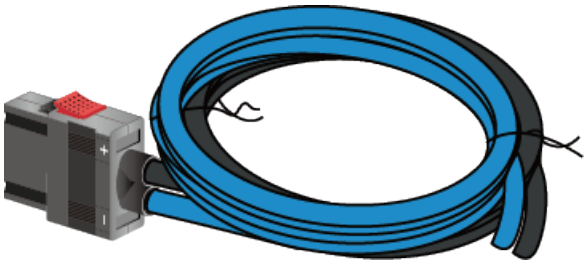
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- One end is connected to the matching DC power terminal on the DC power module of the USG.
- The other end is connected to an external power source.

## Appearance and Structure

The DC power cables include the RTN cable, and NEG (-48 V) cable. Both the RTN and NEG (-48 V) power cables consist of the terminal blocks and cables, as shown in **Figure 3-200**. For details about the power cable appearance, see **A.1 Quick Reference Tables of Power Cables**. You can view the photo of a power cable according to the corresponding BOM.

**Figure 3-200** Appearance of the DC power cable



| Cable Type | Color |
| --- | --- |
| NEG cable | Blue |
| RTN cable | Black |

The USG provides cord end terminals and OT terminals. You can select one of them based on the actual condition.

**Figure 3-201** shows the appearance of the cord end terminal.

**Figure 3-201** Appearance of the cord end terminal

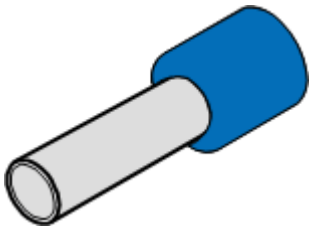


**Figure 3-202** shows the appearance of the OT terminal without a cable.

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**Figure 3-202** Appearance of the OT terminal



## Technical Specifications

**Table 3-341** Technical specifications of the DC Power Cable

| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Power Cable,3m,12AWG,PS2F,12UL1015B+12UL1015BL,2*T4^2GY | 04150695 | DCWIREA-3M | 14190215: PS2F | 14170114: 2*T4^2GY | 3 m | 24 mm |
| Power Cable,3m,4^2mm,PS2F,(H07Z-K-4^2B+H07Z-K-4^2BL),2*T4^2GY | 04152883 | PS2F-T4-3M | 14191175: PS2F | 14170114: 2*T4^2GY | 3 m | 24 mm |

# 3.6.4 Console Cable

## 3.6.4.1 Console Cable

## Connection

A console cable is connected as follows:

- The RJ45 connector is connected to the console port of the USG.

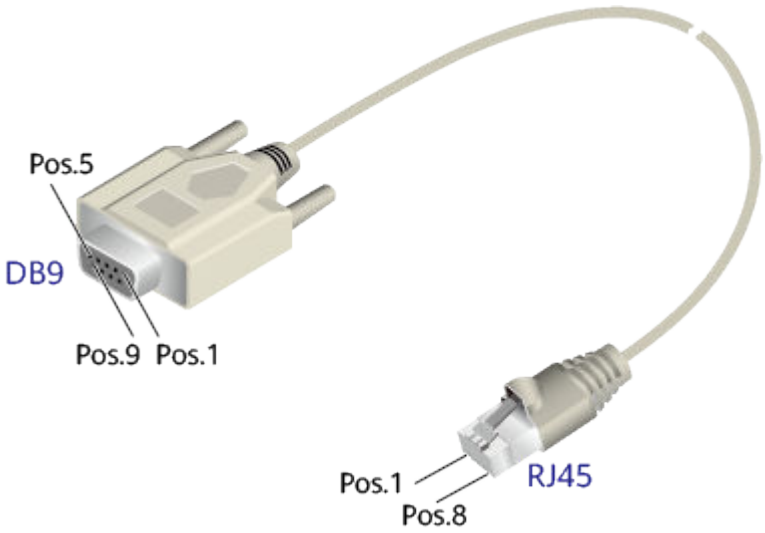
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- The DB9 connector is connected to the COM port on a maintenance terminal.

## Appearance and Structure

A console cable is an 8-wire shielded cable. **Figure 3-203** shows the console cable.

**Figure 3-203** Appearance of the Console Cable




| Name | Description |
| --- | --- |
| DB9 | DB9 connector(female). The connector has 9 pin holes with IDs from 1 to 9. It connects to the COM port of a PC. |
| RJ45 | RJ45 connector |
| Pos | Pin position |

## Pin Assignments

**Table 3-342** lists the pin assignments of a console cable.

**Table 3-342** Pin assignments of a console cable

| USG | Signal Description | RJ45 Pin ID | Signal Direction | DB9 Pin ID |
| --- | --- | --- | --- | --- |
| NA | RTS (Request to Send) | 1 | ← | 7 |
| NA | DTR (Data Terminal Ready) | 2 | ← | 4 |
| TXD | TXD (Transmit Data) | 3 | → | 2 |

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| USG | Signal Description | RJ45 Pin ID | Signal Direction | DB9 Pin ID |
|---|---|---|---|---|
| NA | DCD (Data Carrier Detect) | 4 | → | 1 |
| GND | GND | 5 | - | 5 |
| RXD | RXD (Receive Data) | 6 | ← | 3 |
| NA | DSR (Data Set Ready) | 7 | → | 6 |
| NA | CTS (Clear to Send) | 8 | → | 8 |

**NOTE**

Pins in the same row are connected to each other using a cable.

The symbol → indicates that the signal travels from the RJ45 connector to the DB9 connector, whereas the symbol ← indicates that the signal travels from the DB9 connector to the RJ45 connector.

Pos. 9 of the DB9 connector is empty.

### Technical Specifications

Table 3-343 shows the technical specifications of the console cable.

**Table 3-343** Technical specifications of the console cable

| Item | Description |
|---|---|
| Connector 1 | DB9 connector - D model - 9PIN - female |
| Connector 2 | RJ45 Network Interface Connector, Crystal Connector, 8PIN, 8BIT, Shielded, Plug |
| Cable | Single Cable, Console Serial Port Cable, 3 m, D9F, CC4P0.48B(S), MP8-VI |
| Wire | 8 |

## 3.6.5 Ethernet Cable

### 3.6.5.1 Ethernet Cable

### Connection

The Ethernet cable, also called the network cable, is used for carrying signals on Gigabit networks or networks with a lower packet rate.

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Ethernet cables are classified into straight-through and crossover cables according to the wiring scheme used at the two ends of the cables. The interfaces of the USG can dynamically adapt to straight-through cables and crossover cables.

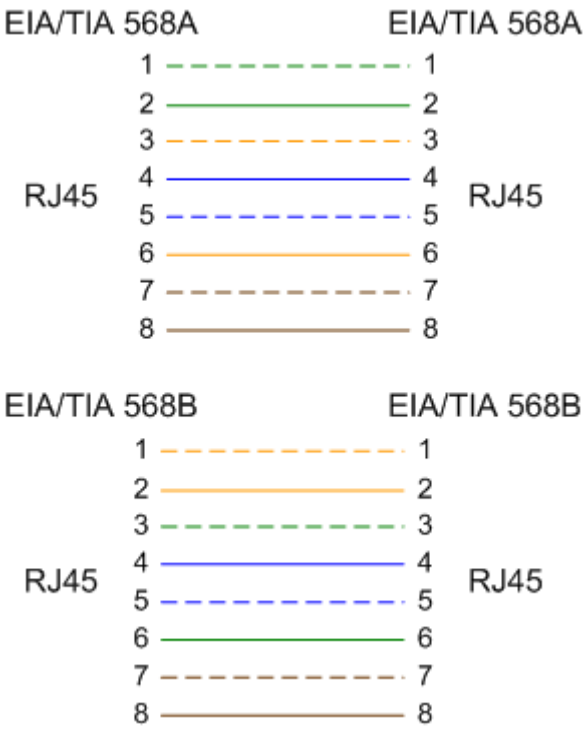
**NOTICE**

**Figure 3-204** and **Figure 3-205** show how to distinguish and make straight-through cables and crossover cables. You need to strictly observe the wiring scheme to ensure the quality of services.

- Straight-through cable

  The straight-through cable is also called the standard cable. Both ends of the cable use wiring scheme EIA/TIA T568A or EIA/TIA T568B. You are advised to use wiring scheme EIA/TIA T568B when you make straight-through cables. **Figure 3-204** shows the detail.

  **Figure 3-204** Schematic diagram of the connection scheme of straight-through cables

  You can use straight-through cables to connect:

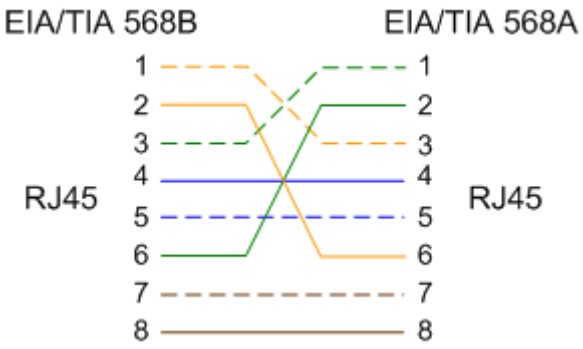
  - A switch or hub with an USG6000E
  - Computers (including servers and workstations) with a switch or hub

- Crossover cable

  The crossover cable uses wiring scheme EIA/TIA T568A for one end and EIA/TIA T568B for the other, as shown in **Figure 3-205**.

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**Figure 3-205** Schematic diagram of the connection scheme of crossover cables




You can use the crossover cables to connect:

- Two USG6000Es
- An USG6000E and a computer
- Two hubs
- A switch and hubs
- Two switches
- Two computers

## Appearance and Structure

Network cables are classified into shielded and unshielded ones based on whether the twisted pair and RJ45 connector are shielded, as shown in **Figure 3-206** and **Figure 3-207**.

**Figure 3-206** Appearance of a shielded cable



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**Figure 3-207** Appearance of an unshielded cable

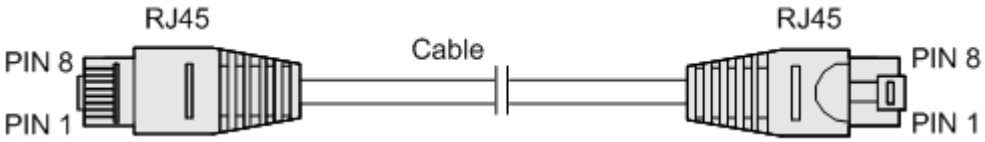


NOTE

The figure is for reference only.

An Ethernet cable consists of twisted pairs and RJ45 connectors, as shown in **Figure 3-208**.

**Figure 3-208** Structure of the Ethernet Cable



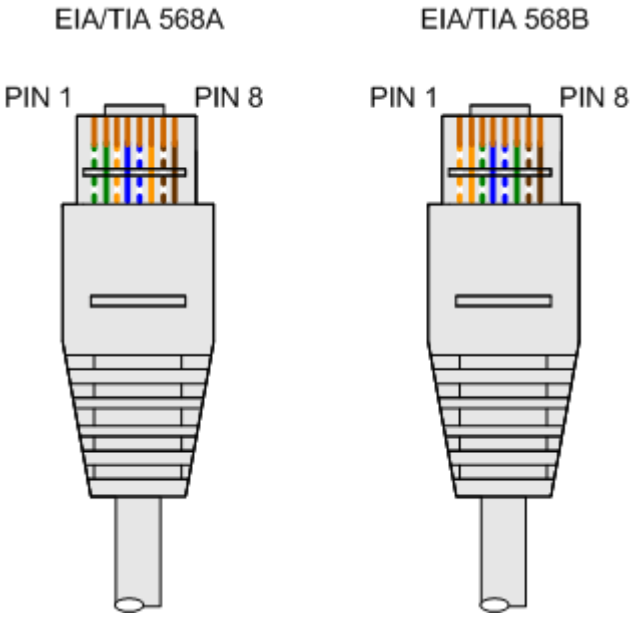

| Name | Description |
| --- | --- |
| RJ45 | RJ45 connector |
| Cable | Category 5 or higher twisted pair network cable |
| PIN1 to PIN8 | 8 pins. Each pin has an ID ranging from 1 to 8. |

## Pin Assignments

The twisted pair network cable has eight wires. The connection between wires and RJ45 connector pins complies with wiring scheme EIA/TIA 568A or EIA/TIA 568B, as shown in **Figure 3-209**.

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**Figure 3-209** Wiring scheme EIA/TIA 568A or EIA/TIA 568B




| EIA/TIA 568A | | EIA/TIA 568B | |
|---|---|---|---|
| **PIN[a]** | **Wire[b]** | **PIN** | **Wire** |
| 1 | Green/White | 1 | Orange/White |
| 2 | Green | 2 | Orange |
| 3 | Orange/White | 3 | Green/White |
| 4 | Blue | 4 | Blue |
| 5 | Blue/White | 5 | Blue/White |
| 6 | Orange | 6 | Green |
| 7 | Brown/White | 7 | Brown/White |
| 8 | Brown | 8 | Brown |

**NOTE**

- a. An RJ45 connector has eight pins.
- b. The wire of a twisted pair connects to the RJ45 connector pin. The color of the plastic jacket is considered as the name of the wire.

## Technical Specifications

Table 3-344 lists the technical specifications of the Ethernet cable.

**Table 3-344** Technical specifications of the Ethernet cable

| Item | Description |
| --- | --- |
| Connector 1 and 2 | Shielded RJ45/Unshielded RJ45 |
| Cable | Category 5 or higher shielded/unshielded twisted pair network cable |
| Wire | 8 |

# 3.6.6 Optical Fiber

## 3.6.6.1 Optical Fiber

### Connection

Optical fibers are used for carrying signals on Gigabit networks or networks with higher packet rates. An optical fiber is a carrier of optical signals and transmits optical signals over a short distance. An optical fiber is connected as follows:

- One end is connected to the optical port on the USG through the corresponding optical module.
- The other end is connected to the optical port on the peer device.

Select optical modules and fibers based on peer device interfaces.

**NOTICE**

You are advised to use Huawei optical modules. Optical modules from other vendors may cause incompatibility issues and lead to faults on the USG6000E.

When using optical modules and optical fibers, pay attention to the following to ensure proper communication between devices:

- Use single-mode and multi-mode optical fibers as required.
- Tx (sending) of the local device corresponds to Rx (receiving) of the peer device.
- The wavelengths of the optical modules on both ends must be the same.
- Do not overbend optical fibers. The bend radius must not be shorter than 40 mm.

**⚠ CAUTION**

To avoid eye injuries, do not look straight at the optical transmit interface of the laser or the optical connector.

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## Appearance and Structure

Optical fibers are classified into single-mode and multi-mode optical fibers. Single-mode optical fibers are used for long-distance transmission whereas multi-mode optical fibers are used for short-distance transmission. Indoor single-mode optical fibers usually have a yellow jacket, as shown in **Figure 3-210**. The jacket of the LC/PC-LC/PC multi-mode optical fiber is orange, as shown in **Figure 3-211**, and that of the MPO/PC-MPO/PC multi-mode optical fiber is cyan, as shown in **Figure 3-212**. Optical fibers are not delivered with the USG6000E. Purchase optical fibers separately as required.

**Figure 3-210** Appearance of the LC/PC-LC/PC single-mode optical fiber



**Figure 3-211** Appearance of the LC/PC-LC/PC multi-mode optical fiber



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**Figure 3-212** Appearance of the MPO/PC-MPO/PC multi-mode optical fiber



**NOTICE**

- To ensure the normal transmission of optical signals, do not use single-mode optical fibers to connect multi-mode optical modules, nor multi-mode optical fibers to connect single-mode optical modules.
- To connect extension optical fibers, purchase LC/PC-LC/PC adapters.

Optical fibers have multiple types of connectors. Common connectors types include LC/PC, SC/PC, FC/PC and MPO/PC.

- LC/PC connector

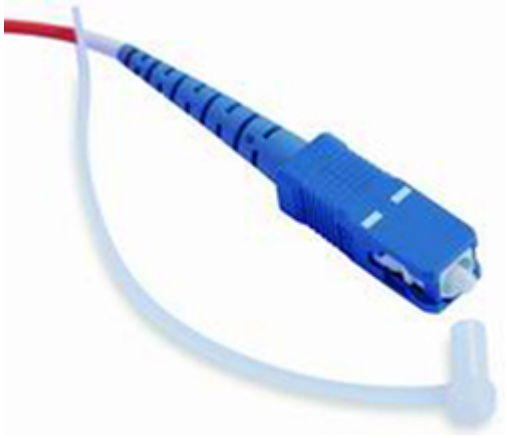


The procedure for inserting and removing LC/PC connectors is as follows:

  - Insert and remove the connector along the axis without rotating the connector.
  - When you insert the fiber, carefully insert the fiber head into the optical interface on the interface board.
  - To remove the optical fiber, squeeze the pinch on the fiber connector, gently push on the fiber connector, and then pull it out.

- SC/PC connector

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The procedure for inserting and removing SC/PC connectors is as follows:

- Insert and remove the connector along the axis without rotating the connector.
- When you insert the fiber, carefully insert the fiber head into the optical interface on the interface board.
- To remove the optical fiber, gently push on the fiber connector, and then pull out the connector.

- FC/PC connector



The procedure for inserting and removing FC/PC connectors is as follows:

- When you insert an optical fiber, aim the FC/PC connector at the optical interface on the interface board to avoid damaging the inner layer of the optical interface. Then insert the fiber into the optical interface, rotate the outer screw jacket clockwise, and then fasten the optical connector.
- To remove an optical fiber, rotate the outer screw jacket counter-clockwise on the optical interface until the screw loosens. Then gently pull out the optical fiber.

- MPO/PC optical fiber connector



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Note the following points when removing and inserting an MPO/PC fiber connector:

- When inserting the connector, hold the shell labeled "PUSH" and feed the male connector into the female connector until you hear a clicking sound. The male and female connectors are securely installed.
- To disassemble the connector, hold the shell labeled "PULL" and remove the male connector.

There are four types of optical fibers for the USG6000E: LC/PC-LC/PC, LC/PC-SC/PC, LC/PC-FC/PC, and MPO/PC-MPO/PC, which are based on the types of connectors on both ends of the fibers, as listed in **Table 3-345**.

**Table 3-345** Common optical fiber types

| Type | Transmission Mode | Connector | |
|---|---|---|---|
| | | On the USG6000E | On the peer device |
| LC/PC-LC/PC | Single-mode/ Multi-mode | LC/PC | LC/PC |
| LC/PC-SC/PC | | | SC/PC |
| LC/PC-FC/PC | | | FC/PC |
| MPO/PC-MPO/PC | Multi-mode | MPO/PC | MPO/PC |

## Technical Specifications

**Table 3-346** Technical specifications of the Optical Fiber

| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Patch Cord,SC/PC,SC/PC,Single-mode,5m,G.652D,3mm | 14130098 | SS-OP-D-SC-S-5 | SC/PC | SC/PC | 5 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Single-mode,5m,G.652D,2mm | 14130193 | SS-OP-LC-SC-S-5 | LC/PC | SC/PC | 5 m | 30 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Patch Cord,LC/PC-LC/PC,Single mode,G.652D,2mm,5m | 14130195 | SS-OP-D-LC-S-5 | LC/PC | LC/PC | 5 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Single-mode,10m,G.652D,2mm | 14130196 | SS-OP-LC-SC-S-10 | LC/PC | SC/PC | 10 m | 30 mm |
| Patch Cord,LC/PC,FC/PC,Single-mode,10m,G.652D,2mm | 14130197 | SS-OP-LC-FC-S-10 | LC/PC | FC/PC | 10 m | 30 mm |
| Patch Cord,LC/PC,LC/PC,Single-mode,10m,G.652D,2mm | 14130199 | SS-OP-D-LC-S-10 | LC/PC | LC/PC | 10 m | 30 mm |
| Patch Cord,LC/PC,FC/PC,Multi-mode,10m,A1b,2mm | 14130221 | SS-OP-LC-FC-M-10 | LC/PC | FC/PC | 10 m | 30 mm |
| Patch Cord,LC/PC,LC/PC,Multi-mode,10m,A1B,2mm | 14130222 | SS-OP-D-LC-M-10 | LC/PC | LC/PC | 10 m | 30 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Patch Cord,LC/PC,SC/PC,Multi-mode,10m,A1b,2mm | 14130223 | SS-OP-LC-SC-M-10 | LC/PC | SC/PC | 10 m | 30 mm |
| Optical adapter-LC/PC-LC/PC-Blue-Shell:Plastic-Sleeve:Zirconia-Square | 14130248 | QW1P0FIBER06 | LC/PC | LC/PC | - | - |
| Patch Cord,LC/PC,LC/PC,Single-mode,20m,G.652D,2mm | 14130251 | SS-OP-D-LC-S-20 | LC/PC | LC/PC | 20 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Multi-mode,30m,A1b,2mm | 14130275 | SS-OP-LC-SC-M-30 | LC/PC | SC/PC | 30 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Single-mode,30m,G.652D,2mm | 14130276 | SS-OP-LC-SC-S-30 | LC/PC | SC/PC | 30 m | 30 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
|---|---|---|---|---|---|---|
| Patch Cord,LC/PC,SC/PC,Multi-mode,20m,A1b,2mm | 14130279 | SS-OP-LC-SC-M-20 | LC/PC | SC/PC | 20 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Single-mode,20m,G.652D,2mm | 14130280 | SS-OP-LC-SC-S-20 | LC/PC | SC/PC | 20 m | 30 mm |
| Patch Cord,LC/PC,LC/PC,Multi-mode,20m,A1b,2mm | 14130295 | SS-OP-D-LC-M-20 | LC/PC | LC/PC | 20 m | 30 mm |
| Patch Cord,LC/PC,LC/PC,Multi-mode,5m,A1b,2mm | 14130296 | SS-OP-D-LC-M-5 | LC/PC | LC/PC | 5 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Multi-mode,50m,A1b,2mm | 14130311 | SS-OP-LC-SC-M-50 | LC/PC | SC/PC | 50 m | 30 mm |
| Patch Cord,LC/PC,SC/PC,Single-mode,50m,G.652D,2mm | 14130312 | SS-OP-LC-SC-S-50 | LC/PC | SC/PC | 50 m | 30 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
| --- | --- | --- | --- | --- | --- | --- |
| Patch Cord,LC/PC,LC/PC,Single-mode,6m,G.652D,2mm | 14130360 | SS-OP-D-LC-S-6 | LC/PC | LC/PC | 6 m | 30 mm |
| Optical Cable Parts,MPO/PC,MPO/PC,MULTI-MODE,2m,8 CORES,0m/0m,GJFH-8A1A.2(OM3),3.5MM,LSZH,43MM SHORT MPO,BENDING INSENSITIVE | 14130823 | F00MPMP01 | MPO/PC | MPO/PC | 2 m | 18 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
| --- | --- | --- | --- | --- | --- | --- |
| Optical Cable Parts,MPO/PC,MPO/PC,Multimode(OM3,Bending Insensitive),30m,8 Cores,0m/0m,GJFH8A1a,3.5mm,LSZH,42mm Short MPO,Indoor | 14130823-002 | FMMM03001 | MPO/PC | MPO/PC | 30 m | 17.5 mm |
| Optical Cable Parts,MPO/PC,MPO/PC,Multimode(OM3,Bending Insensitive),100m,8 Cores,0m/0m,GJFH8A1a,3.5mm,LSZH,42mm Short MPO,Indoor | 14130823-003 | FMMM10001 | MPO/PC | MPO/PC | 100 m | 17.5 mm |

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| Description | Part Number | Model | Connector X1 | Connector X2 | Cable length | Bend radius |
| --- | --- | --- | --- | --- | --- | --- |
| Optical Cable Parts,MPO/PC,MPO/PC,MULTI-MODE,15m,8 CORES,0m/0m,GJFH-8A1A.2(OM3),3.5MM,LSZH,60mm MPO,Bending insensitive | 14130923 | SS-OP-D-MPO12-M-15 | MPO/PC | MPO/PC | 15 m | 18 mm |
| Optical Cable Parts,MPO/PC,MPO/PC,Multi-mode,10m,8 cores,GJFH-8A1a.3(OM4),3.5mm,0mm,LSZH,Bending Insensitive | 14134486 | MPO12-MPO12-M4-10 | MPO/PC | MPO/PC | 10 m | 17.5 mm |

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# 4 Hardware Installation

This chapter describes hardware installation procedures and precautions.

4.1 Installation Procedure
This section describes the installation procedure of the USG6000E.

4.2 Installation Preparation
This section describes the safety precautions that you must observe and the tools that must be prepared before you install the USG.

4.3 Installing a Desktop Device
This chapter provides the cabinet-mounting, workbench-mounting, wall-mounting, and cable connection methods of the USG6510E, USG6510E-POE, USG6530E.

4.4 Installing a 420 mm Deep Device
This chapter provides the cabinet-mounting, workbench-mounting, and cable connection methods of the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E series.

4.5 Installing a 600 mm Deep Device
This chapter provides the cabinet-mounting, workbench-mounting, and cable connection methods of the USG6680E, USG6712E, USG6716E series.

## 4.1 Installation Procedure

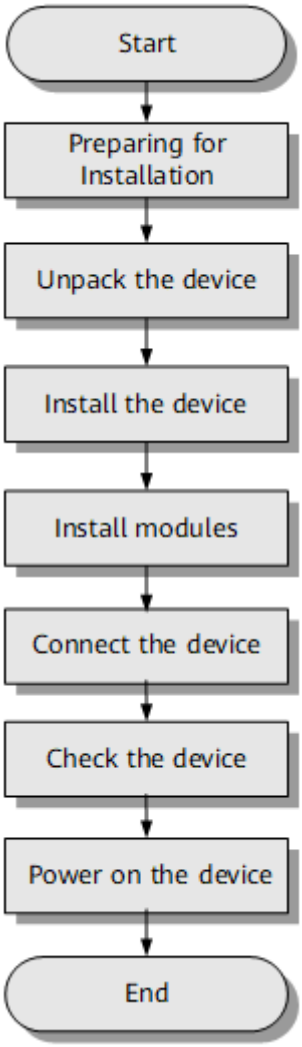
This section describes the installation procedure of the USG6000E.

procedure and precaution for unpacking and checking the products before installation has started.

**Figure 4-1** shows the USG devices installation procedure.

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**Figure 4-1** Installation flowchart




# 4.2 Installation Preparation

This section describes the safety precautions that you must observe and the tools that must be prepared before you install the USG.

## 4.2.1 Precautions

This section describes the precautions that you must observe before installing the USG. Misoperation may cause personal injury or damage to the USG.

NOTE

This section describes common precautions related to installation. For more precautions, see *Safety and Regulatory Compliance Information*.

### Safety Precautions

To ensure your own personal safety and to help protect your device from damage, observe the safety warnings on device labels and in the operation manual.

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Information marked **NOTICE**, **CAUTION**, **WARNING**, and **DANGER** in the operation manual is not exhaustive, but supplements safety precautions.

### Local Laws and Regulations

Comply with local laws and regulations while performing operations on the device. All safety precautions in the operation manual only supplement local safety regulations.

### Basic Installation Requirements

Requirements for Huawei installation and maintenance personnel before they can perform related operations are as follows:

- Only qualified and trained engineers can install, operate, and maintain Huawei equipment.
- Only qualified professionals can remove safety facilities and troubleshoot Huawei equipment.
- Only authorized or certified personnel can replace Huawei components (including software).
- Installation and maintenance personnel must report all faults and errors that may cause safety issues to the person in charge.

### Personal Safety Warnings

- Do not operate or cable the device during electrical storms.
- To avoid electric shock, do not connect the safety extra-low voltage (SELV) terminal to the telephone-network voltage (TNV) terminal.
- To avoid possible eye damage, do not look into the optical cable outlet without eye protection.
- Wear an antistatic suit, ESD gloves, and ESD wrist strap and remove any jewelry and watches before entering the equipment room to avoid possible electrical shock or injury.
- In the event of fire, evacuate the equipment room and nearby areas and pull the fire alarm or call your local emergency number.

### Device Security Precautions

- Fix the device on the ground or other secure places, such as against the wall or on the mounting shelf.
- Do not block the air vent when the device is running.
- Tighten the screws with proper tools when installing the panels.
- Remove all plastic packing materials from the equipment room after the installation is complete.

## 4.2.2 Installation Environment Check

Before you install a USG, verify that the installation environment meets requirements to ensure the normal running and extended life time of the USG.

**Table 4-1** lists installation environment check items.

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**Table 4-1** Installation environment checklist

| Check Item | Requirement | Compliance |
| --- | --- | --- |
| Ventilation and heat dissipation | Complies with requirements in **A.2.1 Device Position**. | □ Y □ N □ N/A |
| Stability | | □ Y □ N □ N/A |
| Grounding | | □ Y □ N □ N/A |
| Temperature | Complies with requirements in **A.2.2 Humidity, Temperature, and Cleanness**. | □ Y □ N □ N/A |
| Relative humidity | | □ Y □ N □ N/A |
| Cleanness | | □ Y □ N □ N/A |
| Electrostatic discharging | Complies with requirements in **A.2.3 ESD Requirements**. | □ Y □ N □ N/A |
| Surge protection | Complies with requirements in **A.2.4 Lightning Protection and Grounding**. | □ Y □ N □ N/A |
| Power supply facility | Complies with requirements in **A.2.5 Power Supply**. | □ Y □ N □ N/A |
| Electromagnetic shielding | Complies with requirements in **A.2.6 Electromagnetic Protection**. | □ Y □ N □ N/A |

## 4.2.3 Instruments Required for the Installation

This section describes the instruments and meters for installing the USG.

**Table 4-2** lists the required instruments and meters for USG installation.

**Table 4-2** Required instruments and meters

| Category | Instrument/Meter |
| --- | --- |
| Measuring and lineation instruments | • Ruler: used to measure length.<br>• Marker: used to indicate specific lengths with drawn lines. |
| Fastening tools | • Flat-head screwdriver: used to fasten small screws and bolts.<br>• Phillips screwdriver: used to fasten small screws and bolts. |

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| Category | Instrument/Meter |
| --- | --- |
| Drilling tools | • Hammer drill: used to drill mounting holes during wall-mounting.<br>• Vacuum cleaner: used to remove dust and debris produced while drilling holes.<br>• Hammer: used to drive the hollow wall anchors into mounting holes. |
| Pliers | • Needle-nose pliers: used to hold small fittings and twist fine wires in a narrow workplace.<br>• Diagonal cutting pliers: used to cut insulated sleeves and tie wraps. |
| Auxiliary tools | • Knife: used to peel the insulated sleeve from the cable.<br>• Ladder: used to access overhead cabling. |
| Special tools | • Wire stripper: used to peel the insulated sleeve of the communication cable.<br>• RJ-45 crimping tool: used to crimp RJ-45 cables for telephony and Ethernet applications.<br>• Crimping tool: used to crimp the metal sleeve at the ends of coaxial cables.<br>• Floating nut mounting bar: used to install floating nuts to the mounting bars of the cabinet.<br>• ESD gloves: used to protect the device from being damaged by static electricity.<br>• ESD wrist strip: used to protect the device from being damaged by static electricity.<br>• Protective gloves: used to shield hands from being injury by sharp objects. |
| Meters | • Network cable tester: used to test whether a network cable is connected and check the connection sequence in a network cable.<br>• Optical power meter: used to test the optical power.<br>• Optical attenuator: used to measure optical attenuation.<br>• Multimeter: used to test insulation within the cabinet, cable connections, and electric performance specifications of the device, such as the voltage, current, and resistance.<br>• Ground resistance tester: used to measure the ground resistance.<br>• Configuration terminal (A common PC is also applicable). |

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# 4.3 Installing a Desktop Device

This chapter provides the cabinet-mounting, workbench-mounting, wall-mounting, and cable connection methods of the USG6510E, USG6510E-POE, USG6530E.

## 4.3.1 Mounting a Device to a Specified Location

The USG6510E, USG6510E-POE, USG6530E can be mounted in a 19-inch standard cabinet. When no cabinet is available, install the USG6510E, USG6510E-POE, USG6530E on a workbench or a wall.

### 4.3.1.1 Mounting a Device in a Cabinet

The USG6510E, USG6510E-POE, USG6530E can be mounted in a 19-inch standard cabinet using mounting ears.

#### Precautions

Before installing the USG, check the following items:

- Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.
- The cabinet is stable.
- The position for the USG in the cabinet is well arranged.
- Ensure that the USG is 1 U of clearance from any devices above and below and 150 mm of clearance from any devices on the right or left.
- The device to be installed is staged near the cabinet for convenience.

You can place either end of the USG chassis towards the front door of the cabinet. In this manual, the front panel of the USG is towards the front door of the cabinet.

#### Tools and Accessories

- Phillips screwdriver
- Floating nuts and matching screws
- Floating mounting bar
- Mounting ears and matching screw (purchased separately, BOM numbers: 21240477)
- Holding plate and matching screw (purchased separately, BOM numbers: 21243646)

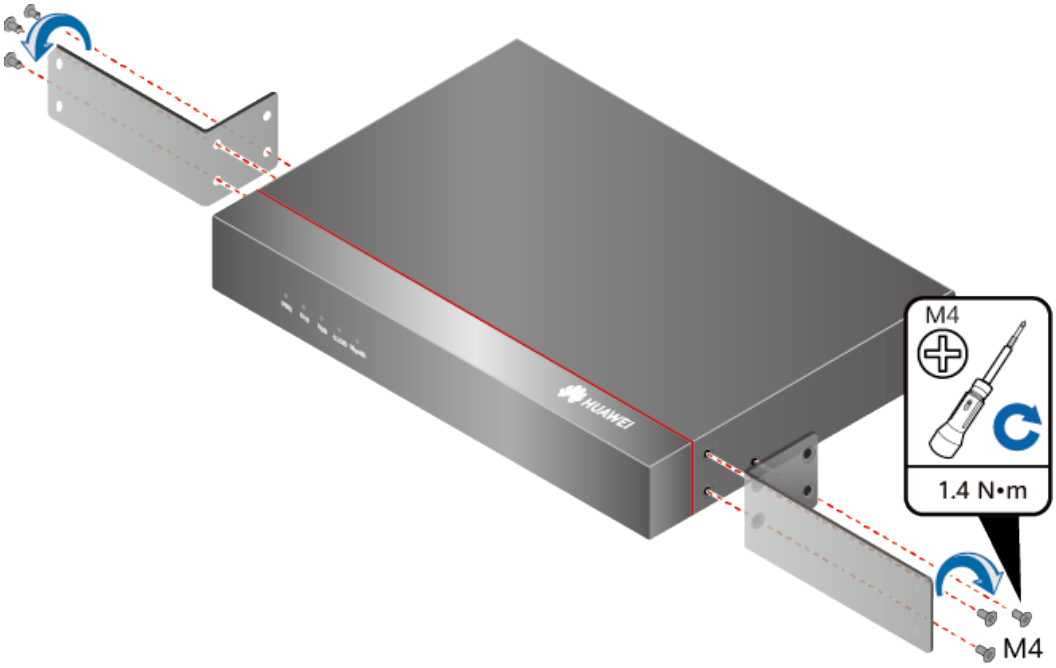
#### Procedure

**Step 1** Install mounting ears on the chassis.

Use a Phillips screwdriver to fix the mounting ears to both sides of the chassis with M4 screws, as shown in **Figure 4-2**.

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**Figure 4-2** Installing the mounting ears on the chassis



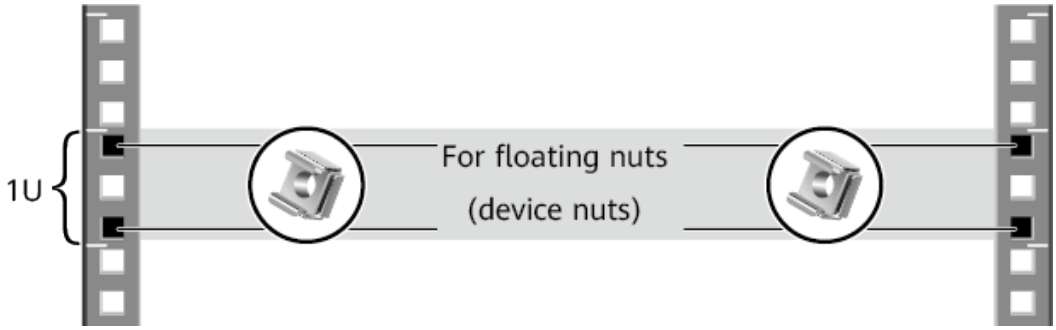

NOTE

If the tray installation mode is used, skip this step.

**Step 2** Install floating nuts.

**Figure 4-3** illustrates the positions of floating nuts.

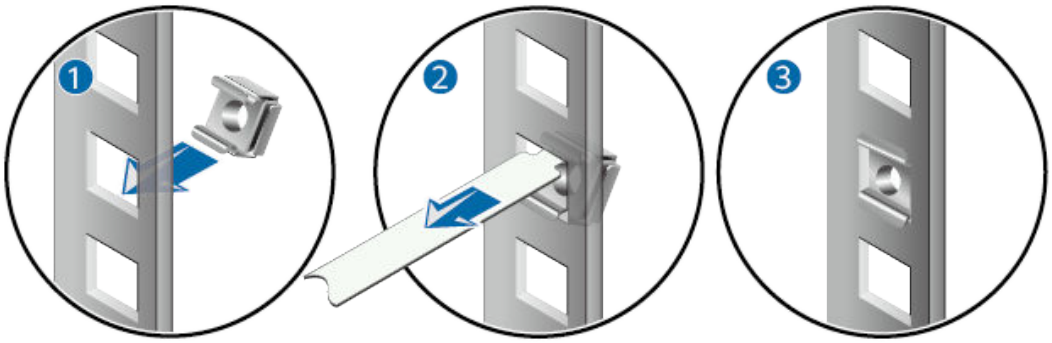
**Figure 4-3** Positions of floating nuts




Use M6 screws to fix the floating nuts at the positions specified in **Figure 4-3**, as shown in **Figure 4-4**.

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**Figure 4-4** Installing floating nuts

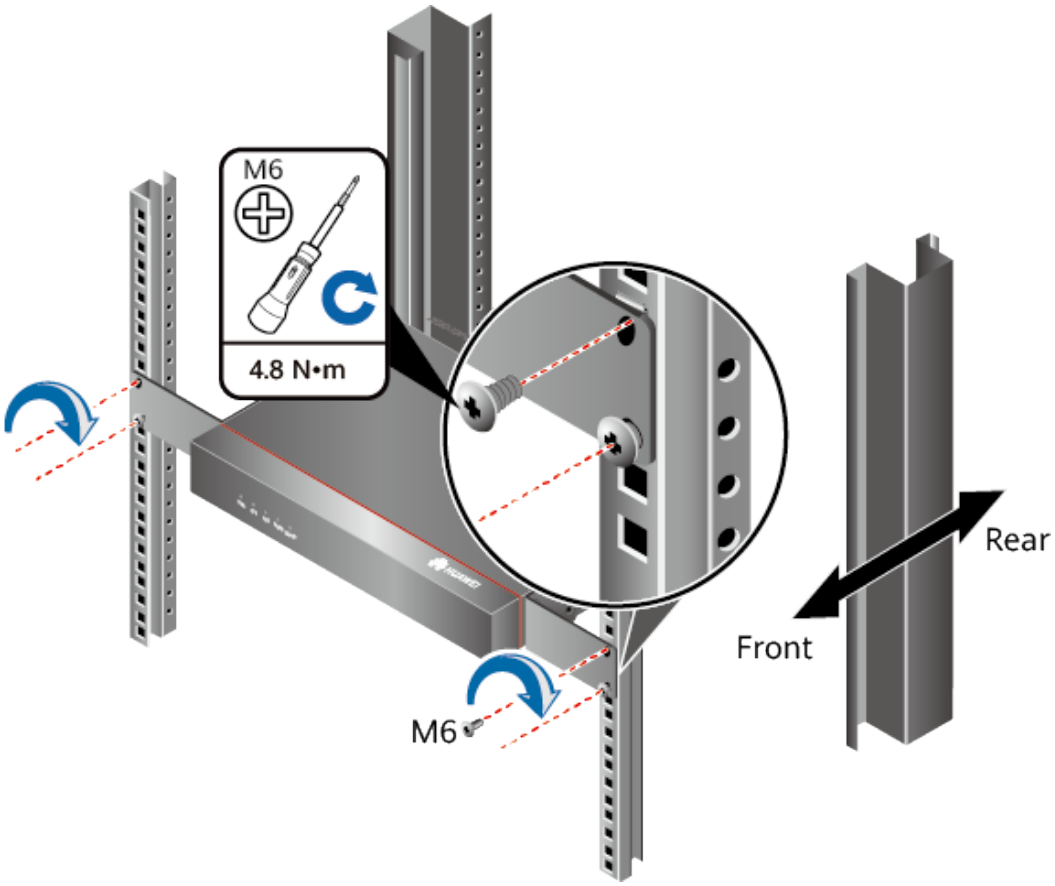



**Step 3** (Optional) Install the device to a tray. For details, see the installation guide delivered with the tray.

**Step 4** Mount the USG in the cabinet.

1. Lift the USG and move it to the cabinet.
2. Use a Phillips screwdriver to install M6 screws and fix the USG into the cabinet through mounting ears, as shown in **Figure 4-5**.

**Figure 4-5** Mounting the USG in a cabinet




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**----End**

## Follow-up Procedure

Perform the following checks after the installation:

- Ensure that the USG is placed securely inside the cabinet.
- Ensure that the exhaust of the USG is not blocked by other objects.

## 4.3.1.2 Mounting a Device on a Workbench

If you do not have a cabinet, you can mount the USG6510E, USG6510E-POE, USG6530E on a workbench.

## Precautions

Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.

The workbench must be:

- Reliably grounded.
- Clean, firm, and securely installed.

## Accessories

Four rubber feet

## Procedure

**Step 1** Fix the rubber feet to the round notches at the bottom of the USG.

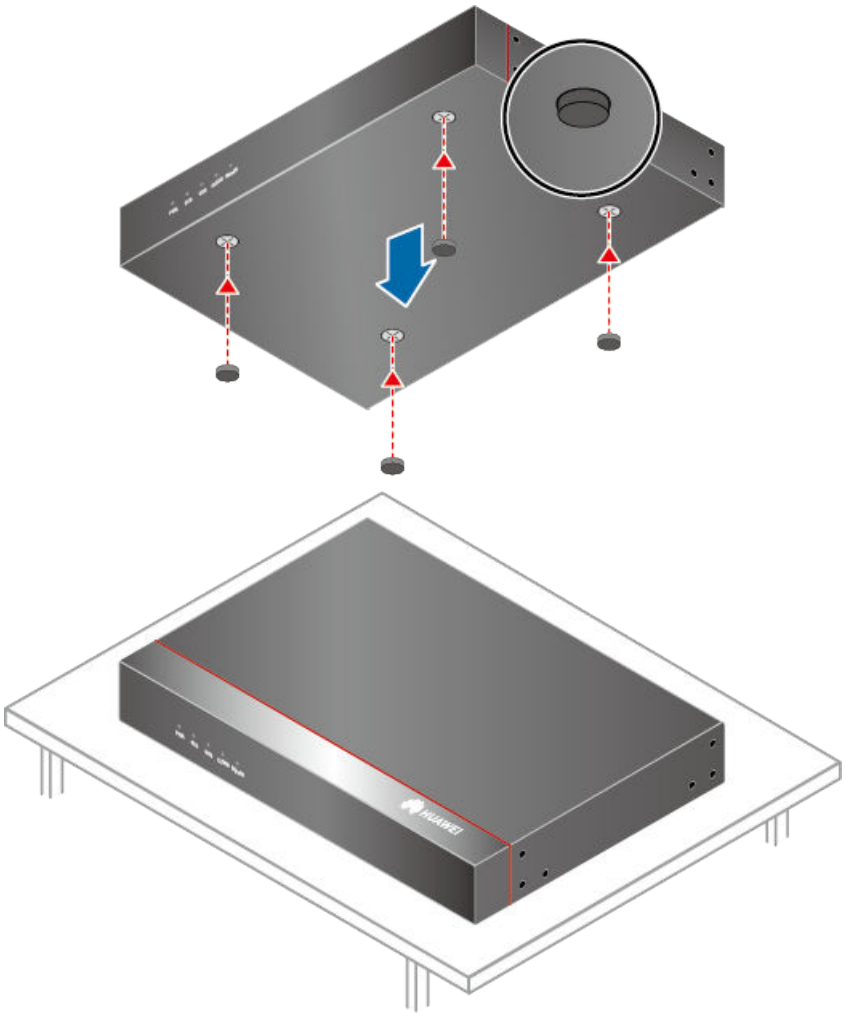
NOTE

Install foot pads at the bottom of the USG to ensure smooth contact between the USG and the workbench and avoid friction between the surface of the USG and the workbench.

**Step 2** Place the USG on the workbench.

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**Figure 4-6** Placing the USG with rubber feet on a workbench



**----End**

## Follow-up Procedure

Verify the following after the installation:

- The USG is securely placed on the workbench.
- No object blocks the exhaust of the USG, and there is at least 10 cm of distance between the USG and surrounding devices.
- There are no heavy objects on the USG.

## 4.3.1.3 Mounting a Device Against a Wall

When no cabinet is available, you can mount the USG6510E, USG6510E-POE, USG6530E on a wall. The customer must have expansion screws for wall-mounting.

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## Precautions

Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.

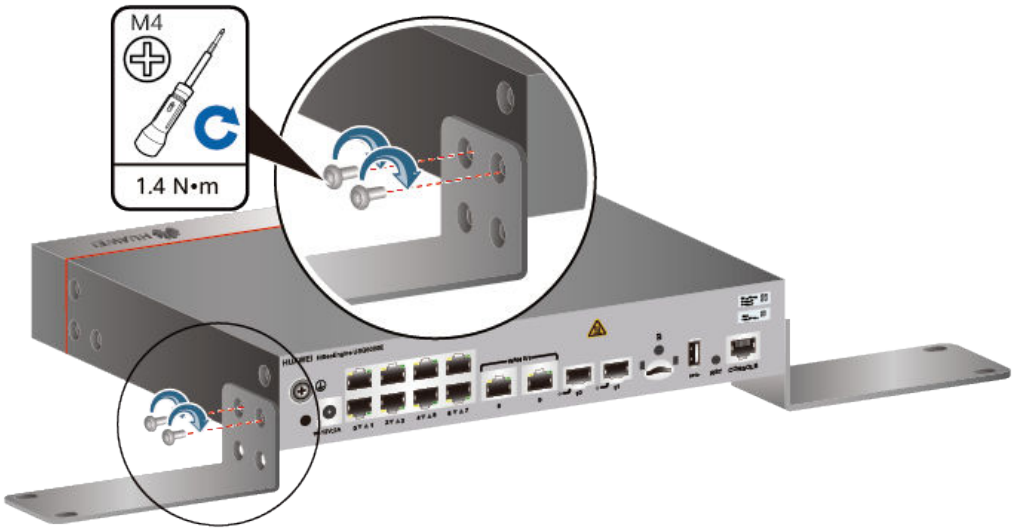
## Tools and Accessories

- Ruler
- Marker
- Hammer drill
- Vacuum cleaner
- Hammer
- Phillips screwdriver
- Mounting ears and matching screw (purchased separately, BOM numbers: 21240477)

## Procedure

**Step 1** Fix mounting ears to both sides of the panel with ports using M4 screws.

**Figure 4-7** Installing the mounting ears on the chassis



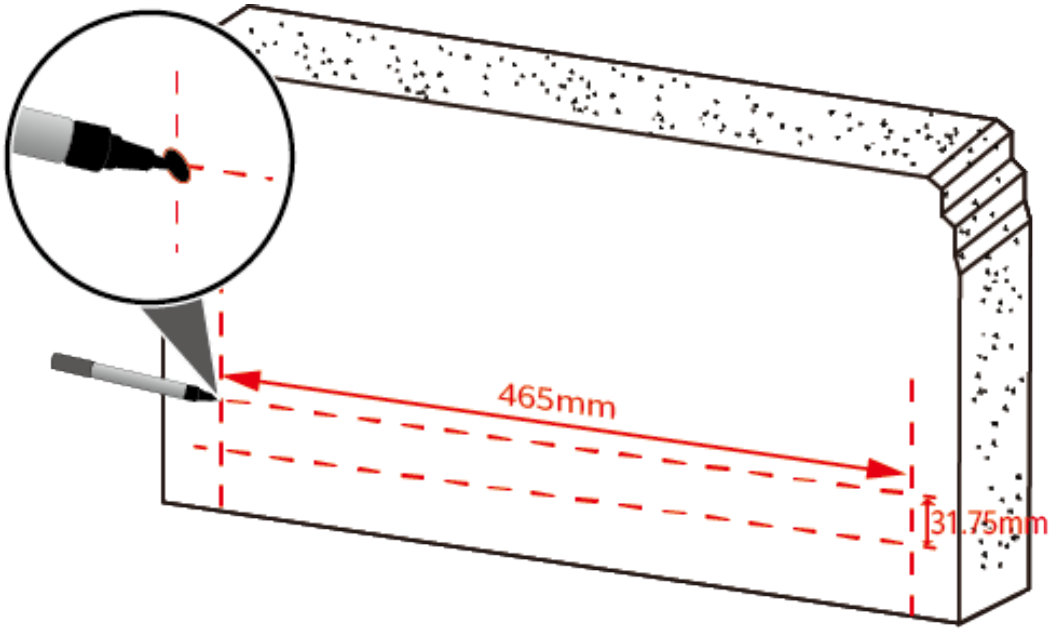

**Step 2** Determine the locations of four mounting holes on the wall using a ruler and mark the mounting holes with a marker.

NOTE

- The wall must be a bearing wall. Otherwise, the wall is not suitable for wall-mounting.
- Ensure that the height of mounting holes is proper so that the indicators are easy to view.

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**Figure 4-8** Spacing between mounting holes




**Step 3** Drill holes and install expansion bolts.

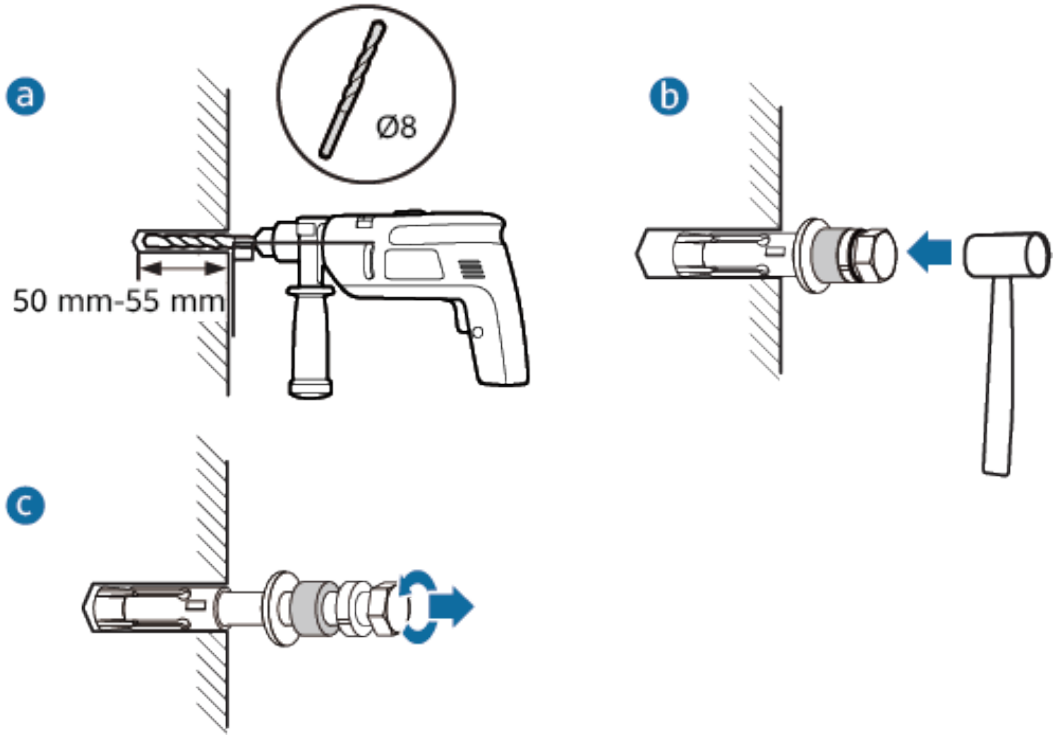
**NOTICE**

Ensure that the expansion bolts are secure and reliable. Otherwise, the tension after cables are connected may cause the USG6510E, USG6510E-POE, USG6530E to fall.

1. Use a ϕ8 drill bit to drill holes in the marked positions.
2. Insert expansion bolts into the holes and screw the nuts to fasten the expansion bolts.
3. Remove the nut, flat washer, and spring washer.

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**Figure 4-9** Drilling holes and installing expansion bolts



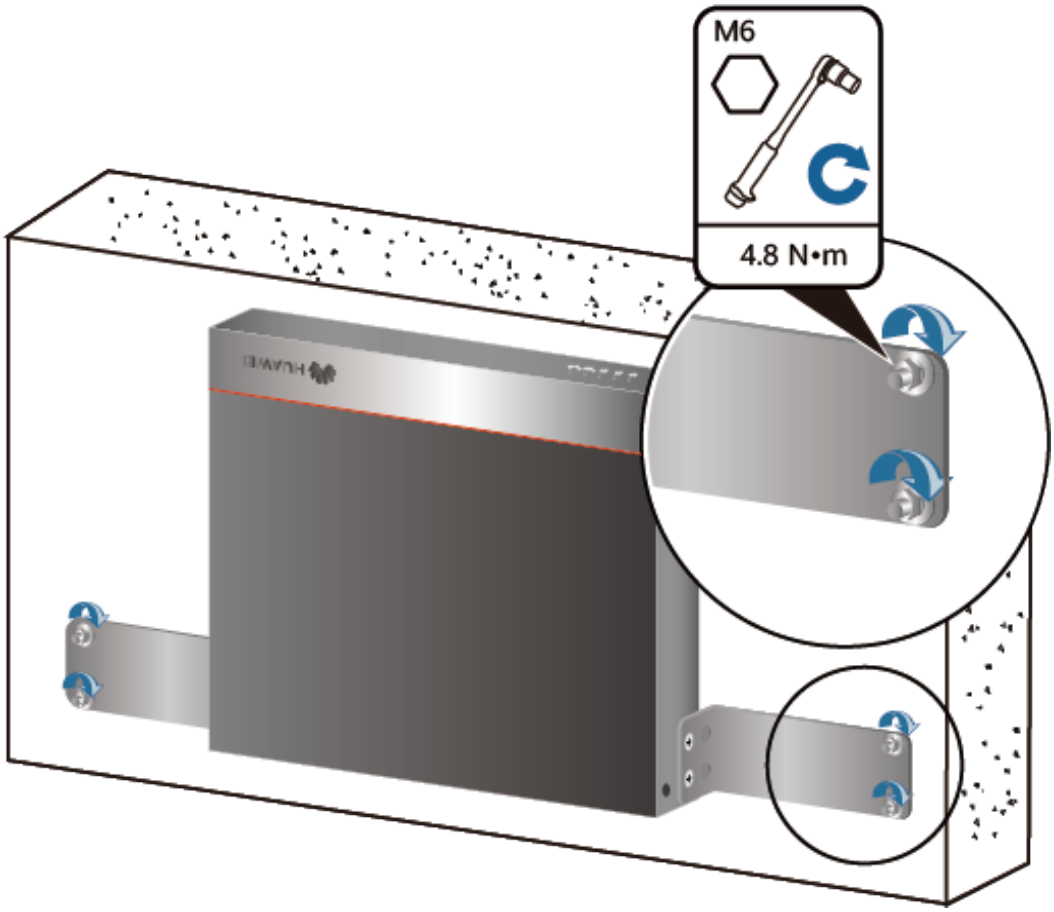

**Step 4** Secure the USG on the wall and fasten the flat washers, spring washers, and nuts in order.

NOTE

The USG supports upward mounting and downward mounting. To prevent water from entering into ports and causing device damage, you are advised to mount the USG with ports facing downward.

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**Figure 4-10** Mounting the USG on a wall




**----End**

### Follow-up Procedure

After wall-mounting is complete, verify that:

- The USG is securely fixed on the wall.
- A clearance of 10 cm is maintained around the USG and the air flow is not blocked.

## 4.3.2 Connecting a PGND Cable

Connecting the PGND cable of a USG6510E, USG6510E-POE, USG6530E correctly is a key measure of surge protection and resistance to interference. Before using the USG6510E, USG6510E-POE, USG6530E, correctly connect the PGND cable. Otherwise, the USG may be damaged.

### Precautions

The USG has been installed inside a cabinet.

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## Tools

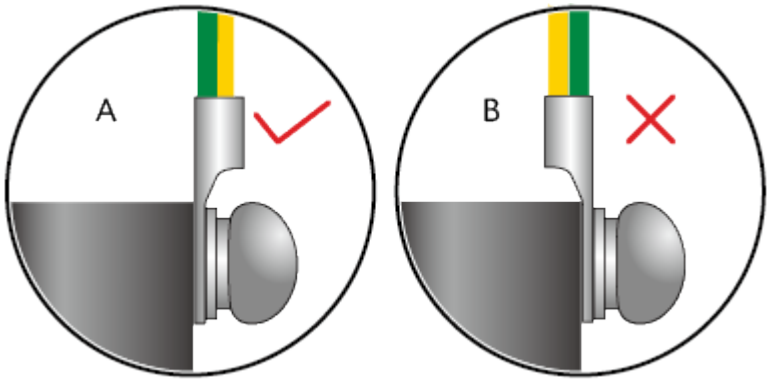
- Phillips screwdriver
- Multimeter

## Procedure

**Step 1** Loosen and remove the screw of the ground terminal on the USG rear panel.

**Step 2** Connect the OT terminal at one end of the PGND cable to the connection hole of the USG, with the conducting wire upward, and tighten the M4 screw, as shown in A of **Figure 4-11**, and tighten the M4 screw. The tightening torque is 1.4 N·m.

**Figure 4-11** Installing the OT terminal



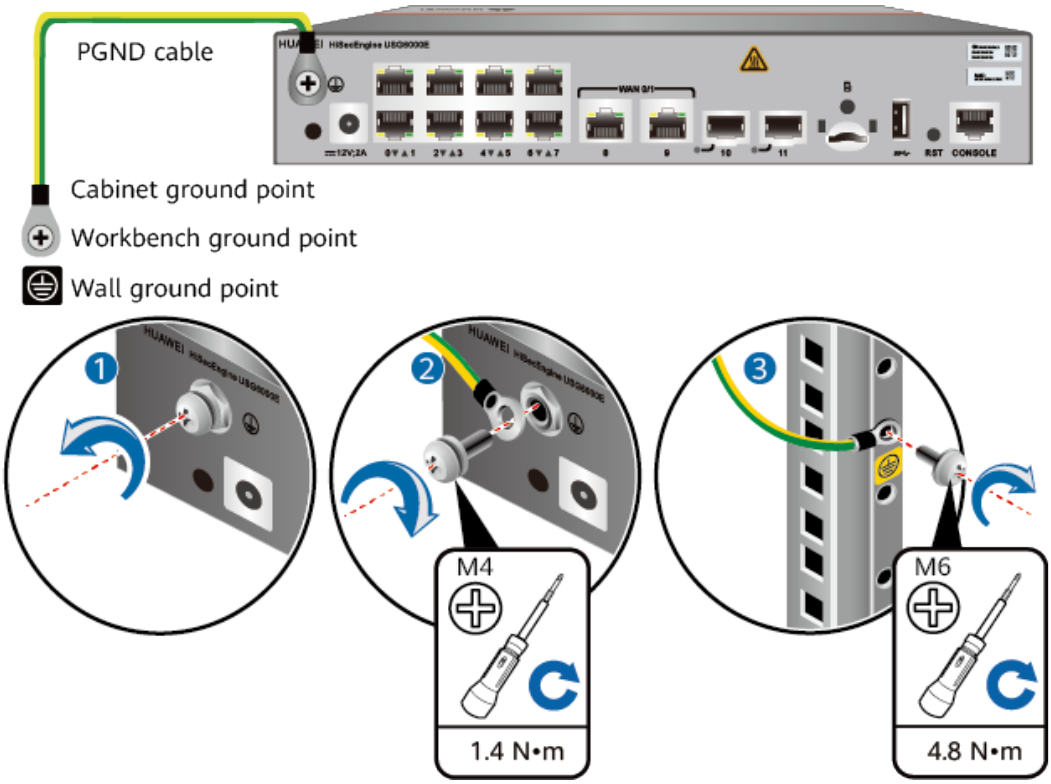

NOTE

The ground cable of the USG6510E, USG6510E-POE, USG6530E must be routed upwards.

**Step 3** Connect the M6 end of the PGND cable to the ground terminal of the cabinet, workbench, or wall. The tightening torque of the M6 screw is 4.8 N·m.

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**Figure 4-12** Connecting a PGND cable




**NOTICE**

The OT terminal may rotate and result in device damage. Make sure that it is independent of the adjacent metal mechanical part or other terminals.

**----End**

### Follow-up Procedure

Verify the following after the cabling is complete:

- The PGND cable is securely connected to the ground terminal.
- The electrical resistance between the ground terminal and ground point is less than 0.1 ohm on a multimeter.

## 4.3.3 Installing a Micro SD Card

This section describes how to install a micro SD card for the first time to avoid damages.

### Precautions

- Micro SD cards are optional and are not delivered with the device. If required, purchase the micro SD card (part number: 06010308) from Huawei. The micro

SD card model is SDSDQAE-064G, the capacity is 64 GB, and dimensions (H x W x D) are 1 mm x 15 mm x 11 mm (0.04 in. x 0.59 in. x 0.43 in.).

- The micro SD card can be installed no matter the device is powered off or the device is running. The installation methods are the same. In this section, the micro SD card is installed when the device is powered off.
- To replace the micro SD card when the USG is powered on, you must run the **sd-card offline** command in the system view first. After the system displays a message indicating that the micro SD card is offline, remove the micro SD card. Otherwise, the micro SD card might be damaged, and the data may be lost. For details, see **5.4 Replacing a Micro SD Card (USG6510E, USG6510E-POE, USG6530E)**.
- Make sure that you have worn an ESD wrist strap and the strap is well grounded before you hold the micro SD card. Otherwise, the micro SD card may be damaged.

## Tools

- Phillips screwdriver
- ESD wrist strap

## Procedure

**Step 1** Determine the slot (the slot with a "micro SD" mark on the rear panel) for installing the micro SD card.

**Step 2** Install the micro SD card and anti-theft board.

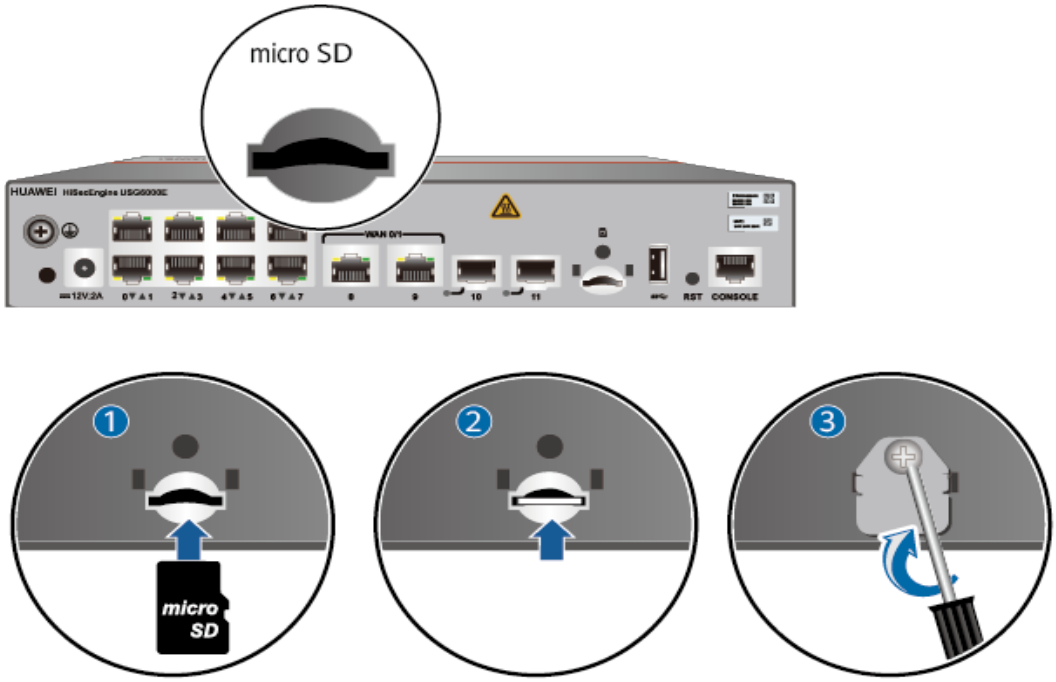
NOTE

- Note that the micro SD card must be installed with the face with words upwards.
- Do not use too much force; otherwise the micro SD or micro SD card slot might be damaged.

1. Insert the micro SD along the guide rail to the micro SD card slot.
2. When you hear a click, the micro SD card is in position.
3. Hook the locating hook on the anti-theft board to the locating hole of the rear panel and tighten the captive screw on the anti-theft board.

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**Figure 4-13** Installing the micro SD card and anti-theft board




**----End**

### Follow-up Procedure

After the micro SD card is installed, power on the USG and run the **display sd-card information** command in any view to check the micro SD card installation and file system mounting status. If **SD Card Physical State** is **Present** and **SD Card File System State** is **Mounted**, the micro SD card is working properly.

- If **SD Card Physical State** is **Absent**, re-install the micro SD card and try again. If **SD Card Physical State** remains the same, the USG may fail to identify the micro SD card. You are advised to use another micro SD card.
- If **SD Card File System State** is **Unmounted**, the micro SD card format might not be **ext4.** You need to run the **reset sd-card** command in the system view to format the micro SD card.

## 4.3.4 Connecting a Console Cable

After connecting a PC to the console port of a USG6510E, USG6510E-POE, USG6530E with a console cable, you can use the terminal emulation program on the PC to access the command configuration interface of the USG6510E, USG6510E-POE, USG6530E.

### Precautions

Before connecting a console cable, perform the following operations:

- Check preparations.

  A PC is ready, a USG has been installed, and the ports to be connected are planned.
- Prepare cable labels.

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Before cable connection, labels must be prepared for the cable.

**NOTICE**

- Make sure that the PC and the USG are connected to the same ground point. Otherwise, the console port of the USG may be damaged.
- Pay attention to port numbering and make sure that the cable is connected to the correct port, preventing damage to ports or the device.

## Tools

Console cable (prepared by the user)

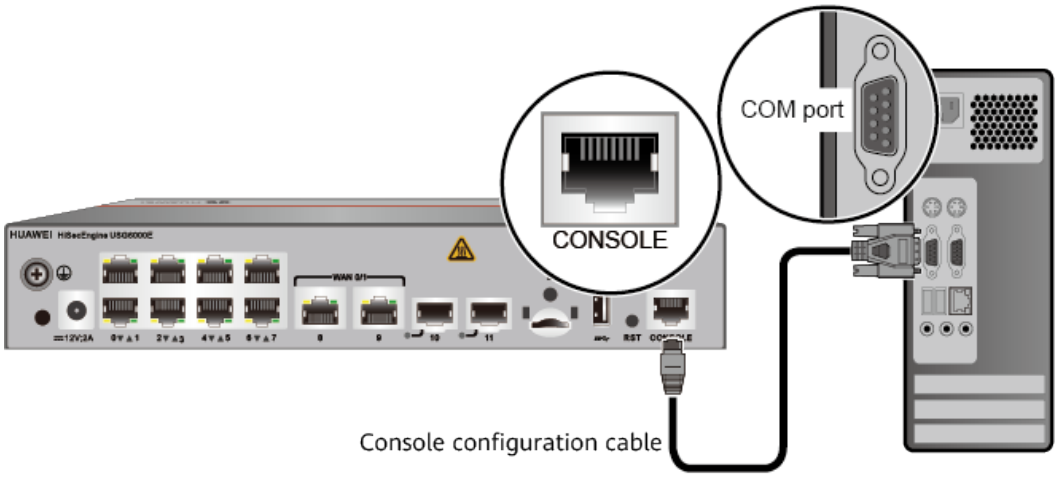
## Procedure

**Step 1** Before connecting a console cable, attach temporary labels to both ends of the cable for identification.

**Step 2** Connect the RJ45 connector of the console cable to the console port (RJ45) of the USG.

**Step 3** Connect the DB9 connector of the console cable to the COM port of the management PC.

**Figure 4-14** Connecting a console cable to the USG




**Step 4** Remove the temporary labels and attach labels 2 cm away from connectors at both ends of the console cable.

**----End**

## Follow-up Procedure

After the cable connection is complete, verify that:

- The labels at both ends of a cable are correct, clear, neat, and facing the same direction.

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- Cables and connectors are free of damage or breakage and are connected properly.

For details on the console login, refer to the Configuration Guide.

## 4.3.5 Connecting an Ethernet Cable

Based on the network plan, you can connect one end of an Ethernet cable to the Ethernet port of a USG6510E, USG6510E-POE, USG6530E and the other end to the Ethernet port of the peer device.

### Context

Before connecting the Ethernet cable, perform the following operations:

- Check construction conditions.

  The peer device has been installed in the equipment room, and the port to which the Ethernet cable is to be connected has been determined.
- Check the cabling route.

  The engineering document should specify the cabling route from the cabinet to the peer device in the equipment room, and the length of the cable is calculated based on the cabling path.
- Label the cable.

  The cable must be labeled before being connected to the devices.

**NOTICE**

- Only shielded cables are supported on the USG.
- Before connecting a cable, note the label on the port and make sure that the cable is inserted into the correct port. Otherwise, the port module or the device might be damaged.

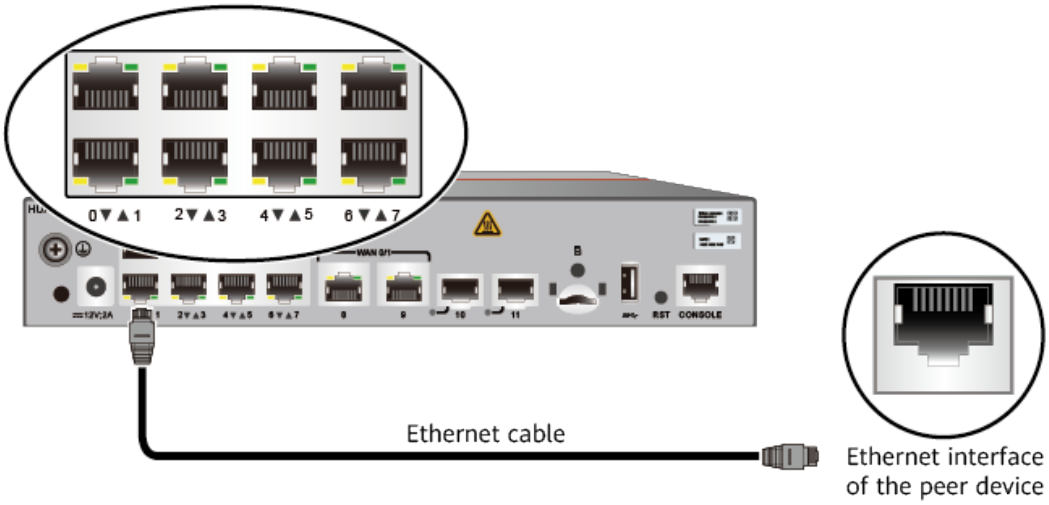
### Procedure

**Step 1** If multiple network cables need to be connected, attach temporary labels to both ends of each cable for identification.

**Step 2** Connect one end of an Ethernet cable to the Ethernet port of the USG and the other end to the Ethernet port of the peer device based on the network plan.

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**Figure 4-15** Connecting an Ethernet cable




**Step 3** Lay out the Ethernet cable along a cabinet and route the cable through the cable hole for the signal cables at the top (overhead cabling) or bottom (underfloor cabling) of the cabinet.

**Step 4** Remove the temporary labels and attach labels (2 cm away from connectors) at both ends of the Ethernet cable.

**----End**

## Follow-up Procedure

Verify the following after the installation:

- The labels at both ends of the cable are correct, clear, neat, and facing the same direction.
- The cables and connectors are free of any damage or breakage and are connected properly and reliably.

# 4.3.6 Installing Optical Transceivers and Connecting Optical Fibers

This section describes how to install optical transceivers on the SFP or SFP+ ports and connect them to the ports of the peer device using optical fibers according to the network plan.

## Context

The USG supports both 1 Gbit/s , 10 Gbit/s, and 40 Gbit/s optical modules. The optical modules at both ends are the same, including the optical fiber type (single-mode or multi-mode), optical fiber connector type (LC/PC, SC/PC, FC/PC, or MPO/PC-MPO/PC), and transmission rate. If different optical modules are used at the two ends, the communication may fail.

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**NOTICE**

Huawei optical modules are recommended. The optical modules from other vendors may cause faults on the USG due to incompatibility.

**⚠ CAUTION**

Do not look into the optical interface of the optical module or the optical fiber connector without eye protection.

Before connecting optical fiber cables, read the following precautions:

- Do not overbend optical fibers, and the radius should not be shorter than 40 mm.
- Do not bundle the optical fibers too tight. Otherwise, the transmission performance of the optical fibers and the communication between devices might be adversely affected.

Before connecting optical ensure the following:

- The optical module has been installed.

## Procedure

**Step 1** Insert an optical transceiver into the SFP or SFP+ port of the USG.

**Step 2** Remove the dust cap from the optical transceiver.

**NOTE**

Set aside the dust cap properly for future use. After optical fiber are disconnected for maintenance, use the dust cap to prevent the optical transceiver from dust.

**Step 3** Before connecting an optical fiber, attach temporary labels to both ends of the optical fiber for identification.

**Step 4** Remove protective caps from optical fiber connectors, insert optical fibers into the optical transceiver, and connect the fiber to the peer device.

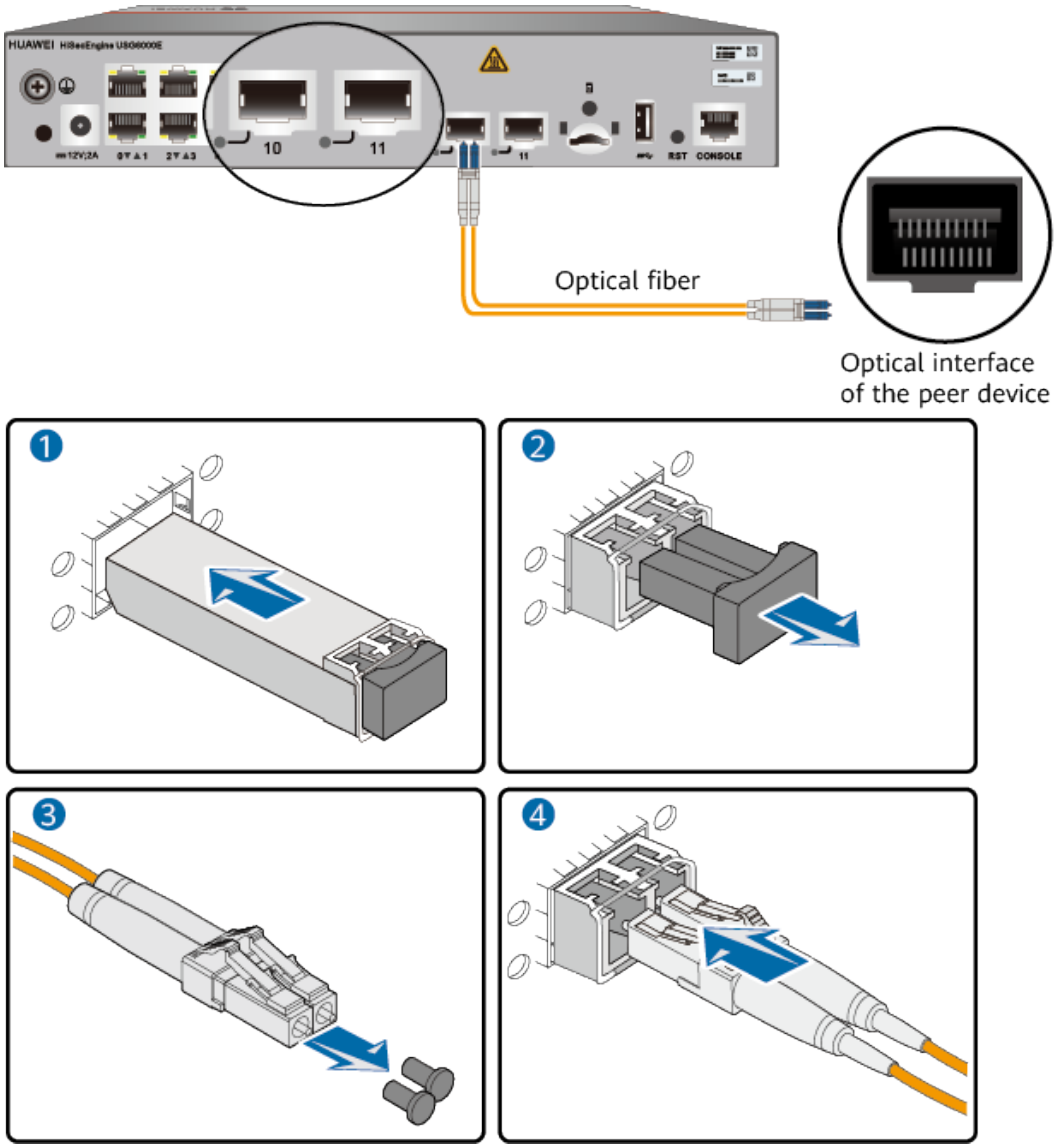
**NOTE**

Ensure that the Tx and Rx ports are correctly connected.

Ensure that the TX and RX ports on one end of the optical fiber cable are connected to the RX and TX ports (respectively) on the other end.

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**Figure 4-16** Installing optical transceivers and connecting optical fibers




**Step 5** Repeat **Step 1** to **Step 4** to install all optical transceivers and connect all optical fibers.

**----End**

## Follow-up Procedure

After you power on the USG, check the connection by observing the optical port indicator. If the indicator is on or blinks, the link is connected or data is being transmitted. If the indicator is off, the link is disconnected. Possible causes for the disconnection are as follows:

- The optical fiber is improperly inserted. Pull out the optical fiber and re-insert it.
- The RX and TX optical ports are inserted reversely. Pull out the optical fibers, change their position, and re-insert them.

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- The optical module is damaged or the optical fiber is broken. Replace the optical module or the optical fiber.

## 4.3.7 Connecting a Power Adapter

A power adapter can be used to supply power for a USG6510E, USG6510E-POE, USG6530E. Prepare an AC power cable to connect the power connector and the power source of the equipment room.

### Context

Before connecting a power connector, ensure that the power source of the equipment room meets the input requirements of the USG power module.

### Procedure

**Step 1** Ensure that the PGND cable is properly grounded.

**Step 2** Insert a cable-retention clip into the jack next to the power socket.

**Step 3** Connect a power adapter.

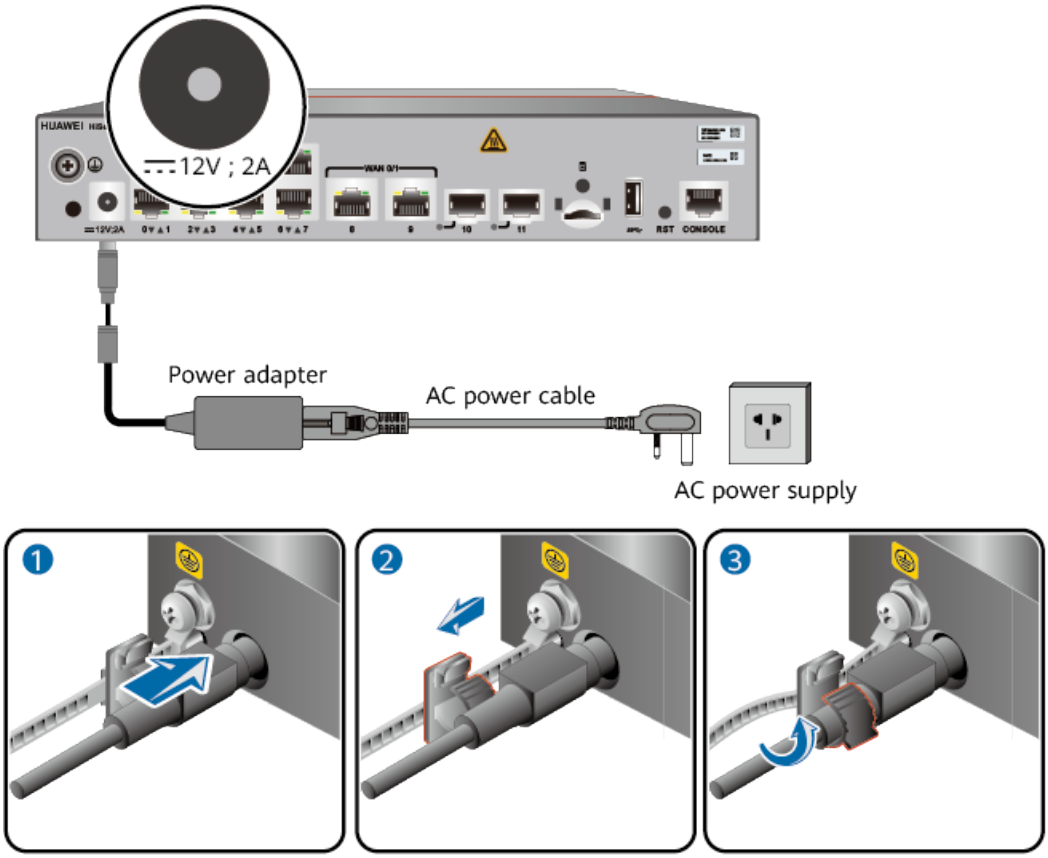
1. Plug the C7 plug of the AC power cable into the C14 socket of the power adapter.
2. Insert the tuning fork plug connector of the power adapter into the power socket on the USG rear panel and adjust the cable-retention clip to an appropriate position.
3. Use a cable-retention clip to bundle the cable of the power adapter and adjust the cable-retention clip to fasten the cable.
4. Plug the other end of the AC power cable to the AC power socket or the output of the AC power supply device.

   NOTE

   The USG does not have any power switch. The power supply switch determines the power-on and power-off of the USG.

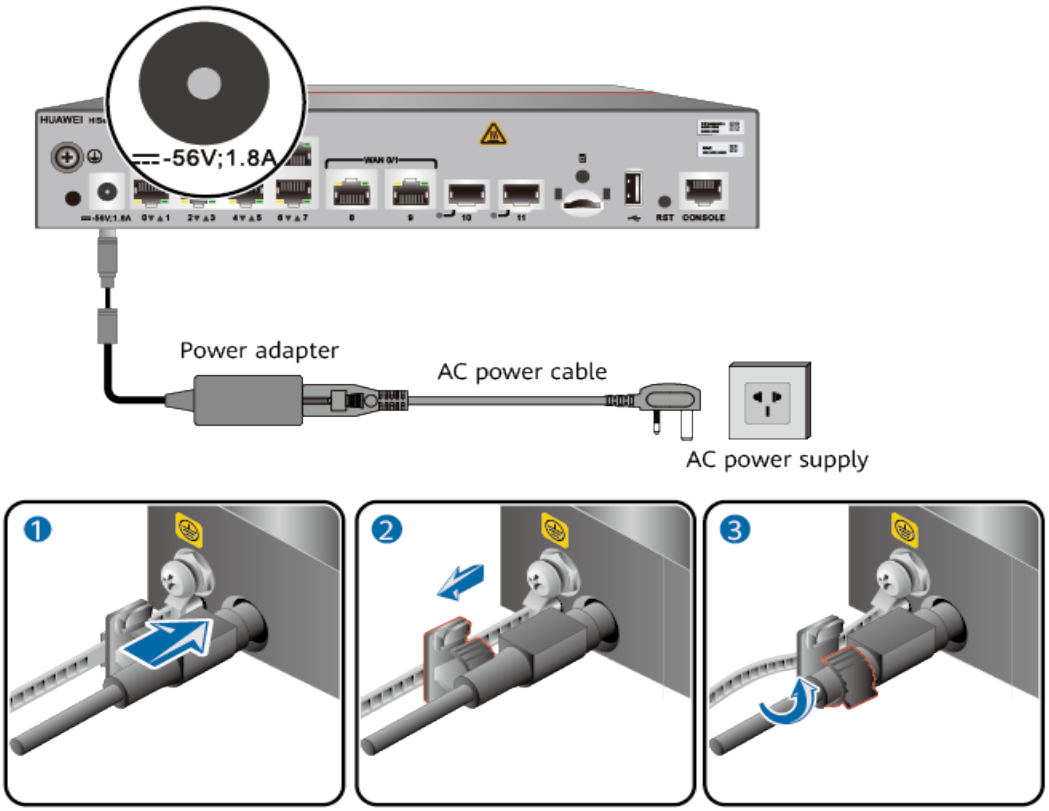
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**Figure 4-17** Connecting a power adapter to the (USG6510E, USG6530E)




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**Figure 4-18** Connecting a power adapter to the (USG6510E-POE)




**----End**

## Follow-up Procedure

Verify the following after the connection is complete:

- The power cable is firmly connected to the power supply socket.
- If multiple USGs are deployed, the power cables of each USG are correctly labeled for distinction.

# 4.3.8 Powering On or Off the USG6510E, USG6510E-POE, USG6530E

This section describes how to power on or off the USG6510E, USG6510E-POE, USG6530E. To ensure the normal start and security of the USG6510E, USG6510E-POE, USG6530E, strictly follow the operation guide to power on or off the USG6510E, USG6510E-POE, USG6530E.

## Context

Before you power on the USG, ensure that:

- The power cable and PGND cable are properly connected.
- The power switch in the equipment room is easy to locate so that you can power off devices in the case of accidents.

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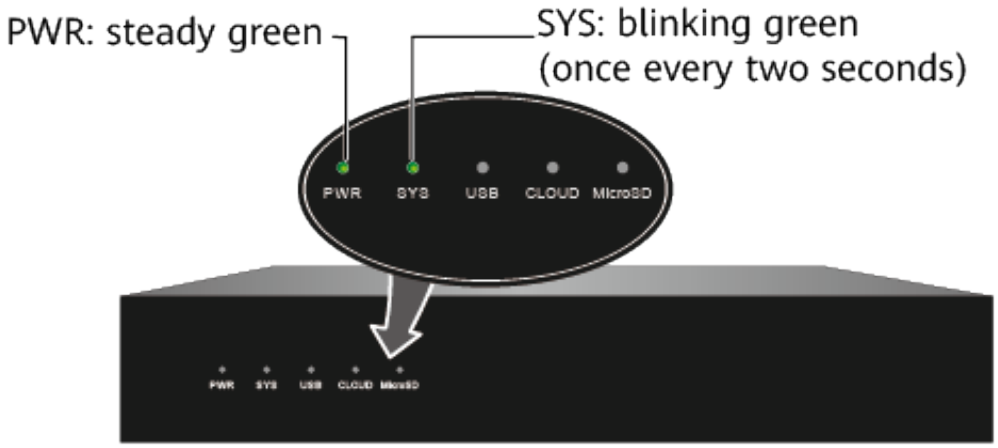
## Procedure

- Power on the USG.

  The USG starts after the switch of the power supply device is turned on.

  You can identify the USG status based on indicators on the front panel. Indicators shown in **Figure 4-19** indicate that the USG is running normally.

  **Figure 4-19** Indicators when the USG runs normally

- Power off the USG.

  **NOTICE**

  Before powering off the USG, ensure that configuration data is saved. Otherwise, the configuration data may be lost.

  If the USG will be administratively shut down for a long time, turn off the power switch. After powering off the USG, set it aside properly according to storage requirements.

**----End**

## Follow-up Procedure

After the USG is powered on, you can log in to the configuration page for management and maintenance. For details, refer to the Configuration Guide.

# 4.4 Installing a 420 mm Deep Device

This chapter provides the cabinet-mounting, workbench-mounting, and cable connection methods of the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E series.

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# 4.4.1 Mounting a Device to a Specified Location

The USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E can be mounted in a 19-inch standard cabinet. When no cabinet is available, you can mount the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E on a workbench.

## 4.4.1.1 Mounting a Device in a Cabinet

Install the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E to the cabinet through the expandable rear mounting ears or adjustable guide rails.

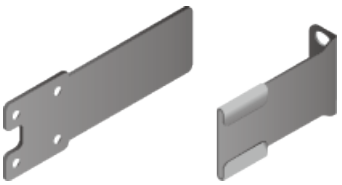
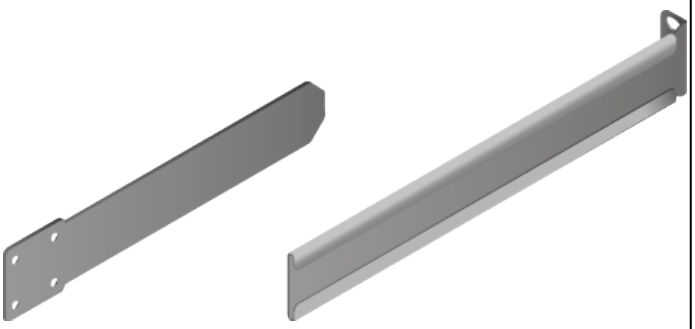
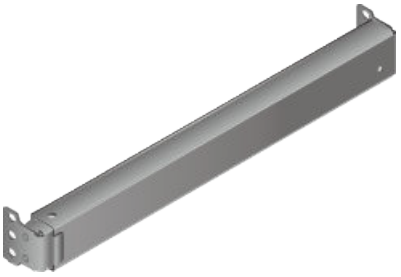
### Precautions

Before installing the USG, check the following items:

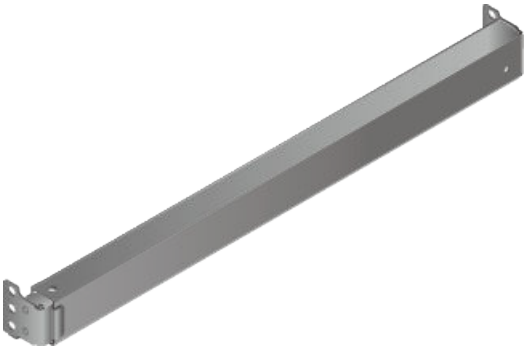
- Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.
- The cabinet is stable.
- Before installing the expandable rear mounting ears or adjustable guide rails on the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E, adjust the distance between the front and rear angle gauges (rack mounting rails/ vertical columns) to satisfy the installation requirements. The specific scope of application is as follows:

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**Table 4-3** Expandable rear mounting ears and adjustable guide rails

| Distance Between the Front and Rear Mounting Bars and Cabinet Type | Expandable Rear Mounting Ears and Adjustable Guide Rails |
| --- | --- |
| Expandable rear mounting ears (BOM number: 21240492)<br>• Distance between the front and rear mounting bars: 310 mm to 538 mm<br>• Cabinet depth: 600 mm to 800 mm |  |
| Expandable rear mounting ears (BOM number: 21240537)<br>• Distance between the front and rear mounting bars: 638 mm to 835 mm<br>• Cabinet depth: 1000 mm to 1200 mm |  |
| Adjustable guide rails (BOM number: 21242247)<br>• Distance between the front and rear mounting bars: 370 mm to 585 mm<br>• Cabinet depth: 600 mm to 800 mm |  |

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| Distance Between the Front and Rear Mounting Bars and Cabinet Type | Expandable Rear Mounting Ears and Adjustable Guide Rails |
| --- | --- |
| Adjustable guide rails (BOM number: 21242246)<br>• Distance between the front and rear mounting bars: 500 mm to 850 mm<br>• Cabinet depth: 1000 mm to 1200 mm |  |

- The position for the USG in the cabinet is well arranged. Ensure that the USG is 1 U of clearance from any devices above and below and 150 mm of clearance from any devices on the right or left.
- The USG to be installed is staged near the cabinet for convenience.
- In the front view, the air flow of the USG is front to rear. If the USG and a device whose air flow is rear to front must be mounted in the same cabinet, maintain certain spacing between them to prevent hot air recirculation.

You can place either end of the USG chassis towards the front door of the cabinet. In this manual, the front panel of the USG is towards the front door of the cabinet.

**NOTE**

The method of installing expandable rear mounting ears (BOM number: 21240492) is the same as that of installing expandable rear mounting ears (BOM number: 21240537). The following uses the expandable rear mounting ears (BOM number: 21240492) as examples.

## Tools and Accessories

- Phillips screwdriver
- Floating nuts and matching screws
- Floating mounting bar
- Front mounting ears and corresponding screws
- Expandable rear mounting ears and corresponding screws (purchased separately, BOM numbers: 21240492 and 21240537)
- Adjustable guide rails and matching screws (purchased separately, BOM numbers: 21242247 and 21242246)

## Procedure

**Step 1** Install mounting ears on the chassis.

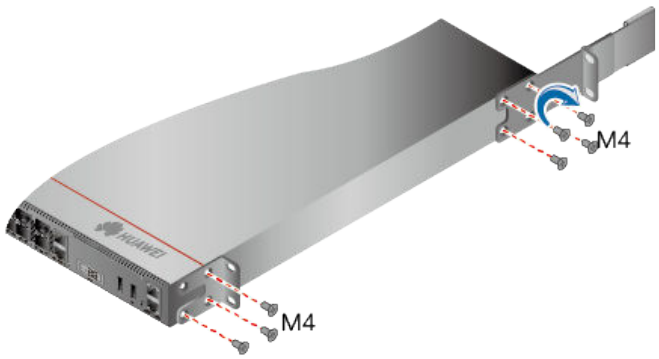
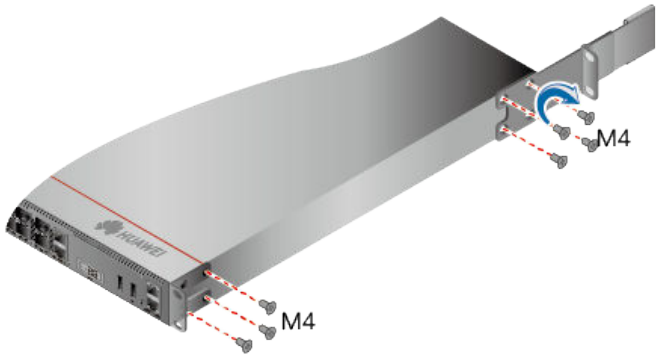
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Use a Phillips screwdriver to fix the mounting ears to both sides of the chassis with M4 screws. The maximum torque is 1.4 N m. For details about how to install the mounting ears, see the installation methods in **Table 4-4**.

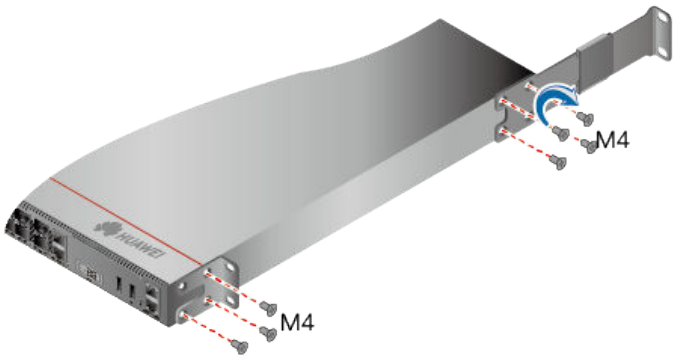
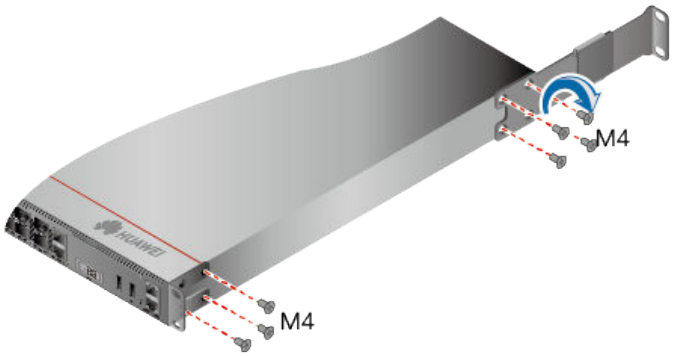
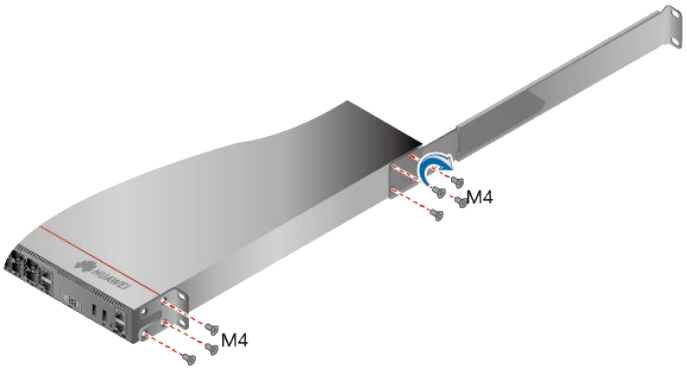
NOTE

- The front mounting brackets, rear mounting brackets, and rear mounting bracket guide rails must be installed on both sides of the USG. **Table 4-4** shows the installation on one side. Install them on the other side in the same way.
- To ensure sufficient cabling space, reserve at least 130 mm of vertical distance from the USG's port side to the interior side of the cabinet door.
- After the installation is complete, the rear mounting brackets and rear mounting bracket guide rails will not block cables routed from the ports or touch the cabinet door.

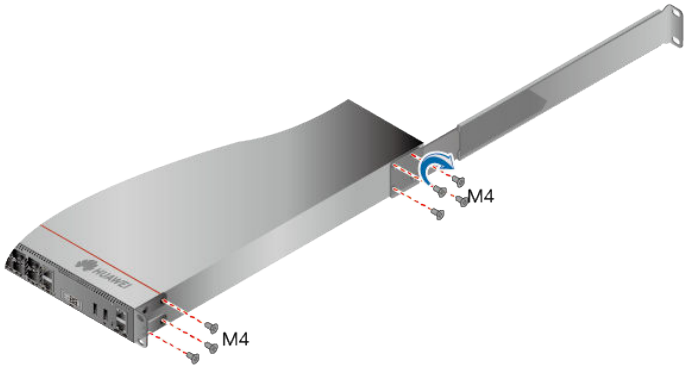
**Table 4-4** Installation of front and rear mounting ears

| Distance Between Front and Rear Mounting Rails | Installation of Front Mounting Brackets, Rear Mounting Brackets, and Rear Mounting Bracket Guide Rails |
| --- | --- |
| 310 mm to 351 mm |  |
| 369 mm to 410 mm |  |

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| Distance Between Front and Rear Mounting Rails | Installation of Front Mounting Brackets, Rear Mounting Brackets, and Rear Mounting Bracket Guide Rails |
| --- | --- |
| 438 mm to 479 mm |  |
| 497 mm to 538 mm |  |
| 638 mm to 696 mm |  |

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| Distance Between Front and Rear Mounting Rails | Installation of Front Mounting Brackets, Rear Mounting Brackets, and Rear Mounting Bracket Guide Rails |
| --- | --- |
| 697 mm to 835 mm |  |

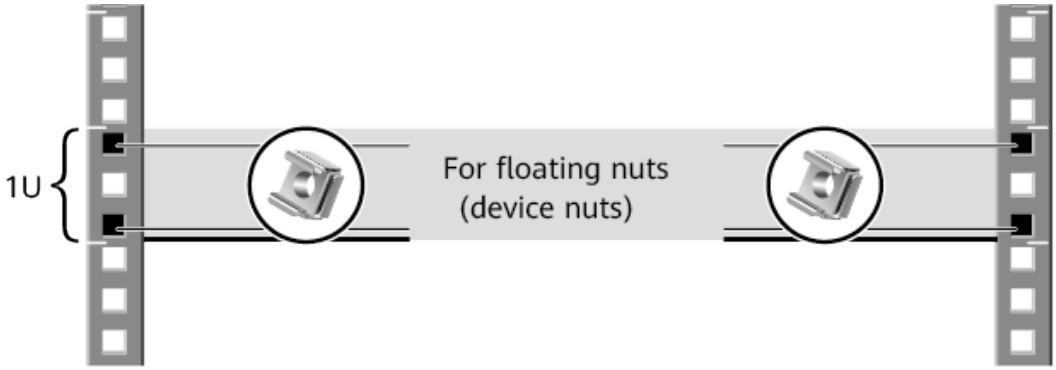
**Step 2** Install floating nuts and cabinet guide rails.

1. Determine the positions for installing the guide rails and floating nuts, as shown in **Figure 4-20**.

**NOTICE**

Install the two guide rails at the same height to ensure that the device is level.

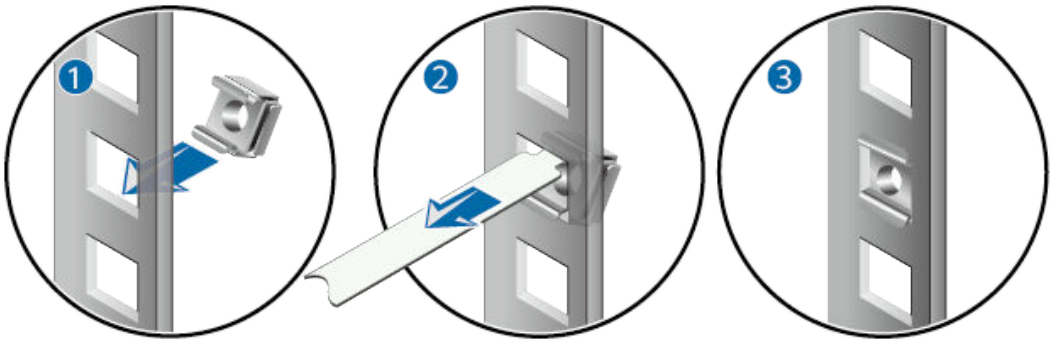
**Figure 4-20** Positions of floating nuts




2. Install the floating nuts that match M6 screws at the positions marked in **Figure 4-20**. **Figure 4-21** illustrates how to install a floating nut.

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**Figure 4-21** Installing floating nuts

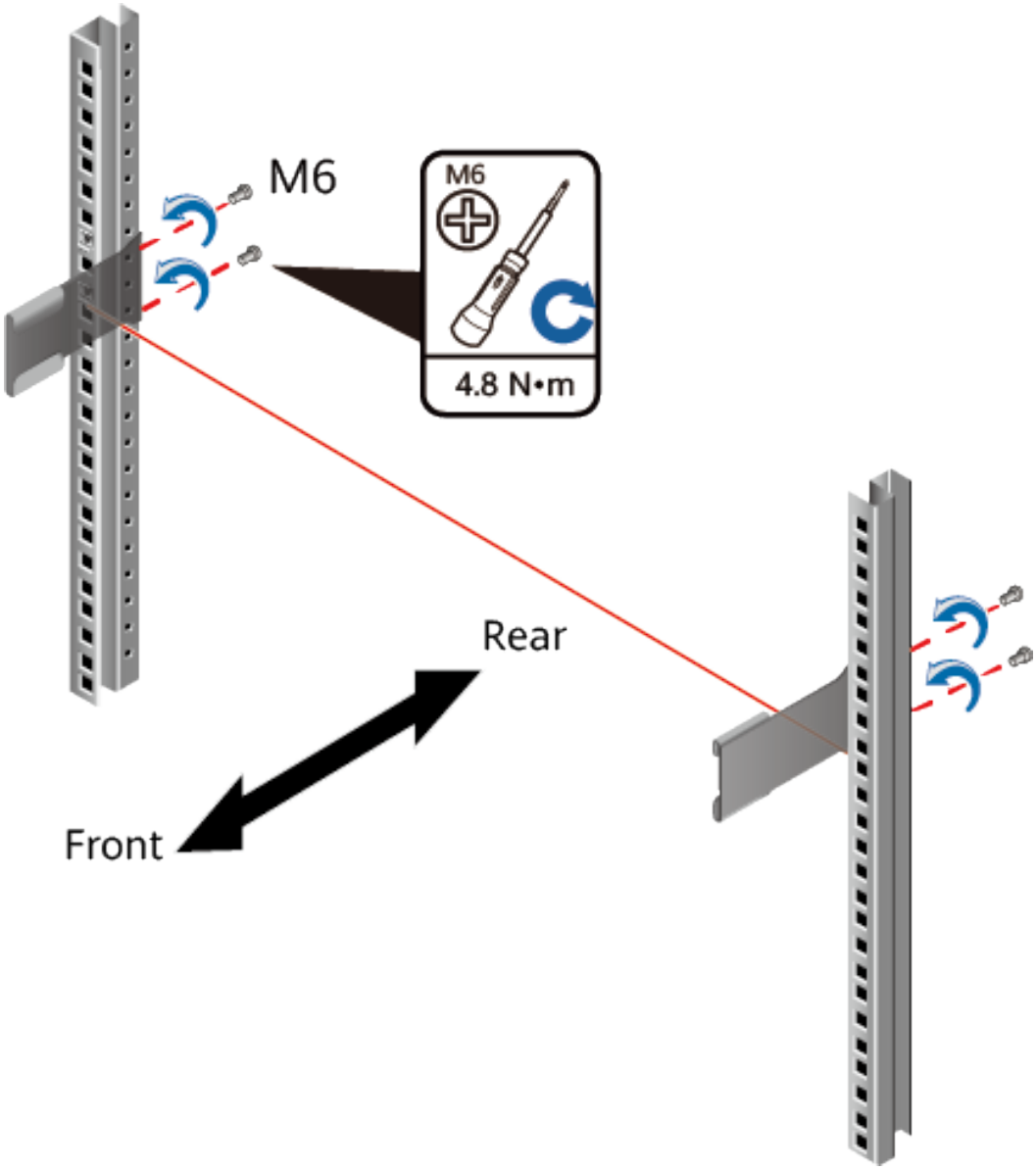



3. Install guide rails in the correct direction. Identify the left and right guide rails, and the front and rear ends of each guide rail.

   Fix expandable rear mounting ears and adjustable guide rails (at the positions marked in **Figure 4-20**) to the left and right of the cabinet with M6 screws, as shown in **Figure 4-22**.

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**Figure 4-22** Installing guide rails



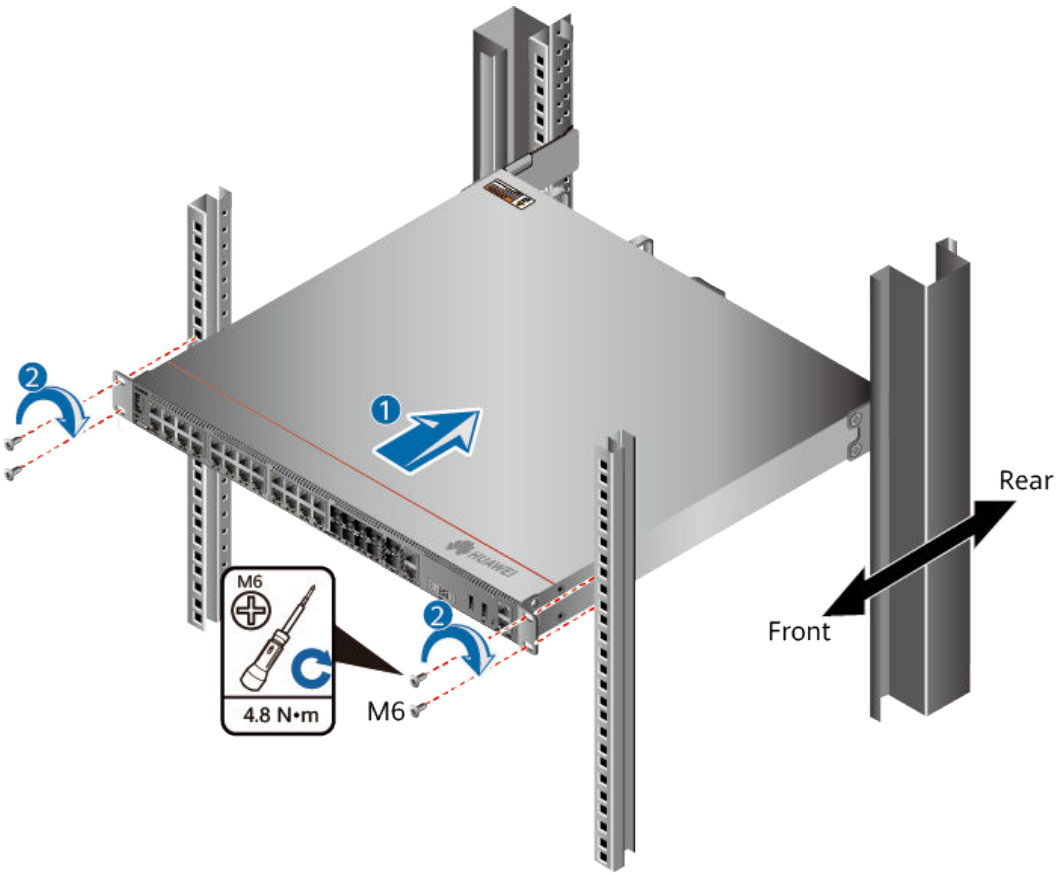

4. (Optional) Install adjustable guide rails (BOM numbers: 21242247 and 21242246). For details, see the installation guide delivered with the adjustable guide rails.

**Step 3** Mount the USG in the cabinet.

1. Hold the bottom of the USG with both hands and move the USG into the cabinet. Align the rear mounting ears with the rear mounting ear guide rails and slowly slide the USG along the guide rails.
2. Place the device on the guide rails, slowly push it into the cabinet, and then tighten screws with the Phillips screwdriver to secure the device.Insert the device into the rear mounting ear guide rails and slide it into the cabinet.
3. Use M6 screws to fix the mounting ears of the USG to the mounting rack, as shown in **Figure 4-23**.

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**Figure 4-23** Mounting the USG in a cabinet




**----End**

## Follow-up Procedure

Perform the following checks after the installation:

- Ensure that the USG is placed securely inside the cabinet.
- Ensure that the exhaust of the USG is not blocked by other objects.

## 4.4.1.2 Mounting a Device on a Workbench

If you do not have a cabinet, you can mount the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B on a workbench.

## Precautions

Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.

The workbench must be:

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- Reliably grounded.
- Clean, firm, and securely installed.

## Accessories

Four rubber feet

## Procedure

**Step 1** Fix the rubber feet to the round notches at the bottom of the USG.

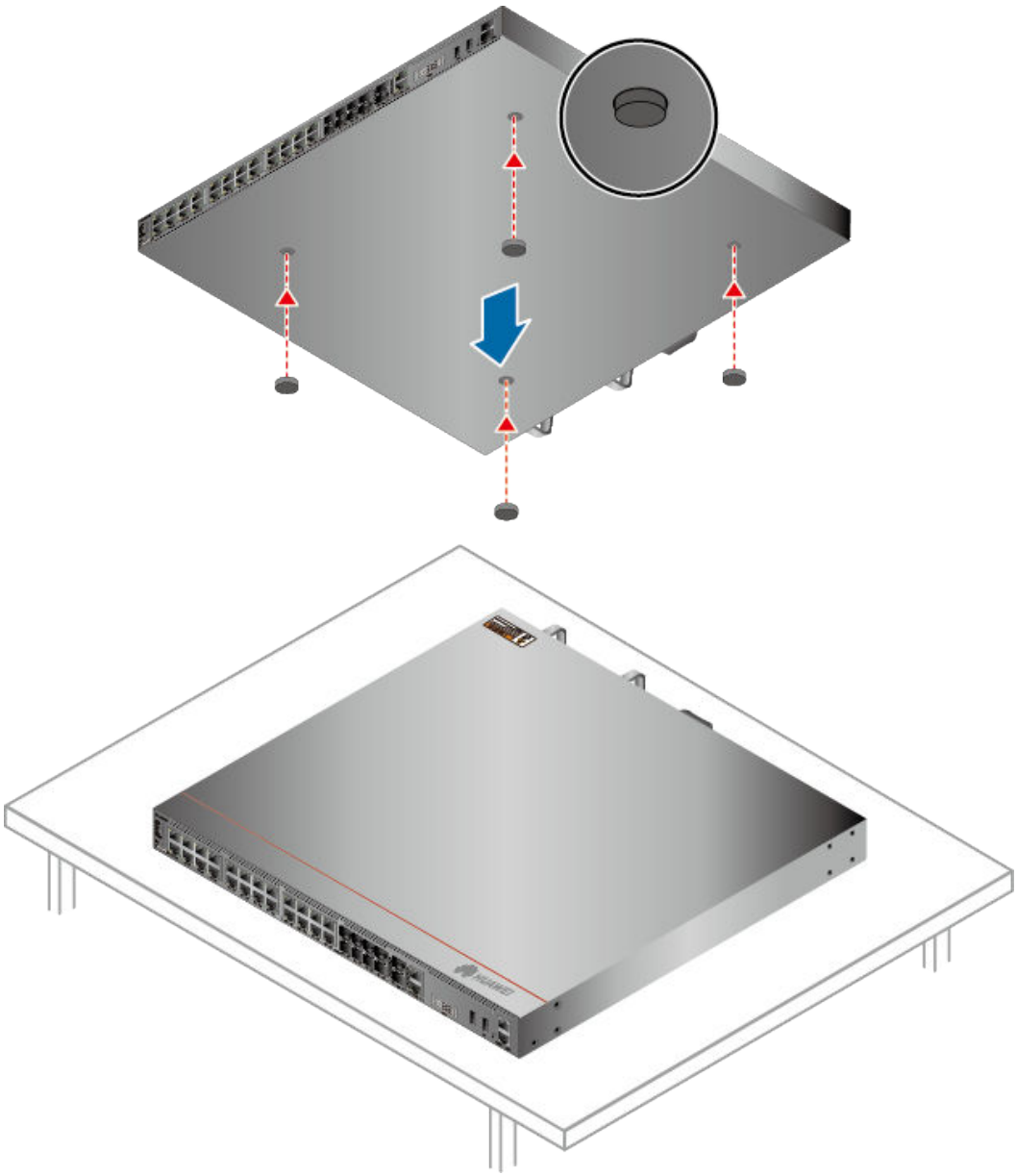
NOTE

Install foot pads at the bottom of the USG to ensure smooth contact between the USG and the workbench and avoid friction between the surface of the USG and the workbench.

**Step 2** Place the USG on the workbench.

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**Figure 4-24** Placing the USG with rubber feet on a workbench



**----End**

## Follow-up Procedure

Verify the following after the installation:

- The USG is securely placed on the workbench.
- No object blocks the exhaust of the USG, and there is at least 10 cm of distance between the USG and surrounding devices.
- There are no heavy objects on the USG.

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# 4.4.2 Connecting a PGND Cable

Connecting the PGND cable of a USG correctly is a key measure of surge protection and resistance to interference. Before using the USG, correctly connect the PGND cable. Otherwise, the USG may be damaged.

## Precautions

The USG has been installed inside a cabinet.

## Tools

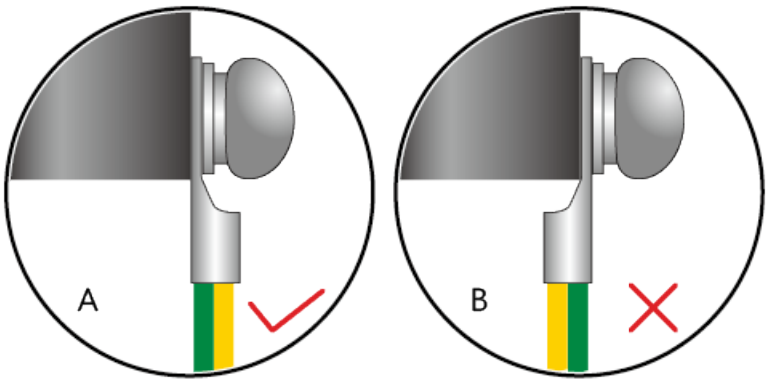
- Phillips screwdriver
- Multimeter

## Procedure

**Step 1** Loosen and remove the screw of the ground terminal on the USG rear panel.

**Step 2** Connect the OT terminal at one end of the PGND cable to the connection hole of the USG, with the conducting wire upward, and tighten the M4 screw, as shown in A of **Figure 4-25**, and tighten the M4 screw. The tightening torque is 1.4 N·m.

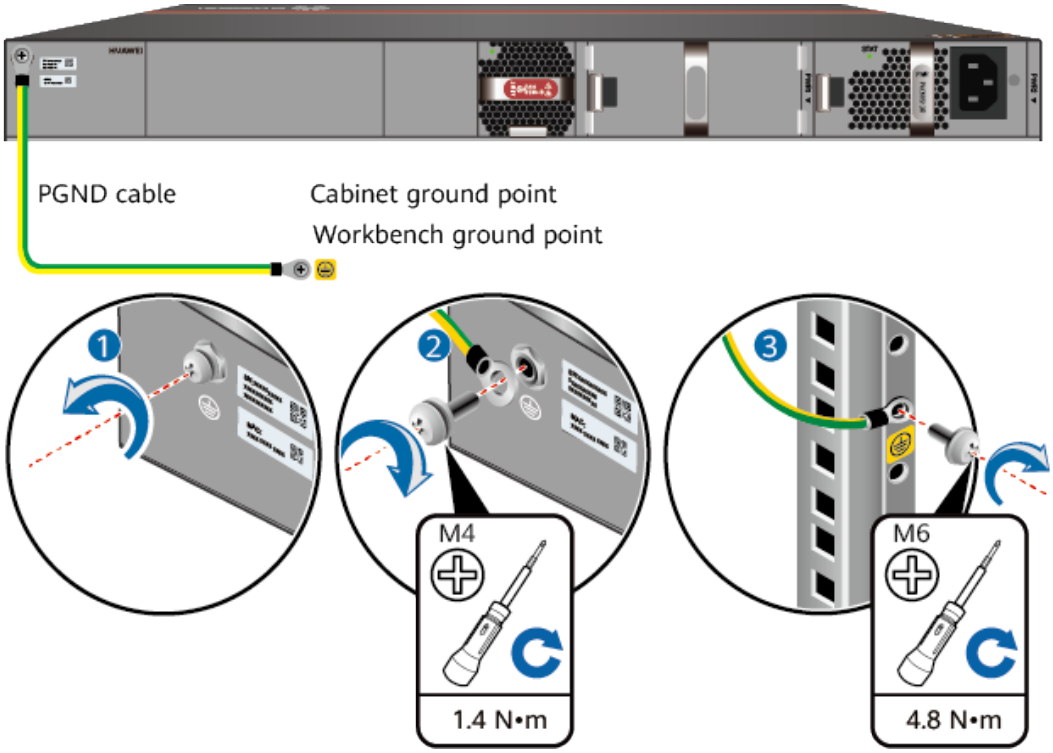
**Figure 4-25** Installing the OT terminal




**Step 3** Connect the M6 end of the PGND cable to the ground terminal of the cabinet, workbench, or wall. The tightening torque of the M6 screw is 4.8 N·m.

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**Figure 4-26** Connecting a PGND cable

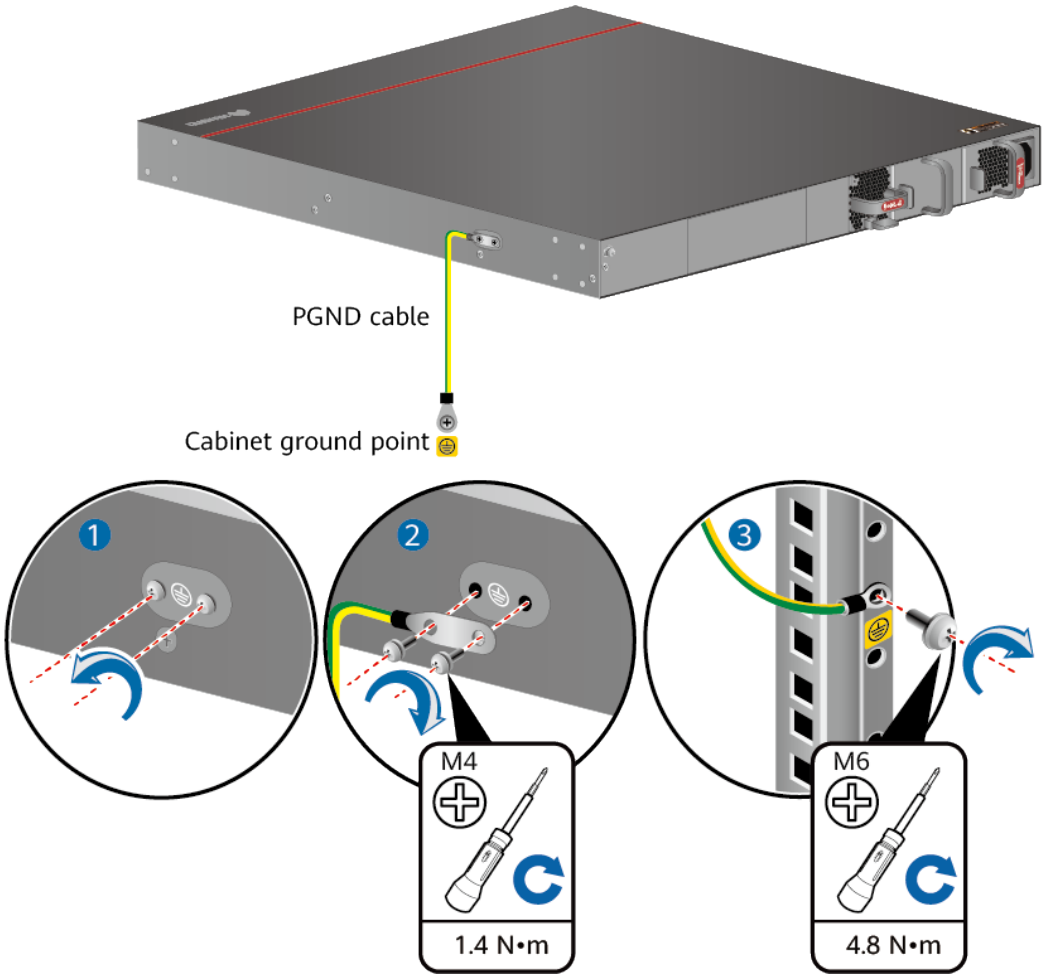



NOTE

In the dual-OT scenario, connect the ground terminal on the side panel of the USG.

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**Figure 4-27** Connecting the ground terminal in the dual-OT scenario




**NOTICE**

The OT terminal may rotate and result in device damage. Make sure that it is independent of the adjacent metal mechanical part or other terminals.

**----End**

## Follow-up Procedure

Verify the following after the cabling is complete:

- The PGND cable is securely connected to the ground terminal.
- The electrical resistance between the ground terminal and ground point is less than 0.1 ohm on a multimeter.

# 4.4.3 Installing a Hard Disk

This section describes how to install a hard disk units for the first time to avoid hard disk damage.

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## Precautions

Precautions for the use of hard disks

- Use Huawei hard disks. The system cannot recognize the hard disks provided by other vendors.
- Wear an ESD wrist strap to protect the USG and hard disks from electrostatic damage.
- Hold the two side surfaces of a hard disk, do not touch the PCB board or squeeze the hard disk, and do not vibrate, bump, or stack hard disks.
- For mapping relationships between the USG and hard disks, see **3 Hardware Overview**.

## Tool

ESD bag

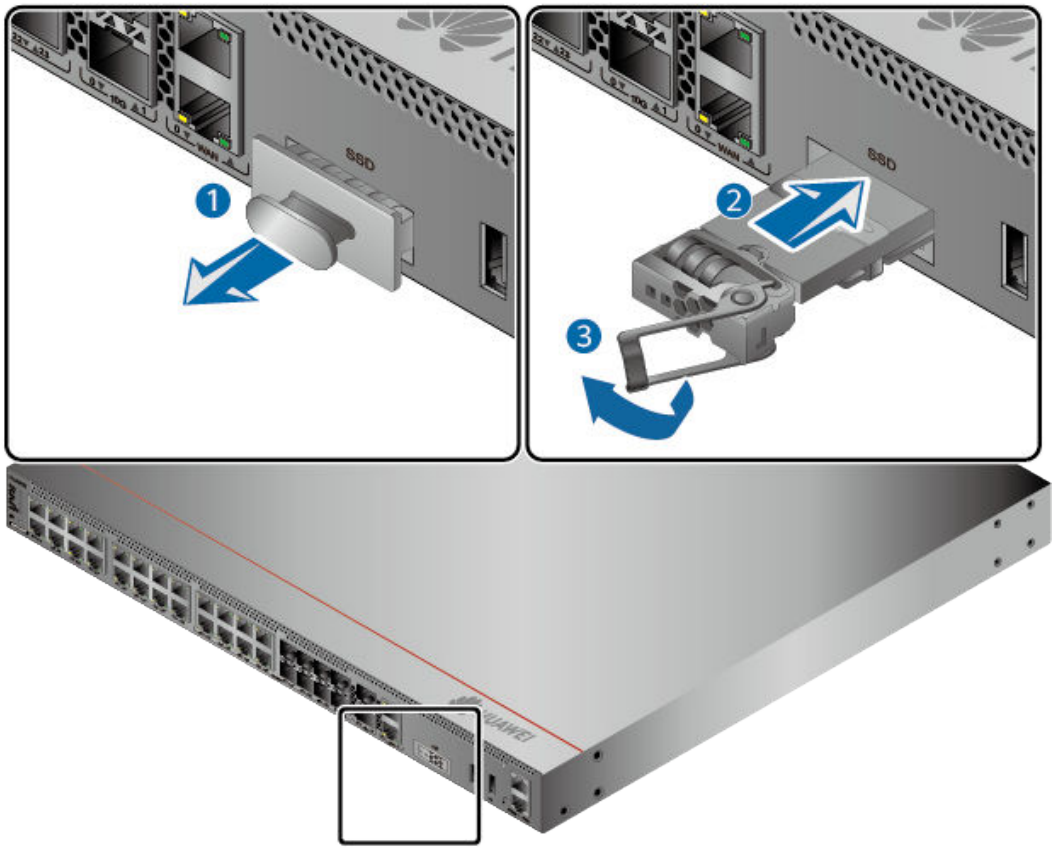
## Installing The Hard Disk (USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6580E, USG6585E)

**Step 1** Remove the baffle plate from the hard disk slot.

**Step 2** Hold the handlers of the hard disk and push the hard disk along the guide rails till the hard disk aligns with the panel of the USG.

**Step 3** Press the hard disk handler inwardly to restore the module.

**Figure 4-28** Installing the hard disk (USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6580E, USG6585E)




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**Step 4** After the installation is complete, perform the following operations:

NOTE

Operations after the USG power-on must be performed after all installation tasks are complete.

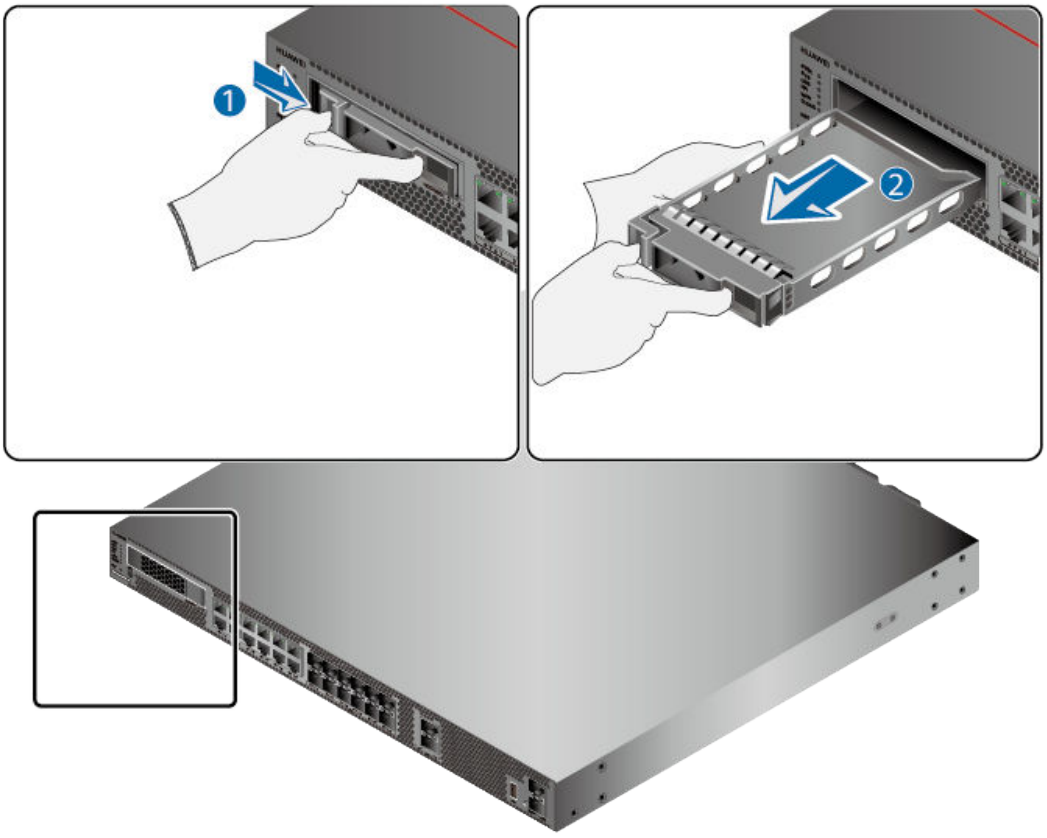
- Before the USG is powered on, ensure that the hard disk is correctly installed.
- After the USG is powered on, run the **display disk information** command in the user view to check whether the **Filesystem Status** value is **Mounted**.

**----End**

## Installing The Hard Disk (USG6575E-B, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E)

**Step 1** Remove the filler panel from a hard disk slot.

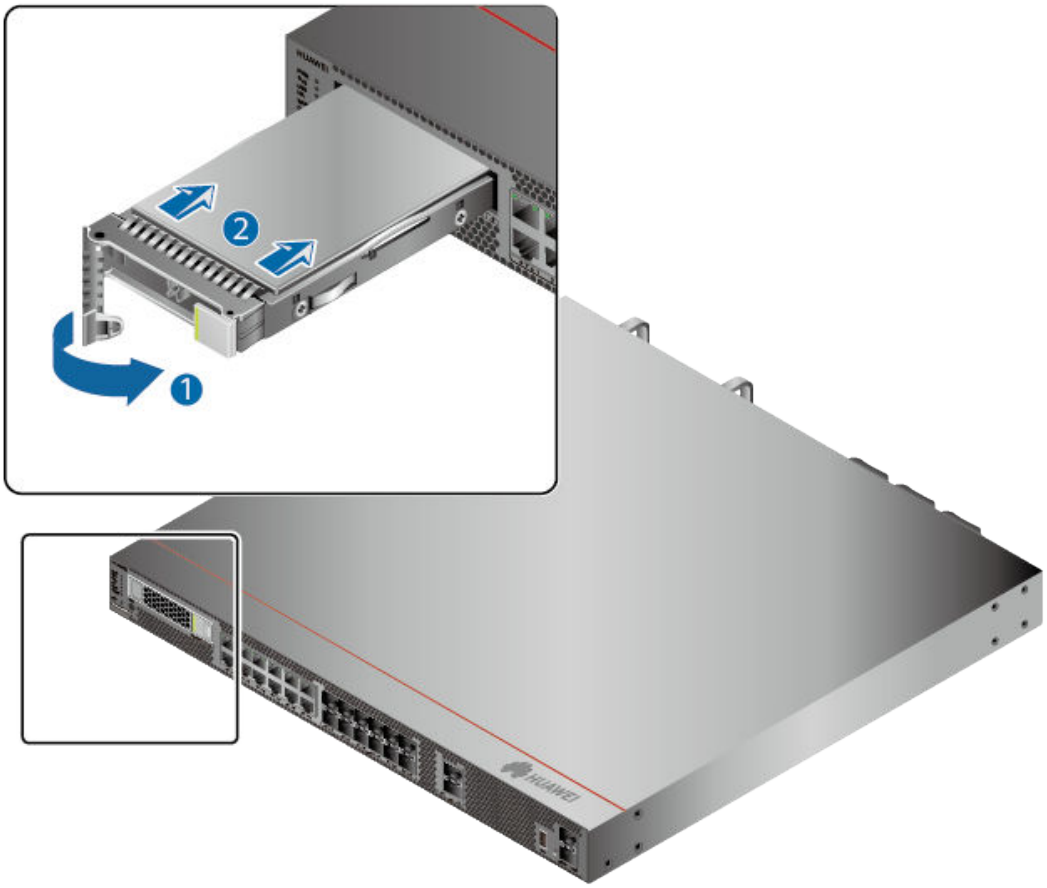
**Figure 4-29** Removing the filler panel from a hard disk slot (USG6575E-B, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E)




**Step 2** Pull the ejector lever on the hard disk holder, keep it in pulled-out state, and push the hard disk holder with two hands along the guide rails of the slot till the front surface of the hard disk holder touches the USG panel.

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**Figure 4-30** Installing a hard disk (USG6575E-B, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E)




**Step 3** After the ejector lever of the hard disk is fastened to the chassis, close the hard disk ejector lever.

**Step 4** After the installation is complete, perform the following operations:

NOTE

Operations after the USG power-on must be performed after all installation tasks are complete.

- Before the USG is powered on, ensure that the hard disk is correctly installed.
- After the USG is powered on, run the **display disk information** command in the user view to check whether the **Filesystem Status** value is **Mounted**.

**----End**

## 4.4.4 Connecting a Console Cable

After connecting a PC to the console port of a USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E with a console cable, you can use the terminal emulation program on the PC to access the command configuration interface of the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E.

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## Precautions

Before connecting a console cable, perform the following operations:

- Check preparations.

  A PC is ready, a USG has been installed, and the ports to be connected are planned.
- Prepare cable labels.

  Before cable connection, labels must be prepared for the cable.

**NOTICE**

- Make sure that the PC and the USG are connected to the same ground point. Otherwise, the console port of the USG may be damaged.
- Pay attention to port numbering and make sure that the cable is connected to the correct port, preventing damage to ports or the device.

## Tools

Console cable (prepared by the user)

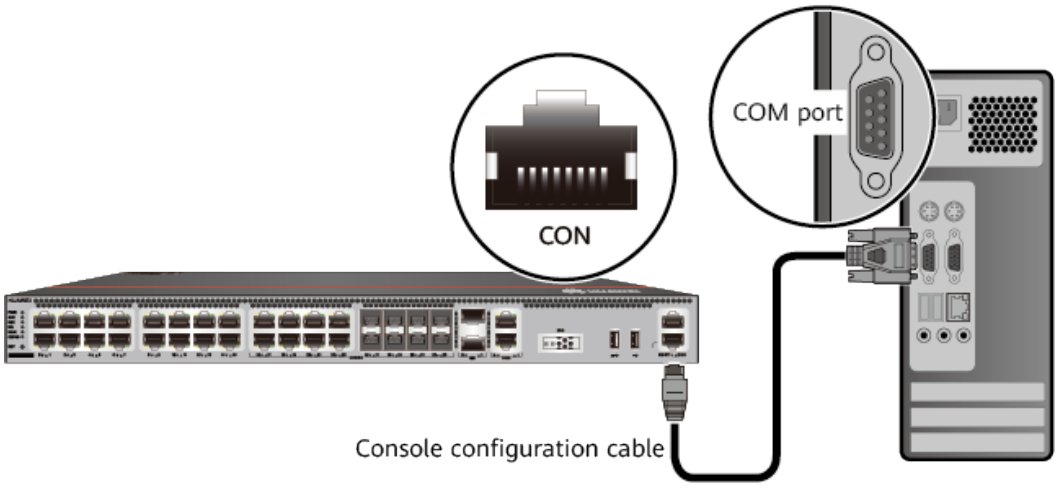
## Procedure

**Step 1** Before connecting a console cable, attach temporary labels to both ends of the cable for identification.

**Step 2** Connect the RJ45 connector of the console cable to the console port (RJ45) of the USG.

**Step 3** Connect the DB9 connector of the console cable to the COM port of the management PC.

**Figure 4-31** Connecting a console cable to the USG




**Step 4** Remove the temporary labels and attach labels (2 cm away from the connectors) at both ends of the console cable.

**----End**

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### Follow-up Procedure

After the cable connection is complete, verify that:

- The labels at both ends of a cable are correct, clear, neat, and facing the same direction.
- Cables and connectors are free of damage or breakage and are connected properly.

For details on the console login, refer to the Configuration Guide.

## 4.4.5 Connecting an Ethernet Cable

Based on the network plan, you can connect one end of an Ethernet cable to the Ethernet port of a USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E and the other end to the Ethernet port of the peer device.

### Context

Before connecting the Ethernet cable, perform the following operations:

- Check construction conditions.

  The peer device has been installed in the equipment room, and the port to which the Ethernet cable is to be connected has been determined.
- Check the cabling route.

  The engineering document should specify the cabling route from the cabinet to the peer device in the equipment room, and the length of the cable is calculated based on the cabling path.
- Label the cable.

  The cable must be labeled before being connected to the devices.

**NOTICE**

- Only shielded cables are supported on the USG.
- Before connecting a cable, note the label on the port and make sure that the cable is inserted into the correct port. Otherwise, the port module or the device might be damaged.

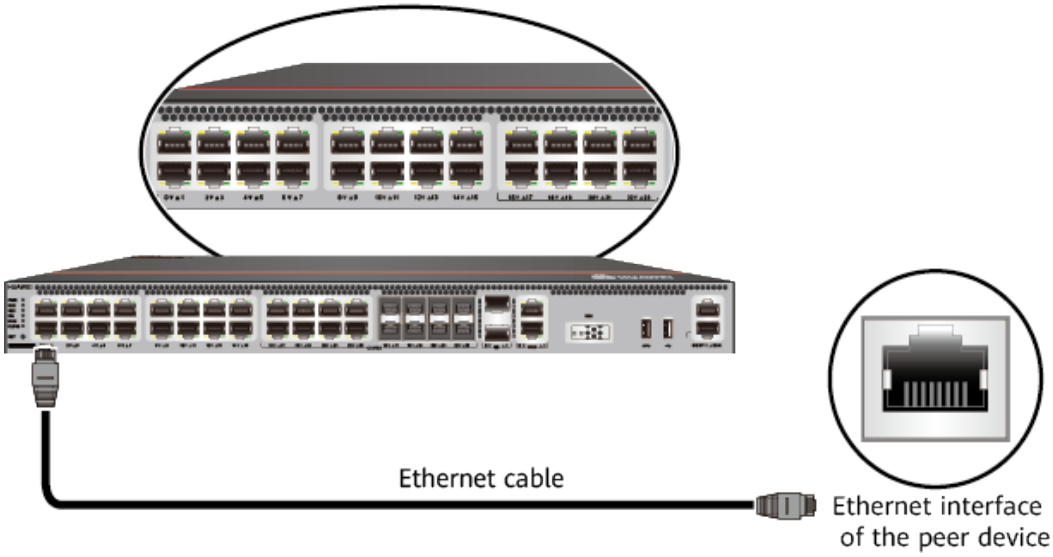
### Procedure

**Step 1** If multiple network cables need to be connected, attach temporary labels to both ends of each cable for identification.

**Step 2** Connect one end of an Ethernet cable to the Ethernet port of a USG and the other end to the Ethernet port of the peer device based on the network plan.

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**Figure 4-32** Connecting an Ethernet cable




**Step 3** Lay out the Ethernet cable along a cabinet and route the cable through the cable hole for the signal cables at the top (overhead cabling) or bottom (underfloor cabling) of the cabinet.

**Step 4** Remove the temporary labels and attach labels (2 cm away from the connectors) at both ends of the Ethernet cable.

**----End**

### Follow-up Procedure

Verify the following after the installation:

- The labels at both ends of the cable are correct, clear, neat, and facing the same direction.
- The cables and connectors are free of any damage or breakage and are connected properly and reliably.

## 4.4.6 Installing Optical Transceivers and Connecting Optical Fibers

This section describes how to install optical transceivers on the GE , 10GE, or 40GE optical ports of the USG and connect them to the ports of the peer device using optical fibers according to the network plan.

### Context

The USG supports both 1 Gbit/s , 10 Gbit/s, and 40 Gbit/s optical modules. The optical modules at both ends are the same, including the optical fiber type (single-mode or multi-mode), optical fiber connector type (LC/PC, SC/PC, FC/PC, or MPO/PC-MPO/PC), and transmission rate. If different optical modules are used at the two ends, the communication may fail.

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**NOTICE**

Huawei optical modules are recommended. The optical modules from other vendors may cause faults on the USG due to incompatibility.

**CAUTION**

Do not look into the optical interface of the optical module or the optical fiber connector without eye protection.

Before connecting optical fiber cables, read the following precautions:

- Do not overbend optical fibers, and the radius should not be shorter than 40 mm.
- Do not bundle the optical fibers too tight. Otherwise, the transmission performance of the optical fibers and the communication between devices might be adversely affected.

Before connecting optical ensure the following:

- The optical module has been installed.

## Procedure

**Step 1** Insert an optical transceiver into the GE , 10GE, or 40GE optical port of the USG.

**Step 2** Remove the dust cap from the optical transceiver.

**NOTE**

Set aside the dust cap properly for future use. After optical fiber are disconnected for maintenance, use the dust cap to prevent the optical transceiver from dust.

**Step 3** Before connecting an optical fiber, attach temporary labels to both ends of the optical fiber for identification.

**Step 4** Remove protective caps from optical fiber connectors, insert optical fibers into the optical transceiver, and connect the fiber to the peer device.

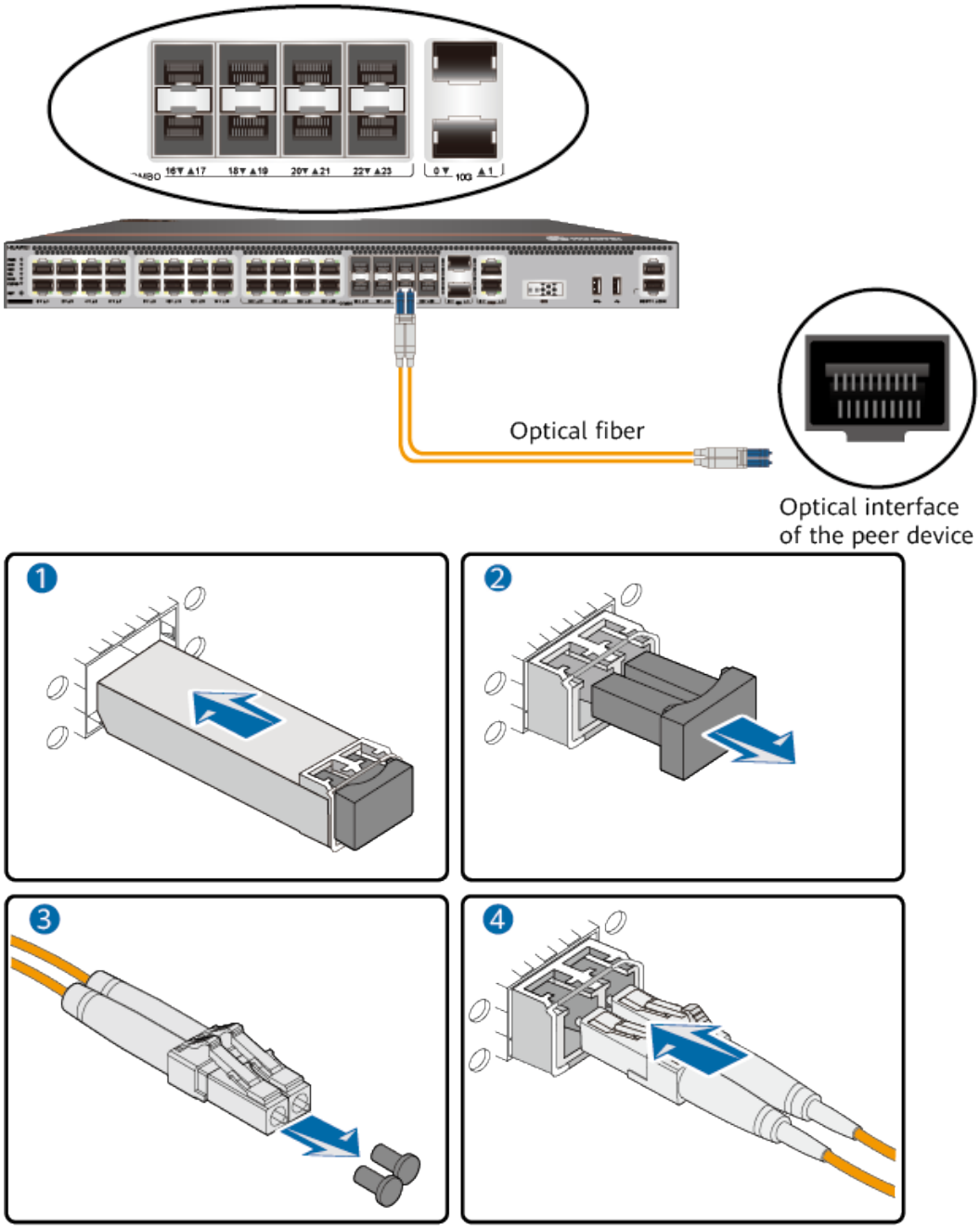
**NOTE**

Ensure that the Tx and Rx ports are correctly connected.

Ensure that the TX and RX ports on one end of the optical fiber cable are connected to the RX and TX ports (respectively) on the other end.

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**Figure 4-33** Installing optical transceivers and connecting optical fibers




**Step 5** Repeat **Step 1** to **Step 4** to install all optical transceivers and connect all optical fibers.

**----End**

## Follow-up Procedure

After you power on the USG, check the connection by observing the optical port indicator. If the indicator is on or blinks, the link is connected or data is being

transmitted. If the indicator is off, the link is disconnected. Possible causes for the disconnection are as follows:

- The optical fiber is improperly inserted. Pull out the optical fiber and re-insert it.
- The RX and TX optical ports are inserted reversely. Pull out the optical fibers, change their position, and re-insert them.
- The optical module is damaged or the optical fiber is broken. Replace the optical module or the optical fiber.

## 4.4.7 Connecting Power Cables

Follow the operation guide to connect power cables from the power module of the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E to the output of the power supply device.

### 4.4.7.1 Connecting AC Power Cables

By default, the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E comes with one AC power module. However, two AC power modules are supported. The USG6630E, USG6635E, USG6650E, USG6655E comes with two AC power module. Each AC power module requires an AC power cable. When two AC power modules are used, connect power cables to both AC power modules.

#### Context

Before connecting the power cables, ensure that the AC power supply in the equipment room meets the input requirements of the USG.

**DANGER**

Do not connect or disconnect the power cables when the USG is powered on.

#### Procedure

**Step 1** Ensure that the PGND cable is adequately grounded.

**Step 2** Connect AC power cables to AC power modules.

**NOTE**

When two power modules are used, connect them to different power sources to improve availability.

1. Plug one end of the C13 power cable to the power socket of the USG power module and adjust the cable-retention clip to an appropriate position.
2. Bundle the AC power cable using a cable-retention clip and adjust the cable-retention clip to fasten the AC power cable.

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3. Plug the other end of the power cable to the AC power socket or the output of the AC power supply device.

**Figure 4-34** Connecting AC power cables (USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E)

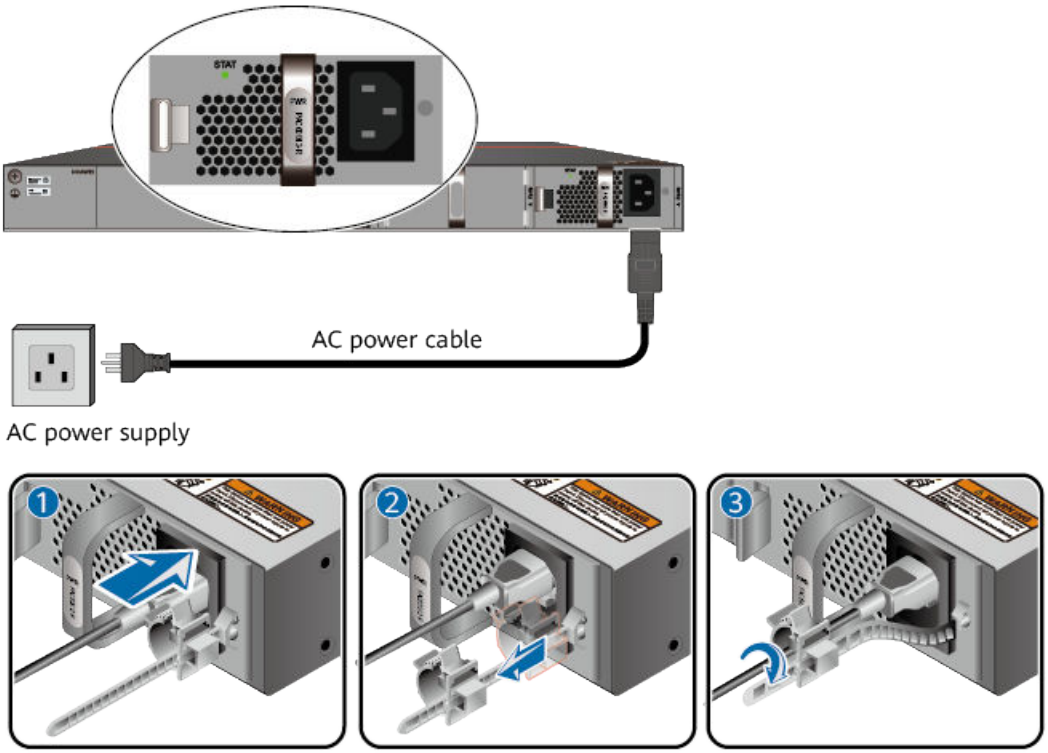
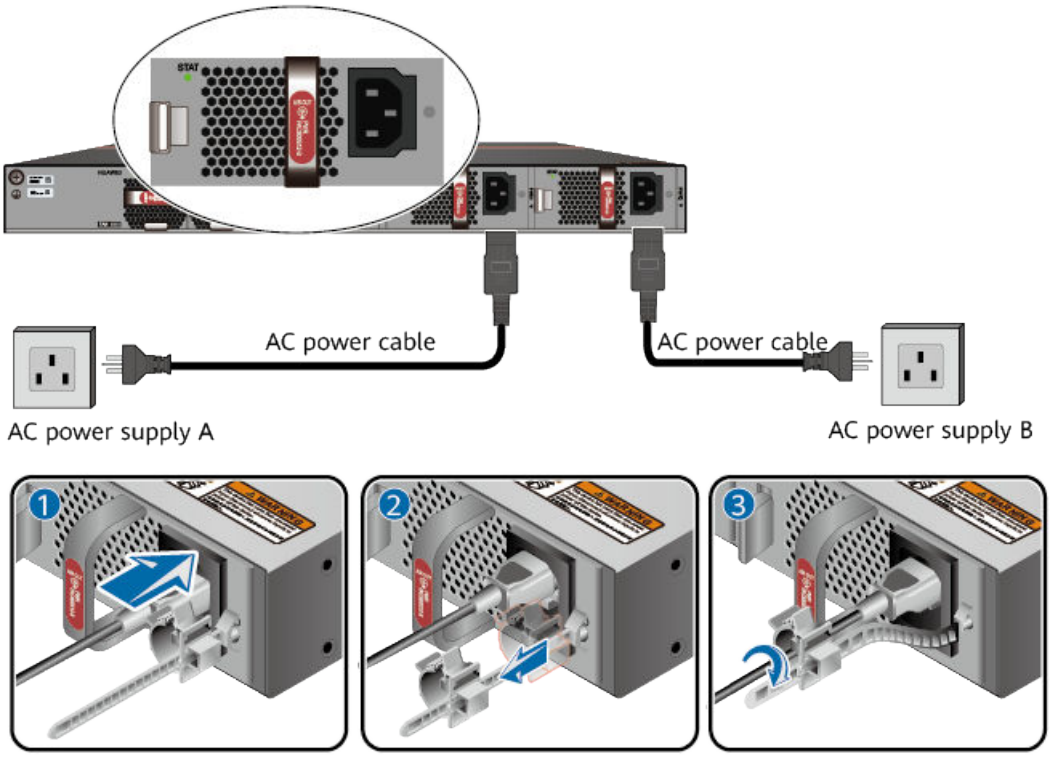


**Figure 4-35** Connecting AC power cables (USG6630E, USG6635E, USG6650E, USG6655E)




**----End**

## Follow-up Procedure

Verify the following after the connection is complete:

- The power cable is firmly connected to the power supply socket.
- If multiple USGs are deployed, the power cables of each USG are correctly labeled for distinction.

## 4.4.7.2 Connecting DC Power Cables

The USG6630E, USG6635E come with two DC power module. Each DC power module requires an NEG power cable and an RTN power cable. Connect power cables to both DC power modules.

## Context

Before connecting power cables, ensure that the DC power supply in the equipment room meets the input requirements of the USG.

**DANGER**

Do not connect or disconnect the power cables when the USG is powered on.

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## Procedure

**Step 1** Ensure that the PGND cable is adequately grounded.

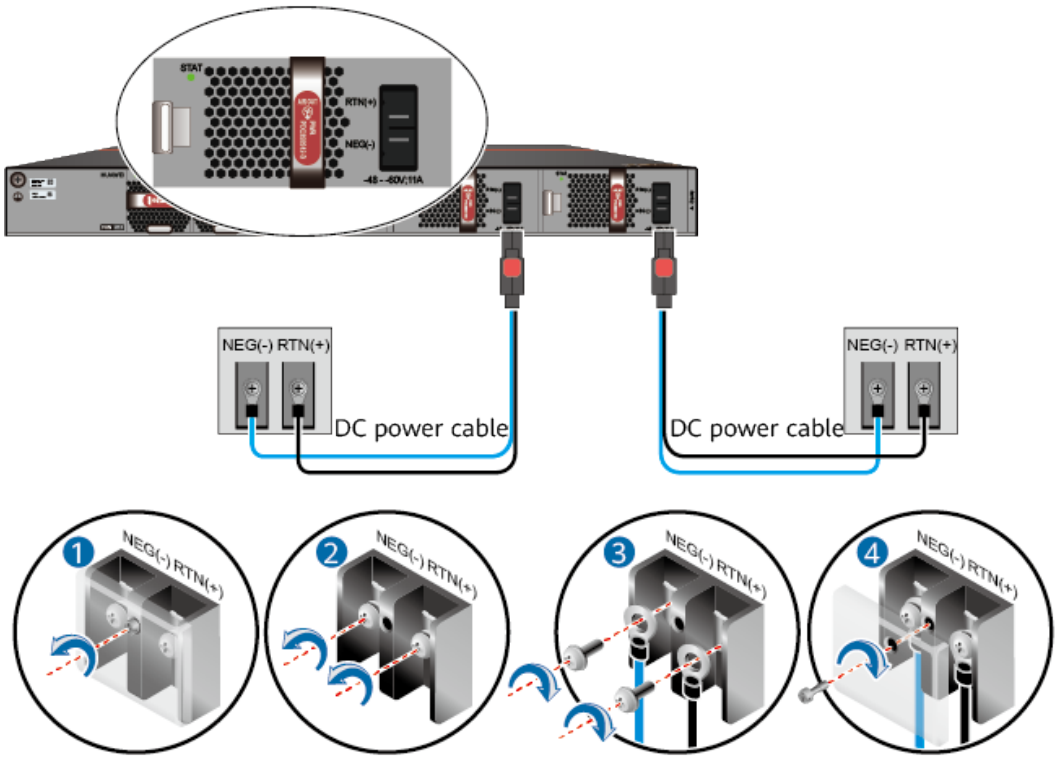
**Step 2** Connect power cables for the two DC power modules one after the other.

**NOTICE**

- Note the labels on the DC power cables to avoid misoperations.
- To improve availability, connect two power modules to different power sources.

1. Plug the connector end of the DC power cable to the DC power module.
2. Connect the other end of the DC power cable to the output of the power supply device in the equipment room.

**Figure 4-36** Connecting DC power cables




**----End**

## Follow-up Procedure

Verify the following after the connection is complete:

- The power cables are correctly and firmly connected to the power supply device.
- If multiple USGs are deployed, the power cables of each USG are correctly labeled for distinction.

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# 4.4.8 Powering On or Off the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E

This section describes how to power on or off the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E. To ensure the normal start and security of the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E, strictly follow the operation guide to power on or off the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E.

## Context

Before you power on the USG, ensure that:

- The power cable and PGND cable are properly connected.
- The power switch in the equipment room is easy to locate so that you can power off devices in the case of accidents.

## Procedure

- Power on the USG.

  Turn on the switch of the power supply device. The USG starts.

  You can identify the USG status based on indicators on the front and rear panels. Indicators shown in **Figure 4-37** indicate that the USG runs normally.

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**Figure 4-37** Indicators when the USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6575E-B, USG6580E, USG6585E, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E (AC model) runs normally

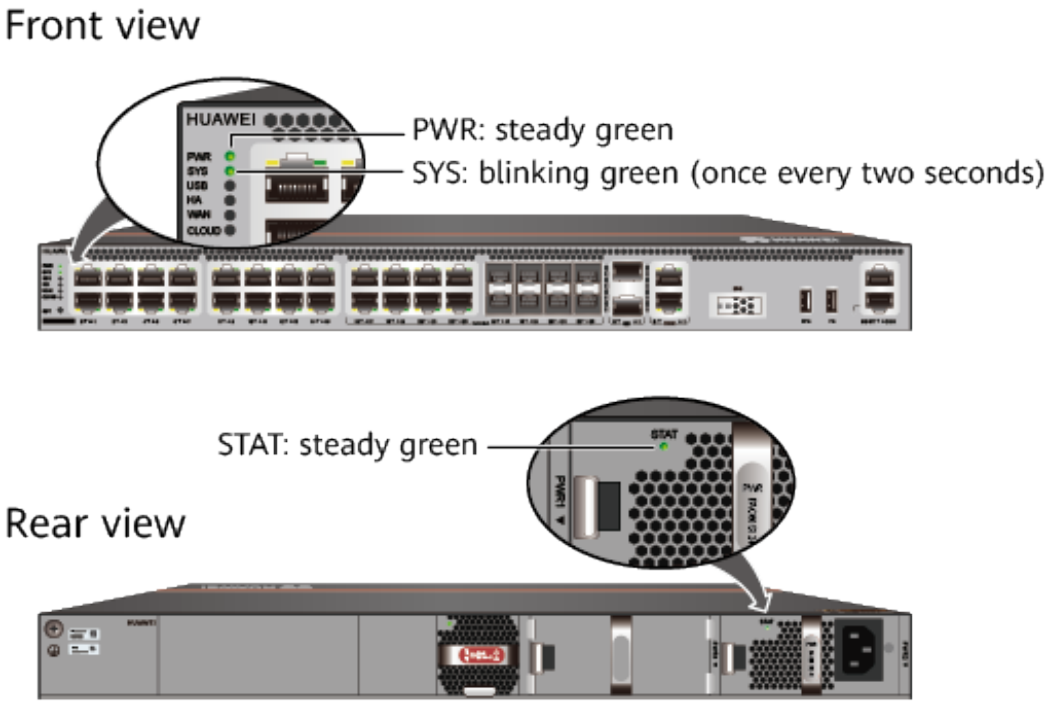
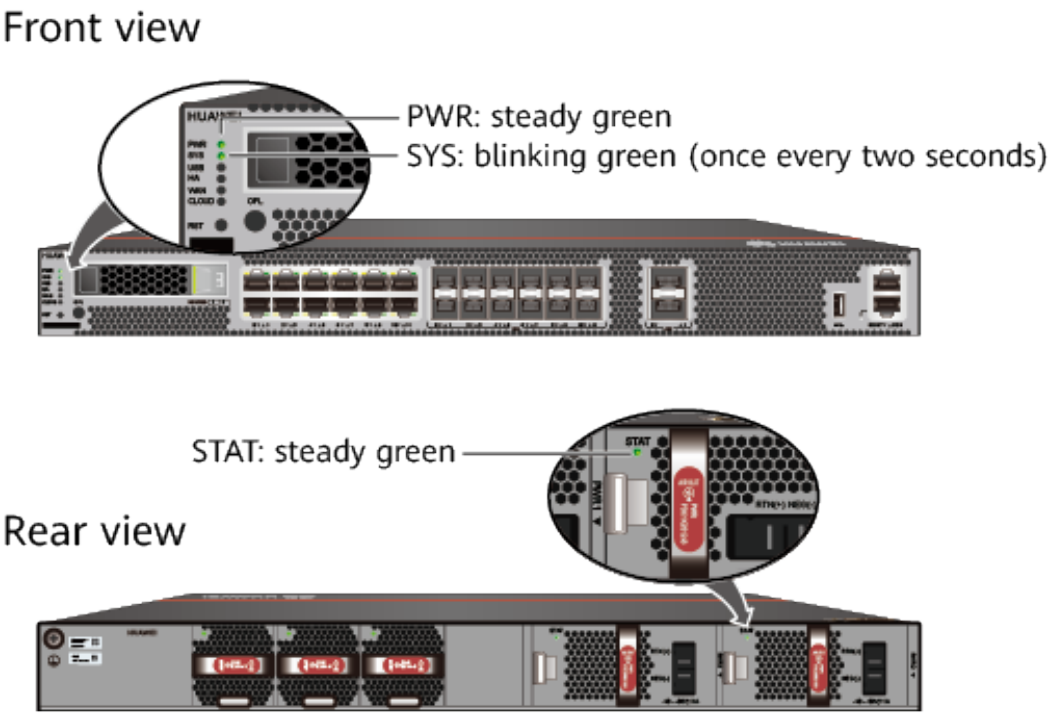



**Figure 4-38** Indicators when the USG (DC model) runs normally




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- Power off the USG.

**NOTICE**

Before powering off the USG, ensure that configuration data is saved. Otherwise, the configuration data may be lost.

If the USG will be administratively shut down for a long time, turn off the power switch. After powering off the USG, set it aside properly according to storage requirements.

**⚠ CAUTION**

If the USG has hard disks, do not power off the USG by removing the power cable or switching off the power supply to prevent data loss and hard disk damage. You must run the **disk offline** command in the system view first and wait for about 30 seconds. After the system prompts you with a message indicating that hard disks have stopped, turn off the power switch of the USG.

**----End**

### Follow-up Procedure

After the USG is powered on, you can log in to the configuration page for management and maintenance. For details, refer to the Configuration Guide.

## 4.5 Installing a 600 mm Deep Device

This chapter provides the cabinet-mounting, workbench-mounting, and cable connection methods of the USG6680E, USG6712E, USG6716E series.

## 4.5.1 Mounting a Device in a Cabinet

Install the USG6680E, USG6712E, USG6716E to the 19-inch standard cabinet through the expandable rear mounting ears.

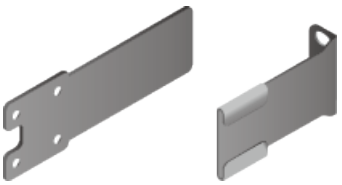
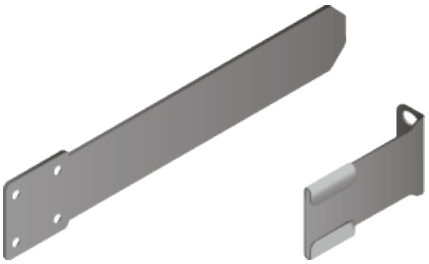
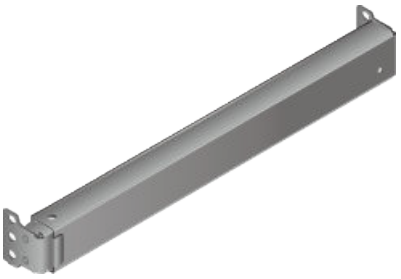
### Precautions

Before installing the USG, check the following items:

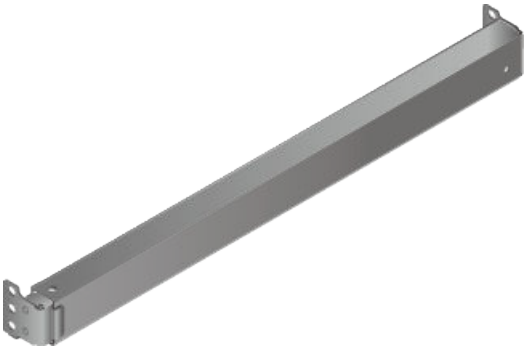
- Before unpacking the carton, ensure that the packing carton is intact and not damaged or soaked. Stop unpacking if the USG is rusted or soggy. Then, investigate causes and contact the supplier.
- The cabinet is stable.
- Before installing the expandable rear mounting ears on the USG6680E, USG6712E, USG6716E, adjust the distance between the front and rear angle gauges (rack mounting rails/vertical columns) to satisfy the installation requirements. The specific scope of application is as follows:

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**Table 4-5** Expandable rear mounting ears and adjustable guide rails

| Distance Between the Front and Rear Mounting Bars and Cabinet Type | Expandable Rear Mounting Ears and Adjustable Guide Rails |
| --- | --- |
| Expandable rear mounting ears (BOM number: 21240492)<br>• Distance between the front and rear mounting bars: 310 mm to 538 mm<br>• Cabinet depth: 800 mm |  |
| Expandable rear mounting ears (BOM number: 21240538)<br>• Distance between the front and rear mounting bars: 493 mm to 815 mm<br>• Cabinet depth: 1000 mm to 1200 mm |  |
| Adjustable guide rails (BOM number: 21242247)<br>• Distance between the front and rear mounting bars: 370 mm to 585 mm<br>• Cabinet depth: 800 mm |  |

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| Distance Between the Front and Rear Mounting Bars and Cabinet Type | Expandable Rear Mounting Ears and Adjustable Guide Rails |
| --- | --- |
| Adjustable guide rails (BOM number: 21242246)<br>• Distance between the front and rear mounting bars: 500 mm to 850 mm<br>• Cabinet depth: 1000 mm to 1200 mm |  |

- The position for the USG in the cabinet is well arranged. Ensure that the USG is 1 U of clearance from any devices above and below and 150 mm of clearance from any devices on the right or left.
- The USG to be installed is staged near the cabinet for convenience.
- In the front view, the air flow of the USG is front to rear. If the USG and a device whose air flow is rear to front must be mounted in the same cabinet, maintain certain spacing between them to prevent hot air recirculation.

You can place either end of the USG chassis towards the front door of the cabinet. In this manual, the front panel of the USG is towards the front door of the cabinet.

**NOTE**

The method of installing expandable rear mounting ears (BOM number: 21240492) is the same as that of installing expandable rear mounting ears (BOM number: 21240538). The following uses the expandable rear mounting ears (BOM number: 21240492) as examples.

## Tools and Accessories

- Phillips screwdriver
- Floating nuts and matching screws
- Floating mounting bar
- Front mounting ears and corresponding screws
- Expandable rear mounting ears and corresponding screws (purchased separately, BOM numbers: 21240492 and 21240538)
- Adjustable guide rails and matching screws (purchased separately, BOM numbers: 21242247 and 21242246)

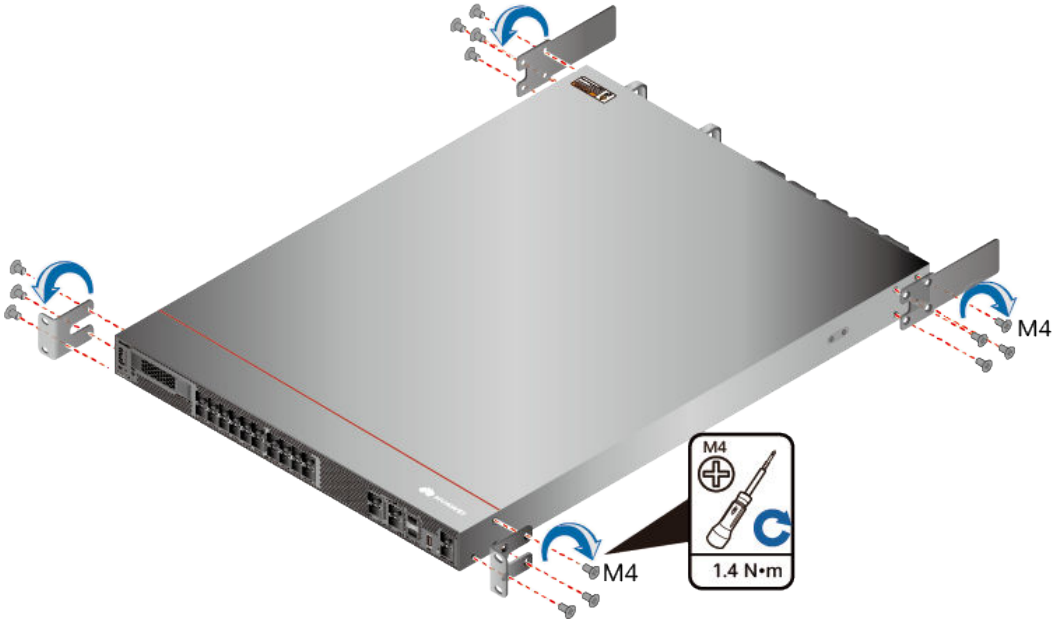
## Procedure

**Step 1** Install mounting ears on the chassis.

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Use a Phillips screwdriver to fix the mounting ears to both sides of the chassis with M4 screws. For details about how to install the mounting ears, see the installation methods in **Figure 4-39**.

**Figure 4-39** Installation of front and rear mounting ears



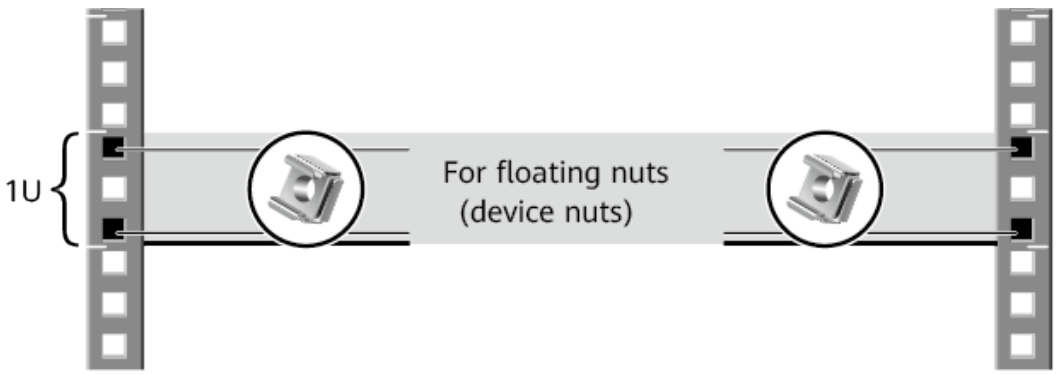

**Step 2** Install floating nuts and cabinet guide rails.

1. Determine the positions for installing the guide rails and floating nuts, as shown in **Figure 4-40**.

**NOTICE**

Install the two guide rails at the same height to ensure that the device is level.

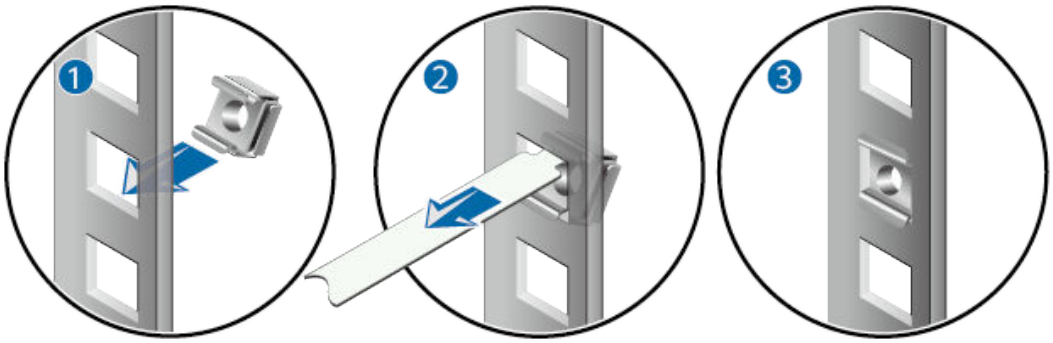
**Figure 4-40** Positions of floating nuts




2. Install the floating nuts that match M6 screws at the positions marked in **Figure 4-40**. **Figure 4-41** illustrates how to install a floating nut.

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**Figure 4-41** Installing floating nuts

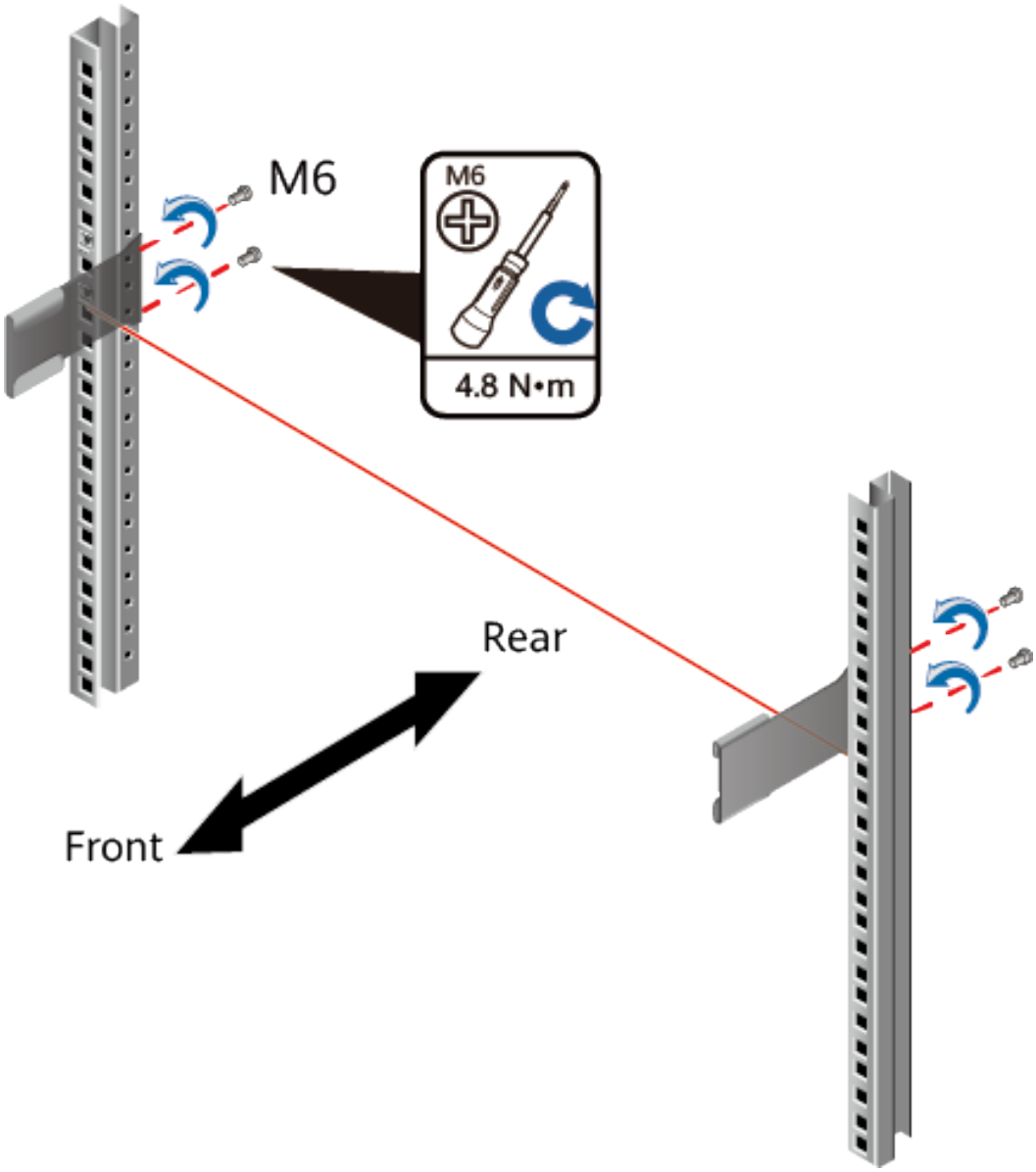



3. Install guide rails in the correct direction. Identify the left and right guide rails, and the front and rear ends of each guide rail.

   Fix expandable rear mounting ears and adjustable guide rails (at the positions marked in **Figure 4-40**) to the left and right of the cabinet with M6 screws, as shown in **Figure 4-42**.

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**Figure 4-42** Installing guide rails



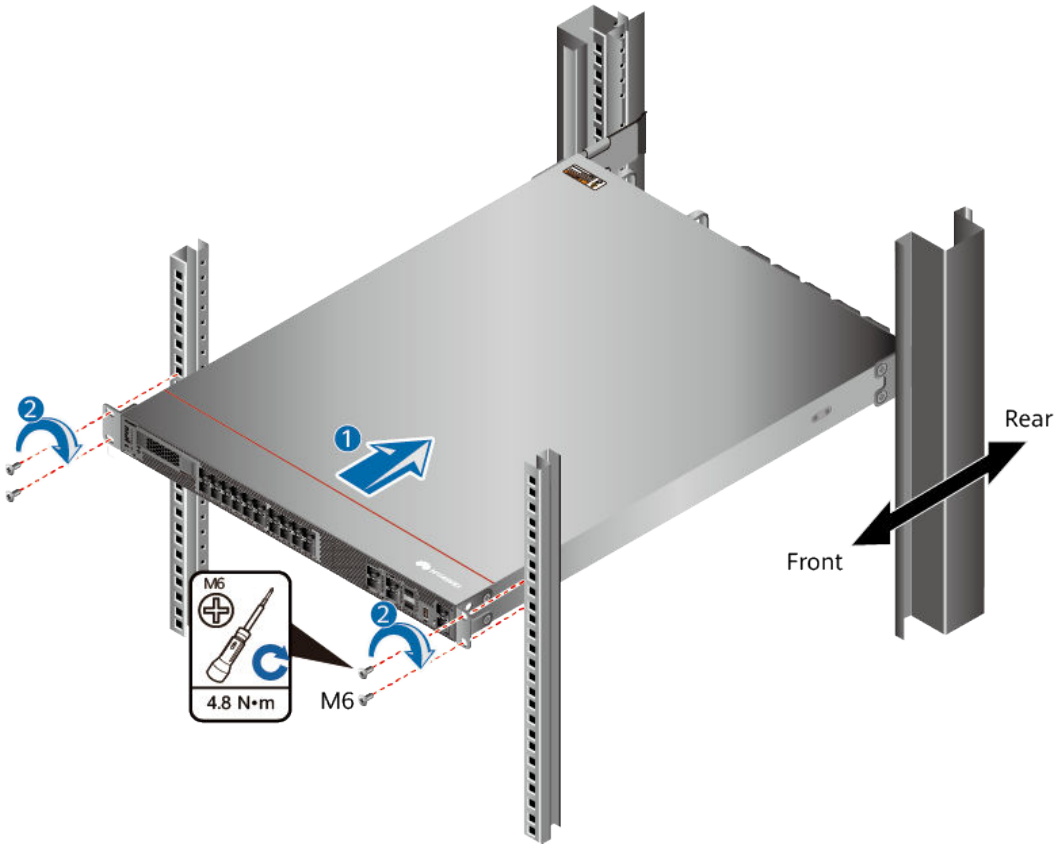

4. (Optional) Install adjustable guide rails (BOM numbers: 21242247 and 21242246). For details, see the installation guide delivered with the adjustable guide rails.

**Step 3** Mount the USG in the cabinet.

1. Hold the bottom of the USG with both hands and move the USG into the cabinet. Align the rear mounting ears with the rear mounting ear guide rails and slowly slide the USG along the guide rails.
2. Use M6 screws to fix the mounting ears of the USG to the mounting rack, as shown in **Figure 4-43**.

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**Figure 4-43** Mounting the USG in a cabinet




**----End**

## Follow-up Procedure

Perform the following checks after the installation:

- Ensure that the USG is placed securely inside the cabinet.
- Ensure that the exhaust of the USG is not blocked by other objects.

# 4.5.2 Connecting a PGND Cable

Connecting the PGND cable of a USG6680E, USG6712E, USG6716E correctly is a key measure of surge protection and resistance to interference. Before using the USG6680E, USG6712E, USG6716E, correctly connect the PGND cable. Otherwise, the USG may be damaged.

## Precautions

The USG has been installed inside a cabinet.

## Tools

- Phillips screwdriver

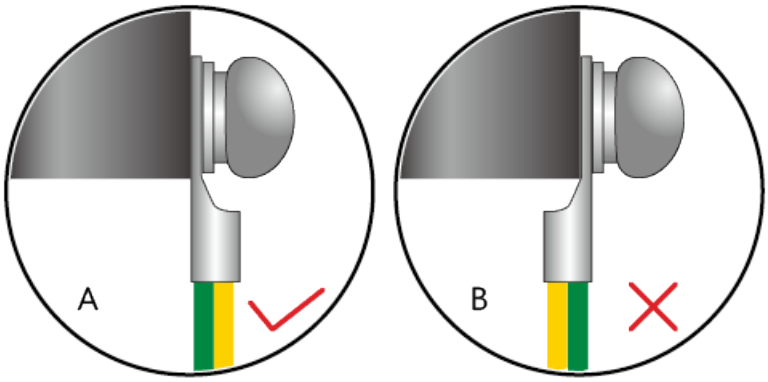
- Multimeter

## Procedure

**Step 1** Loosen and remove the screw of the ground terminal on the lower right of the USG rear panel.

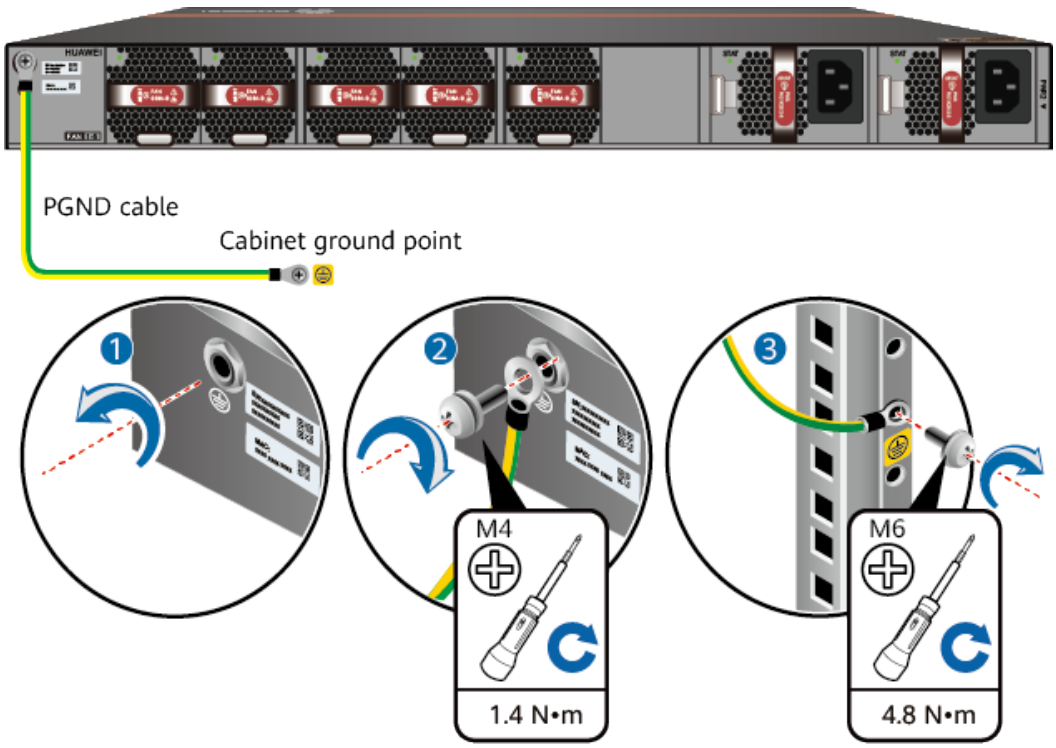
**Step 2** Connect the OT terminal at one end of the PGND cable to the connection hole of the USG, with the conducting wire upward, as shown in A of **Figure 4-44**, and tighten the M4 screw. The tightening torque is 1.4 N·m.

**Figure 4-44** Installing the OT terminal




**Step 3** Connect the M6 end of the PGND cable to the ground terminal of the cabinet, workbench, or wall. The tightening torque of the M6 screw is 4.8 N·m.

**Figure 4-45** Connecting a PGND cable

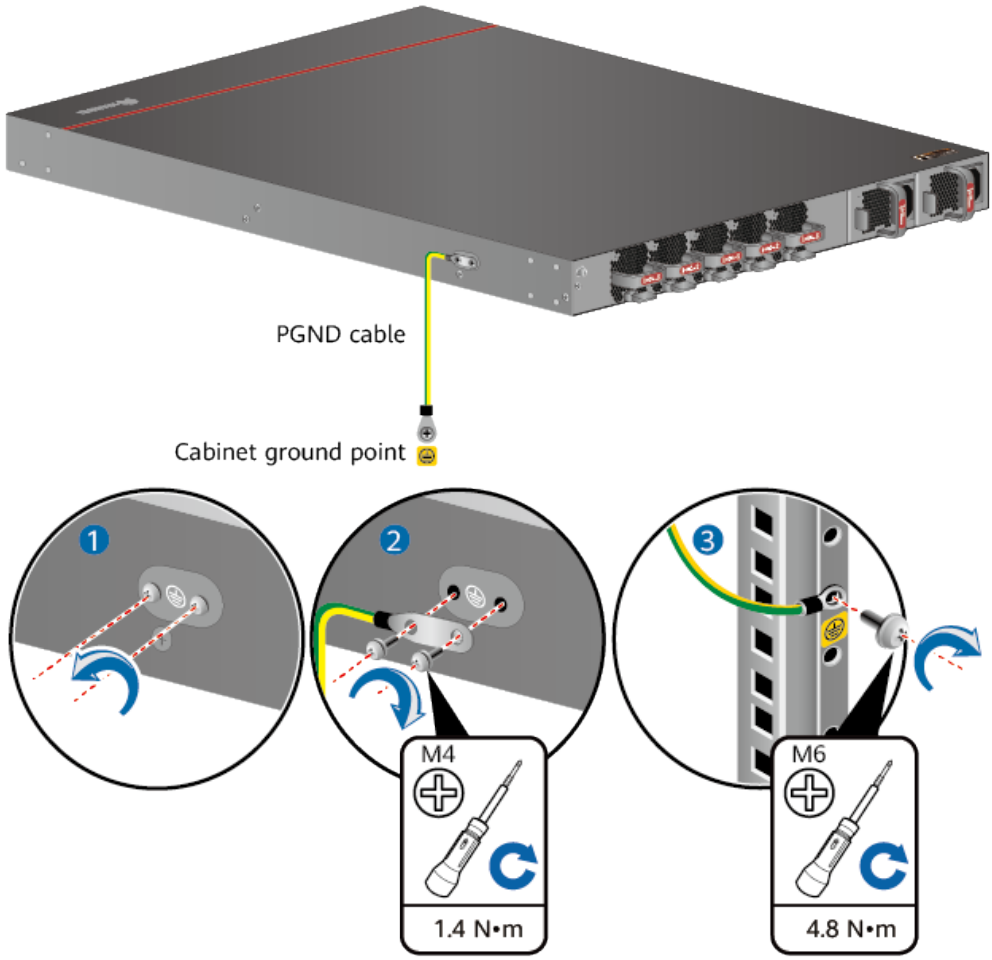



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NOTE

In the dual-OT scenario, connect the ground terminal on the side panel of the USG.

**Figure 4-46** Connecting the ground terminal in the dual-OT scenario




**NOTICE**

The OT terminal may rotate and result in device damage. Make sure that it is independent of the adjacent metal mechanical part or other terminals.

**----End**

## Follow-up Procedure

Verify the following after the cabling is complete:

- The PGND cable is securely connected to the ground terminal.

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- The electrical resistance between the ground terminal and ground point is less than 0.1 ohm on a multimeter.

## 4.5.3 Installing a Hard Disk

This section describes how to install a hard disk units for the first time to avoid hard disk damage.

### Precautions

Precautions for the use of hard disks

- Use Huawei hard disks. The system cannot recognize the hard disks provided by other vendors.
- Wear an ESD wrist strap to protect the USG and hard disks from electrostatic damage.
- Hold the two side surfaces of a hard disk, do not touch the PCB board or squeeze the hard disk, and do not vibrate, bump, or stack hard disks.
- For mapping relationships between the USG and hard disks, see **3 Hardware Overview**.

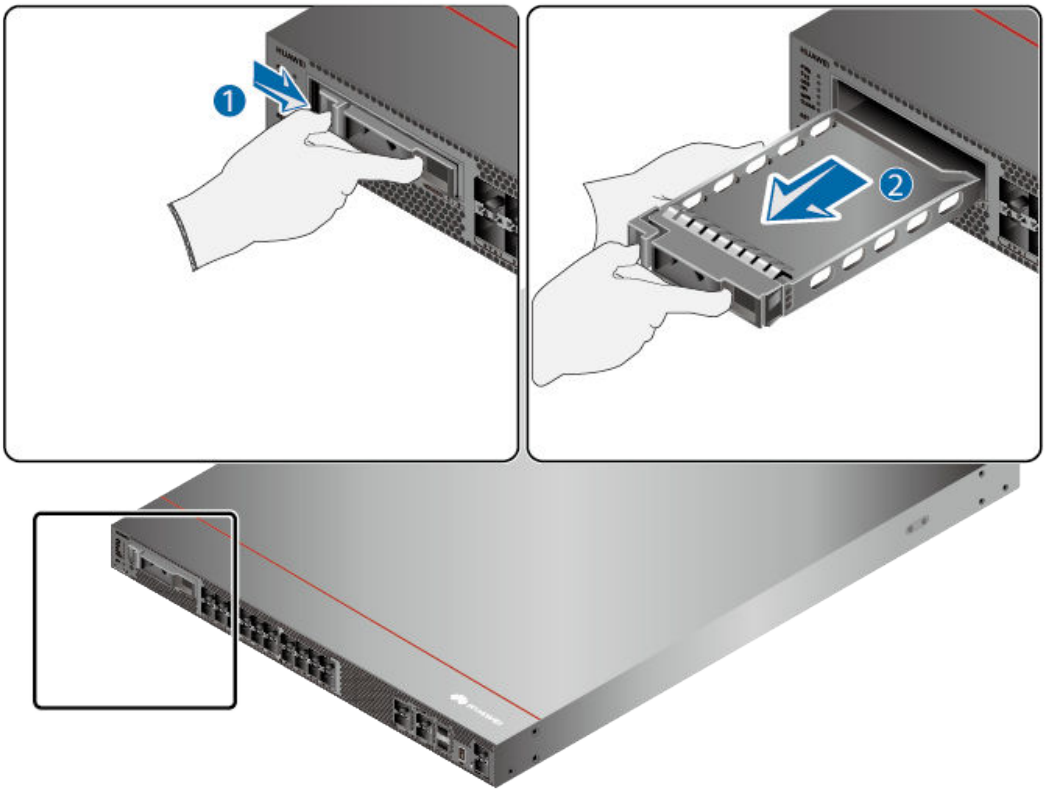
### Tool

ESD bag

### Procedure

**Step 1** Remove the filler panel from a hard disk slot.

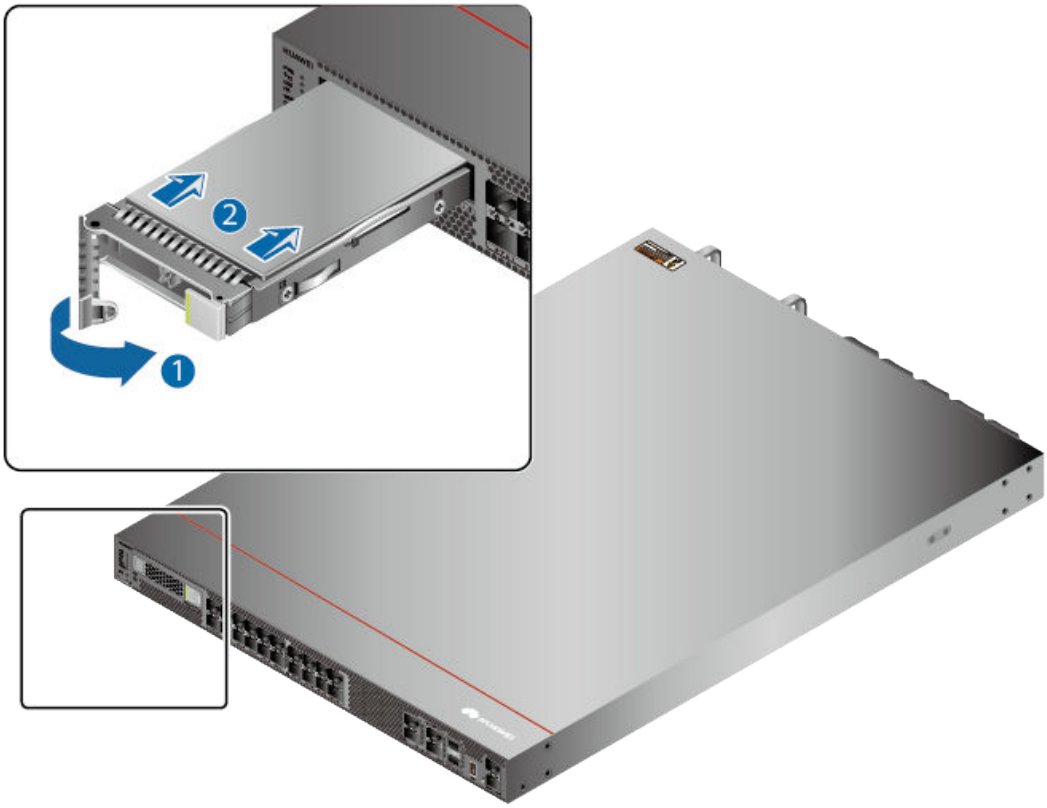
**Figure 4-47** Removing the filler panel from a hard disk slot




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**Step 2** Pull the ejector lever on the hard disk holder, keep it in pulled-out state, and push the hard disk holder with two hands along the guide rails of the slot till the front surface of the hard disk holder touches the USG panel.

**Figure 4-48** Installing a hard disk




**Step 3** After the ejector lever of the hard disk is fastened to the chassis, close the hard disk ejector lever.

**Step 4** After the installation is complete, perform the following operations:

NOTE

Operations after the USG power-on must be performed after all installation tasks are complete.

- Before the USG is powered on, ensure that the hard disk is correctly installed.
- After the USG is powered on, run the **display disk information** command in the user view to check whether the **Filesystem Status** value is **Mounted**.

**----End**

## 4.5.4 Connecting a Console Cable

After connecting a PC to the console port of a USG6680E, USG6712E, USG6716E with a console cable, you can use the terminal emulation program on the PC to access the command configuration interface of the USG6680E, USG6712E, USG6716E.

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## Precautions

Before connecting a console cable, perform the following operations:

- Check preparations.

  A PC is ready, a USG has been installed, and the ports to be connected are planned.
- Prepare cable labels.

  Before cable connection, labels must be prepared for the cable.

**NOTICE**

- Make sure that the PC and the USG are connected to the same ground point. Otherwise, the console port of the USG may be damaged.
- Pay attention to port numbering and make sure that the cable is connected to the correct port, preventing damage to ports or the device.

## Tools

Console cable (prepared by the user)

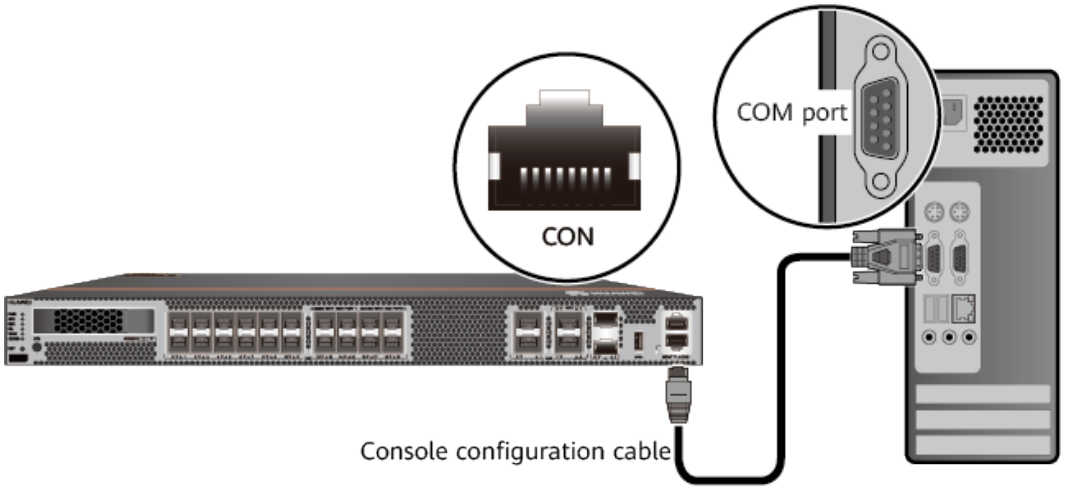
## Procedure

**Step 1** Before connecting a console cable, attach temporary labels to both ends of the cable for identification.

**Step 2** Connect the RJ45 connector of the console cable to the console port (RJ45) of the USG.

**Step 3** Connect the DB9 connector of the console cable to the COM port of the management PC.

**Figure 4-49** Connecting a console cable to the USG




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**Step 4** Remove the temporary labels and attach labels (2 cm away from the connectors) at both ends of the console cable.

**----End**

## Follow-up Procedure

After the cable connection is complete, verify that:

- The labels at both ends of a cable are correct, clear, neat, and facing the same direction.
- Cables and connectors are free of damage or breakage and are connected properly.

For details on the console login, refer to the Configuration Guide.

# 4.5.5 Installing Optical Transceivers and Connecting Optical Fibers

This section describes how to install optical transceivers on the 10GE, 40GE, 100GE optical ports of the USG and connect them to the ports of the peer device using optical fibers according to the network plan.

## Context

The USG supports 1 Gbit/s, 10 Gbit/s, 40 Gbit/s, 100 Gbit/s optical modules. The optical modules at both ends are the same, including the optical fiber type (single-mode or multi-mode), optical fiber connector type (LC/PC, SC/PC, FC/PC, or MPO/PC-MPO/PC), and transmission rate. If different optical modules are used at the two ends, the communication may fail.

**NOTICE**

Huawei optical modules are recommended. The optical modules from other vendors may cause faults on the USG due to incompatibility.

**⚠ CAUTION**

Do not look into the optical interface of the optical module or the optical fiber connector without eye protection.

Before connecting optical fiber cables, read the following precautions:

- Do not overbend optical fibers, and the radius should not be shorter than 40 mm.
- Do not bundle the optical fibers too tight. Otherwise, the transmission performance of the optical fibers and the communication between devices might be adversely affected.

Before connecting optical ensure the following:

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- The optical module has been installed.

## Procedure

**Step 1** Insert an optical transceiver into the 10GE, or 40GE, or 100GE optical port of the USG.

**Step 2** Remove the dust cap from the optical transceiver.

NOTE

Set aside the dust cap properly for future use. After optical fiber are disconnected for maintenance, use the dust cap to prevent the optical transceiver from dust.

**Step 3** Before connecting an optical fiber, attach temporary labels to both ends of the optical fiber for identification.

**Step 4** Remove protective caps from optical fiber connectors, insert optical fibers into the optical transceiver, and connect the fiber to the peer device.

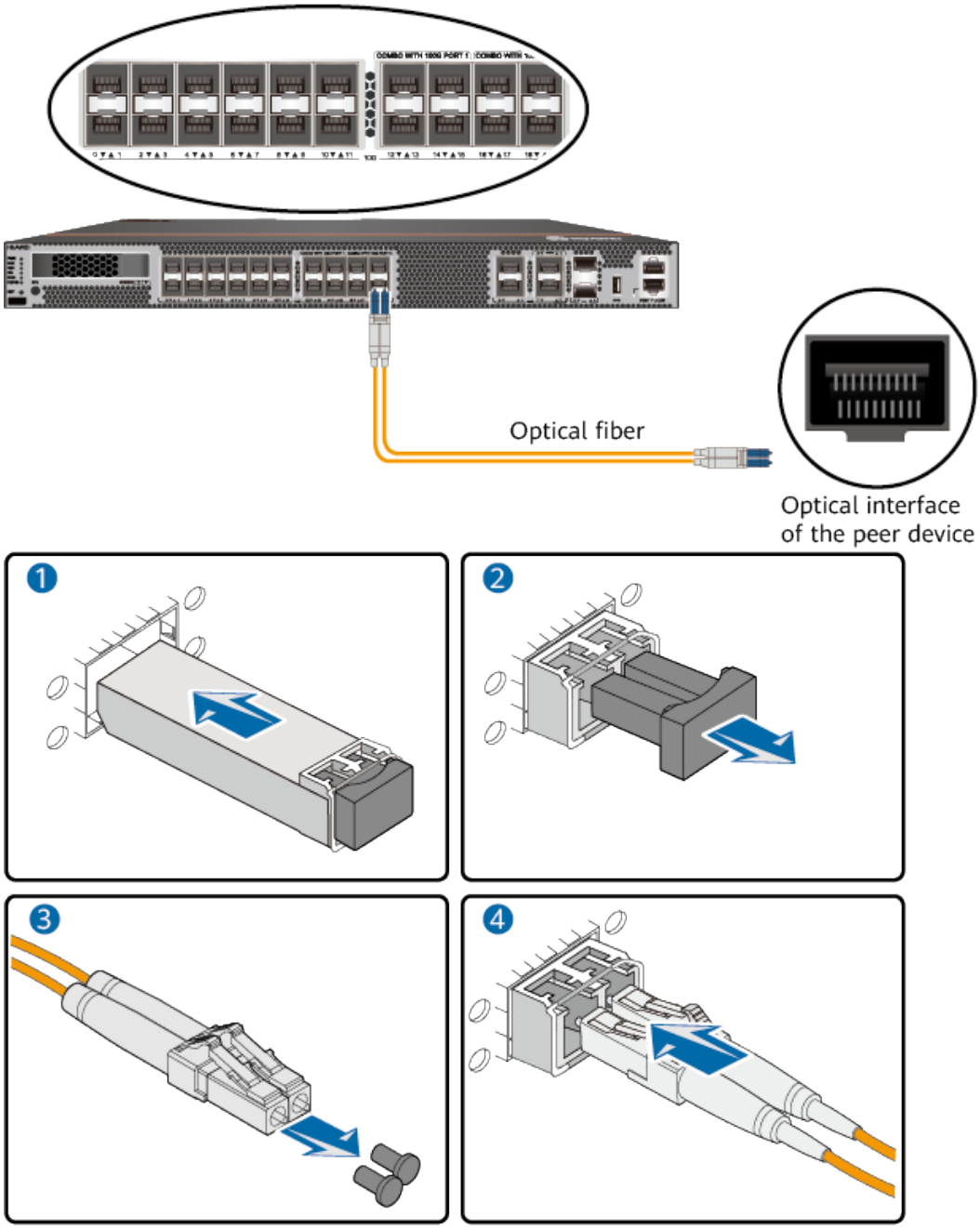
NOTE

Ensure that the Tx and Rx ports are correctly connected.

Ensure that the TX and RX ports on one end of the optical fiber cable are connected to the RX and TX ports (respectively) on the other end.

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**Figure 4-50** Installing optical transceivers and connecting optical fibers




**Step 5** Repeat **Step 1** to **Step 4** to install all optical transceivers and connect all optical fibers.

**----End**

## Follow-up Procedure

After you power on the USG, check the connection by observing the optical port indicator. If the indicator is on or blinks, the link is connected or data is being transmitted. If the indicator is off, the link is disconnected. Possible causes for the disconnection are as follows:

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- The optical fiber is improperly inserted. Pull out the optical fiber and re-insert it.
- The RX and TX optical ports are inserted reversely. Pull out the optical fibers, change their position, and re-insert them.
- The optical module is damaged or the optical fiber is broken. Replace the optical module or the optical fiber.

## 4.5.6 Connecting AC Power Cables

The USG6680E, USG6712E, USG6716E comes with two AC power module. Each AC power module requires an AC power cable. When two AC power modules are used, connect power cables to both AC power modules.

### Context

Before connecting the power cables, ensure that the AC power supply in the equipment room meets the input requirements of the USG.

**DANGER**

Do not connect or disconnect the power cables when the USG is powered on.

### Procedure

**Step 1** Ensure that the PGND cable is adequately grounded.

**Step 2** Connect AC power cables to AC power modules.

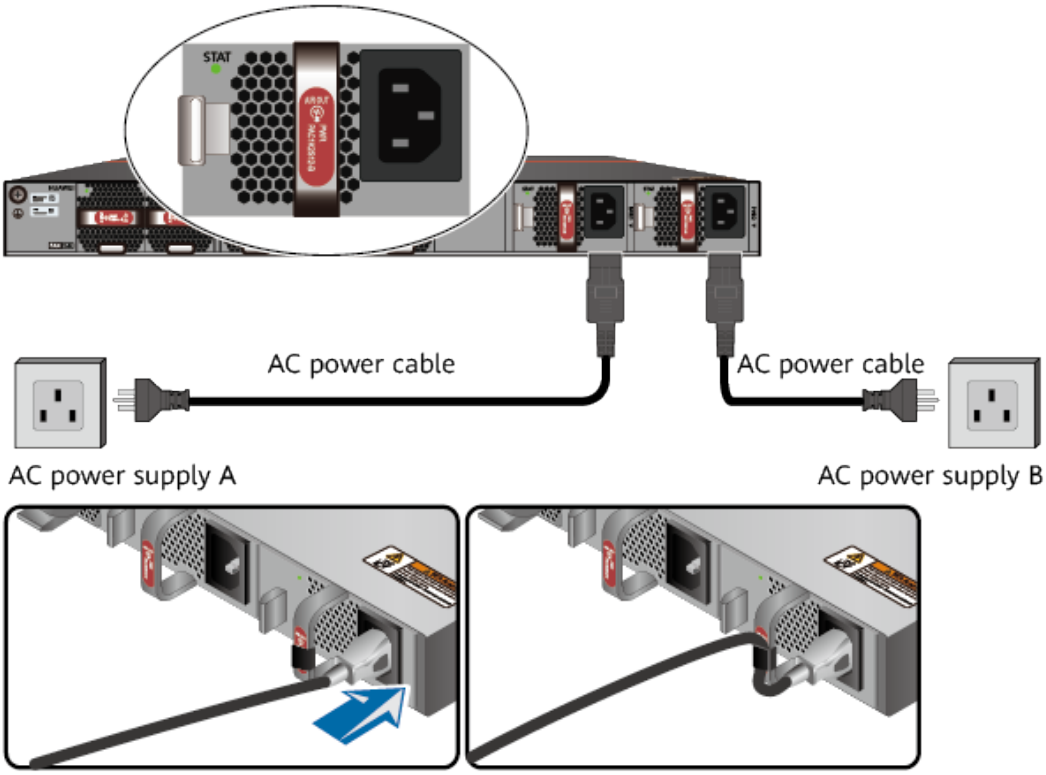
**NOTE**

When two power modules are used, connect them to different power sources to improve availability.

1. Plug one end of the C13 power cable to the power socket of the USG power module and adjust the cable-retention clip to an appropriate position.
2. Use a velcro strap to bundle the power cable onto the handle of the power module.
3. Connect the other end of the power cable to the external power supply system.
   - If the external power supply is AC, plug the other end of the power cable to the AC power socket or the output of the AC power supply device.

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**Figure 4-51** Connecting AC power cables




- If a 240 V high-voltage DC PDU is used, directly connect the straight male connector to the PDU.
- If a 240 V high-voltage DC power distribution box is used, make OT or cord end terminals for the cable. Cut the straight male connector off and crimp OT or cord end terminals on the bare wires. Connect the brown wire to a negative terminal on the DC power distribution box, the blue wire to a positive terminal, and the yellow-green wire to a protection ground. If the switch fails to be powered on after you connect the power cable, swap the wires on the positive and negative terminals.

**----End**

### Follow-up Procedure

Verify the following after the connection is complete:

- The power cable is firmly connected to the power supply socket.
- If multiple USGs are deployed, the power cables of each USG are correctly labeled for distinction.

## 4.5.7 Powering On or Off the USG6680E, USG6712E, USG6716E

This section describes how to power on or off the USG6680E, USG6712E, USG6716E. To ensure the normal start and security of the USG6680E, USG6712E, USG6716E, strictly follow the operation guide to power on or off the USG6680E, USG6712E, USG6716E.

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## Context

Before you power on the USG, ensure that:

- The power cable and PGND cable are properly connected.
- The power switch in the equipment room is easy to locate so that you can power off devices in the case of accidents.

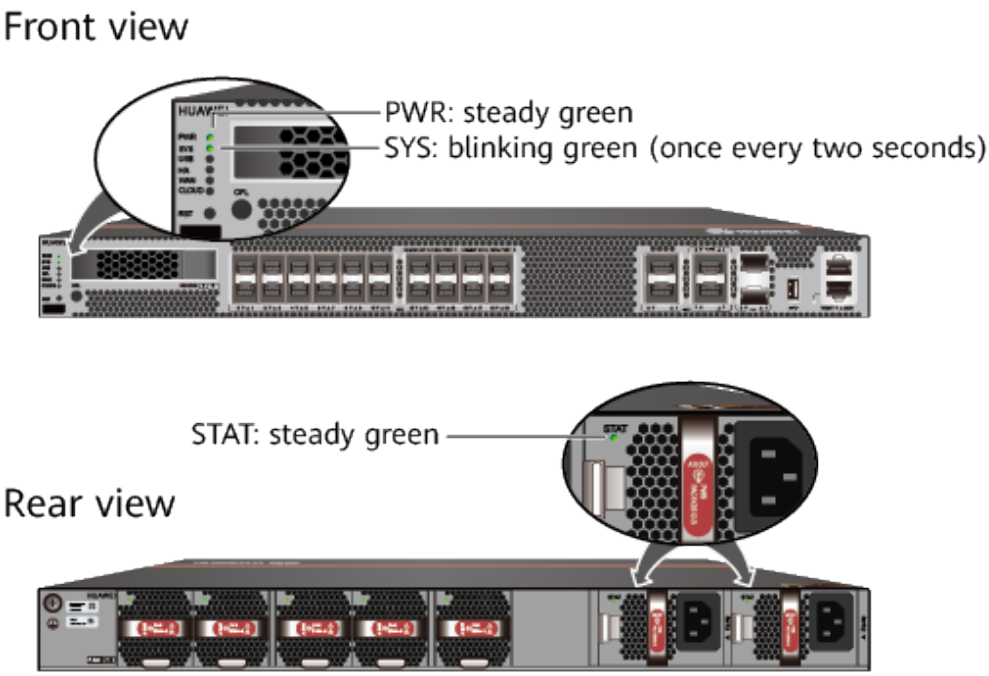
## Procedure

- Power on the USG.

  Turn on the switch of the power supply device. The USG starts.

  You can identify the USG status based on indicators on the front and rear panels. Indicators shown in USG**Figure 4-52** indicate that the USG runs normally.

  **Figure 4-52** Indicators when the USG runs normally

- Power off the USG.

  **NOTICE**

  Before powering off the USG, ensure that configuration data is saved. Otherwise, the configuration data may be lost.

  If the USG will be administratively shut down for a long time, turn off the power switch. After powering off the USG, set it aside properly according to storage requirements.

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**CAUTION**

If the USG has hard disks, do not power off the USG by removing the power cable or switching off the power supply to prevent data loss and hard disk damage. You must run the **disk offline** command in the system view first and wait for about 30 seconds. After the system prompts you with a message indicating that hard disks have stopped, turn off the power switch of the USG.

**----End**

## Follow-up Procedure

After the USG is powered on, you can log in to the configuration page for management and maintenance. For details, refer to the Configuration Guide.

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# 5 Maintaining the Hardware

The hard disks, power modules, and fan modules of the USG6000E are replaceable. You can replace them as instructed.

5.1 Replacing the Hard Disk
This section describes how to replace the hard disk as well as the precautions.

5.2 Replacing a Power Module
This section describes how to replace a faulty power module.

5.3 Replacing a Fan Module
If the only fan of the USG fails, replace the fan module immediately. Otherwise, heat dissipation of the device is affected. If the USG has two or more fans and two or more fans fail, replace the faulty fan modules immediately. Otherwise, heat dissipation of the device is affected. This section is irrelevant to other models that do not have independent fan modules.

5.4 Replacing a Micro SD Card (USG6510E, USG6510E-POE, USG6530E)
This section describes how to replace the micro SD card and the precautions.

5.5 Replacing a Device

## 5.1 Replacing the Hard Disk

This section describes how to replace the hard disk as well as the precautions.

### Precautions

Precautions for using the hard disks

- Use hard disks purchased from Huawei. Otherwise, the system may not identify them.
- Powering off the USG interrupts services. Install or replace the hard disks during off-peak hours.
- Wear the ESD wrist strap while working on the USG to avoid possible damages to the USG and hard disks.
- While holding a hard disk, do not touch the PCB board or squeeze the disk. Carry only one hard disk at a time. Do not vibrate, crash, or pile multiple hard disks up.

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- Put the hard disks inside ESD bags when they are not used.
- For mapping relationships between the USG and hard disks, see **3 Hardware Overview**.

## Tools

ESD bag

## Replacing the Hard Disk (USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6580E, USG6585E)

If a hard disk is faulty (for example, a hard disk failure log is generated on the USG), you can replace the hard disk. The hard disk can be replaced no matter when the USG is powered on or off. If the USG is powered off, skip **Step 1** and do not run the **disk offline** command. The following procedure guides you through hard disk replacement while the USG is powered on:

**Step 1** Ensure that all configurations are saved.

**Step 2** Run the **disk offline** command in the system view. Wait about 30 seconds until the system indicates that the hard disk stops working and then remove the hard disk to be replaced.

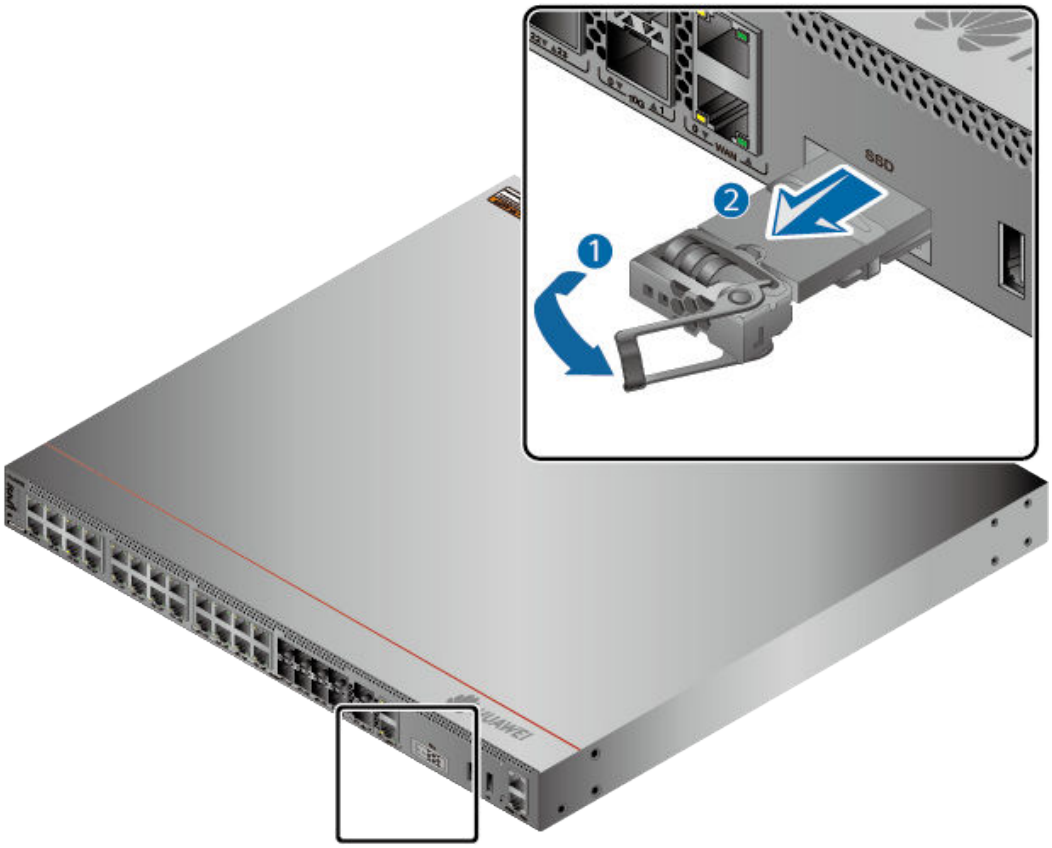
NOTE

If you remove and insert a hard disk without running the **disk offline** command first, the hard disk may be damaged, data in the hard disk may be lost, or the system may stop responding to services.

1. Use the handler to pull the hard disk out of the slot.
2. Put the replaced hard disk into an ESD bag.

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**Figure 5-1** Removing the hard disk (USG6515E, USG6525E, USG6550E, USG6555E, USG6560E, USG6565E, USG6580E, USG6585E)




**Step 3** Install the new hard disk, see **Installing the Hard Disk**.

**Step 4** Run the **disk online** command in the system view and then the **display disk information** command. If **Filesystem Status** is **Mounted**, the hard disk works properly.

**----End**

## Replacing the Hard Disk (USG6575E-B, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E, USG6680E, USG6712E, USG6716E)

If a hard disk is faulty (for example, a hard disk failure log is generated on the USG), you can replace the hard disk. The hard disk can be replaced no matter when the USG is powered on or off. If the USG is powered off, skip **Step 1** and do not run the **disk offline** command or press the **OFL** button. The following procedure guides you through hard disk replacement while the USG is powered on:

**Step 1** Ensure that all configurations are saved.

**Step 2** Run the **disk offline** command in the system view. Wait about 30 seconds until the system indicates that the hard disk stops working and then remove the hard disk to be replaced. Press and hold the **OFL** button for about four seconds until the **OFL** indicator is on. Then remove the hard disk unit to be replaced.

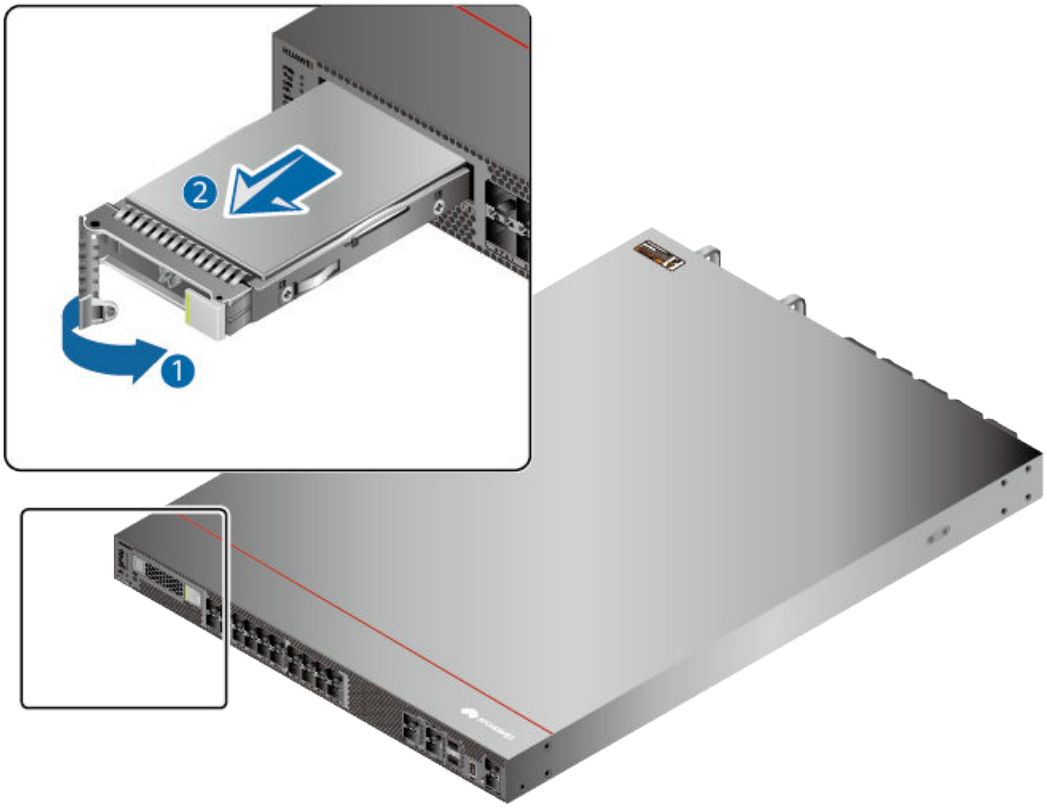
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NOTE

If you remove and insert a hard disk without running the **disk offline** command or press the **OFL** button first, the hard disk may be damaged, data in the hard disk may be lost, or the system may stop responding to services.

1. Press the button on the right of the hard disk tray to eject the handler.
2. Use the handler to pull the hard disk out of the slot.
3. Put the replaced hard disk into an ESD bag.

**Figure 5-2** Removing the hard disk (USG6575E-B, USG6605E-B, USG6610E, USG6615E, USG6620E, USG6625E, USG6630E, USG6635E, USG6650E, USG6655E, USG6680E, USG6712E, USG6716E)




**Step 3** Install the new hard disk, see **Installing a Hard Disk**.

**Step 4** Run the **disk online** command in the system view and then the **display disk information** command. If **Filesystem Status** is **Mounted**, the hard disk works properly.

**----End**

## Follow-up Procedure

After replacing the hard disk, collect all tools. If the replaced hard disk is faulty, fill in the **Repair Transmission Sheet** and send the faulty module with the Repair Transmission Sheet to the equipment supplier or the specified repair service provider.

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# 5.2 Replacing a Power Module

This section describes how to replace a faulty power module.

## Context

When two power modules are installed on the USG, one of them is hot swappable. Pay attention to the following items during hot swapping:

- Power off the power module to be replaced.
- Do not power off the other power module during the replacement.
- Do not touch the other power module either by body or by tools, preventing human injury or device short circuit.
- DC and AC power modules are incompatible with each other and cannot be used in substitution.

## Tools

ESD wrist strap

## Replacing the Power Module

**Step 1** Determine the location of the power module to be replaced and attack a replacement label on the panel to identify the power module.

**Step 2** Disconnect the power supply device from the power module to be replaced.

**Step 3** Wear an ESD wrist strap.

**Step 4** Remove the power cable connected to the power module.

Refer to the following instructions to remove the power cables:

- To remove the AC power cable:
  a. Press the baffle on the cable retention clip and remove the cable retention clip from the AC power cable. (Skip this step if the power module does not have any cable retention clip.)
  b. Pull out the power cable from the socket on the power module.
- To remove the DC power cable:
  a. Remove the fastening screws from the OT terminals of the DC power cable.
  b. Pull out the power cable from the socket on the power module.

**Step 5** Replace the power module.

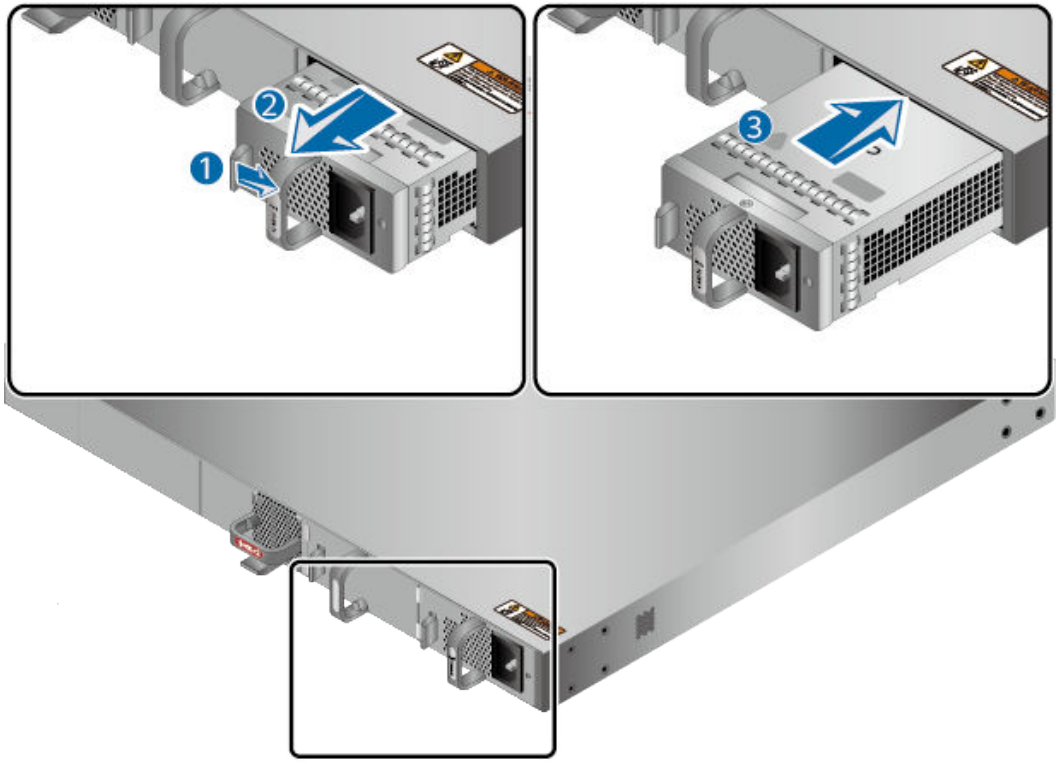
The methods to replace the DC power module and the AC power module are the same. The following uses AC power module replacement as an example.

1. Press the cable retention clip of the power module to the right, hold the front panel of the power module, and pull out the power module.
2. Note down the cause and time of the replacement as well as the bar code of the replaced power module and then put away the replaced power module.

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3. Take the new power module from the packing box and check whether its model is the same as the replaced one.
4. Gently push the new power module into the chassis along the power slot. After the latch on the power module is tightened, pull the power module gently to check whether it can be removed.
5. Connect the power cable to the socket on the new power module. For details, see **Connecting AC Power Cables (for a 420 mm Deep Device)** or **Connecting AC Power Cables (for a 600 mm Deep Device)**.

**Figure 5-3** Replace the power module




6. Switch on the power supply device. If the STATUS indicator on the new power module of the USG6000E is steady green, the power module works properly.

**----End**

### Follow-up Procedure

After replacing the power module, collect all tools. If the replaced power module is faulty, fill in the **Repair Transmission Sheet** and send the faulty module with the Repair Transmission Sheet to the equipment supplier or the specified repair service provider.

## 5.3 Replacing a Fan Module

If the only fan of the USG fails, replace the fan module immediately. Otherwise, heat dissipation of the device is affected. If the USG has two or more fans and two or more fans fail, replace the faulty fan modules immediately. Otherwise, heat dissipation of the device is affected. This section is irrelevant to other models that do not have independent fan modules.

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## Context

NOTE

The fan module of the USG is hot swappable. Therefore, when the fan module is faulty, you can replace the fan module without powering the USG off.

**NOTICE**

To prevent USG overheating, replace the fan module within 1 minute.

## Tools

ESD wrist strap

## Procedure

**Step 1** Wear an ESD wrist strap.

**Step 2** Remove the faulty fan module.

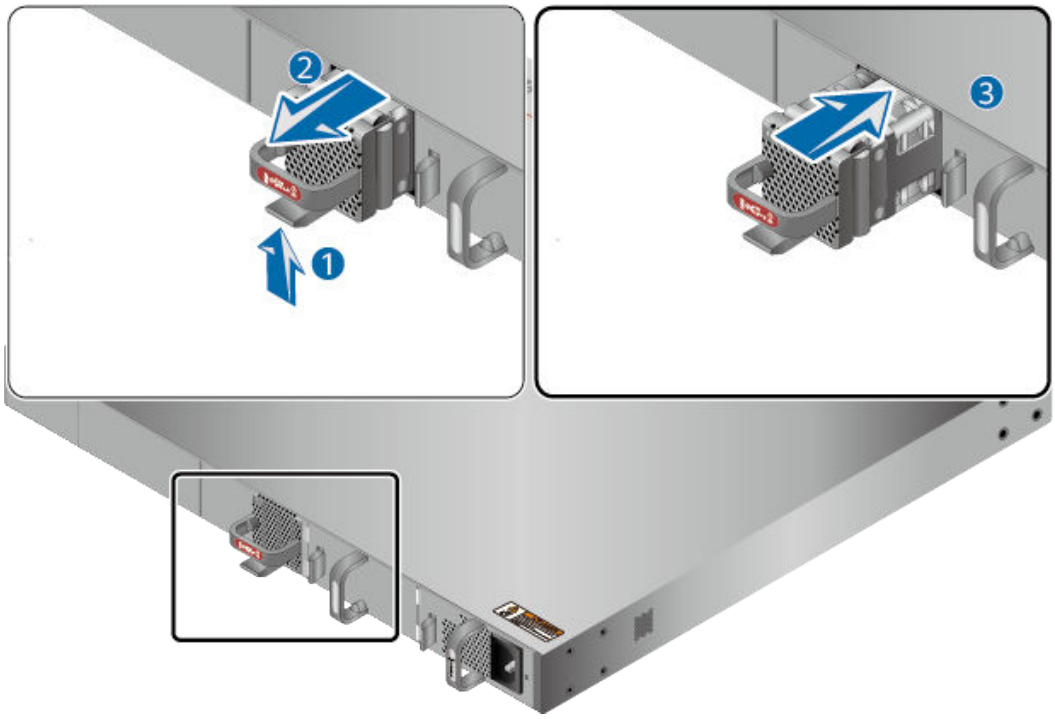
Press upwards the fan module latch, hold the handle of the fan module, and pull out part of the fan module. After the fan stops rotating, slowly pull out the fan module from the chassis.

**Step 3** Install a new fan module.

Hold the handler of the fan module with one hand and the bottom of the fan module with the other hand, insert the fan module along the guide rail of the fan slot until the back of the fan module gets in good contact with the chassis backplane. After the latch on the power module is tightened, pull the fan module gently to check whether it can be removed.

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**Figure 5-4** Replacing the fan module




**Step 4** Check the FAN STATUS indicator on the panel. If the indicator is blinking green every two seconds (0.5 Hz), the new fan module works properly.

**----End**

## Follow-up Procedure

After replacing the fan module, collect all tools. If the replaced fan module is faulty, fill in the **Repair Transmission Sheet** and send the faulty module with the Repair Transmission Sheet to the equipment supplier or the specified repair service provider.

# 5.4 Replacing a Micro SD Card (USG6510E, USG6510E-POE, USG6530E)

This section describes how to replace the micro SD card and the precautions.

## Precautions

If the micro SD card has one of the following conditions, replace the micro SD card:

- The micro SD card is damaged. That is, the USG generates the micro SD card damage alarm: ENTEXT_1.3.6.1.4.1.2011.5.25.31.2.0.16 hwEntityExtDiskDamaged.
- The micro SD card is full. That is, the USG generates the micro SD card full alarm: ENTEXT_1.3.6.1.4.1.2011.5.25.31.2.0.17 hwEntityExtDiskFull.

Read the precautions on using the micro SD card:

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- Micro SD cards are optional and are not delivered with the device. If required, purchase the micro SD card (part number: 06010308) from Huawei. The micro SD card model is SDSDQAE-064G, the capacity is 64 GB, and dimensions (H x W x D) are 1 mm x 15 mm x 11 mm (0.04 in. x 0.59 in. x 0.43 in.).
- Make sure that you have worn an ESD wrist strap and the strap is well grounded before you hold the micro SD card. Otherwise, the micro SD card may be damaged.

The micro SD card can be replaced no matter when the USG is powered on or off. If the USG is powered off, skip the **sd-card offline** command. The following procedure guides you through micro SD card replacement while the USG is powered on.

## Tools

- Phillips screwdriver
- ESD wrist strap

## Procedure

**Step 1** Ensure that all configurations are saved.

**Step 2** In the system view, run the **sd-card offline** command. When the system displays a message indicating that the SD card is offline, remove the micro SD card to be replaced.

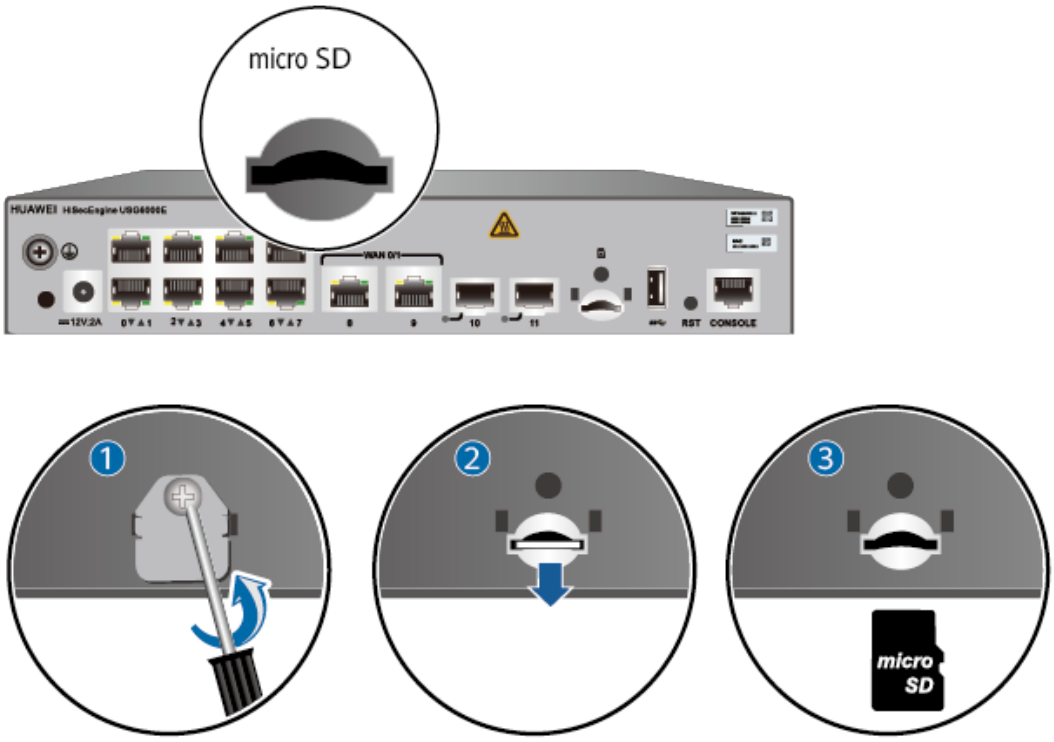
NOTE

- Before replacing a micro SD card, run the **sd-card offline** command to prevent micro SD card damage and data loss.
- Do not use too much force; otherwise the micro SD or micro SD card slot might be damaged.

1. Loosen the screws on the anti-theft board and remove the anti-theft board.
2. Press the micro SD card along the guide rail to loosen the internal card clip. Then the micro SD card is ejected from the slot. You can then remove the micro SD card.
3. Place the replaced micro SD card properly.

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**Figure 5-5** Removing a micro SD card




**Step 3** Install the new micro SD card and anti-theft board.

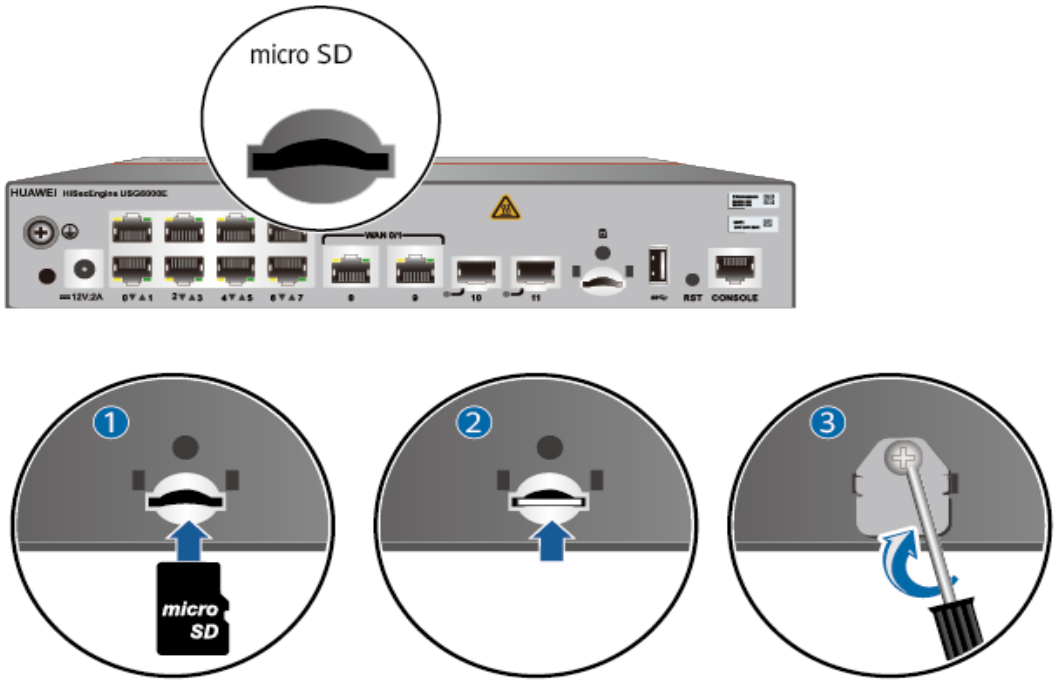
NOTE

- Note that the micro SD card must be installed with the face with words upwards.
- Do not use too much force; otherwise the micro SD or micro SD card slot might be damaged.

1. Insert the micro SD along the guide rail to the micro SD card slot.
2. When you hear a click, the micro SD card is in position.
3. Hook the locating hook on the anti-theft board to the locating hole of the rear panel and tighten the captive screw on the anti-theft board.

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**Figure 5-6** Installing the new micro SD card and anti-theft board




**Step 4** In any view, run the **display sd-card information** command to check the micro SD card installation and file system mounting status. In the command output, **SD Card Physical State** should be **Present**, and **SD Card File System State** should be **Mounted**.

- If **SD Card Physical State** is **Absent**, re-install the micro SD card and try again. If **SD Card Physical State** remains the same, the USG may fail to identify the micro SD card. You are advised to use another micro SD card.
- If **SD Card File System State** is **Unmounted**, the micro SD card format might not be **ext4.** You need to run the **reset sd-card** command in the system view to format the micro SD card.

**----End**

### Follow-up Procedure

After replacing the micro SD card, collect all tools. If the replaced micro SD card is faulty, fill in the **Repair Transmission Sheet** and send the faulty module with the Repair Transmission Sheet to the equipment supplier or the specified repair service provider.

## 5.5 Replacing a Device

### Context

When a fault occurs on the USG device, such as a memory fault, CPLD fault, or startup failure, you can replace the USG device to rectify the fault.

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## Precautions

Pay attention to the following during device replacement:

- If a device can still be started, save and back up the configuration data before powering off the device. Otherwise, the configuration will be lost.
- If a device with a hard disk can still be started, run the **disk offline** command in the system view. Wait for about 30 seconds. After the system displays a message indicating that the hard disk stops working, remove the hard disk unit to be replaced. Alternatively, hold down the OFL button for about 4 seconds until the OFL indicator turns on, and then remove the hard disk unit. Do not directly remove the power cable or disconnect the power supply to power off the USG device. Otherwise, data loss and hard disk damage may occur.
- To prevent a USG device from falling down, hold the device tightly with both hands while removing or installing it.
- Be careful of the heat when replacing the USG device.
- In a hot standby scenario, ensure that the model and software version of the new USG device are the same as those of the standby USG device.

## Preparing Tools

- ESD wrist strap or ESD gloves
- Phillips screwdriver

## Replacing a device

**Step 1** Determine the installation position of the USG device to be replaced and attach a replacement label on the panel of the device to avoid misoperations.

**Step 2** Wear an ESD wrist strap or ESD gloves.

**Step 3** Disconnect the power supply device from the USG device to be replaced. When the PWR indicator on the USG device is off, you can proceed with subsequent operations.

**Step 4** Remove the cables from the USG device to be replaced and attach labels that specify interface numbers to the cables.

NOTE

Remove the power cables, service cables, and ground cables from the USG device to be replaced in sequence.

**Step 5** Remove the USG device to be replaced.

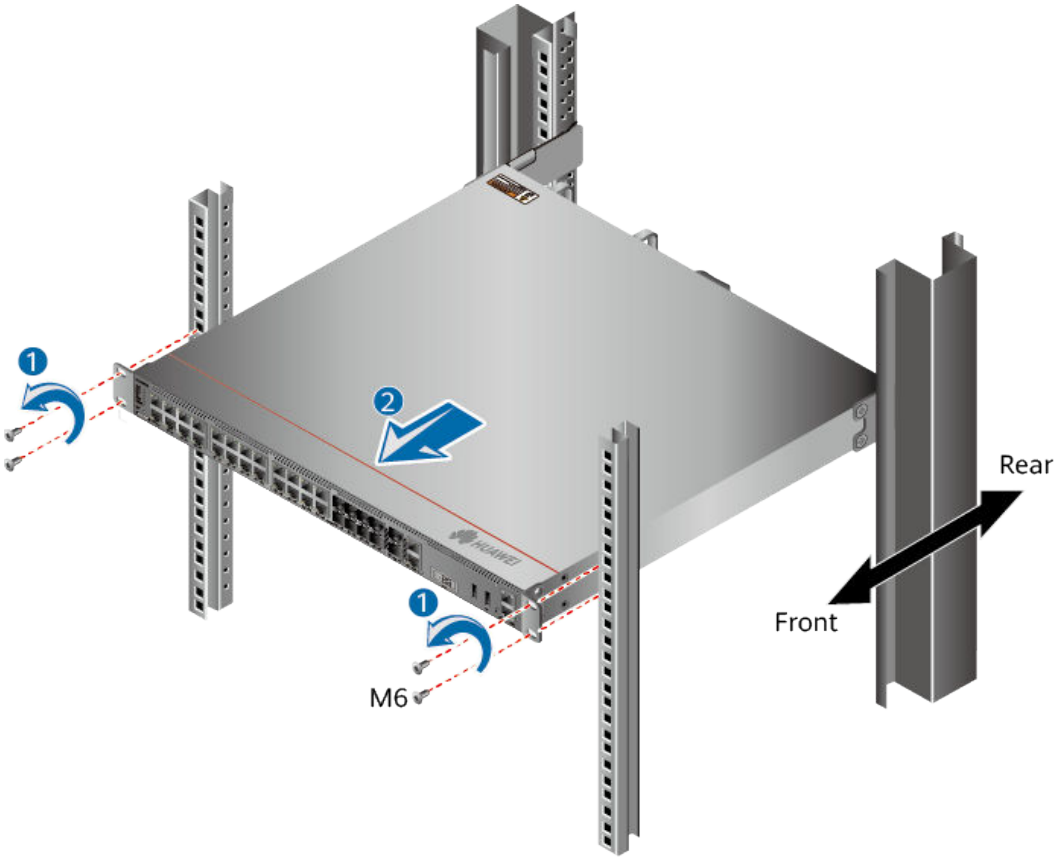
For a device installed on a workbench, you only need to move the device away. For a device installed on a wall, use a Phillips screwdriver to loosen the M6 screws on the mounting ears at both ends of the device, and remove the device from the wall with both hands. For a device installed in a cabinet, perform the following steps to remove it:

1. Use a Phillips screwdriver to loosen the M6 screws on the mounting ears at both ends of the USG device.

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2. Hold the USG device with both hands and pull it out along the guide rails of the rear mounting ears. Place the removed USG device in an ESD bag or on an ESD pad.

**Figure 5-7** Removing a device from the cabinet




**Step 6** Install the new USG device, connect the cables according to the labels, and then power on the USG device. For details, see **4 Hardware Installation**.

**----End**

## Follow-up Procedure

After replacing the device, pack up all the tools. If the old USG device is faulty, fill in the **Repair Transmission Sheet** and send the faulty device with the Repair Transmission Sheet to the equipment supplier or the specified repair service provider.

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# A Appendix

This section describes the requirements for the USG installation environment and **Fault Tag** template.

## A.1 Quick Reference Tables of Power Cables

### AC Power Cables for the Power Adapters Directly Connected to the Socket

**Table A-1** AC power cables for the power adapters directly connected to the socket

| Part Number | Description | Image |
| --- | --- | --- |
| 04043491 | External Power Cable, Power Cable 250V2.5A,2m,2X0.75mm^2,Black,PASM,227IEC53-0.75^2(2C)-I,C7SF |  |
| 04044216 | Power cord,Europe AC Power 250V2,2.0m,PCSM,(H03VVH2F-0.5^2(2C)),C7SF,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04044496 | Power Cords Cable,Japan AC Power 125V7A,2.0m,PASM,VCTFK-0.75^2(2C),C7SF,Black |  |
| 04044497 | Power Cable,America AC Power Cable 125V7A,2.0m,PASM,18NISPT-1(2C),C7SF,Black |  |
| 04045094 | Power Cords Cable,India AC Power 250V2.5A,2.0m,PC-IISM,IS 694-0.75^2(2C),C7S,250V,2.5A,Black |  |
| 04050956 | Power Cord Cable,Brazil AC Power Cable,250V2.5A,2m,PC-IIISM,H05VVH2-F 2*0.75^2(2C),C7SF,250V,2.5A,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04051081 | Power cord,BS546 250V2.5A,2m,PD-IAM,H03VVH2-F-0.75mm^2(2C),C7SF,250V,2.5A,Black |  |
| 04051997 | Power cord,2m,IRAM2063 Straight Male,H05VVH2-F 2*0.75^2(2C),C7 Straight Female,250V,2.5A,BLACK |  |
| 0405G025 | Power Cords Cable,South Africa AC Power 250V2.5A,2m,PDAM,H03VVH2-F-0.75^2(2C),C7SF |  |

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## AC Power Cables for the Power Modules Directly Connected to the Socket

**Table A-2** AC power cables for the power modules directly connected to the socket

| Part Number | Description | Image |
| --- | --- | --- |
| 04020728 | Power Cable,America AC Power Cable,125V10A,3.0m,PBSM,18 SJT(3C),C13SF,Black |  |
| 04040887 | Power Cable,Japan AC Power Cable 125V12A,3.0m,PBSM,HVCTF-1.25mm^2(3C),C13SF,Black |  |
| 04040888 | Power Cords Cable,Australia AC Power Cable,250V 10A,3.0m,PISM,H05VV-F-1.0mm^2(3C),C13SF,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04040889 | Power cord,BS546 250V10A,3.0m,PM-IAM,H05VV-F-1.5mm^2(3C),C13SF,250V,10A,Black |  |
| 04040890 | Power Cable,Britain AC Power Cable 250V10A,3.0m,PGAM ,H05VV-F-1.0mm^2(3C),C13SF,Black |  |
| 04041056 | Power cord,Europe AC Power Cable,250V10A,3.0m,PFSM,(H05VVF 1.0^2(3C)),C13SF,250V,10A,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04041117 | Power Cable,Britain AC Power Cable 250V10A,3.0m,PGAM ,H05VV-F-1.0mm^2(3C),C13AF-L,Black |  |
| 04041120 | Power Cable,Italy AC Power Cable 250V10A,3.0m,PLSM,H05VV-F-1.0mm^2(3C),C13SF,Black |  |
| 04042697 | Power Cords Cable,China AC Power 250V10A,10.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04042699 | Power Cords Cable,China AC Power 250V10A,20.0m,PISM,227IEC53-1.0^2(3C),C13SF,Black |  |
| 04047785 | Power Cords Cable,Argentina AC Power 250V10A,3.0m,PISM,H05VV-F-1.0mm^2(3C),C13SF,Black |  |
| 04050139 | Power Cords Cable,China AC Power Cable,250V10A,3m,PISM,227IEC53-1.0^2(3C),C13SF,Black |  |
| 04050206 | Power Cable,China AC Power Cable 250V10A,1.0m,PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,Black |  |
| 04050517 | Power Cable,China AC Power Cable 250V10A,2.0m,PISM,227IEC53(RVV)1.0mm^2(3C),C13SF,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 04050955 | Power Cords Cable,China AC Power,250V10A,1.5m,PISM,227 IEC53(RVV)1.0mm^2(3C),C13SF,250V,10A,Black |  |
| 04051035 | Power cord,India AC Power 250V6A,3m,PM-IIAM,IS 694-1.0^2(3C),C13 SF,250V,6A,Black |  |
| 04051080 | Power cord,South Africa AC Power 250V10A,3m,PMAM,H05VV-F-1.0mm^2(3C),C13SF,250V,10A,Black |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 0405G028 | Power Cords Cable,Korea AC Power 250V10A,3m,PFSM,H05VV-F 3*1.0^2(3C),C13SF,Black |  |
| 0405G02K | Power Cords Cable,Denmark AC Power 250V10A,3m,PKSM,H05VV-F-3*1.0^2(3C),C13SF,Black |  |

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## AC Power Cables for the Power Modules Directly Connected to the PDU

**Table A-3** AC power cables for the power modules directly connected to the PDU

| Part Number | Description | Image |
| --- | --- | --- |
| 04050188 | Power Cords Cable,China AC Power 250V10A,1.5m,C14SM,227IEC53(RVV)1.0mm^2(3C),C13SF,PDU Cable |  |
| 04050846 | Power cord,China AC Power Cable,250V10A,3.0m,C14SM,(227IEC53-1.0^2(3C)),C13SF,250V,10A,Black,PDU Cable |  |
| 04050847 | Power cord,China AC Power Cable,250V10A,6.0m,C14SM,(227IEC53-1.0^2(3C)),C13SF,250V,10A,Black,PDU Cable |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 0405G019 | Power Cords Cable,Europe AC 250V10A,1.8m,C14SM,H05VV-F- 3*1.00^2,C13SF,PDU Cable |  |
| 0405G029 | Power Cords Cable,North America AC Power 250V10A,1.8m,C14SM,SJT 18AWG(3C),C13SF,PDU Cable |  |
| 0405G02D | Power Cords Cable,Japan AC Power 250V12A,1.8m,C14SM,HVCTF 1.25^2(3C),C13SF,PDU Cable |  |

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| Part Number | Description | Image |
| --- | --- | --- |
| 0405G02F | Power Cords Cable,Australia AC Power 250V10A,1.8m,C14SM,H05VV-F-1.0^2(3C),C13SF,PDU Cable |  |
| 0405G02H | Power Cords Cable,Korea AC Power 250V10A,1.8m,C14SM,H05VV-F-1.0^2(3C),C13SF,PDU Cable |  |

## DC Power Cables

**Table A-4** DC power cables

| Part Number | Description | Image |
| --- | --- | --- |
| 04152883 | Power Cable,3m,4^2mm,PS2F,(H07Z-K-4^2B+H07Z-K-4^2BL),2*T4^2GY |  |

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# A.2 Requirements for Installation Environment

This section describes the requirements for the USG installation environment, such as the device position, humidity and temperature, cleanness, antistatic measures, lightning protection measures, power supply, and anti-magnetic measures.

## A.2.1 Device Position

This section describes the USG position requirements to ensure its security.

Table A-5 lists the requirements for the device position.

**Table A-5** Device position requirements

| Item | Requirement |
| --- | --- |
| Ventilation and heat dissipation | • To ensure good heat dissipation, keep the air vent of the device at least 150 mm away from other devices.<br>• Ensure that the ventilation and heat dissipation system is available at the position where the device is to be installed. |
| Stability | The cabinet or mounting table must be firm enough to support the weight of the device and its accessories. |
| Grounding | Ensure that the chassis or mounting table is properly grounded. |

## A.2.2 Humidity, Temperature, and Cleanness

This section describes the requirements for the humidity, temperature, and cleanness of the equipment room. To ensure the stability and life cycle of the USG and its components, check that the equipment room meets the requirements.

Ensure that there is no explosive, conductive, magnetic, or corrosive dust or debris in the equipment room. Dust that settle on the device may cause electrostatic adsorption, resulting in poor contact of the metal socket connectors and metal contacts. This shortens the life cycle of the device and causes faults.

In addition to dust and debris, toxic gases, such as $SO_2$, $H_2S$, and $NH_3$ must be cleaned out of the equipment room.

**Table A-6** Requirements for humidity, temperature, and cleanness in the equipment room

| Item | Description | | | | | |
| --- | --- | --- | --- | --- | --- | --- |
| Cleanness | Dust particle | Maximum diameter (μm) | 0.05 | 1 | 3 | 5 |

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| Item | Description | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Maximum density (number of dust particles per cubic meter) | 1.4 x $10^7$ | | 7 x $10^5$ | | 2.4 x $10^5$ | | 1.3 x $10^5$ | |
| | Noxious gas density | Gas | $SO_2$ | $H_2S$ | $Cl_2$ | HCl | HF | $NH_3$ | $O_3$ | $NO_2$ |
| | | Average ($mg/m^3$) | 0.3 | 0.1 | 0.1 | 0.1 | 0.01 | 1.0 | 0.05 | 0.5 |
| | | Maximum ($mg/m^3$) | 1.0 | 0.5 | 0.3 | 0.5 | 0.03 | 3.0 | 0.1 | 1.0 |
| Humidity and temperature | Temperature | Long-term operating temperature | • Without hard disk: 0 °C to 45 °C<br>• With hard disk(s): 5 °C to 40 °C | | | | | | | |
| | | Short-term operating temperature | • Without hard disk: -5 °C to 55 °C<br>• With hard disk(s): 5 °C to 40 °C | | | | | | | |
| | Operating humidity | | • Without hard disk: 5% RH to 95% RH, non-condensing<br>• With hard disk(s): 5% RH to 90% RH, non-condensing | | | | | | | |
| | Storage humidity | | • Without hard disk: 5% RH to 95% RH, non-condensing<br>• With hard disk(s): 5% RH to 90% RH, non-condensing | | | | | | | |

To meet the requirements listed in **Table A-6**, take the following measures in the equipment room:

- Install a permanent temperature controller regardless of the climate conditions.
- In dry regions, use humidifiers or regularly mop the floor to ensure proper humidity in the equipment room.
- In regions with high humidity, use dehumidifiers.
- Use dust-free materials for the floor, walls, and ceilings.
- Screen exterior doors and windows. The external windows of the equipment room should be sealed for anti-dust purposes.
- Clean the equipment room and air filters on the devices once every three months.

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- Wear the ESD uniform, ESD gloves, and ESD shoes before entering the equipment room.
- Locate the equipment room far away from areas with dense corrosive gases, such as chemical plants.
- The air intake vent of the equipment room must not face any pollution source.
- Place batteries in a different room from the devices.
- Invite professionals to measure the temperature, humidity, and other factors periodically.

## A.2.3 ESD Requirements

This section describes the ESD requirements. Ensure that the equipment room meets these requirements because static electricity may damage the USG components and cause the USG to malfunction.

The absolute value of the static voltage must be less than 1000 V. To meet the requirement, take the following measures in the equipment room:

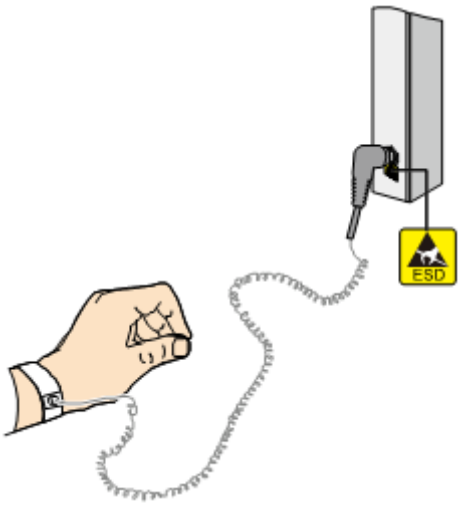
- Provide operators with ESD protection training.
- Adjust the humidity to reduce the impact of static electricity.
- Install an ESD floor in the equipment room.
- Wear an antistatic suit, ESD gloves, and antistatic shoes before entering the equipment room.
- Use ESD tools, such as ESD wrist straps, ESD tweezers, and extractors.
- All the conductors in the equipment room, including computer terminals, must be properly grounded. Set up an antistatic workbench.
- Electrostatic sources, such as non-ESD plastic bags, non-ESD foam, and rubber objects must be kept at least 30 cm away from ESD-sensitive components and boards.

To protect the USG boards from damage caused by static electricity discharge, take the following measures:

- Ensure that the USG is properly grounded according to the grounding requirements.
- Wear an ESD wrist strap before performing any operations on the USG.
- Ensure proper contact between the metal buckle of the ESD wrist strap and the operator's skin. Ensure that the other end of the ESD wrist strap is already connected to the ESD jack on the USG, as shown in **Figure A-1**. In addition, wearing ESD gloves is recommended.

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**Figure A-1** Wearing an ESD wrist strap properly




- Ensure that the ESD wrist strap works properly and its resistance ranges from 0.75 Mohm to 10 Mohm. Generally, the service life of an ESD wrist strap is two years. If the ESD wrist strap resistance does not meet the requirement within its service life, replace it with a new one.
- Avoid contact between the boards and clothes because the ESD wrist strap cannot protect operators from the static electricity caused by this type of contact.
- Use an anti-static pad when replacing boards or chips. In addition to wearing the ESD wrist strap, use ESD tweezers and extractors when inserting and removing boards and chips. Do not touch chips and their pins with bare hands.
- Keep all boards and components in ESD bags until they are to be installed. Place temporarily demounted boards and components on the anti-static pad or other effective antistatic materials. Do not use non-ESD materials, such as foam, plastic bags, and paper bags to wrap or make contact with the boards.
- Wear an ESD wrist strap before working on board terminals. Discharge cables and terminal protection jackets using either contact discharge or air discharge before connecting them to device terminals.
- Save board packing materials, such as plastic boxes and ESD bags, for future use.

## A.2.4 Lightning Protection and Grounding

This section describes the lightning protection and grounding requirements. Ensure that the equipment room meets the requirements because lightning is one of the major factors that causes damage to the USG.

**Table A-7** lists the lightning protection and grounding requirements.

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**Table A-7** Lightning protection and grounding requirements

| Item | Requirement |
| --- | --- |
| Civil construction of the equipment room | The equipment room should be built of reinforced concrete.<br>The equipment room should be equipped with lightning protection devices, such as a lightning arrester.<br>The lightning protection ground for the equipment room (the grounding of the lightning arrester) should share the same grounding conductor with the protection ground of the equipment room. |
| AC power system (TN-S power supply recommended) | A dedicated transformer should be used at a communications station. Power cables that are connected to the communications station should be buried with metal jackets or insulated jackets passing through steel pipes. Both sides of the metal jackets or steel pipes are connected to the nearest ground bar. The length of the buried power cables should be no less than 15 meters.<br>The three phase lines at the low-voltage side of the AC transformer at the communications station should each be installed with a closed zinc-oxide lightning arrester for grounding. The enclosure of the transformer, AC zero wire of the low-voltage side of the transformer, and the metal outer protector of the power cable connected to the transformer enclosure must be connected to the nearest grounding post. |
| Incoming power cables | AC and DC power cables should not be led into or out of the communications station through overhead lines.<br>After low-voltage power cables are led into the equipment room, in the AC voltage regulator and AC power distribution box (PDB), install a lightning arrester for power cables and connect the lightning arrester to the nearest grounding post.<br>If the equipment room is located in an urban area, the AC power system of the equipment room should have a lightning protection unit with a nominal discharge current of no less than 20 kA. If the equipment room is located in a suburban area that is classified as a medium or high level lightning zone, install a lightning protection unit with a nominal discharge current of greater than 60 kA. If the equipment room is located in a mountainous area that is classified as a high-level lightning zone or in an isolated building in an urban area, install a lightning protection unit with a nominal discharge current of greater than 100 kA.<br>The ground cable of the lightning arrester used for the power supply must be shorter than one meter. |

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| Item | Requirement |
| --- | --- |
| DC distribution grounding | The DC working ground of the communications station (the positive pole of the -48 V DC power supply or the negative pole of the 24 V DC power supply) should be led in from the nearest indoor main earthing conductor.<br>The device that supplies power to the communications station should provide the DC working ground that connects from the collective ground cable of the building (or from the protection ground bar in the equipment room) to the power supply. |
| Equipotential bonding | The communications devices and auxiliary facilities in the equipment room must be properly grounded. These devices and facilities include mobile base transceiver stations (BTSs), transmission devices, switching devices, power supply, and distribution frames. All the devices in the communications station should be grounded to the same ground busbar. All devices in the equipment room should be grounded to the same protection ground bar in the equipment room.<br>A ground grid must be shared by the working grounds and protection grounds of all devices in the equipment room.<br>The cable tray, iron suspension racks, racks, chassis, metal ventilation pipes, and metal doors and windows must be grounded. |
| Common grounding requirements | The neutral of the AC power cables must not be connected to the protection ground of any communications device in the equipment room.<br>Do not install switches or fuses on the ground cables.<br>Ground cables should be as short and straight as possible and should not be coiled. |
| Grounding resistance | Less than 1 Ohm<br>The upper end of the ground body must be no less than 0.7 m from the ground. In cold regions, the ground body should be buried under the frozen soil layer.<br>Measure the grounding resistance periodically to ensure that the grounding works properly. |
| Signal cable layout | No overhead signal cable should exist in the communications station. Signal cables should be led into the station from underground.<br>The communications cables led into or out of the communication station should be protected with metal sheaths or laid out in metal pipes.<br>The ground cable of a lightning arrester should be as short as possible. The unused wire pairs in the cables should be grounded in the equipment room. |

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| Item | Requirement |
| --- | --- |
| Collective ground cable | The main earthing conductor can be a cable grounding ring or busbar.<br>Do not use aluminum materials as ground cables. Avoid electrochemical corrosion when different types of metals are interconnected.<br>Generally, a copper busbar with a cross-sectional area of no less than 120 $mm^2$ or galvanized steel with the same resistance is used as the main earthing conductor. The main earthing conductor must be insulated from the reinforcing steel bar in the building. |
| Grounding lead-in cable | The grounding lead-in cable must be no longer than 30 m. The grounding lead-in cable should use galvanized steel with a cross-sectional area of 40 mm x 4 mm or 50 mm x 5 mm. |

## A.2.5 Power Supply

Normal power supply is the prerequisite for proper running of the USG. This section describes the requirements for power supply.

### Basic AC Power Supply

The AC power supply system that consists of mains, uninterruptible power supply (UPS), and self-provided generators should supply power in centralized mode. The power supply system should meet the needs of the communications station, and its cabling should be as simple as possible to facilitate operation and maintenance. The low-voltage AC power supply system should use 3-phase 5-wire or 1-phase 3-wire for power supply. **Table A-8** lists the low-voltage AC nominal voltage and frequency.

**Table A-8** AC voltage and frequency

| Nominal Voltage | Rated Frequency |
| --- | --- |
| 110/127/220 (V) | 50Hz/60Hz |

In common cases, the UPS serves as the backup power supply for network products. The backup power supply must be the same as the mains in phases, and the duration for the switchover between the UPS and the mains should be less than 10 ms. Otherwise, the device might restart or be reset. Each device must have an independent AC surge protector. The capacity of the surge protector for the power distribution room must be greater than the sum of the operating current and fault current on the devices to be powered. The DC power supply system must be able to safely withstand the maximum load of the device, regardless of whether the device is in working or standby state. The wire type and gauge of each outlet of the power distribution panel must be able to withstand

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the maximum power load of the devices. The specifications for the AC power supply voltage of all communications and power supply devices are as follows:

- Communications devices must be equipped with AC power supply with the rated voltage ranging from +5% to -10%.
- Communications power supply devices and key constructions must be equipped with AC power supply with the rated voltage ranging from +10% to -15%.
- The frequency of the AC power current ranges from +4% to -4%, and the sinusoidal distortion rate of voltage waveform must be 5% or less.

The self-provided generator sets in the communications station must be automatic in activation, deactivation, and replenishment, be equipped with remote signaling, remote measurement, and remote control, and provide standard interfaces and communications protocols.

The power cables used for AC and DC power distribution should comply with the following specifications:

- The AC neutral for communications purposes must be a conductive wire that has the same cross section as the phase cable.
- The selection of the DC power feeder depends on the long-term load. If the cross-sectional area is greater than 95 $mm^2$, use the rigid busbar. If the short-term load differs greatly from long-term load, lay out the cables by stages.
- The DC and AC conducting wires must be fire resistant and the wiring must comply with the *Class A Fire Resistance Design Norm for High-Rise Civil Buildings* (GB50045-95). The low-voltage power distribution facilities must comply with the *Low-Voltage Power Distribution Facilities and Wiring Design Norm* (GB50054-95)

In addition, the basic AC power supply system should meet the following requirements:

- Use voltage regulating or stabilization facilities to restrict the voltage fluctuation within a reasonable scale if:
  - The communications devices are powered by mains, and the voltage exceeds the rated voltage by +5% to -10% or the allowed voltage range.
  - The communications devices are not powered directly by the mains and the voltage of the mains exceeds the rated voltage by +10% to -15% or the AC voltage range allowed by the DC power supply device.
- Use the UPS or inverter power supply system if the communication load requires non-interruptible and non-transient AC power supply.
- Equip the site with the electric generator set to ensure normal communication in case of mains failure. The capacity of the generator set is greater than or equal to 1.5 to 2 times the capacity of the UPS.
- A UPS usually has only one batter set. Connect two or more UPSs in parallel or series for redundancy. If the inverter or UPS is used, configure the inverter or UPS that provides the maximum amount of power as the active one, and configure another inverter or UPS as the standby.

## Basic DC Power Supply

Ensure the reliability of the DC power supply system at the communications site. Deploy the power supply device as close as possible to the communications

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devices so as to shorten the power feeder and lower the circuit voltage drop between the battery port and device port to less than 3.2 V, thereby reducing installation costs and power consumption.

Use two or more independent power supply systems if the communication volume is large or if more than two switching systems are deployed at the site.

For large communications hubs, deploy an independent power supply system on each floor, each providing power for the communications equipment room on this floor. Medium-sized communications stations can be centrally powered by a power room or storage battery room, or powered in a decentralized manner. For small-sized communications stations, the power supply system can be deployed in the same equipment room as the communications devices, but you must take appropriate measures to ensure that the corrosive gases discharged by the batteries in the equipment room do not corrode the circuit boards of communications devices.

**Table A-9** lists the DC power supply specifications.

**Table A-9** DC power supply specifications

| Item | Specifications |
| --- | --- |
| Voltage fluctuation range allowed for the -48 V input end | -48 V to -60 V |
| Surge current tolerance capability in DC power supply | At least 1.5 times higher than load rated current capability |
| Regulated voltage precision | The regulated voltage precision is less than or equal to 1% when the AC input voltage fluctuates between 85% and 110% of the rated voltage, the load current varies between 5% and 100% of the rated current, and the output voltage of the rectifier is any fixed value in the -46.0 V to -56.4 V range. |
| On/Off overshoot magnitude | Within the 95% to 105% range of the DC rated voltage value |
| Peak-to-peak noise voltage | Less than or equal to 200 mV |
| Dynamic response | The recovery time should be less than 200 ms, and the overshoot must be within the 95% to 105% range of the DC voltage set value. |

The following are suggestions for the basic DC power supply system:

- Decentralized power supply is recommended. Use multiple DC power supply systems and multiple power sources.
- Use a standard-compliant DC power supply. The output voltage of the communications power supply must be in the voltage range of the device to be powered.

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- Improve the reliability of the AC power supply system to properly reduce the battery capacity. When it is difficult to improve the reliability of the AC power supply system at a small communications station, properly increase the battery capacity.
- The total capacity configuration of the high-frequency switch rectifier should match the communications load power and battery charging power. Rectifier modules should use a redundancy configuration. If the number of active modules is less than or equal to 10, one standby module should be deployed. If the number of active modules is greater than 10, one standby module should be deployed for every 10 active modules.
- Storage batteries should be installed in two or multiple sets. The total capacity depends on the duration in which the storage battery sets independently supply power to the load. For most communications stations, storage battery sets should supply power for at least one hour.

## A.2.6 Electromagnetic Protection

This section describes the electromagnetic protection requirements. Check that the equipment room meets these requirements so as to ensure the normal running of the USG.

Possible interference sources are as follows:

- Capacitive coupling
- Inductance coupling
- Electromagnetic radiation
- Common impedance (including the PGND system) coupling

To reduce as much interference as possible, complete the following steps:

- Take effective measures to avoid possible power grid interference to the power supply system.
- Do not use the power line ground or lightning protection ground as the working ground of the device, and leave as much space as possible between the working ground of the device and the power line ground or lightning protection ground.
- Ensure that no high-power radio transmitter, radar transmitter, or high-frequency high-current device is deployed nearby.
- Take electromagnetic protection measures if necessary.

# A.3 Fault Tag

| | |
|---|---|
| *Customer name: | |
| Address: | |
| Contact person: | |
| Tel.: | Fax: |
| Category*: □ Replacement □ Repair □ Analysis | |

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| RMA code/SR* | Bom Code* | SN | Goods Description | Whether the Data in the Parts Removed | Fault Occurring Date* | Description of the Fault Phenomena* | Category No.* | Software Version | Customer Name | Property Owner* |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | □ Yes □ No | | | | | | |
| | | | | □ Yes □ No | | | | | | |
| | | | | □ Yes □ No | | | | | | |
| | | | | □ Yes □ No | | | | | | |

Please help confirm the information below:

1. If the above returned BOM Code has the function of saving data, please delete the data saved in this BOM Code before returning to Huawei or removing the storage medium, like HD or CF Card. □ Yes □ No

2. Record any data that should be remained as follows:

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Note:

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